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AK3918AV100 Series Processors Programmer's Guide

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Contact Information

Guangzhou Anyka Microelectronics Co., Ltd.
Anyka H Building, No.107 Bowen Rd.,
Guangzhou, Guangdong 510555
P.R. China
Tel: (86)-20-3221 9000
Fax: (86)-20-3221 9258

Sales Hotline:

(86)-20-3221 9499

E-mail:

sales@anyka.com

Home Page:

<http://www.anyka.com>

About This Manual

This Programmer's Guide describes the features and operation of the AK3918AV100 series Processors. It also provides the details on how to initialize, configure, and program the AK3918AV100 series processors.

Conventions

- *#Abc* is used to indicate a signal that is active when pulled low: for example, #Reset.
- *Logic level one* is a voltage that corresponds to Boolean true (1) state.
- *Logic level zero* is a voltage that corresponds to Boolean false (0) state.
- To *set* a bit or bits means to establish logic level one.
- To *clear* a bit or bits means to establish logic level zero.
- A *signal* is an electronic construct whose state conveys or changes in state convey information.
- A *pin* is an external physical connection. The same pin can be used to connect a number of signals.
- *Asserted* means that a discrete signal is in active logic state.
- *Negated* means that an asserted discrete signal changes logic state.
- LSB means *least significant bit or bits*, and MSB means *most significant bit or bits*. References to low and high bytes or words are spelled out.
- Numbers with 'b are binary. Numbers with 'd are decimal. Numbers with 'h or preceded by 0x are hexadecimal.
- "-" is used to indicate the cell is blank or reserved.

Register Attributes

Symbol	Description
RW	Register field is read and write.
RO	Register field is read only. Writes to read-only field have no effect.
R	Register field is read only. Writes to read-only field have no effect.
WO	Register field is write only.
SC	Register field can be cleared to 1'h0 by the core.
W1C	Register field can be cleared by writing 1.
W0C	Register field can be cleared by writing 0.
RC	Register field can be cleared by read.

Definitions, Acronyms, and Abbreviations

Unless otherwise specified, all the acronyms and abbreviations used in this manual are defined hereunder.

ADC	Analog to Digital Converter
ADE	Customized Analog Delay Element
AHB	Advanced High-performance Bus
CMOS	Complimentary Metal-Oxide Semiconductor
CIS	CMOS Image Sensor
CNN	Convolutional Neural Networks
CRC	Cyclic Redundancy Check

DAC	Digital to Analog Converter
DDE	2-NAND gate Digital Delay Element
DDR	Double Data Rate
DMA	Direct Memory Access
DNN	Deep Neural Network
DTE	Data Terminal Equipment
DPI	Short for dots per inch
FIFO	First In First Out
GPIO	General Purpose Input/Output
I2S	Inter-IC Sound
JPEG	Joint Picture Expert Group
LSB	Least Significant Bit
MAC	Media Access Control
MDLU	Master Delay Line Unit
MMC	Multimedia Card
MMU	Memory Management Unit
NPU	Netural-network Processing Unit
MSB	Most Significant Bit
ODT	On Die Terminator
PCM	Pulse Code Modulation
PGA	Programmable Gain Amplifier
PReLU	Parameteric Rectified Linear Unit
PWM	Pulse-Width Modulator
PLL	Phase Locked Loop
RAM	Random Access Memory
ROM	Read Only Memory
RTC	Real Time Clock
SD	Secure Digital
SDLU	Slave Delay Line Unit
SDIO	Secure Digital Input/Output
SDRAM	Synchronous Dynamic Random Access Memory
SPI	Serial Peripheral Interface
SRAM	Static Random Access Memory
TBD	To be determined
TWI	Two-wire Interface
UART	Universal Asynchronous Receiver/Transmitter
USB	Universal Serial Bus
XTAL	Crystal

Product Documentation

The following document(s) is (are) required for a complete description of the AK3918AV100 series processors and is (are) necessary to design properly with the device.

- *AK3918AV100 Series Processors Specification*
- *AK3918AV100 应用须知 (AK3918AV100 Application Notes)*
- *Secure Boot User Guide (《安全启动使用指南》)*

Document Revision History

The following table provides revision history for this release. This history includes technical content revisions only and not stylistic or grammatical changes.

VERSION	DESCRIPTION	DATE COMPLETED
0.1.0	Initial release	November, 2021
0.2.0	1) Section 1.1 Corrected the description of DDR2. 2) Section 3.4 Modified the description of <i>PAD slew rate configuration register</i> and <i>PAD Drive Configuration Register</i>. 3) Section 3.5.1 Updated the data in <i>Table 3-1 Driving Strength Setting Condition</i>. 4) Section 4.2.5 Modified the description of bit[29] and bit[19] in register <i>CSI_SCLK/MAC_OPCLK Clock Control (Address: 0x0800,0018)</i>. 5) Section 4.7.4 Added detailed description of bits[30:0] in <i>Polynomial Length Register (Address: 0x0800,00AC)</i>. 6)Section 4.11.4 a. Added the description of bit[30], bit [19:10], bit and [9:0] in <i>Tsensor Control Register (Address: 0x0800,024C)</i>. b. Corrected the description of bit [30], bit [29], bit [2], bit [1] in <i>Tsensor Interrupt Register (Address: 0x0800,0250)</i>. 7)Section 4.11.5.2 Modified the note in <i>Cyclic Mode Configuration</i>. 8)Section 5.1.1 Updated the memory type of address space from 0x0000,0000 to 0x0800,0000 in <i>Figure 5-1 Memory Map</i>. 9)Section 5.3.3 Updated the <i>Table 5-5 Definition of AHB Channels</i>. 10)Section 5.4 Optimized the description of Overview, Functional Description and Programmer's Guide parts in <i>Fast SPI(SPI0)</i> module. 11) Section 6.1.6 Added the note of bit[22] <i>MIPI DPHY Power Control Register (Address: 0X0800,0218)</i> and bit[30] <i>MIPI DPHY Swap Selection Register (Address: 0X0800,0220)</i>. 12)Section 6.6.5 Modified the note of bit [14] in <i>Encryption Control Register (Address: 0x2018,0000)</i>.	January, 2022

VERSION	DESCRIPTION	DATE COMPLETED
	13)Section 6.6.6.2 Modified the description about register setting in 6.6.6.2. 14)Section 7.6 Modified the description of bits[17:15] in <i>Analog Control Register 1 (Address: 0x0800,00A0)</i>. 15) Section 8.1&8.2&8.3&8.4 Optimized the description of Overview, Functional Description and Programmer's Guide parts in <i>I2S</i>, <i>TWI</i>, <i>UART</i> and <i>SPI1/SPI2</i> modules.	
0.3.0	1) Section 1.1 & 4.2.3.2 & 4.2.6.1.1 & 5.3.2 Updated the working frequency of CPU. 2) Section 2 Added the detailed description of SAR ADC. 3) Section 4.2.5 Updated the address of <i>PLL2/PLL3 Bandwidth Adjustment Configuration (Address: 0x0800,00E0)</i>.	February, 2022
0.4.0	1) Section 2 Updated the operating temperature of the chip in <i>Table 2-1</i>. 2) Section 3.5 a. Corrected the negative values to positive values in <i>Table 3-1</i>. b. Corrected the Note description of <i>Table 3-2</i>. 3) Section 4.2.5 Modified the description of bit[16] and bit[19] in <i>Clock Gate Control (Address: 0x0800,001C)</i>. 4) Section 6.1.6 Modified the description of bits[11:10] in <i>MIPI DPHY Reset and Control Register (Address: 0x0800,0224)</i> and bits[19:8] in <i>MIPI DPHY Swap Selection Register (Address: 0x0800,0220)</i>. 5) Section 6.5.2 Added the encoding output resolution in Features part. 6) Section 4.1 & 4.6 & 4.8 & 7 & 8.5& 8.8 Optimized the description of Overview, Functional Description and Programmer's Guide parts in <i>Reset</i>, <i>Interrupt</i>, <i>Timer/PWM</i>, <i>Analog</i>, <i>USB</i>, and <i>GPIO</i>.	April, 2022
0.5.0	1) Section 1 Updated <i>Figure 1-1 AK3918AV100 Series Processors Functional Block Diagram</i>. 2) Section 4.1.2 Updated <i>Figure 4-1 AK3918AV100 Series Processors Power on Reset Timing</i>. 3) Section 4.4 Updated descriptions of bootstrap module and <i>Figure 4-13 AK3918AV100 Series Processors Boot</i>	September, 2022

VERSION	DESCRIPTION	DATE COMPLETED
	Flow. 4) Section 4.9.5 Corrected the description of bits[2:1] in RTC Signals to Analog Configuration Register (Address: 0xB). 5) Section 5.3.5 Corrected the descriptions of bits[7:4] & bits[3:0] in Memory Controller Configuration Register 2 (Address: 0x2100,0008). 6) Section 8.6.5 Corrected attributes of bits[17:16] & bits[8:0] in Status Register (Address: 0x0034).	

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1 Introduction

AK3918AV100 series processors are specially designed for Artificial Intelligence & Internet of Things camera (AIoT Camera) applications, one of the key components of electronic surveillance system.

With the intelligent NPU (Neural-network Processing Unit), the optimized image signal processing algorithm and hardware H.265/H.264 encoder, AK3918AV100 series processors provide an enhanced object detection/tracking and face detection/recognition ability with high quality pictures and low bit rate video encoding at minimal power consumption. It also supports security boot for a better security level. A set of peripheral interfaces, such as UART, SPI, MMC/SD/SDIO, Ethernet MAC and USB2.0, feature AK3918AV100 series processors with high extensibility and high flexibility. Meanwhile, the integrated Fast Ethernet PHY transceiver can reduce the cost of bill-of-materials (BOM) of the final products.

Product Development Kit, including hardware development kit (HDK), software development kit (SDK) and tools, is available for customers to develop in a most convenient way.

AK3918AV100 series processors functional block diagram is shown as follows.

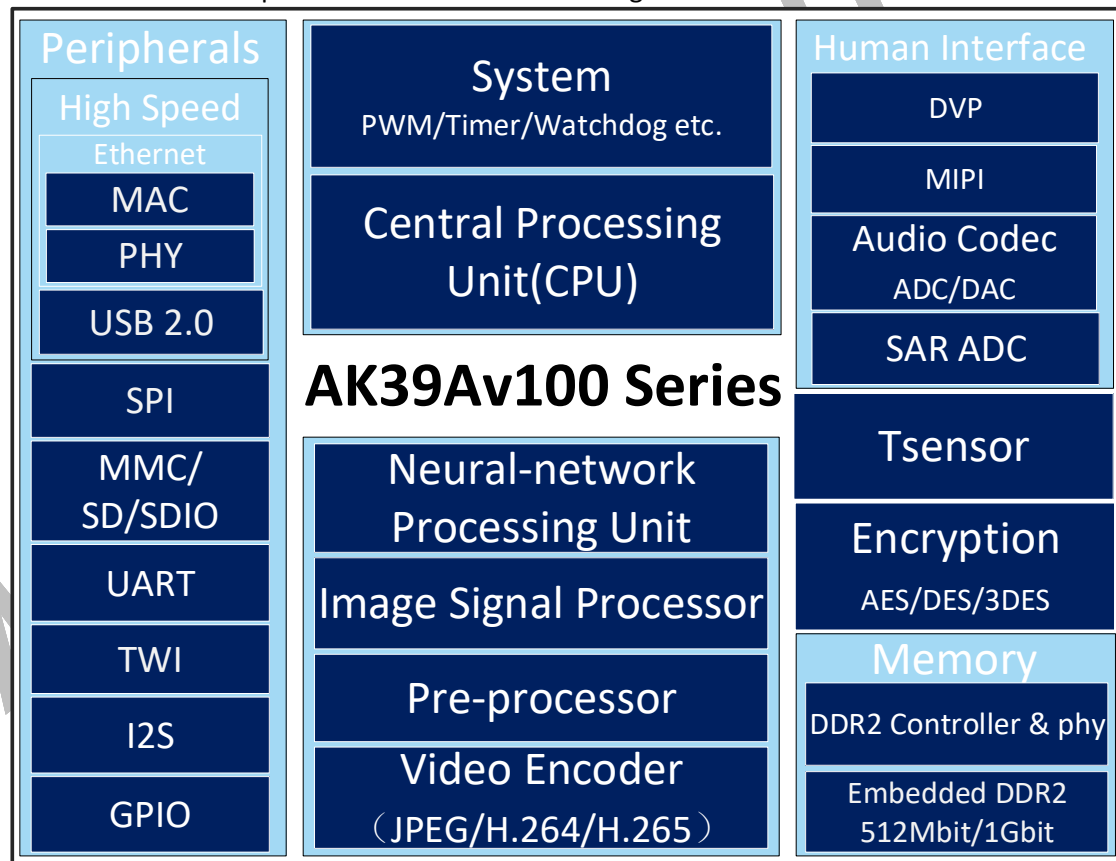


Figure 1-1 AK3918AV100 Series Processors Functional Block Diagram

1.1 Features

- ARM926EJ-S core, 64 KB I cache and 32 KB D cache
- 900MHz CPU working frequency
- Supports little endian only

- Neural-network Processing Unit (NPU) supports object detection/tracking and face detection/recognition
- BT.601/BT.656/BT.1120 CMOS Image Sensor interfaces with programmable image size and smart scaling capability
- Video Processor, supporting H.264/H.265/MJPEG/JPEG encoding
- 1 Sigma-Delta ADC(16-bit) with built-in microphone pre-amplifier
- 1 Sigma-Delta DAC(22-bit) with headphone driver
- 1 SAR ADC(12-bit) for analog keypad and general analog-to-digital conversion
- Stacked 512Mbit/1Gbit DDR2 SDRAM
- 1 Fast SPI Controller(SPI0)
 - Specially integrated for SPI NOR Flash or SPI NAND Flash
 - Up to 100MHz working frequency
- Supports Normal boot and Security boot
 - Serial NOR Flash Boot
 - SPI NAND Flash Boot
 - eMMC Boot
 - USB Mass Storage Boot
 - UART Boot
- Security engine supports AES/DES/3DES
- Tsensor for chip temperature monitoring
- Flexible clock sources
- 32.768KHz RTC
- Internal 32KHz RC oscillator
- Up to 5 PWMs
- 10 General Purpose timers
- 1 watchdog timer
- 3 UARTs (one with hardware flow control)
- 2 general SPI interfaces (SPI1 & SPI2), supporting master & slave mode
- 1 I2S master & slave interface
- 3 TWI master & slave interfaces, compatible with SCCB protocol
- 3 MMC/SD/SDIO controllers, each controller supporting:
 - MMC interface, version 4.2
 - SD interface, version 2.0
 - SDIO interface, version 2.0
- USB 2.0 HS Host/Slave controller
- Ethernet
 - 10/100Mbps full-duplex/half-duplex mode
 - Compliant with IEEE 802.3az-2010 (EEE)/10Base-T IEEE 802.3/ 100Base-TX IEEE 802.3u
 - Supports Auto-Negotiation/ Auto-MDIX
- Up to 59 GPIOs (3 dedicated and 56 shared pins)
- Package: 88-pin QFN

1.2 Applications

- Artificial Intelligence & Internet of Things camera (AIoT camera)

1.3 Part Number Information

As shown in [Figure 1-2](#), the Part Number information consists of three lines. The first line symbolizes ANYKA, the second line indicates the product ID, and the third line is production lot number plus the secondary model number, the former of which is reserved by the producer for specific purposes.

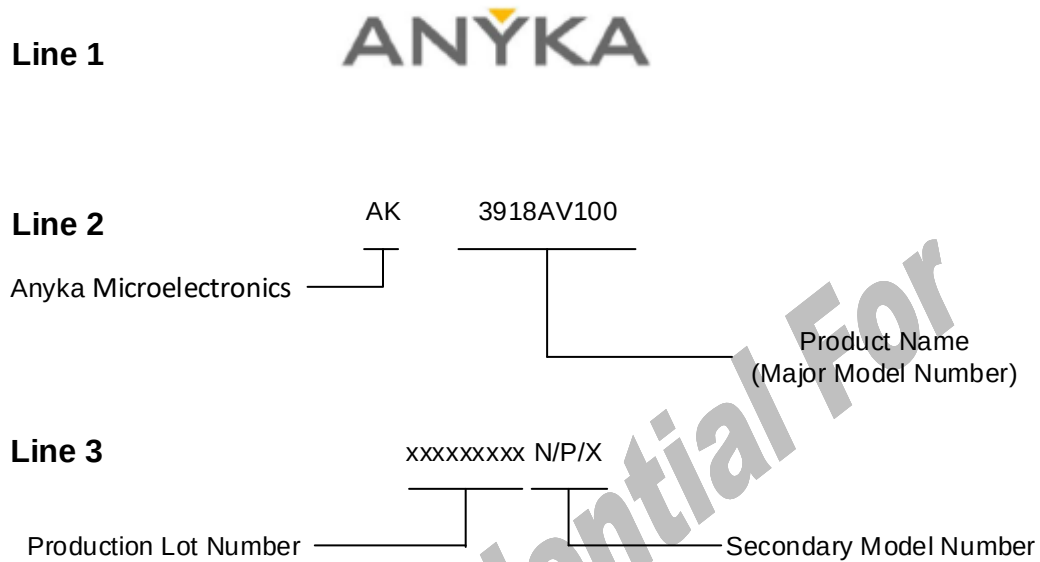


Figure 1-2 Part Number Information

2 Device Characteristics

Table 2-1 shows the differences of AK3918AV100 series devices, in which the sign “√” means supporting and “-” means not supporting.

Table 2-1 Characteristics of the AK3918AV100 Series Devices

Items			AK3918AV100N	AK3918AV100P	AK3918AV100X
CORE	CPU		ARM926EJ-S		
	I Cache (KB)		64		
	D Cache (KB)		32		
Storage	External	SPI NOR Flash	√	√	√
		SPI NAND Flash	√	√	√
	Internal	DDR2 SDRAM	512Mbit (4 banks x 8M words x 16 bits)	512Mbit (4 banks x 8M words x 16 bits)	1Gbit (8 banks x 8M words x 16 bits)
System Control	Bootstrap	SPI NOR Flash Boot	√	√	√
		SPI NAND Flash Boot	√	√	√
		USB Mass Storage Boot	√	√	√
		UART Boot	√	√	√
		eMMC/SD/SDIO Boot	√	√	√
	Timer		10	10	10
	PWM		5 (PWM[0:4])	4 (PWM[0:3])	4 (PWM[0:3])

Items		AK3918AV100N	AK3918AV100P	AK3918AV100X
	RTC	✓	-	-
	Encryption	Hardware AES, DES and 3DES engine	✓	✓
	Tsensord		✓	✓
	Camera Interface	DVP interface (BT.601/BT.656/BT.1120)	✓	✓
		MIPI interface	1/2-data lane	1/2-data lane
	ISP		✓	✓
	Pre-processor		✓	✓
	Video Processor (H.265/H.264/JPEG encoder)		✓	✓
	NPU		✓	✓
Analog	SAR ADC		3 channels (AIN0, AIN1, VCC_RTC)	3 channels (AIN1, VCC_RTC)
	Sigma-Delta ADC		1 (16bits)	1 (16bits)
	Sigma-Delta DAC		1 (22bits)	1 (22bits)
Peripherals	I2S		1	1
	TWI		3	3

Items		AK3918AV100N	AK3918AV100P	AK3918AV100X
	UART	3 (UART2 supports hardware flow control)	3 (UART2 supports hardware flow control)	3 (UART2 supports hardware flow control)
	Fast SPI0	1	1	1
	SPI	2	2	2
	USB2.0 HS Host & Slave	1	1	1
	MMC/SD/SDIO	3	3	3
	Ethernet	MAC	✓	✓
		PHY	✓	✓
	GPIO	59 (GPIO[0:11], GPIO[16:27], GPIO[31:37], GPIO[42:52], GPIO[55:56], GPIO[58], GPIO[65:71], GPIO[78:79], GPIO[98:102])	51 (GPIO[0:11], GPIO[17:23], GPIO[31:37], GPIO[42:52], GPIO[58], GPIO[65:70], GPIO[78:79], GPIO[98:102])	51 (GPIO[0:11], GPIO[17:23], GPIO[31:37], GPIO[42:52], GPIO[58], GPIO[65:70], GPIO[78:79], GPIO[98:102])
Power	Core Voltage (V)	0.8		
	IO Voltage (V)	3.3		
	DDR2 Voltage (V)	1.5		
	CIS (V)	1.8/3.3		
Package	Operating Temperature (°C)	-20 ~ 70		

Items		AK3918AV100N	AK3918AV100P	AK3918AV100X
	Package	88-pin QFN 9.00mm x 9.00mm x 0.90mm 0.35mm pitch		

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3 IO Control

3.1 Overview

The IO of the chip provides a wealth of programmability, which can be easily and flexibly adapted to different application scenario.

3.2 Features

The chip opens the configuration interface of the IO to the software, and each IO may have the following attributes depending on its type:

- Supports enabling or disabling of internal IO pull-up/pull-down resistor
- Configurable pull-up or pull-down function of certain IOs
- Configurable 4-level output driving strength of certain IOs
- Supports enabling or disabling of input function
- Configurable slew rate of IO output
- Supports digital-analog multiplexing function of certain IOs
- Supports power down retention of certain IOs
- Supports flexible multi-power domain of certain IOs

3.3 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Register Address	Register Name and Description
1	0x0800,0178	<i>Shared-pin Control Register 0</i> IO multiplexing selection register, each IO corresponding to 3 bits.
2	0x0800,017C	<i>Shared-pin Control Register 1</i> IO multiplexing selection register, each IO corresponding to 3 bits.
3	0x0800,0180	<i>Shared-pin Control Register 2</i> IO multiplexing selection register, each IO corresponding to 3 bits.
4	0x0800,0184	<i>Shared-pin Control Register 3</i> IO multiplexing selection register, each IO corresponding to 3 bits.
5	0x0800,0188	<i>Shared-pin Control Register 4</i> IO multiplexing selection register, each IO corresponding to 3 bits.
6	0x0800,018C	<i>Shared-pin Control Register 5</i> IO multiplexing selection register, each IO corresponding to 3 bits.
7	0x0800,0190	<i>Shared-pin Control Register 6</i> IO multiplexing selection register, each IO corresponding to 3 bits.
8	0x0800,0194	<i>Shared-pin Control Register 7</i> IO multiplexing selection register, each IO corresponding

Register No.	Register Address	Register Name and Description
		to 3 bits.
9	0x0800,0198	<i>Shared-pin Control Register 8</i> IO multiplexing selection register, each IO corresponding to 3 bits.
10	0x0800,019C	<i>Shared-pin Control Register 9</i> IO multiplexing selection register, each IO corresponding to 3 bits.
11	0x0800,01A0	<i>Shared-pin Control Register 10</i> IO multiplexing selection register, each IO corresponding to 3 bits.
12	0x0800,01A4	<i>PAD Drive Configuration Register 0</i>
13	0x0800,01A8	<i>PAD Drive Configuration Register 1</i>
14	0x0800,01AC	<i>PAD Drive Configuration Register 2</i>
15	0x0800,01B0	<i>PAD Drive Configuration Register 3</i>
16	0x0800,01B4	<i>PAD Drive Configuration Register 4</i>
17	0x0800,01B8	<i>PAD Drive Configuration Register 5</i>
18	0x0800,01BC	<i>PAD Drive Configuration Register 6</i>
19	0x0800,01C0	<i>PAD IE Configuration Register 0</i>
20	0x0800,01C4	<i>PAD IE Configuration Register 1</i>
21	0x0800,01C8	<i>PAD IE Configuration Register 2</i>
22	0x0800,01CC	<i>PAD IE Configuration Register 3</i>
22	0x0800,01D0	<i>PAD Slew Rate Configuration Register 0</i>
23	0x0800,01D4	<i>PAD Slew Rate Configuration Register 1</i>
24	0x0800,01D8	<i>PAD Slew Rate Configuration Register 2</i>
25	0x0800,01DC	<i>PAD Slew Rate Configuration Register 3</i>
26	0x0800,0264	<i>Pull-up/Pull-down Enable Configuration 0</i>
27	0x0800,0268	<i>Pull-up/Pull-down Enable Configuration 1</i>
28	0x0800,026C	<i>Pull-up/Pull-down Enable Configuration 2</i>
29	0x0800,0270	<i>Pull-up/Pull-down Enable Configuration 3</i>
30	0x0800,0274	<i>Pull-up/Pull-down Selection Configuration 0</i>
31	0x0800,0278	<i>Pull-up/Pull-down Selection Configuration 1</i>
32	0x0800,027C	<i>Pull-up/Pull-down Selection Configuration 2</i>
33	0x0800,0280	<i>Pull-up/Pull-down Selection Configuration 3</i>
34	0x0800,0284	<i>PAD ST Configuration Register 0</i>
35	0x0800,0288	<i>PAD ST Configuration Register 1</i>
36	0x0800,028C	<i>PAD ST Configuration Register 2</i>
37	0x0800,0290	<i>PAD ST Configuration Register 3</i>
38	0x0800,00A0	<i>Analog Control Register 1</i> Digital-analog multiplexing function configuration
39	0x0800,00A4	<i>Analog Control Register 2</i> Digital-analog multiplexing function configuration

3.4 Register Definition

- Shared-pin Control Register 0 (Address: 0x0800,0178)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	funcmux_reg0[31:0] See the table IO Function Multiplexing and Attribute Configuration

- Shared-pin Control Register 1 (Address: 0x0800,017C)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	funcmux_reg1[31:0] See the table IO Function Multiplexing and Attribute Configuration

- Shared-pin Control Register 2 (Address: 0x0800,0180)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h600	funcmux_reg2[31:0] See the table IO Function Multiplexing and Attribute Configuration

- Shared-pin Control Register 3 (Address: 0x0800,0184)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	funcmux_reg3[31:0] See the table IO Function Multiplexing and Attribute Configuration

- Shared-pin Control Register 4 (Address: 0x0800,0188)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	funcmux_reg4[31:0] See the table IO Function Multiplexing and Attribute Configuration

- Shared-pin Control Register 5 (Address: 0x0800,018C)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	funcmux_reg5[31:0] See the table IO Function Multiplexing and Attribute Configuration

- Shared-pin Control Register 6 (Address: 0x0800,0190)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h1248,0000	funcmux_reg6[31:0] See the table IO Function Multiplexing and Attribute Configuration

● Shared-pin Control Register 7 (Address: 0x0800,0194)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0924,924A	funcmux_reg7[31:0] See the table IO Function Multiplexing and Attribute Configuration

● Shared-pin Control Register 8 (Address: 0x0800,0198)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0020,0001	funcmux_reg8[31:0] See the table IO Function Multiplexing and Attribute Configuration

● Shared-pin Control Register 9 (Address: 0x0800,019C)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	funcmux_reg9[31:0] See the table IO Function Multiplexing and Attribute Configuration

● Shared-pin Control Register 10 (Address: 0x0800,01A0)

Bits	Attributes	Reset	Description
[31]	RO	1'h0	Reserved
[30:29]	RW	2'h0	GPIO[9], GPIO[10], and GPIO[18]~GPIO23 delay time configuration bits 2'h0: No delay 2'h1: 0.3 ns 2'h2: 0.6 ns 2'h3: 0.9 ns NOTE: When pin GPIO[9]/GPIO[10]/GPIO[18]~GPIO23 are multiplexed as DVP ports, configuration of this field can make DVP- CIS more stable.
[28:9]	RO	20'h0	Reserved
[8:0]	RW	9'h0	funcmux_reg10[8:0] See the table IO Function Multiplexing and Attribute Configuration

● PAD Drive Configuration Register 0 (Address: 0x0800,01A4)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hAAAAA,AAAAA	pad_drv_reg0[31:0] Each IO corresponds to 2 bits and details are as follows. 2'h0: 2 mA 2'h1: 4 mA 2'h2: 8 mA 2'h3: 12 mA NOTE: It is suggested to choose the lowest level of driving strength. See the table

Bits	Attributes	Reset	Description
			IO Function Multiplexing and Attribute Configuration

● PAD Drive Configuration Register 1 (Address: 0X0800,01A8)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hAAAA,A AAA	pad_drv_reg1[31:0] Each IO corresponds to 2 bits and details are as follows. 2'h0: 2 mA 2'h1: 4 mA 2'h2: 8 mA 2'h3: 12 mA NOTE: It is suggested to choose the lowest level of driving strength. See the table IO Function Multiplexing and Attribute Configuration

● PAD Drive Configuration Register 2 (Address: 0X0800,01AC)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hAAAA,A AAA	pad_drv_reg2[31:0] Each IO corresponds to 2 bits and details are as follows. 2'h0: 2 mA 2'h1: 4 mA 2'h2: 8 mA 2'h3: 12 mA NOTE: It is suggested to choose the lowest level of driving strength. See the table IO Function Multiplexing and Attribute Configuration

● PAD Drive Configuration Register 3 (Address: 0X0800,01B0)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hAAAA,A AAA	pad_drv_reg3[31:0] Each IO corresponds to 2 bits and details are as follows. 2'h0: 2 mA 2'h1: 4 mA 2'h2: 8 mA 2'h3: 12 mA NOTE: It is suggested to choose the lowest level of driving strength. See the table IO Function Multiplexing and Attribute Configuration

● PAD Drive Configuration Register 4 (Address: 0X0800,01B4)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hAAAA,A	pad_drv_reg4[31:0]

Bits	Attributes	Reset	Description
		AAA	Each IO corresponds to 2 bits and details are as follows. 2'h0: 2 mA 2'h1: 4 mA 2'h2: 8 mA 2'h3: 12 mA NOTE: It is suggested to choose the lowest level of driving strength. See the table IO Function Multiplexing and Attribute Configuration

● PAD Drive Configuration Register 6 (Address: 0X0800,01BC)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hAAAA,A AAA	pad_drv_reg6[31:0] Each IO corresponds to 2 bits and details are as follows. 2'h0: 2 mA 2'h1: 4 mA 2'h2: 8 mA 2'h3: 12 mA NOTE: It is suggested to choose the lowest level of driving strength. See the table IO Function Multiplexing and Attribute Configuration

● PAD IE Configuration Register 0 (Address: 0X0800,01C0)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hFFFF,FFF F	pad_ie_reg0[31:0] 1: Enable 0: Disable See the table IO Function Multiplexing and Attribute Configuration

● PAD IE Configuration Register 1 (Address: 0X0800,01C4)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hFFFF,FFF F	pad_ie_reg1[31:0] 1: Enable 0: Disable See the table IO Function Multiplexing and Attribute Configuration

● PAD IE Configuration Register 2 (Address: 0X0800,01C8)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hFFFF,FFF F	pad_ie_reg2[31:0] 1: Enable 0: Disable See the table

Bits	Attributes	Reset	Description
			IO Function Multiplexing and Attribute Configuration

● PAD IE Configuration Register 3 (Address: 0X0800,01CC)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hFFFF,FFF F	pad_ie_reg3[31:0] 1: Enable 0: Disable See the table IO Function Multiplexing and Attribute Configuration

● PAD Slew Rate Configuration Register 0 (Address: 0x0800,01D0)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	pad_sl_reg0[31:0] 0: Fast 1: Slow NOTE: It is recommended to use Slow mode. see the table IO Function Multiplexing and Attribute Configuration

● PAD Slew Rate Configuration Register 1 (Address: 0x0800,01D4)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	pad_sl_reg2[31:0] 0: Fast 1: Slow NOTE: It is recommended to use Slow mode. see the table IO Function Multiplexing and Attribute Configuration

● PAD Slew Rate Configuration Register 2 (Address: 0x0800,01D8)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	pad_sl_reg2[31:0] 0: Fast 1: Slow NOTE: It is recommended to use Slow mode. see the table IO Function Multiplexing and Attribute Configuration

● PAD Slew Rate Configuration Register 3 (Address: 0x0800,01DC)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	pad_sl_reg2[31:0] 0: Fast 1: Slow NOTE: It is recommended to use Slow mode. see the table IO Function Multiplexing and Attribute Configuration

● Pull-up/Pull-down Enable Configuration 0 (Address: 0X0800,0264)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hFFFF,FFF F	pupd_en_reg0[31:0] 1: Enable 0: Disable See the table IO Function Multiplexing and Attribute Configuration

● Pull-up/Pull-down Enable Configuration 1 (Address: 0X0800,0268)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hFFFF,FFF F	pupd_en_reg1[31:0] 1: Enable 0: Disable See the table IO Function Multiplexing and Attribute Configuration

● Pull-up/Pull-down Enable Configuration 2 (Address: 0X0800,026C)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hFFFF,FFF F	pupd_en_reg2[31:0] 1: Enable 0: Disable See the table IO Function Multiplexing and Attribute Configuration

● Pull-up/Pull-down Enable Configuration3 (Address: 0X0800,0270)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hFFFF,FFF F	pupd_en_reg2[31:0] 1: Enable 0: Disable See the table IO Function Multiplexing and Attribute Configuration

● Pull-up/Pull-down Selection Configuration 0 (Address: 0X0800,0274)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hBFFF,FFF B	pupd_sel_reg0[31:0] 1: Pull-up 0: Pull-down See the table IO Function Multiplexing and Attribute Configuration

● Pull-up/Pull-down Selection Configuration 1 (Address: 0X0800,0278)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hFFFF,FFD B	pupd_sel_reg1[31:0] 1: Pull-up 0: Pull-down See the table IO Function Multiplexing and Attribute Configuration

● Pull-up/Pull-down Selection Configuration 2 (Address: 0X0800,027C)

Bits	Attributes	Reset	Description
[31:0]	RW	32'hF023,DF18	pupd_sel_reg2[31:0] 1: Pull-up 0: Pull-down See the table IO Function Multiplexing and Attribute Configuration

● Pull-up/Pull-down Selection Configuration 3 (Address: 0X0800,0280)

Bits	Attributes	Reset	Description
[31]	RW	1'h1	xgpio4_rto_cfg Retention from GPIO [11:4] and GPIO 15 NOTE: Only when this bit is set as 0, can the pads in DVDD3318_VIVO domain work well.
[30:0]	RW	31'h7FFF,FFF F	pupd_sel_reg3[30:0] 1: Pull-up 0: Pull-down See the table IO Function Multiplexing and Attribute Configuration

● PAD ST Configuration Register 0 (Address: 0x0800,0284)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	pad_st_reg0[31] GPIO31 schmitt trigger enable control 1: Enable 0: Disable
[30:28]	RO	3'h0	Reserved
[27]	RW	1'h0	pad_st_reg0[27] GPIO27 schmitt trigger enable control 1: Enable 0: Disable
[26]	RW	1'h0	pad_st_reg0[26] GPIO26 schmitt trigger enable control 1: Enable 0: Disable
[25]	RW	1'h0	pad_st_reg0[25] GPIO25 schmitt trigger enable control 1: Enable 0: Disable
[24]	RW	1'h0	pad_st_reg0[24] GPIO24 schmitt trigger enable control 1: Enable 0: Disable
[23]	RW	1'h0	pad_st_reg0[23] GPIO23 schmitt trigger enable control 1: Enable 0: Disable
[22]	RW	1'h0	pad_st_reg0[22]

Bits	Attributes	Reset	Description
			GPIO22 schmitt trigger enable control 1: Enable 0: Disable
[21]	RW	1'h0	pad_st_reg0[21] GPIO21 schmitt trigger enable control 1: Enable 0: Disable
[20]	RW	1'h0	pad_st_reg0[20] GPIO20 schmitt trigger enable control 1: Enable 0: Disable
[19]	RW	1'h0	pad_st_reg0[19] GPIO19 schmitt trigger enable control 1: Enable 0: Disable
[18]	RW	1'h0	pad_st_reg0[18] GPIO18 schmitt trigger enable control 1: Enable 0: Disable
[17]	RW	1'h0	pad_st_reg0[17] GPIO17 schmitt trigger enable control 1: Enable 0: Disable
[16]	RW	1'h0	pad_st_reg0[16] GPIO16 schmitt trigger enable control 1: Enable 0: Disable
[15:12]	RO	4'h0	Reserved
[11]	RW	1'h0	pad_st_reg0[11] GPIO11 schmitt trigger enable control 1: Enable 0: Disable
[10]	RW	1'h0	pad_st_reg0[10] GPIO10 schmitt trigger enable control 1: Enable 0: Disable
[9]	RW	1'h0	pad_st_reg0[9] GPIO9 schmitt trigger enable control 1: Enable 0: Disable
[8]	RW	1'h0	pad_st_reg0[8] GPIO8 schmitt trigger enable control 1: Enable 0: Disable
[7]	RW	1'h0	pad_st_reg0[7] GPIO7 schmitt trigger enable control 1: Enable 0: Disable

Bits	Attributes	Reset	Description
[6]	RW	1'h0	pad_st_reg0[6] GPIO6 schmitt trigger enable control 1: Enable 0: Disable
[5]	RW	1'h0	pad_st_reg0[5] GPIO5 schmitt trigger enable control 1: Enable 0: Disable
[4]	RW	1'h0	pad_st_reg0[4] GPIO4 schmitt trigger enable control 1: Enable 0: Disable
[3]	RW	1'h0	pad_st_reg0[3] GPIO3 schmitt trigger enable control 1: Enable 0: Disable
[2]	RW	1'h0	pad_st_reg0[2] GPIO2 schmitt trigger enable control 1: Enable 0: Disable
[1]	RW	1'h0	pad_st_reg0[1] GPIO1 schmitt trigger enable control 1: Enable 0: Disable
[0]	RW	1'h0	pad_st_reg0[0] GPIO0 schmitt trigger enable control 1: Enable 0: Disable

● PAD ST Configuration Register 1 (Address: 0x0800,0288)

Bits	Attributes	Reset	Description
[31:27]	RO	5'h0	Reserved
[26]	RW	1'h0	pad_st_reg1[26] GPIO58 schmitt trigger enable control 1: Enable 0: Disable
[25]	RO	1'h0	Reserved
[24]	RW	1'h0	pad_st_reg1[24] GPIO56 schmitt trigger enable control 1: Enable 0: Disable
[23]	RW	1'h0	pad_st_reg1[23] GPIO55 schmitt trigger enable control 1: Enable 0: Disable
[22:21]	RO	2'h0	Reserved
[20]	RW	1'h0	pad_st_reg1[20] GPIO52 schmitt trigger enable control

Bits	Attributes	Reset	Description
			1: Enable 0: Disable
[19]	RW	1'h0	pad_st_reg1[19] GPIO51 schmitt trigger enable control 1: Enable 0: Disable
[18]	RW	1'h0	pad_st_reg1[18] GPIO50 schmitt trigger enable control 1: Enable 0: Disable
[17]	RW	1'h0	pad_st_reg1[17] GPIO49 schmitt trigger enable control 1: Enable 0: Disable
[16]	RW	1'h0	pad_st_reg1[16] GPIO48 schmitt trigger enable control 1: Enable 0: Disable
[15]	RW	1'h0	pad_st_reg1[15] GPIO47 schmitt trigger enable control 1: Enable 0: Disable
[14]	RW	1'h0	pad_st_reg1[14] GPIO46 schmitt trigger enable control 1: Enable 0: Disable
[13]	RW	1'h0	pad_st_reg1[13] GPIO45 schmitt trigger enable control 1: Enable 0: Disable
[12]	RW	1'h0	pad_st_reg1[12] GPIO[44] schmitt trigger enable control 1: Enable 0: Disable
[11]	RW	1'h0	pad_st_reg1[11] GPIO43 schmitt trigger enable control 1: Enable 0: Disable
[10]	RW	1'h0	pad_st_reg1[10] GPIO42 schmitt trigger enable control 1: Enable 0: Disable
[9:6]	RO	4'h0	Reserved
[5]	RW	1'h0	pad_st_reg1[5] GPIO37 schmitt trigger enable control 1: Enable 0: Disable
[4]	RW	1'h0	pad_st_reg1[4]

Bits	Attributes	Reset	Description
			GPIO36 schmitt trigger enable control 1: Enable 0: Disable
[3]	RW	1'h0	pad_st_reg1[3] GPIO35 schmitt trigger enable control 1: Enable 0: Disable
[2]	RW	1'h0	pad_st_reg1[2] GPIO34 schmitt trigger enable control 1: Enable 0: Disable
[1]	RW	1'h0	pad_st_reg1[1] GPIO33 schmitt trigger enable control 1: Enable 0: Disable
[0]	RW	1'h0	pad_st_reg1[0] GPIO32 schmitt trigger enable control 1: Enable 0: Disable

● PAD ST Configuration Register 2 (Address: 0x0800,028C)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15]	RW	1'h0	pad_st_reg2[15] GPIO79 schmitt trigger enable control 1: Enable 0: Disable
[14]	RW	1'h0	pad_st_reg2[14] GPIO78 schmitt trigger enable control 1: Enable 0: Disable
[13:8]	RO	6'h0	Reserved
[7]	RW	1'h0	pad_st_reg2[7] GPIO71 schmitt trigger enable control 1: Enable 0: Disable
[6]	RW	1'h0	pad_st_reg2[6] GPIO70 schmitt trigger enable control 1: Enable 0: Disable
[5]	RW	1'h0	pad_st_reg2[5] GPIO69 schmitt trigger enable control 1: Enable 0: Disable
[4]	RW	1'h0	pad_st_reg2[4] GPIO68 schmitt trigger enable control 1: Enable 0: Disable

Bits	Attributes	Reset	Description
[3]	RW	1'h0	pad_st_reg2[3] GPIO67 schmitt trigger enable control 1: Enable 0: Disable
[2]	RW	1'h0	pad_st_reg2[2] GPIO66 schmitt trigger enable control 1: Enable 0: Disable
[1]	RW	1'h0	pad_st_reg2[1] GPIO65 schmitt trigger enable control 1: Enable 0: Disable
[0]	RO	1'h0	Reserved

● PAD ST Configuration Register 3 (Address: 0x0800,0290)

Bits	Attributes	Reset	Description
[31:7]	RO	25'h0	Reserved
[6]	RW	1'h0	pad_st_reg3[5] GPIO102 schmitt trigger enable control 1: Enable 0: Disable
[5]	RW	1'h0	pad_st_reg3[5] GPIO101 schmitt trigger enable control 1: Enable 0: Disable
[4]	RW	1'h0	pad_st_reg3[4] GPIO100 schmitt trigger enable control 1: Enable 0: Disable
[3]	RW	1'h0	pad_st_reg3[3] GPIO99 schmitt trigger enable control 1: Enable 0: Disable
[2]	RW	1'h0	pad_st_reg3[2] GPIO98 schmitt trigger enable control 1: Enable 0: Disable
[1:0]	RO	2'h0	Reserved

● Analog Control Register 1 (Address: 0x0800,00A0)

The register table only shows the register bits associated with digital-analog hybrid function IO multiplexing. For other complete configuration signals, please refer to the section of the analog circuit description.

Bits	Attributes	Reset	Description
-	-	-	<i>Not described</i>
[6]	RW	1'h0	ain1_sel 1: Select AIN1 to SAR ADC input for measurement, active

Bits	Attributes	Reset	Description
			high. 0: AIN1 for SAR ADC input measurement is not selected.
[5]	RW	1'h0	ain0_sel 1: Select AIN0 to SAR ADC input for measurement, active high. 0: AIN0 for SAR ADC input measurement is not selected.
[4]	RW	1'h0	VCC_RTC_sel 1: Select VCC_RTC to SARADC input for measurement, active high. 0: VCC_RTC for SAR ADC input measurement is not selected.
-	-	-	<i>Not described</i>

● Analog Control Register 2 (Address: 0x0800,00A4)

The register table only shows the register bits associated with digital-analog hybrid function IO multiplexing. For other complete configuration signals, please refer to the section of the analog circuit description.

Bits	Attributes	Reset	Description
-	-	-	<i>Not described</i>
[26]	RW	1'h0	micn_gpi_en Enable MICN pin to be multiplexed as GPI, active high
[25]	RW	1'h0	micp_gpi_en Enable MICP pin to be multiplexed as GPI, active high.
-	-	-	<i>Not described</i>

3.5 Programmer's Guide

3.5.1 Driving Strength

The driving strength is related to the PVT parameter, and the actual value may have some deviation from the one listed in the table.

Table 3-1 Driving Strength Setting Condition

DS[1:0]	@SS, 125℃ 2.97V/0.72V		@SS, 125℃ 2.25V/0.72V		@SS,125℃ 1.62V/0.72V		Unit
	IOL	IOH	IOL	IOH	IOL	IOH	
Level0	5.9	7.9	6.3	5.8	2.7	3.1	mA
Level1	11.8	15.7	12.6	11.5	5.4	6.2	mA
Level2	23.5	31.2	25.3	22.8	10.8	12.5	mA
Level3	35.2	46.1	37.9	33.9	16.2	18.6	mA

DS[1:0]	@TT, 25℃ 3.30V/0.90V		@TT, 25℃ 2.50V/0.90V		@TT, 25℃ 1.80V/0.90V		Unit
	IOL	IOH	IOL	IOH	IOL	IOH	
Level0	9.3	16.5	10.7	11.3	5.1	5.0	mA
Level1	17.7	32.9	21.0	22.6	9.9	9.9	mA
Level2	32.7	65.2	40.2	45.0	19.1	19.7	mA
Level3	45.7	96.9	57.7	67.0	27.7	29.5	mA

3.5.2 IO Function Multiplexing and Attribute Configuration

The following table lists each IO attribute that can be configured by the software, and identifies some special cases. See the NOTE description below the table for details.

Table 3-2 IO Function Multiplexing and Attribute Configuration

IO Name	IO Function	Function MUX	Output Driving and Reset Value	Input Enable and Reset Value	PUPD Enable and Reset Value	PUPD Selection and Reset Value	Slew Rate and Reset value
XGPIO_000	4-MUX	<i>funcmux_reg0</i> [2:0]	<i>pad_drv_reg0</i> [1:0] Default value: 2'h2	<i>pad_ie_reg0</i> [0] 1: Enable	<i>pupd_en_reg0</i> [0] 1: Enable	<i>pupd_sel_reg0</i> [0] 1: Pull-up	<i>pad_sl_reg0</i> [0] 0: Slow
	GPIO0	3'h0					
	UART1_TXD	3'h1					
	SPI1_CLK	3'h2					
	OPCLK1	3'h3					
	Others	Reserved					
XGPIO_001	4-MUX	<i>funcmux_reg0</i> [5:3]	<i>pad_drv_reg0</i> [3:2] Default value: 2'h2	<i>pad_ie_reg0</i> [1] 1: Enable	<i>pupd_en_reg0</i> [1] 1: Enable	<i>pupd_sel_reg0</i> [1] 1: Pull-up	<i>pad_sl_reg0</i> [1] 0: Slow
	GPIO1	3'h0					
	UART1_RXD	3'h1					
	SPI1_DIN(IO1)	3'h2					
	RMII_RXER	3'h3					
	Others	Reserved					
XGPIO_002	4-MUX	<i>funcmux_reg0</i> [8:6]	<i>pad_drv_reg0</i> [5:4] Default value: 2'h2	<i>pad_ie_reg0</i> [2] 1: Enable	<i>pupd_en_reg0</i> [2] 1: Enable	<i>pupd_sel_reg0</i> [2] 0: Pull-down	<i>pad_sl_reg0</i> [2] 0: Slow
	GPIO2	3'h0					
	PWM3	3'h2					
	SPI1_DOUT(IO0)	3'h3					
	RMII_RXDV	3'h4					
	Others	Reserved					
XGPIO_003	3-MUX	<i>funcmux_reg0</i> [11:9]	<i>pad_drv_reg0</i> [7:6] Default value: 2'h2	<i>pad_ie_reg0</i> [3] 1: Enable	<i>pupd_en_reg0</i> [3] 1: Enable	<i>pupd_sel_reg0</i> [3] 1: Pull-up	<i>pad_sl_reg0</i> [3] 0: Slow
	GPIO3	3'h0					
	SPI1_CS	3'h1					
	RMII_MDIO	3'h2					

IO Name	IO Function	Function MUX	Output Driving and Reset Value	Input Enable and Reset Value	PUPD Enable and Reset Value	PUPD Selection and Reset Value	Slew Rate and Reset value
	Others	Reserved					
XGPIO_004	5-MUX	<i>funcmux_reg0</i> [14:12]	<i>pad_drv_reg0</i> [9:8] Default value: 2'h2	<i>pad_ie_reg0</i> [4] 1: Enable	<i>pupd_en_reg0</i> [4] 1: Enable	<i>pupd_sel_reg0</i> [4] 1: Pull-up	<i>pad_sl_reg0</i> [4] 0: Slow
	GPIO4	3'h0					
	TWI2_SCL	3'h1					
	SDIO1_DATA3	3'h2					
	SPI2_CS1	3'h3					
	RMII_TXEN	3'h4					
	Others	Reserved					
XGPIO_005	4-MUX	<i>funcmux_reg0</i> [17:15]	<i>pad_drv_reg0</i> [11:10] Default value: 2'h2	<i>pad_ie_reg0</i> [5] 1: Enable	<i>pupd_en_reg0</i> [5] 1: Enable	<i>pupd_sel_reg0</i> [5] 1: Pull-up	<i>pad_sl_reg0</i> [5] 0: Slow
	GPIO5	3'h0					
	TWI2_SDA	3'h1					
	SDIO1_DATA2	3'h2					
	RMII_MDC	3'h3					
	Others	Reserved					
XGPIO_006	4-MUX	<i>funcmux_reg0</i> [20:18]	<i>pad_drv_reg0</i> [13:12] Default value: 2'h2	<i>pad_ie_reg0</i> [6] 1: Enable	<i>pupd_en_reg0</i> [6] 1: Enable	<i>pupd_sel_reg0</i> [6] 1: Pull-up	<i>pad_sl_reg0</i> [6] 0: Slow
	GPIO6	3'h0					
	TWI0_SDA	3'h1					
	SDIO1_DATA1	3'h2					
	RMII_RXD1	3'h3					
	Others	Reserved					
XGPIO_007	4-MUX	<i>funcmux_reg0</i> [23:21]	<i>pad_drv_reg0</i> [15:14] Default value: 2'h2	<i>pad_ie_reg0</i> [7] 1: Enable	<i>pupd_en_reg0</i> [7] 1: Enable	<i>pupd_sel_reg0</i> [7] 1: Pull-up	<i>pad_sl_reg0</i> [7] 0: Slow
	GPIO7	3'h0					
	TWI0_SCL	3'h1					
	SDIO1_DATA0	3'h2					
	RMII_TXD0	3'h3					
	Others	Reserved					

IO Name	IO Function	Function MUX	Output Driving and Reset Value	Input Enable and Reset Value	PUPD Enable and Reset Value	PUPD Selection and Reset Value	Slew Rate and Reset value
XGPIO_008	4-MUX	funcmux_reg0 [26:24]	pad_drv_reg0 [17:16] Default value: 2'h2	pad_ie_reg0 [8] 1: Enable	pupd_en_reg0 [8] 1: Enable	pupd_sel_reg0 [8] 1: Pull-up	pad_sl_reg0 [8] 0: Slow
	GPIO8	3'h0					
	SDIO1_MCLK	3'h3					
	SPI2_CLK	3'h4					
	RMII_TXD1	3'h5					
	Others	Reserved					
XGPIO_009	4-MUX	funcmux_reg0 [29:27]	pad_drv_reg0 [19:18] Default value: 2'h2	pad_ie_reg0 [9] 1: Enable	pupd_en_reg0 [9] 1: Enable	pupd_sel_reg0 [9] 1: Pull-up	pad_sl_reg0 [9] 0: Slow
	GPIO9	3'h0					
	CSI_HSYNC	3'h1					
	SDIO1_CMD	3'h2					
	SPI2_DIN(IO1)	3'h3					
	Others	Reserved					
XGPIO_010	4-MUX	funcmux_reg1 [2:0]	pad_drv_reg0 [21:20] Default value: 2'h2	pad_ie_reg0 [10] 1: Enable	pupd_en_reg0 [10] 1: Enable	pupd_sel_reg0 [10] 1: Pull-up	pad_sl_reg0 [10] 0: Slow
	GPIO10	3'h0					
	CSI_VSYNC	3'h1					
	SPI2_DOUT(IO0)	3'h2					
	TWI1_SCL	3'h3					
	Others	Reserved					
XGPIO_011	6-MUX	funcmux_reg1 [5:3]	pad_drv_reg0 [23:22] Default value: 2'h2	pad_ie_reg0 [11] 1: Enable	pupd_en_reg0 [11] 1: Enable	pupd_sel_reg0 [11] 1: Pull-up	pad_sl_reg0 [11] 0: Slow
	GPIO11	3'h0					
	CSI_SCLK	3'h2					
	PWM2	3'h3					
	SPI2_CS0	3'h4					
	RMII_RXD0	3'h5					
	TWI1_SDA	3'h6					
	Others	Reserved					

IO Name	IO Function	Function MUX	Output Driving and Reset Value	Input Enable and Reset Value	PUPD Enable and Reset Value	PUPD Selection and Reset Value	Slew Rate and Reset value
XGPIO_016	4-MUX	funcmux_reg1 [23:21]	pad_drv_reg1 [1:0] Default value: 2'h2	pad_ie_reg0 [16] 1: Enable	pupd_en_reg0 [16] 1: Enable	pupd_sel_reg0 [16] 1: Pull-up	pad_sl_reg0 [16] 0: Slow
	GPIO16	3'h0					
	SDIO1_MCLK	3'h1					
	PWM0	3'h2					
	UART2_RXD	3'h3					
	Others	Reserved					
XGPIO_017	6-MUX	funcmux_reg1 [26:24]	pad_drv_reg1 [3:2] Default value: 2'h2	pad_ie_reg0 [17] 1: Enable	pupd_en_reg0 [17] 1: Enable	pupd_sel_reg0 [17] 1: Pull-up	pad_sl_reg0 [17] 0: Slow
	GPIO17	3'h0					
	CSI_D0	3'h1					
	SPI1_CS	3'h2					
	SDIO1_CMD	3'h3					
	PWM3	3'h4					
	UART2_TXD	3'h5					
	Others	Reserved					
XGPIO_018	4-MUX	funcmux_reg1 [29:27]	pad_drv_reg1 [5:4] Default value: 2'h2	pad_ie_reg0 [18] 1: Enable	pupd_en_reg0 [18] 1: Enable	pupd_sel_reg0 [18] 1: Pull-up	pad_sl_reg0 [18] 0: Slow
	GPIO18	3'h0					
	TWI2_SCL	3'h1					
	CSI_D1	3'h2					
	SPI1_CLK	3'h3					
	Others	Reserved					
XGPIO_019	4-MUX	funcmux_reg2 [2:0]	pad_drv_reg1 [7:6] Default value: 2'h2	pad_ie_reg0 [19] 1: Enable	pupd_en_reg0 [19] 1: Enable	pupd_sel_reg0 [19] 1: Pull-up	pad_sl_reg0 [19] 0: Slow
	GPIO19	3'h0					
	TWI2_SDA	3'h1					
	CSI_D2	3'h2					
	SPI1_DOUT(IO0)	3'h3					
	Others	Reserved					

IO Name	IO Function	Function MUX	Output Driving and Reset Value	Input Enable and Reset Value	PUPD Enable and Reset Value	PUPD Selection and Reset Value	Slew Rate and Reset value
XGPIO_020	3-MUX	funcmux_reg2 [5:3]	pad_drv_reg1 [9:8] Default value: 2'h2	pad_ie_reg0 [20] 1: Enable	pupd_en_reg0 [20] 1: Enable	pupd_sel_reg0 [20] 1: Pull-up	pad_sl_reg0 [20] 0: Slow
	GPIO20	3'h0					
	TWI0_SDA	3'h1					
	CSI_D3	3'h2					
	Others	Reserved					
XGPIO_021	3-MUX	funcmux_reg2 [8:6]	pad_drv_reg1 [11:10] Default value: 2'h2	pad_ie_reg0 [21] 1: Enable	pupd_en_reg0 [21] 1: Enable	pupd_sel_reg0 [21] 1: Pull-up	pad_sl_reg0 [21] 0: Slow
	GPIO21	3'h0					
	TWI0_SCL	3'h1					
	CSI_D5	3'h2					
	Others	Reserved					
CSI_SCLK	3-MUX	funcmux_reg2 [11:9]	pad_drv_reg1 [13:12] Default value: 2'h2	pad_ie_reg0 [22] 1: Enable	pupd_en_reg0 [22] 1: Enable	pupd_sel_reg0 [22] 1: Pull-up	pad_sl_reg0 [22] 0: Slow
	CSI_SCLK	3'h1					
	CSI_D6	3'h2					
	GPIO22	3'h3					
	Others	Reserved					
XGPIO_023	2-MUX	funcmux_reg2 [14:12]	pad_drv_reg1 [15:14] Default value: 2'h2	pad_ie_reg0 [23] 1: Enable	pupd_en_reg0 [23] 1: Enable	pupd_sel_reg0 [23] 1: Pull-up	pad_sl_reg0 [23] 0: Slow
	GPIO23	3'h0					
	CSI_D4	3'h2					
	Others	Reserved					
XGPIO_024	3-MUX	funcmux_reg2 [20:18]	pad_drv_reg1 [17:16] Default value: 2'h2	pad_ie_reg0 [24] 1: Enable	pupd_en_reg0 [24] 1: Enable	pupd_sel_reg0 [24] 1: Pull-up	pad_sl_reg0 [24] 0: Slow
	GPIO24	3'h0					
	SPI1_CS	3'h1					
	SDIO1_DATA1	3'h2					
	Others	Reserved					
XGPIO_025	3-MUX	funcmux_reg2 [23:21]	pad_drv_reg1 [19:18]	pad_ie_reg0 [25]	pupd_en_reg0 [25]	pupd_sel_reg0 [25]	pad_sl_reg0 [25]
	GPIO25	3'h0					

IO Name	IO Function	Function MUX	Output Driving and Reset Value	Input Enable and Reset Value	PUPD Enable and Reset Value	PUPD Selection and Reset Value	Slew Rate and Reset value
	SPI1_CLK	3'h1	Default value: 2'h2	1: Enable	1: Enable	1: Pull-up	0: Slow
	SDIO1_DATA0	3'h2					
	Others	Reserved					
XGPIO_026	3-MUX	<i>funcmux_reg2</i> [26:24]	<i>pad_drv_reg1</i> [21:20] Default value: 2'h2	<i>pad_ie_reg0</i> [26] 1: Enable	<i>pupd_en_reg0</i> [26] 1: Enable	<i>pupd_sel_reg0</i> [26] 1: Pull-up	<i>pad_sl_reg0</i> [26] 0: Slow
	GPIO26	3'h0					
	SPI1_DOUT(IO0)	3'h1					
	SDIO1_CMD	3'h2					
	Others	Reserved					
XGPIO_027	4-MUX	<i>funcmux_reg2</i> [29:27]	<i>pad_drv_reg1</i> [23:22] Default value: 2'h2	<i>pad_ie_reg0</i> [27] 1: Enable	<i>pupd_en_reg0</i> [27] 1: Enable	<i>pupd_sel_reg0</i> [27] 1: Pull-up	<i>pad_sl_reg0</i> [27] 0: Slow
	GPIO27	3'h0					
	SPI1_DIN(IO1)	3'h1					
	SDIO1_MCLK	3'h2					
	PWM1	3'h3					
XGPIO_031	5-MUX	<i>funcmux_reg3</i> [8:6]	<i>pad_drv_reg1</i> [31:30] Default value: 2'h2	<i>pad_ie_reg0</i> [31] 1: Enable	<i>pupd_en_reg0</i> [31] 1: Enable	<i>pupd_sel_reg0</i> [31] 1: Pull-up	<i>pad_sl_reg0</i> [31] 0: Slow
	GPIO31	3'h0					
	SDIO0_DATA2	3'h1					
	eMMC_DATA3	3'h3					
	SDIO1_DATA1	3'h4					
XGPIO_032	5-MUX	<i>funcmux_reg3</i> [11:9]	<i>pad_drv_reg2</i> [1:0] Default value: 2'h2	<i>pad_ie_reg1</i> [0] 1: Enable	<i>pupd_en_reg1</i> [0] 1: Enable	<i>pupd_sel_reg1</i> [0] 1: Pull-up	<i>pad_sl_reg1</i> [0] 0: Slow
	GPIO32	3'h0					
	SDIO0_DATA3	3'h1					
	eMMC_DATA0	3'h3					
	SDIO1_DATA0	3'h4					
	Others	Reserved					

IO Name	IO Function	Function MUX	Output Driving and Reset Value	Input Enable and Reset Value	PUPD Enable and Reset Value	PUPD Selection and Reset Value	Slew Rate and Reset value
XGPIO_033	4-MUX	funcmux_reg3 [14:12]	pad_drv_reg2 [3:2] Default value: 2'h2	pad_ie_reg1 [1] 1: Enable	pupd_en_reg1 [1] 1: Enable	pupd_sel_reg1 [1] 1: Pull-up	pad_sl_reg1 [1] 0: Slow
	GPIO_033	3'h0					
	SDIO0_CMD	3'h1					
	eMMC_CMD	3'h2					
	SDIO1_CMD	3'h3					
	Others	Reserved					
XGPIO_034	5-MUX	funcmux_reg3 [17:15]	pad_drv_reg2 [5:4] Default value: 2'h2	pad_ie_reg1 [2] 1: Enable	pupd_en_reg1 [2] 1: Enable	pupd_sel_reg1 [2] 0: Pull-down	pad_sl_reg1 [2] 0: Slow
	GPIO34	3'h0					
	SDIO0_MCLK	3'h1					
	eMMC_MCLK	3'h3					
	SDIO1_MCLK	3'h4					
	Others	Reserved					
XGPIO_035	5-MUX	funcmux_reg3 [20:18]	pad_drv_reg2 [7:6] Default value: 2'h2	pad_ie_reg1 [3] 1: Enable	pupd_en_reg1 [3] 1: Enable	pupd_sel_reg1 [3] 1: Pull-up	pad_sl_reg1 [3] 0: Slow
	GPIO35	3'h0					
	SDIO0_DATA0	3'h1					
	eMMC_DATA1	3'h3					
	SDIO1_DATA3	3'h4					
	Others	Reserved					
XGPIO_036	5-MUX	funcmux_reg3 [23:21]	pad_drv_reg2 [9:8] Default value: 2'h2	pad_ie_reg1 [4] 1: Enable	pupd_en_reg1 [4] 1: Enable	pupd_sel_reg1 [4] 1: Pull-up	pad_sl_reg1 [4] 0: Slow
	GPIO36	3'h0					
	SDIO0_DATA1	3'h1					
	eMMC_DATA2	3'h3					
	SDIO1_DATA2	3'h4					
	Others	Reserved					

IO Name	IO Function	Function MUX	Output Driving and Reset Value	Input Enable and Reset Value	PUPD Enable and Reset Value	PUPD Selection and Reset Value	Slew Rate and Reset value
XGPIO_037	-	-	<i>pad_drv_reg2</i> [11:10] Default value: 2'h2	<i>pad_ie_reg1</i> [5] 1: Enable	<i>pupd_en_reg1</i> [5] 1: Enable	<i>pupd_sel_reg1</i> [5] 1: Pull-down	<i>pad_sl_reg1</i> [5] 0: Slow
XGPIO_042	3-MUX	<i>funcmux_reg4</i> [11:9]	<i>pad_drv_reg2</i> [21:20] Default value: 2'h2	<i>pad_ie_reg1</i> [10] 1: Enable	<i>pupd_en_reg1</i> [10] 1: Enable	<i>pupd_sel_reg1</i> [10] 1: Pull-up	<i>pad_sl_reg1</i> [10] 0: Slow
	GPIO42	3'h0					
	eMMC_DATA3	3'h1					
	SPI0_WP(IO2)	3'h2					
	Others	Reserved					
XGPIO_043	3-MUX	<i>funcmux_reg4</i> [14:12]	<i>pad_drv_reg2</i> [23:22] Default value: 2'h2	<i>pad_ie_reg1</i> [11] 1: Enable	<i>pupd_en_reg1</i> [11] 1: Enable	<i>pupd_sel_reg1</i> [11] 1: Pull-up	<i>pad_sl_reg1</i> [11] 0: Slow
	GPIO43	3'h0					
	eMMC_DATA0	3'h1					
	SPI0_DIN(IO1)	3'h2					
	Others	Reserved					
XGPIO_044	3-MUX	<i>funcmux_reg4</i> [17:15]	<i>pad_drv_reg2</i> [25:24] Default value: 2'h2	<i>pad_ie_reg1</i> [12] 1: Enable	<i>pupd_en_reg1</i> [12] 1: Enable	<i>pupd_sel_reg1</i> [12] 1: Pull-up	<i>pad_sl_reg1</i> [12] 0: Slow
	GPIO44	3'h0					
	eMMC_DATA1	3'h1					
	SPI0_CS	3'h2					
	Others	Reserved					
XGPIO_045	3-MUX	<i>funcmux_reg4</i> [20:18]	<i>pad_drv_reg2</i> [27:26] Default value: 2'h2	<i>pad_ie_reg1</i> [13] 1: Enable	<i>pupd_en_reg1</i> [13] 1: Enable	<i>pupd_sel_reg1</i> [13] 1: Pull-up	<i>pad_sl_reg1</i> [13] 0: Slow
	GPIO45	3'h0					
	eMMC_DATA2	3'h1					
	SPI0_HOLD(IO3)	3'h2					
	Others	Reserved					
XGPIO_046	3-MUX	<i>funcmux_reg4</i> [23:21]	<i>pad_drv_reg2</i> [29:28]	<i>pad_ie_reg1</i> [14]	<i>pupd_en_reg1</i> [14]	<i>pupd_sel_reg1</i> [14]	<i>pad_sl_reg1</i> [14]
	GPIO46	3'h0					

IO Name	IO Function	Function MUX	Output Driving and Reset Value	Input Enable and Reset Value	PUPD Enable and Reset Value	PUPD Selection and Reset Value	Slew Rate and Reset value
	eMMC_MCLK	3'h1	Default value: 2'h2	1: Enable	1: Enable	1: Pull-up	0: Slow
	SPIO_CLK	3'h2					
	Others	Reserved					
XGPIO_047	3-MUX	funcmux_reg4 [26:24]	pad_drv_reg2 [31:30] Default value: 2'h2	pad_ie_reg1 [15] 1: Enable	pupd_en_reg1 [15] 1: Enable	pupd_sel_reg1 [15] 1: Pull-up	pad_sl_reg1 [15] 0: Slow
	GPIO47	3'h0					
	eMMC_CMD	3'h1					
	SPIO_DOUT(IO0)	3'h2					
	Others	Reserved					
XGPIO_048	4-MUX	funcmux_reg4 [29:27]	pad_drv_reg3 [1:0] Default value: 2'h2	pad_ie_reg1 [16] 1: Enable	pupd_en_reg1 [16] 1: Enable	pupd_sel_reg1 [16] 1: Pull-up	pad_sl_reg1 [16] 0: Slow
	GPIO48	3'h0					
	PWM0	3'h1					
	UART1_TXD	3'h2					
	TWI1_SCL	3'h3					
	Others	Reserved					
XGPIO_049	4-MUX	funcmux_reg5 [2:0]	pad_drv_reg3 [3:2] Default value: 2'h2	pad_ie_reg1 [17] 1: Enable	pupd_en_reg1 [17] 1: Enable	pupd_sel_reg1 [17] 1: Pull-up	pad_sl_reg1 [17] 0: Slow
	GPIO49	3'h0					
	PWM1	3'h1					
	UART1_RXD	3'h2					
	TWI1_SDA	3'h3					
	Others	Reserved					
XGPIO_050	-	-	pad_drv_reg3 [5:4] Default value: 2'h2	pad_ie_reg1 [18] 1: Enable	pupd_en_reg1 [18] 1: Enable	pupd_sel_reg1 [18] 1: Pull-up	pad_sl_reg1 [18] 0: Slow
XGPIO_051	2-MUX	funcmux_reg5 [8:6]	pad_drv_reg3 [7:6] Default value: 2'h2	pad_ie_reg1 [19] 1: Enable	pupd_en_reg1 [19] 1: Enable	pupd_sel_reg1 [19] 1: Pull-up	pad_sl_reg1 [19] 0: Slow
	GPIO51	3'h0					
	UART0_TXD	3'h1					

IO Name	IO Function	Function MUX	Output Driving and Reset Value	Input Enable and Reset Value	PUPD Enable and Reset Value	PUPD Selection and Reset Value	Slew Rate and Reset value
	Others	Reserved					
XGPIO_052	2-MUX	funcmux_reg5 [11:9]	pad_drv_reg3 [9:8] Default value: 2'h2	pad_ie_reg1 [20] 1: Enable	pupd_en_reg1 [20] 1: Enable	pupd_sel_reg1 [20] 1: Pull-up	pad_sl_reg1 [20] 0: Slow
	GPIO52	3'h0					
	UART0_RXD	3'h1					
	Others	Reserved					
XGPIO_055	4-MUX	funcmux_reg5 [20:18]	pad_drv_reg3 [15:14] Default value: 2'h2	pad_ie_reg1 [23] 1: Enable	pupd_en_reg1 [23] 1: Enable	pupd_sel_reg1 [23] 1: Pull-up	pad_sl_reg1 [23] 0: Slow
	GPIO55	3'h0					
	SPI2_CS0	3'h1					
	UART2_RXD	3'h2					
	PWM3	3'h3					
	Others	Reserved					
XGPIO_056	4-MUX	funcmux_reg5 [23:21]	pad_drv_reg3 [17:16] Default value: 2'h2	pad_ie_reg1 [24] 1: Enable	pupd_en_reg1 [24] 1: Enable	pupd_sel_reg1 [24] 1: Pull-up	pad_sl_reg1 [24] 0: Slow
	GPIO56	3'h0					
	SPI2_CLK	3'h1					
	UART2_TXD	3'h2					
	PWM4	3'h3					
	Others	Reserved					
XGPIO_058	5-MUX	funcmux_reg5 [29:27]	pad_drv_reg3 [21:20] Default value: 2'h2	pad_ie_reg2 [26] 1: Enable	pupd_en_reg1 [26] 1: Enable	pupd_sel_reg1 [26] 1: Pull-up	pad_sl_reg1 [26] 0: Slow
	GPIO58	3'h0					
	SPI2_CS1	3'h1					
	UART2_RTS	3'h2					
	TWI1_SDA	3'h3					
	SDIO1_CMD	3'h4					
	Others	Reserved					
SPI2_CS0	6-MUX	funcmux_reg6 [20:18]	pad_drv_reg4 [3:2]	pad_ie_reg2 [1]	pupd_en_reg2 [1]	pupd_sel_reg2 [1]	pad_sl_reg2 [1]
	SPI2_CS0	3'h1					

IO Name	IO Function	Function MUX	Output Driving and Reset Value	Input Enable and Reset Value	PUPD Enable and Reset Value	PUPD Selection and Reset Value	Slew Rate and Reset value
	GPIO65	3'h2	Default value: 2'h2	1: Enable	1: Enable	0: Pull-down	0: Slow
	UART2_RXD	3'h3					
	I2S_MCLK	3'h4					
	SDIO1_MCLK	3'h5					
	Others	Reserved					
SPI2_CLK	5-MUX	funcmux_reg6 [23:21]	pad_drv_reg4 [5:4] Default value: 2'h2	pad_ie_reg2 [2] 1: Enable	pupd_en_reg2 [2] 1: Enable	pupd_sel_reg2 [2] 0: Pull-down	pad_sl_reg2 [2] 0: Slow
	SPI2_CLK	3'h1					
	GPIO66	3'h2					
	UART2_TXD	3'h3					
	I2S_BCLK	3'h4					
	SDIO1_DATA2	3'h5					
	Others	Reserved					
SPI2_DIN(IO1)	7-MUX	funcmux_reg6 [26:24]	pad_drv_reg4 [7:6] Default value: 2'h2	pad_ie_reg2 [3] 1: Enable	pupd_en_reg2 [3] 1: Enable	pupd_sel_reg2 [3] 1: Pull-up	pad_sl_reg2 [3] 0: Slow
	SPI2_DIN(IO1)	3'h1					
	GPIO67	3'h2					
	PWM2	3'h3					
	UART2_CTS	3'h4					
	I2S_LRCLK	3'h5					
	SDIO1_DATA3	3'h6					
	Others	Reserved					
SPI2_CS1	6-MUX	funcmux_reg6 [29:27]	pad_drv_reg4 [9:8] Default value: 2'h2	pad_ie_reg2 [4] 1: Enable	pupd_en_reg2 [4] 1: Enable	pupd_sel_reg2 [4] 1: Pull-up	pad_sl_reg2 [4] 0: Slow
	SPI2_CS1	3'h1					
	GPIO68	3'h2					
	PWM3	3'h4					
	I2S_DOUT	3'h5					
	SDIO1_DATA0	3'h6					

IO Name	IO Function	Function MUX	Output Driving and Reset Value	Input Enable and Reset Value	PUPD Enable and Reset Value	PUPD Selection and Reset Value	Slew Rate and Reset value
	Others	Reserved					
SPI2_DOUT(IO0)	6-MUX	funcmux_reg7[2:0]	<i>pad_drv_reg4</i> [11:10] Default value: 2'h2	<i>pad_ie_reg2</i> [5] 1: Enable	<i>pupd_en_reg2</i> [5] 1: Enable	<i>pupd_sel_reg2</i> [5] 0: Pull-down	<i>pad_sl_reg2</i> [5] 0: Slow
	SPI2_DOUT(IO0)	3'h1					
	GPIO69	3'h2					
	UART2_RTS	3'h3					
	I2S_DIN	3'h4					
	SDIO1_DATA1	3'h5					
	Others	Reserved					
XGPIO_070	2-MUX	funcmux_reg8[26:24]	<i>pad_drv_reg4</i> [13:12] Default value: 2'h2	<i>pad_ie_reg2</i> [6] 1: Enable	<i>pupd_en_reg2</i> [6] 1: Enable	<i>pupd_sel_reg2</i> [6] 0: Pull-down	<i>pad_sl_reg2</i> [6] 0: Slow
	GPIO70	3'h0					
	AIN1	3'h1					
	Others	Reserved					
XGPIO_071	2-MUX	funcmux_reg8 [29:27]	<i>pad_drv_reg4</i> [15:14] Default value: 2'h2	<i>pad_ie_reg2</i> [7] 1: Enable	<i>pupd_en_reg2</i> [7] 1: Enable	<i>pupd_sel_reg2</i> [7] 0: Pull-down	<i>pad_sl_reg2</i> [7] 0: Slow
	GPIO71	3'h0					
	AIN0	3'h1					
	Others	Reserved					
MIPI_RX_L1P	3-MUX	funcmux_reg9[8:6]	<i>pad_drv_reg4</i> [29:28] Default value: 2'h2	<i>pad_ie_reg2</i> [14] 1: Enable	<i>pupd_en_reg2</i> [14] 1: Enable	<i>pupd_sel_reg2</i> [14] 1: Pull-up	<i>pad_sl_reg2</i> [14] 0: Slow
	MIPI_RX_L1P	3'h0					
	CSI_D6	3'h1					
	GPIO78	3'h2					
	Others	Reserved					
MIPI_RX_L1N	3-MUX	funcmux_reg9 [11:9]	<i>pad_drv_reg4</i> [31:30] Default value: 2'h2	<i>pad_ie_reg2</i> [15] 1: Enable	<i>pupd_en_reg2</i> [15] 1: Enable	<i>pupd_sel_reg2</i> [15] 1: Pull-up	<i>pad_sl_reg2</i> [15] 0: Slow
	MIPI_RX_L1N	3'h0					
	CSI_D5	3'h1					
	GPIO79	3'h2					
	Others	Reserved					

IO Name	IO Function	Function MUX	Output Driving and Reset Value	Input Enable and Reset Value	PUPD Enable and Reset Value	PUPD Selection and Reset Value	Slew Rate and Reset value
GPIO98	2-MUX	<i>funcmux_reg8</i> [23:21]	<i>pad_drv_reg6</i> [5:4] Default value: 2'h2	<i>pad_ie_reg3</i> [2] 1: Enable	<i>pupd_en_reg3</i> [2] 1: Enable	<i>pupd_sel_reg3</i> [2] 1: Pull-up	<i>pad_sl_reg3</i> [2] 0: Slow
	GPIO98	3'h1					
	Others	Reserved					
MIPI_RX_LOP	3-MUX	<i>funcmux_reg9</i> [2:0]	<i>pad_drv_reg6</i> [7:6] Default value: 2'h2	<i>pad_ie_reg3</i> [3] 1: Enable	<i>pupd_en_reg3</i> [3] 1: Enable	<i>pupd_sel_reg3</i> [3] 1: Pull-up	<i>pad_sl_reg3</i> [3] 0: Slow
	MIPI_RX_LOP	3'h0					
	CSI_D7	3'h1					
	GPIO99	3'h2					
	Others	Reserved					
MIPI_RX_LON	3-MUX	<i>funcmux_reg9</i> [5:3]	<i>pad_drv_reg6</i> [9:8] Default value: 2'h2	<i>pad_ie_reg3</i> [4] 1: Enable	<i>pupd_en_reg3</i> [4] 1: Enable	<i>pupd_sel_reg3</i> [4] 1: Pull-up	<i>pad_sl_reg3</i> [4] 0: Slow
	MIPI_RX_LON	3'h0					
	CSI_CLK	3'h1					
	GPIO100	3'h2					
	Others	Reserved					
MIPI_RX_L2P	3-MUX	<i>funcmux_reg9</i> [14:12]	<i>pad_drv_reg6</i> [11:10] Default value: 2'h2	<i>pad_ie_reg3</i> [5] 1: Enable	<i>pupd_en_reg3</i> [5] 1: Enable	<i>pupd_sel_reg3</i> [5] 1: Pull-up	<i>pad_sl_reg3</i> [5] 0: Slow
	MIPI_RX_L2P	3'h0					
	CSI_D9	3'h1					
	GPIO101	3'h2					
	Others	Reserved					
MIPI_RX_L2N	3-MUX	<i>funcmux_reg9</i> [17:15]	<i>pad_drv_reg6</i> [13:12] Default value: 2'h2	<i>pad_ie_reg3</i> [6] 1: Enable	<i>pupd_en_reg3</i> [6] 1: Enable	<i>pupd_sel_reg3</i> [6] 1: Pull-up	<i>pad_sl_reg3</i> [6] 0: Slow
	MIPI_RX_L2N	3'h0					
	CSI_D8	3'h1					
	GPIO102	3'h2					
	Others	Reserved					

NOTE

1)The driving strength of some IOs can be configured. For detailed configuration, please refer to [Section 3.5.1](#).

2)AK3918AV100 series supports three pairs of MIPI pins: MIPI_RX_LOP & MIPI_RX_LON, MIPI_RX_L1P & MIPI_RX_L1N, and MIPI_RX_L2P & MIPI_RX_L2N.

●Programers can select any pair of pins as MIPI clock lanes.

●For 1 pair of data lanes plus 1 pair of clock lanes application (1D1C application), users can select any two pairs of these pins as MIPI pins. **It is recommended that the remaining MIPI pins be left unconnected and NOT be multiplexed as GPIO pins.**

●At one time, all of these pins are used as MIPI pins or otherwise as DVP pins/GPIOs. In the former case, these pins are powered by AVDD3325_MIPIRX(2.5V/3.3V); while in the latter, these pins are powered by DVDD3318_CIS and **AVDD3325_MIPIRX should be connected to a 2.5V/3.3V power supply.** The circuit diagram of multiplexing of MIPI pins is given below.

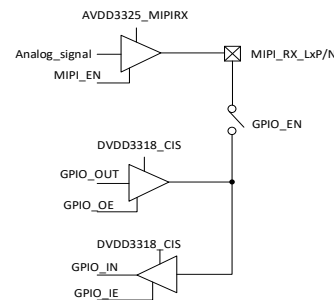


Figure 3-1 The Circuit Diagram of Multiplexing of MIPI pins

4 System

4.1 Reset Module

The RST signal, when asserted low, puts all registers (with the exception of those in RTC module) into predefined values. Once the reset signal is deasserted, CPU will reload the boot program stored in the on-chip ROM.

4.1.1 Features

Features of reset module are as follows:

- Supports two global reset sources
 - Hardware internal power-on reset
 - Watchdog reset (global reset except crystal oscillator stabilization counter in core domain)
- Controls the separate soft reset of each functional module. For more, please refer to Chapter 4.1.4.

4.1.2 Power on Reset Timing

AK3918AV100 series processors support power-on/Brown-out detection of DVDD33 IO ports, based on which the chip generates power-on/Brown-out reset signals. Figure 4-1 shows the reset timing of AK3918AV100 series processors.

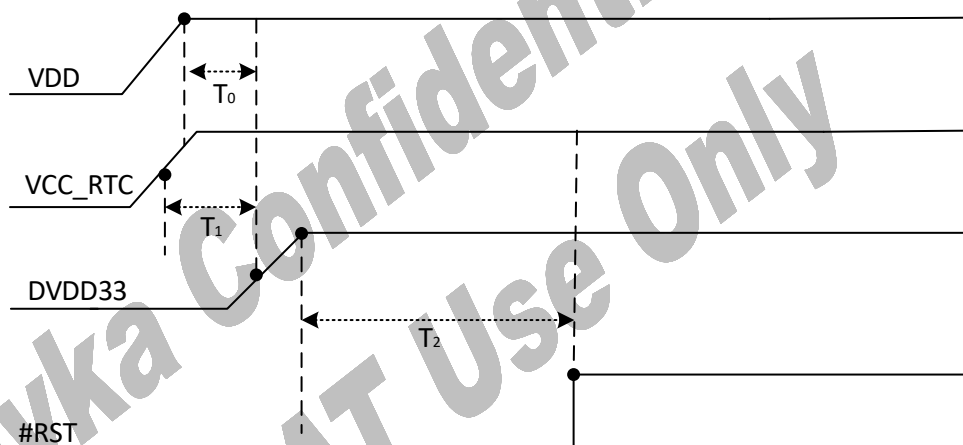


Figure 4-1 AK3918AV100 Series Processors Power on Reset Timing

NOTE

- 1) $T_0 \geq 0\text{ms}$; $T_1 \geq 0\text{ms}$
- 2) T_2 is about 10ms.

4.1.3 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Register Address	Register Name
1	0x0800,0020	Software Reset Control
2	0x0800,011C	Refer to ISP/NPU/ENC VCLK Clock Source Configuration
3	0x0800,00E4	Watchdog Time Control
4	0x0800,00E8	Watchdog Control

4.1.4 Register Definition

- Software Reset Control (Address: 0x0800,0020)

Bits	Attributes	Reset	Description
[31]	RW	1'h1	spi0_hsclock_rst_cfg The software reset control signal applies to the ISP module in spi0_hsclock domain. 1: Reset spi0_hsclock. 0: Do not reset spi0_hsclock.
[30]	RW	1'h1	sar_adc_rst_cfg 1: Reset the module driven by sar_adc_clk. 0: Do not reset the module driven by sar_adc_clk.
[29]	RW	1'h1	sdadc_hsrst_cfg 1: Reset the module driven by sdadc_hsclock. 0: Do not reset the module driven by sdadc_hsclock. NOTE: sdadc_rst_cfg and sdadc_hsrst_cfg must be set or cleared at the same time.
[28]	RW	1'h1	sddac_hsrst_cfg 1: Reset the module driven by sddac_hsclock. 0: Do not reset the module driven by sddac_hsclock. NOTE: sddac_rst_cfg and sddac_hsrst_cfg must be set or cleared at the same time.
[27]	RW	1'h1	sdadc_rst_cfg 1: Reset the module driven by sdadc_clk. 0: Do not reset the module driven by sdadc_clk. NOTE: sdadc_rst_cfg and sdadc_hsrst_cfg must be set or cleared at the same time.
[26]	RW	1'h1	sddac_rst_cfg 1: Reset the module driven by sddac_clk. 0: Do not reset the module driven by sddac_clk. NOTE: sddac_rst_cfg and sddac_hsrst_cfg must be set or cleared at the same time.
[25]	RW	1'h1	ddrphyck_rst_cfg 1: Reset the module driven by dphyck. 0: Do not reset the module driven by dphyck.
[24]	RW	1'h1	dram_rst_cfg 1: Reset the module driven by dclk. 0: Do not reset the module driven by dclk. NOTE: DCLK is the working clock of the following modules: DRAM Controller, DMA Asynchronous Buffer 1~5, AHB2DDR Bridge and Register Bus 1.
[23]	RO	1'h1	Reserved

Bits	Attributes	Reset	Description
[22]	RW	1'h1	gclk_rst_cfg[22] 1: Reset TWI2. 0: Do not reset TWI2.
[21]	RW	1'h1	gclk_rst_cfg[21] 1: Reset MMC/SD/SDIO Controller2. 0: Do not reset MMC/SD/SDIO Controller2.
[20]	RW	1'h1	gclk_rst_cfg[20] 1: Reset SPI2. 0: Do not reset SPI2.
[19]	RO	1'h1	Reserved
[18]	RW	1'h1	gclk_rst_cfg[18] 1: Reset UART2. 0: Do not reset UART2.
[17]	RW	1'h1	gclk_rst_cfg[17] 1: Reset TWI1. 0: Do not reset TWI1.
[16]	RW	1'h1	gclk_rst_cfg[16] 1: Reset MIPI Controller. 0: Do not reset MIPI Controller.
[15]	RW	1'h1	gclk_rst_cfg[15] 1: Reset USB Controller. 0: Do not reset USB Controller.
[14]	RW	1'h1	gclk_rst_cfg[14] 1: Reset Ecryption module. 0: Do not reset Ecryption module.
[13]	RW	1'h1	gclk_rst_cfg[13] 1: Reset MAC, DMA Arbiter and DMA Asynchronous Buffer2. 0: Do not reset MAC, DMA Arbiter and DMA Asynchronous Buffer2.
[12]	RW	1'h1	gclk_rst_cfg[12] 1: Reset GPIO. 0: Do not reset GPIO.
[11]	RW	1'h1	gclk_rst_cfg[11] 1: Reset L2 Buffer1. 0: Do not reset L2 Buffer1
[10]	RW	1'h1	gclk_rst_cfg[10] 1: Reset TWI0. 0: Do not reset TWI0.
[9]	RW	1'h1	gclk_rst_cfg[9] 1: Reset L2 Buffer0. 0: Do not reset L2 Buffer0.
[8]	RW	1'h1	gclk_rst_cfg[8] 1: Reset UART1. 0: Do not reset UART1.
[7]	RW	1'h1	gclk_rst_cfg[7] 1: Reset UART0. 0: Do not reset UART0.

Bits	Attributes	Reset	Description
[6]	RW	1'h1	gclk_rst_cfg[6] 1: Reset SPI1. 0: Do not reset SPI1.
[5]	RW	1'h1	gclk_rst_cfg[5] 1: Reset SPI0. 0: Do not reset SPI0.
[4]	RW	1'h1	gclk_rst_cfg[4] 1: Reset Sigma-delta DAC Controller. 0: Do not reset Sigma-delta DAC Controller.
[3]	RW	1'h1	gclk_rst_cfg[3] 1: Reset Sigma-delta ADC Controller. 0: Do not reset Sigma-delta ADC Controller.
[2]	RW	1'h1	gclk_rst_cfg[2] 1: Reset MMC/SD/SDIO Controller1. 0: Do not reset MMC/SD/SDIO Controller1.
[1]	RW	1'h1	gclk_rst_cfg[1] 1: Reset MMC/SD/SDIO Controller0. 0: Do not reset MMC/SD/SDIO Controller0.
[0]	RO	1'h1	Reserved

● Watchdog Time Control (Address: 0x0800,00E4)

Bits	Attributes	Reset	Description
[31:24]	WO	0x0	Write data protection bits Only when these bits are set to 0x55 Data can data be written to bits[23:0] of this register.
[23:0]	RW	0xFF_FFFF	watchdog_time_cfg Watchdog time configuration bits (unit: 21333.3333ns) NOTE: Bit[3] is always set to 1.

● Watchdog Control (Address: 0x0800,00E8)

Bits	Attributes	Reset	Description
[31:24]	WO	8'h0	wr_guard Write data protection bits Only when these bits are set to 0xAA, can data be written to bits[2:0] of this register.
[23:3]	RO	21'h0	Reserved
[2]	WR	1'h0	watchdog_rst_status Watchdog reset status NOTE: Programmers should write 1 to this bit to clear it.
[1]	WO	1'h0	watchdog_feed_en 1: Feed watchdog. 0: No instruction. NOTE: This bit is write only.
[0]	WR	1'h0	watchdog_en 0: Disable watchdog. 1: Enable watchdog.

4.1.5 Programmer's Guide

4.1.5.1 Hardware Internal Power-on Reset

The global reset control triggered by POR takes place automatically inside the chip once the AK3918AV100 processor is powered on.

4.1.5.2 Watchdog Reset

4.1.5.2.1 Starting Watchdog Function

1. Write *Watchdog Time Control (Address: 0x0800,00E4)*. When data to be written into this register are $((0x55 \ll 24) \mid (0xFFFF \& \text{watchdog_time_cfg}))$, configure the watchdog time.
2. Write *Watchdog Control (Address: 0x0800,00E8)*. when data to be written into this register are $((0xAA \ll 24) \mid 0x1)$, enable watchdog.

4.1.5.2.2 Feeding Watchdog

Write *Watchdog Control (Address: 0x0800,00E8)*. When data to be written into this register are $((0xAA \ll 24) \mid 0x3)$, feed watchdog.

4.2 Clock Generation and Control

4.2.1 Overview

Clock module controls clock inputs, divides clock frequencies, and generates working clocks for all the modules.

4.2.2 Features

The features of clock modules are as follows:

- Two external crystal input: a 32.768 KHz crystal oscillator and a 24MHz crystal oscillator
- Three independent PLLs
- 32.768 KHz crystal oscillator or internal 32KHz RC oscillator generates working clock for RTC
- Supports multiple output clocks for various peripherals, reducing total BOM of end products
- Flexible clock sources for ISP, NPU, Video Processor and CMOS image sensor

4.2.3 Block Diagram

4.2.3.1 Clock Tree of 24MHz Crystal Oscillator

All the modules shown in [Figure 4-2](#) source from the external 24MHz crystal oscillator.

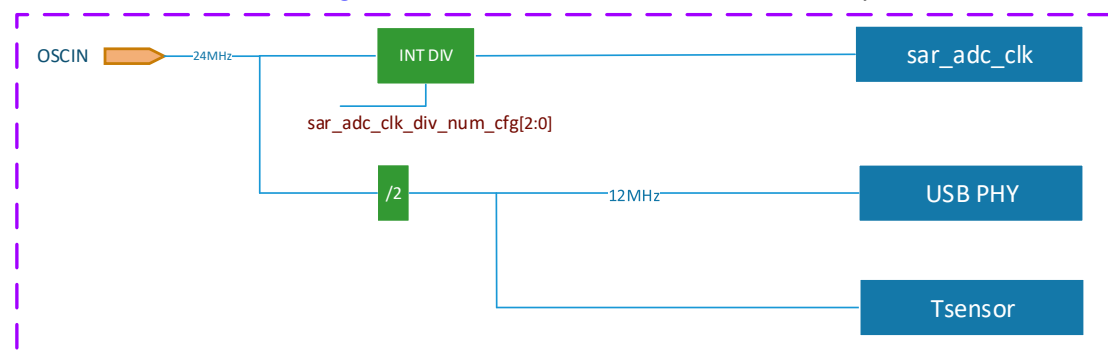


Figure 4-2 Clock Tree of 24MHz

The chip contains 3 PLLs, and the main functions of each PLL are as follows:

PLL ID	PLL Function
PLL1	Generates working clocks for modules such as CPU/DDR2/Memory controller and AMBA primary bus.
PLL2	Mainly used to generate working clocks for internal video processing modules and the working clocks for the peripheral control modules.
PLL3	Generates the main clock for audio modules and some of the peripherals.

NOTE:

Some modules can choose any one of the three PLLs to generate the required Working clocks based on the application needs, including ISP, NPU, Video Processor and CMOS image sensor.

4.2.3.2 Clock Tree of PLL1

The CPU and DDR2/SDRAM memory system adopt a synchronous design. The PLL1's clock tree block diagram allows setting the frequency combination between CPU and DDR2 according to the performance and bandwidth requirements of different application scenarios. The PLL1's clock tree is shown in the figure below.

PLL1 clock tree generates several working clocks:

- DPHY is DDR PHY source clock frequency
- JCLK is the CPU operating clock
- DDR2 is the the DDR2 clock frequency
- HCLK is the CPU bus frequency
- DCLK is the working clock of the memory controller, which has the same frequency with HCLK

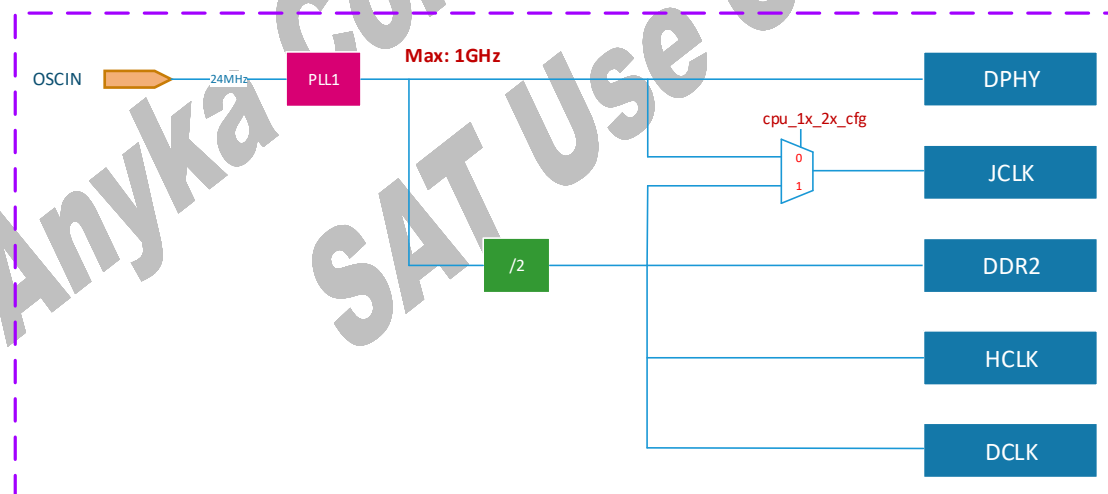


Figure 4-3 Clock generation—PLL1

- The PLL1's clock tree supports 2 working modes, which are shown in the following table:

Mode	PLL Out e.g. (MHz)	JCLK	HCLK/ DCLK	DDR2	Relationships	Description
Mode 1	888	PLL	PLL/2	PLL/2	CPU2X MEM1X	In this mode, CPU has the best processing capability.

Mode	PLL Out e.g. (MHz)	JCLK	HCLK/ DCLK	DDR2	Relationships	Description
						NOTE: Without boost in voltage, CPU can work at a highest frequency of 900MHz.
Mode 2	444	PLL/2	PLL/2	PLL/2	CPU1X MEM1X	In this mode, DDR2 has the ideally highest bandwidth. NOTE: DDR2 can work at a highest frequency of 500MHz.

NOTE

- 1) AK3918AV100 series processors work at mode 1 by default.
- 2) It is **NOT** recommended to boost the voltage of CPU to further improve the working frequency of AK3918AV100 series processors.

4.2.3.3 Clock Tree of PLL2

The figure below is the PLL2's clock tree.

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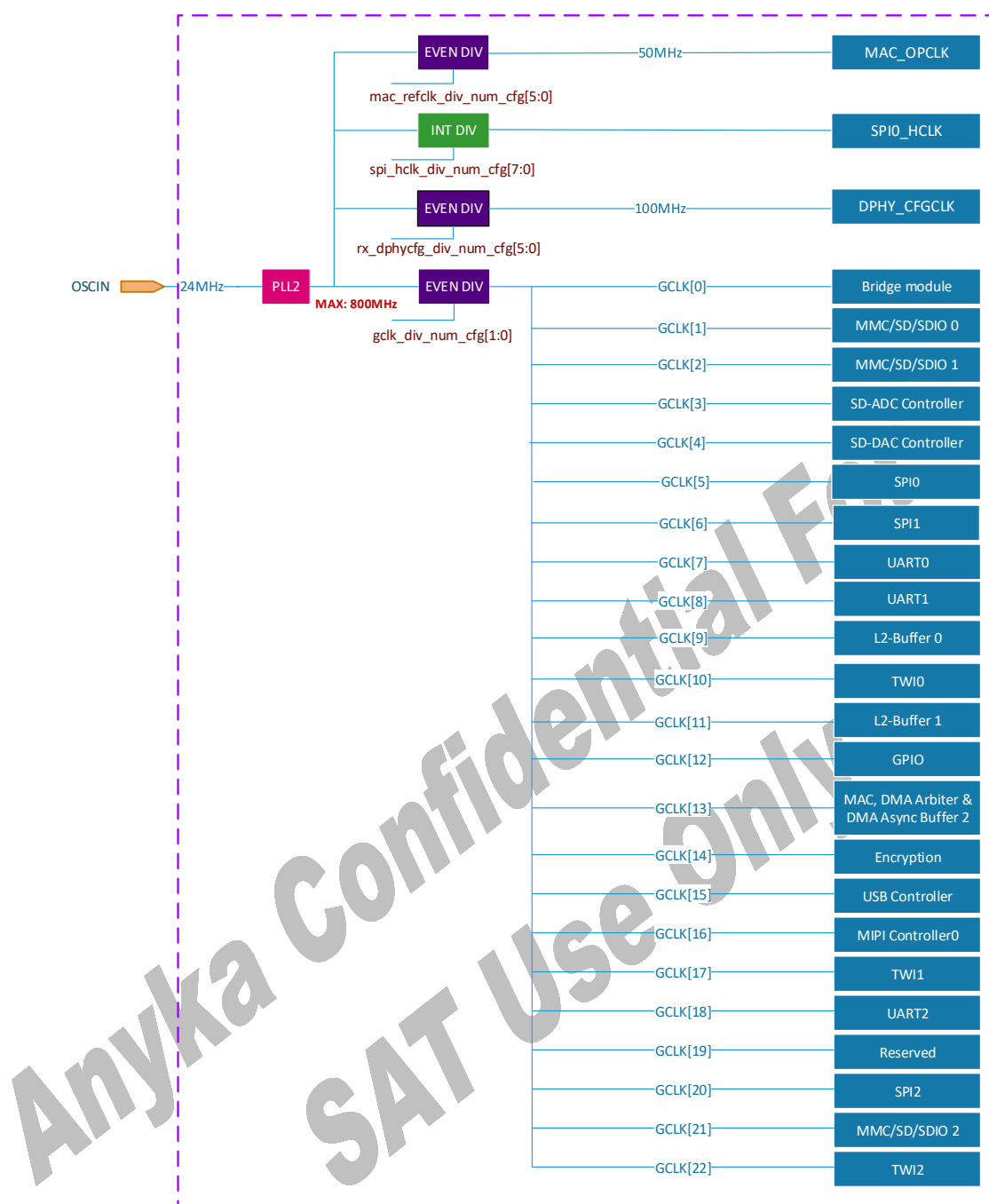


Figure 4-4 Clock Generation—PLL2

GCLKs are for slow processing modules such as peripheral controllers and All GCLKs are homologous. Each working clock has its own independent gate enable control signal in order to reduce dynamic power consumption.

The PLL2's clock tree supports 2 types of dividers: even dividers and integer dividers. If the external MAC peripheral is connected, the 50MHz clock required in RMII mode can be generated by MAC_OPCLK. If the external MAC peripheral is not connected, an ordinary clock can be generated by MAC_OPCLK's even clock divider.

Figures of clock trees of USB, SPI0, and Ethernet MAC module are listed below.

4.2.3.3.1 USB Clock

The figure below is the clock tree of USB.

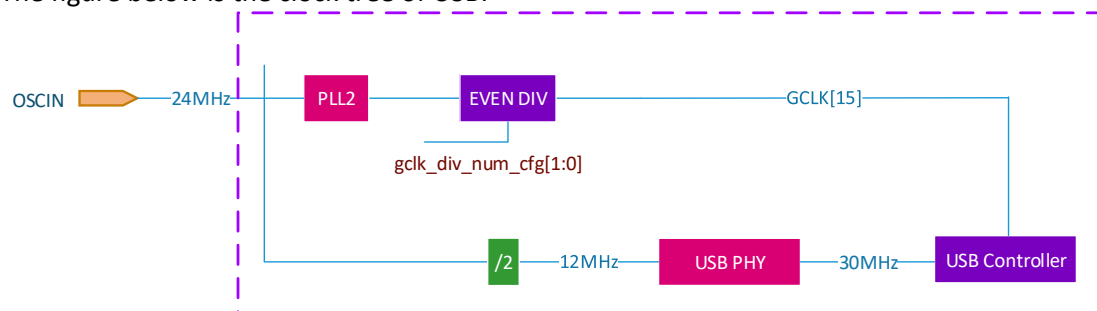


Figure 4-5 Clock Generation—USB

NOTE

- 1) 24MHz crystal oscillator propagates a 12MHz reference clock for USB PHY after being regulated by a divide-by-2 circuit.
- 2) USB PHY generates a 30MHz clock for USB controller for data transmission between USB PHY and USB controller.
- 3) GCLK [15] is the working clock of USB controller, and its frequency should be higher than 48MHz.

4.2.3.3.2 Fast SPI(SPI0)

The chip supports SPI0, one high-speed SPI, which is also generated from the PLL2 clock tree. The maximum frequency of SPI0_CLK is 100MHz, the clock schematic is shown in the figure below.

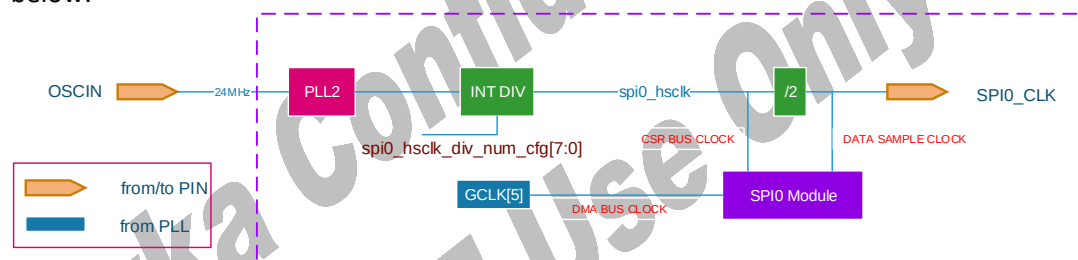


Figure 4-6 Clock Generation—SPI0

NOTE

The highest speed of SPI0 is related to the PCB layout of the SPI Flash peripheral. It is recommended that external traces need to be as short and equal as possible.

4.2.3.3.3 Ethernet MAC

The clock schematic of Ethernet MAC is shown in the figure below.

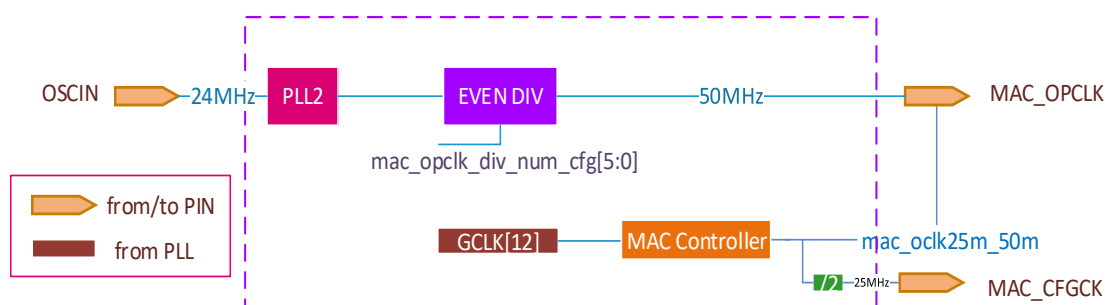


Figure 4-7 Clock Generation—Ethernet MAC

4.2.3.4 Clock Tree of PLL3

The major working clocks are described as follows:

- sddac_clk and sddac_hclk are the working frequency of Sigma-Delta DAC.
- sdadc_clk and sdadc_hclk are the working frequency of Sigma-Delta ADC.
- sddac_clk and sdadc_clk clocks affect the accuracy of the I2S related clocks.

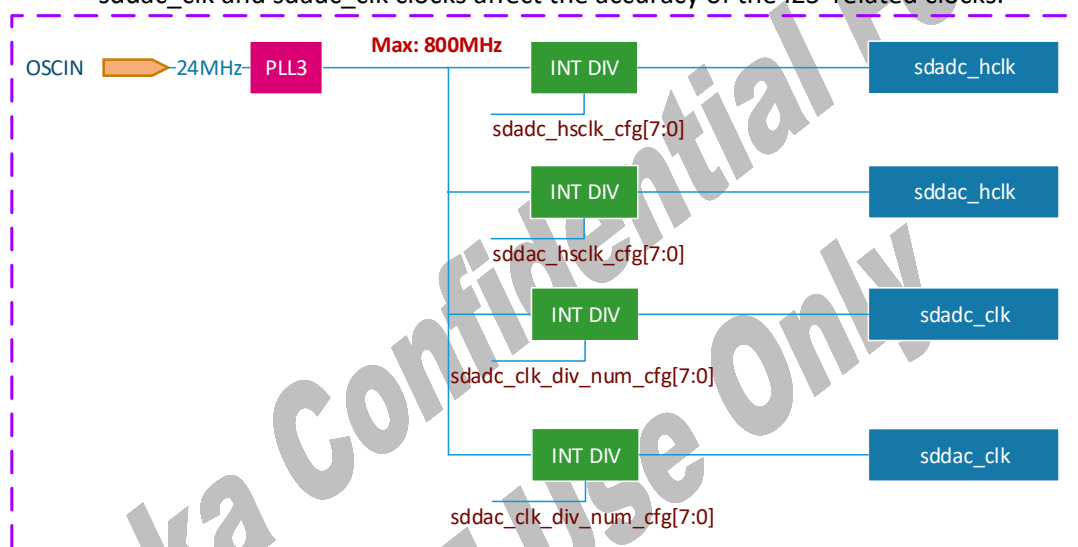


Figure 4-8 Clock Generation—PLL3

I2S module supports master-slave mode. When operating in master mode, PLL3 first generates the sddac_clk from the first-stage clock divider, and then passes through the second-stage clock divider to generate the LRCK and BCLK required by I2S module. Figure 4-9 is a schematic diagram of the I2S related clock.

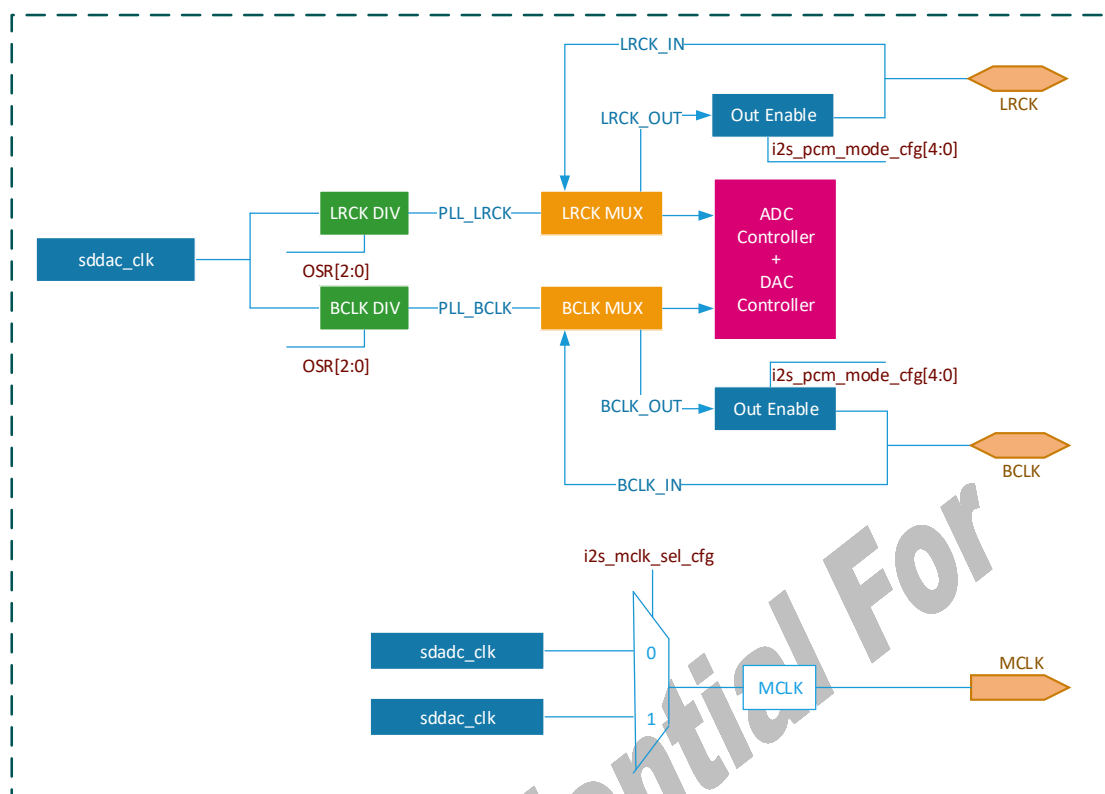


Figure 4-9 Clock Generation—Sigma-Delta ADC/DAC and I2S

4.2.3.5 Multiple Clock Sources for ASICs

AK3918AV100 series processors support multiple clock sources for ASICs (including ISP, NPU, Video Processor and CMOS image sensor) via a glitch-free clock switching structure to satisfy more application needs. Figure 4-10 shows the schematic diagram of clock switching structure for ISP, NPU and Video Processor and Figure 4-11 shows that for CMOS image sensor.

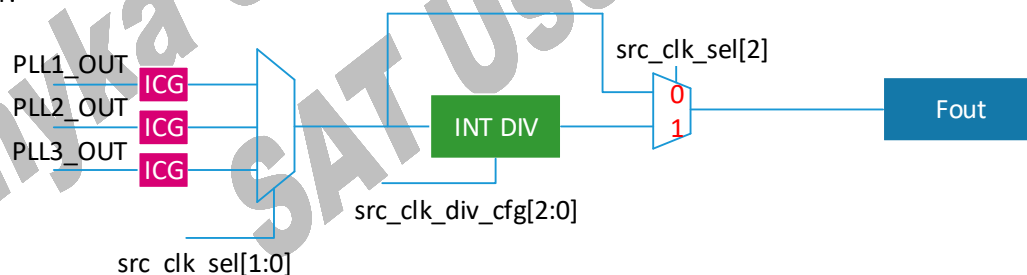


Figure 4-10 Clock Switching Structure for ISP/NPU/Video Processor

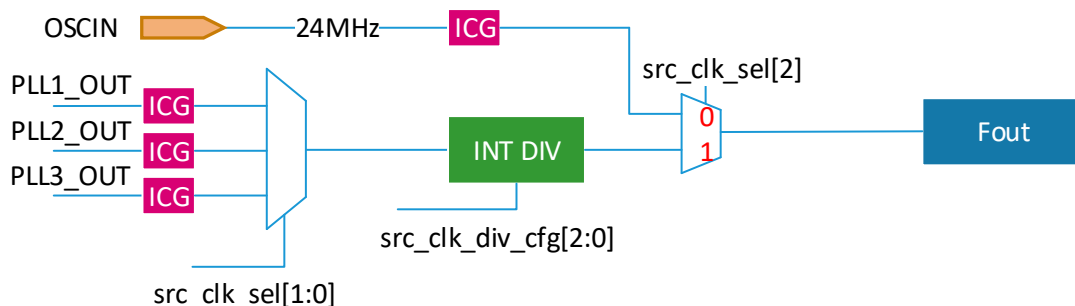


Figure 4-11 Clock Switching Structure for CMOS image sensor

4.2.3.6 Clock Tree of RTC Module

RTC module requires three clocks, including:

- 32.768KHz clock, which can be selected from an external 32.768KHz crystal oscillator or an internal 32KHz RC oscillator. Both internal and external 32.768KHz clock source can be turned on or off to reduce the power consumption.
- 2MHz register bus clock, which is used by the CPU to configure the registers in RTC module. The clock will be automatically closed when there is no communication between CPU and RTC module.
- 24MHz calibration clock, which is sent directly from the external 24MHz crystal oscillator. When the internal 32KHz RC clock is selected, the 24MHz clock can be used as the reference clock for calibration.

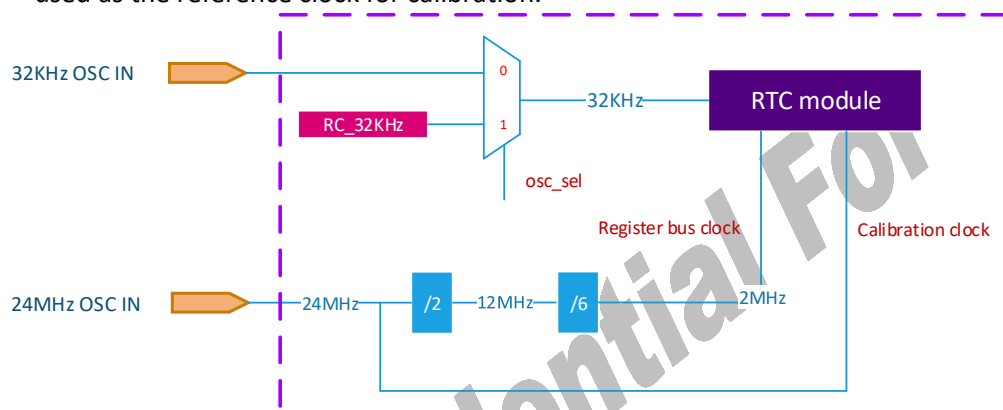


Figure 4-12 Clock Generation—RTC module

4.2.4 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Register Address	Register Name
1	0x0800,0004	<i>PLL1 Channel Clock Control</i>
2	0x0800,0008	<i>PLL2 Channel Clock Control</i>
3	0x0800,000C	<i>SAR_ADC/SPI0/ SD_DAC Clock Control</i>
4	0x0800,0010	<i>SD_ADC/SD_DAC High Speed Clock Control</i>
5	0x0800,0014	<i>PLL3 Channel Clock Control</i>
6	0x0800,0018	<i>CSI_SCLK/MAC_OPCLK Clock Control</i>
7	0x0800,001C	<i>Clock Gate Control</i>
8	0x0800,0058	<i>USB Operate Mode and I2S Mode Control</i>
9	0x0800,00DC	<i>PLL1 Bandwidth Adjustment Configuration</i>
10	0x0800,00E0	<i>PLL2/PLL3 Bandwidth Adjustment Configuration</i>
11	0x0800,011C	<i>ISP/NPU/ENC VCLK Clock Source Configuration</i>

4.2.5 Register Definition

- PLL1 Channel Clock Control (Address: 0x0800,0004)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	cpu_1x_2x_sel_cfg CPU operating clock selection

Bits	Attributes	Reset	Description
			0: CPU2X, HCLK=DCLK=DDR2=1/2JCLK 1: CPU1X, HCLK=DCLK=DDR2=JCLK
[30]	RW/SC	1'h0	pll1_freq_adj_cfg PLL1 output frequency adjustment configuration 1: To update frequency division coefficient of PLL1 and dividers under the PLL1 clock tree. NOTE: This bit will be cleared automatically when the adjustment of PLL1 is completed.
[29]	RW/SC	1'h0	pll3_freq_adj_cfg PLL3 output frequency adjustment configuration 1: To update frequency division coefficient of PLL3 and dividers under the PLL3 clock tree. NOTE: This bit will be cleared automatically when the adjustment of PLL3 is completed.
[28]	RW/SC	1'h0	pll2_freq_adj_cfg PLL2 output frequency adjustment configuration 1: To update frequency division coefficient of PLL2 and dividers under the PLL2 clock tree. NOTE: This bit will be cleared automatically when the adjustment of PLL2 is completed.
[27]	RO	1'h0	st_mode_cfg Standby mode configuration. NOTE: The system will enter standby mode when bit[0] of register (Address: 0x2180,0000) is written into 1.
[26:23]	RO	4'h0	Reserved
[22:19]	RW	4'h5	pll1_clkod_cfg [3:0](odd and 0) PLL1 output frequency setting parameter3 NOTE: For PLL1 output frequency calculation formula, please refer to bits[12:0] of this register.
[18:13]	RW	6'h0	pll1_clkr_cfg [5:0] PLL1 output frequency setting parameter2 NOTE: 1) It is strongly recommended that the setting parameter NR of PLL be set to 1(clkr_cfg as 0), which can reduce the jitter of the PLL output clock, thereby improving the stability of DDR2. 2) For PLL1 output frequency calculation formula, please refer to bits[12:0] of this register.
[12:0]	RW	13'h1D	pll1_clkf_cfg [12:0] PLL1 output frequency setting parameter1 By default, PLL1 output frequency is 120MHz and PLL1 output frequency calculation formula: PLL1 output frequency= 24M*Nf/NR/OD Nf = PLL1_CLKF_CFG [12:0] + 1 NR = PLL1_CLKR_CFG [5:0] + 1 OD = PLL1_CLKOD_CFG [3:0] + 1 NOTE: 1) Since 500MHz <= 24*Nf/NR <=2.5GHz, PLL VCO

Bits	Attributes	Reset	Description
			ranges from 500MHz to 2.5GHz. 2) Changes to these bits are valid only when pll1_freq_adj_cfg is set to 1 (bit[30] of this register) or the system exits standby mode.

● PLL2 Channel Clock Control (Address: 0x0800,0008)

Bits	Attributes	Reset	Description
[31:26]	RO	6'h0	Reserved
[25]	RW/SC	1'h0	gclk_div_vld_cfg 1: Enable even divider of GCLK 0: Disable even divider of GCLK NOTE: This bit will be cleared automatically when the adjustment of gclk is completed.
[24:23]	RW	2'h1	gclk_div_num_cfg[1:0] GCLK frequency division coefficient configuration By default, The output frequency of GCLK is 60MHz. $GCLK = PLL2_CLKOUT / 2(gclk_div_num_cfg[1:0] + 1)$
[22:19]	RW	4'h3	pll2_clkod_cfg[3:0](odd and 0) PLL2 output frequency setting parameter3 NOTE: For PLL2 output frequency calculation formula, please refer to bits[12:0] of this register.
[18:13]	RW	6'h0	pll2_clkr_cfg[5:0] PLL2 output frequency setting parameter2 NOTE: 1) It is strongly recommended that the setting parameter NR of PLL be set to 1(clkr_cfg as 0), which can reduce the jitter of the PLL output clock, thereby improving the stability of DDR2. 2) For PLL2 output frequency calculation formula, please refer to bits[12:0] of this register.
[12:0]	RW	13'h27	pll2_clkf_cfg[12:0] PLL2 output frequency setting parameter1 By default, PLL2 output frequency is 240MHz and PLL2 output frequency calculation formula: $PLL2\ output\ frequency = 24M * NF / NR / OD$ $NF = PLL2_CLKF_CFG[12:0] + 1$ $NR = PLL2_CLKR_CFG[5:0] + 1$ $OD = PLL2_CLKOD_CFG[3:0] + 1$ NOTE: 1) Since $500MHz \leq 24 * NF / NR \leq 2.5GHz$, PLL VCO ranges from 500MHz to 2.5GHz. 2) Changes to these bits are valid only when pll2_freq_adj_cfg(bit[28] of PLL1 Channel Clock Control (Address: 0x0800,0004) is set to 1 or the system exits standby mode.

● SAR_ADC/SPI0/ SD_DAC Clock Control (Address: 0x0800,000C)

Bits	Attributes	Reset	Description
[31:30]	RO	2'h0	Reserved
[29]	RW/SC	1'h0	sddac_div_vld_cfg Sigma-delta DAC clock divider adjustment configuration NOTE: This bit is cleared automatically when the adjustment of Sigma-delta DAC clock divider adjustment is completed.
[28]	RW	1'h0	sddac_clk_en_cfg 1: Enable sddac_clk 0: Disable sddac_clk
[27:20]	RW	8'h09	sddac_clk_div_num_cfg[7:0] $sddac_clk = pll3_clk / (sddac_clk_div_num_cfg + 1)$ sdadc_clk frequency is 12MHz by default.
[19]	RW/SC	1'h0	spi0_hsclock_div_vld_cfg spi0_hsclock clock divider adjustment configuration NOTE: This bit is cleared automatically when the adjustment of SPI0 high speed clock divider adjustment is completed.
[18]	RW	1'h0	spi0_hsclock_en_cfg 1: Enable spi0_hsclock 0: Disable spi0_hsclock
[17:10]	RW	8'h13	spi0_hsclock_div_num_cfg[7:0] $spi0_hsclock = pll2_clk / (spi0_hsclock_div_num_cfg + 1)$ spi0_hsclock frequency is 12MHz by default. spi0 port clk = spi0_hsclock/2
[9]	RW/SC	1'h0	dphy_cfgclk_div_vld_cfg, dphy_cfgclk divider adjustment configuration NOTE: 1) This bit is cleared automatically when the adjustment of dphy clock divider adjustment is completed. 2) The acceptable frequency range is from 80MHz to 120MHz.
[8]	RW	1'h0	dphy_cfgclk_en_cfg 1: Enable dphy_cfgclk 0: Disable dphy_cfgclk NOTE: The acceptable frequency range is from 80 MHz to 120 MHz.
[7:4]	RW	4'h1	dphy_cfgclk_div_num_cfg[3:0] $dphy_cfgclk = pll2_clk / (dphy_cfgclk_div_num_cfg + 1)$ NOTE: dphy_cfgclk frequency is 120MHz by default and the acceptable frequency range is from 80 MHz to 120 MHz.
[3]	RW	1'h0	sar_adc_clk_en_cfg 1: Enable sar_adc_clk 0: Disable sar_adc_clk
[2:0]	RW	3'h5	sar_adc_clk_div_num_cfg[2:0] $sar_adc_clk = 24M / (sar_adc_clk_div_num_cfg + 1)$ sar_adc_clk frequency is 4MHz by default.

● SD_ADC/SD_DAC High Speed Clock Control (Address: 0x0800,0010)

Bits	Attributes	Reset	Description
[31:30]	RO	2'h0	Reserved
[29]	RW/SC	1'h0	sdadc_hsddiv_vld_cfg Sigma-Delta ADC high speed clock divider adjustment configuration NOTE: This bit will be cleared automatically when the adjustment of Sigma-Delta ADC high speed clock divider is completed.
[28]	RW	1'h0	sdadc_hsclock_en_cfg 1: Enable sdadc_hsclock 0: Disable sdadc_hsclock
[27:20]	RW	6'h1	sdadc_hsclock_div_num_cfg[7:0] $sdadc_hsclock = pll3_clk / (sdadc_hsclock_div_num_cfg + 1)$ sdadc_hsclock frequency is 60MHz by default.
[19]	RW/SC	1'h0	sddac_hsddiv_vld_cfg Sigma_delta DAC high speed clock divider adjustment configuration NOTE: This bit will be cleared automatically when the adjustment of Sigma-Delta DAC high speed clock divider is completed.
[18]	RW	1'h0	sddac_hsclock_en_cfg 1: Enable sddac_hsclock 0: Disable sddac_hsclock
[17:10]	RW	8'h01	sddac_hsclock_div_num_cfg[7:0] $sddac_hsclock = pll3_clk / (sddac_hsclock_div_num_cfg + 1)$ The output frequency of sddac_hsclock is 60MHz by default.
[9]	RW/SC	1'h0	sdadc_div_vld_cfg Sigma_delta ADC clock divider adjustment configuration NOTE: This bit will be cleared automatically when Sigma-delta ADC clock divider adjustment is completed.
[8]	RW	1'h0	sdadc_clk_en_cfg 1: Enable sdadc_clk 0: Disable sdadc_clk
[7:0]	RW	8'h9	sdadc_clk_div_num_cfg[7:0] $sdadc_clk = pll3_clk / (sdadc_clk_div_num_cfg + 1)$ sdadc_clk frequency is 12MHz by default.

● PLL3 Channel Clock Control (Address: 0x0800,0014)

Bits	Attributes	Reset	Description
[31]	RO	1'h0	Reserved
[30]	RW	1'h1	csi_pclk_pol csi_pclk polarity selection 0: Negative 1: Positive
[29:27]	RO	3'h0	Reserved
[26]	RW	1'h0	csi_mipi_dvp_sel_cfg CMOS Image sensor interface selection 0: DVP interface

Bits	Attributes	Reset	Description
			1: MIPI interface
[25]	RW	1'h1	csi_prst_cfg Software reset control signal corresponding to the ISP module in pclk domain. 1: Reset image capture PCLK 0: Do not reset image capture PCLK
[24]	RW	1'h0	mac_mem_icg_bypass 0: Enable memory clock gate of Ethernet MAC to save power 1: Bypass memory clock gate of Ethernet MAC
[23]	RW	1'h0	mac_speed_cfg MAC PHY speed configuration. 0: 10MHz 1: 100MHz NOTE: The MAC PHY speed should be configured before setting the related clocks of MAC/PHY interface.
[22:19]	RW	4'h5	pll3_clkod_cfg[3:0](odd and 0) PLL3 output frequency setting parameter3 NOTE: For PLL3 output frequency calculation formula, please refer to bits[12:0] of this register.
[18:13]	RW	6'h0	pll3_clkr_cfg[5:0] PLL3 output frequency setting parameter2 NOTE: 1) It is strongly recommended that the setting parameter NR of PLL be set to 1(clkr_cfg as 0), which can reduce the jitter of the PLL output clock, thereby improving the stability of DDR2. 2) For PLL3 output frequency calculation formula, please refer to bits[12:0] of this register.
[12:0]	RW	13'h1D	pll3_clkf_cfg[12:0] PLL3 output frequency setting parameter1 By default, PLL3 output frequency is 120MHz and PLL3 output frequency calculation formula: $PLL3\ output\ frequency = 24M * NF / NR / OD$ $NF = PLL2_CLKF_CFG[12:0] + 1$ $NR = PLL2_CLKR_CFG[5:0] + 1$ $OD = PLL2_CLKOD_CFG[3:0] + 1$ NOTE: 1) Since $500MHz \leq 24 * NF / NR \leq 2.5GHz$, PLL VCO ranges from 500MHz to 2.5GHz. 2) Changes to these bits are valid only when pll3_freq_adj_cfg(bit[29] of PLL1 Channel Clock Control(Address: 0x0800,0004) is set to 1 or the system exits standby mode.

● CSI_SCLK/MAC_OPCLK Clock Control (Address: 0x0800,0018)

Bits	Attributes	Reset	Description
[31:20]	RO	12'h49	Reserved

Bits	Attributes	Reset	Description
[19]	RW	1'h0	csi_sclk_en_cfg 0: Disable csi_sclk 1: Enable csi_sclk
[18:10]	RO	9'h49	Reserved
[9]	RW/SC	1'h0	mac_opclk_div_vld_cfg MAC_OPCLK divider adjustment configuration Notes: This bit will be cleared automatically when the adjustment of opclk divider is completed.
[8]	RW	1'h0	mac_opclk_en_cfg 1: Enable MAC_OPCLK 0: Disable MAC_OPCLK
[7:6]	RO	2'h0	Reserved
[5:0]	RW	6'h13	mac_opclk_div_num_cfg[5:0] MAC_OPCLK = pll2_clk/(mac_opclk_div_num_cfg+1) The OPCLK frequency is 12MHz by default.

● Clock Gate Control (Address: 0x0800,001C)

Bits	Attributes	Reset	Description
[31:26]	RO	6'h0	Reserved
[25]	RW	1'h1	ddrphy_clk_gat_cfg 0: Enable DPHYCLK(DDR PHY)
[24]	RW	1'h1	dram_clk_gat_cfg 0: Enable the DCLK 1: Disable the DCLK NOTE: DCLK is the working clock of the following modules: DRAM Controller, DMA Asynchronous Buffer 1~5, AHB2DDR Bridge and Register Bus 1.
[23]	RO	1'h1	Reserved
[22]	RW	1'h1	gclk_gat_cfg[22] 0: Enable the working clock for TWI2 1: Disable the working clock for TWI2
[21]	RW	1'h1	gclk_gat_cfg[21] 0: Enable the working clock for MMC/SD/SDIO Controller2 1: Disable the working clock for MMC/SD/SDIO Controller2
[20]	RW	1'h1	gclk_gat_cfg[20] 0: Enable the working clock for SPI2 1: Disable the working clock for SPI2
[19]	RO	1'h1	Reserved
[18]	RO	1'h0	gclk_gat_cfg[18] 0: Enable the working clock for UART2 1: Disable the working clock for UART2
[17]	RW	1'h1	gclk_gat_cfg[17] 0: Enable the working clock for TWI1 1: Disable the working clock for TWI1
[16]	RW	1'h1	gclk_gat_cfg[16]

Bits	Attributes	Reset	Description
			0: Enable the working clock for MIPI controller 1: Disable the working clock for MIPI controller
[15]	RW	1'h1	gclk_gat_cfg[15] 0: Enable the working clock for USB Controller 1: Disable the working clock for USB Controller
[14]	RW	1'h1	gclk_gat_cfg[14] 0: Enable the working clock for Encryption module 1: Disable the working clock for Encryption module
[13]	RW	1'h1	gclk_gat_cfg[13] 0: Enable the working clock for MAC, DMA Arbiter and DMA Asynchronous Buffer 2 1: Disable the working clock for MAC, DMA Arbiter and DMA Asynchronous Buffer 2
[12]	RW	1'h1	gclk_gat_cfg[12] 0: Enable the working clock for GPIO 1: Disable the working clock for GPIO
[11]	RW	1'h1	gclk_gat_cfg[11] 0: Enable the working clock for L2 Buffer1 1: Disable the working clock for L2 Buffer1
[10]	RW	1'h1	gclk_gat_cfg[10] 0 = To enable the working clock for TWIO 1 = To disable the working clock for TWIO
[9]	RW	1'h1	gclk_gat_cfg[9] 0: Enable the working clock for L2 Buffer0 1: Disable the working clock for L2 Buffer0
[8]	RW	1'h1	gclk_gat_cfg[8] 0: Enable the working clock for UART1 1: Disable the working clock for UART1
[7]	RW	1'h1	gclk_gat_cfg[7] 0: Enable the working clock for UART0 1: Disable the working clock for UART0
[6]	RW	1'h1	gclk_gat_cfg[6] 0: Enable the working clock for SPI1 1: Disable the working clock for SPI1
[5]	RW	1'h1	gclk_gat_cfg[5] 0: Enable the working clock for SPI0 1: Disable the working clock for SPI0
[4]	RW	1'h1	gclk_gat_cfg[4] 0: Enable the working clock for Sigma-Delta DAC Controller 1: Disable the working clock for Sigma-Delta DAC Controller
[3]	RW	1'h1	gclk_gat_cfg[3] 0: Enable the working clock for Sigma-Delta ADC Controller0 1: Disable the working clock for Sigma-Delta ADC Controller0
[2]	RW	1'h1	gclk_gat_cfg[2]

Bits	Attributes	Reset	Description
			0: Enable the working clock for MMC/SD/SDIO Controller 1 1: Disable the working clock for MMC/SD/SDIO Controller 1
[1]	RW	1'h1	gclk_gat_cfg[1] 0: Enable the working clock for MMC/SD/SDIO Controller 0 1: Disable the working clock for MMC/SD/SDIO Controller 0
[0]	RO	1'h0	gclk_gat_cfg[0] Software can't write and GCLK[0] is always opened.

● USB Operate Mode and I2S Mode Control (Address: 0x0800,0058)

Bits	Attributes	Reset	Description
[31]	RO	1'h0	Reserved
[30]	RW	1'h0	i2s_mclk_sel_cfg 0: Select sdadc_clk to be i2s_mclk 1: Select sddac_clk to be i2s_mclk
[29]	RW	1'h0	i2sm_adc_mode_enabled 1: To enable external Sigma-Delta ADC (via I2S master mode). 0: To disable external Sigma-Delta ADC (via I2S master mode).
[28]	RW	1'h0	i2sm_mode enabled 1: To enable external Sigma-Delta DAC(via I2S master mode). 0: To disable external Sigma-Delta DAC(via I2S master mode).
[27]	RW	1'h0	i2ssr_mode enabled 1: To enable external Sigma-Delta ADC (via I2S slave mode). 0: To disable external Sigma-Delta ADC (via I2S slave mode).
[26]	RW	1'h0	i2sst_mode enabled 1: To enable external Sigma-Delta DAC (via I2S slave mode). 0: To disable external Sigma-Delta DAC (via I2S slave mode).
[25]	RW	1'h0	i2si_mode enabled 1: To enable internal Sigma-Delta DAC (via I2S master mode). 0: To disable nternal Sigma-Delta DAC (via I2S master mode).
[24:19]	RO	6'h0	Reserved
[18]	RW	1'h0	usbphy_bist_mode 0: Normal operation mode of USB PHY 1: Bist test mode of USB PHY

Bits	Attributes	Reset	Description
[17]	RO	1'h0	otg_phy_bist_ok USB PHY bist test result flag 1: otg phy test pass 0: No instruction
[16:15]	RO	2'h0	usb_linestate_wp USB Host/Slave line state These signals reflect the current state of the single ended receivers. They directly reflect the current state of the DP and DM signals: 2'h0: SE0 2'h1: 'J' State 2'h2: 'K' State 2'h3: SE1
[14]	RW	1'h0	usb_dp_pu_en 1: Enable DP pull up. 0: Disable DP pull up.
[13:12]	RW	2'h0	usb_id_cfg When USB is used as a device, these bits must be set to 2'h3. When USB is used as host, these bits must be set to 2'h1.
[11:6]	RW	6'h0	usb_phy_cfg When USB is used as a device, these bits must be set to 6'h17. When USB is used as host, these bits must be set to 6'h1F.
[5]	RO	1'h0	Reserved
[4]	RW	1'h0	otg_tm1_cfg , only use for simulation NOTE: This bit can not be set to 1.
[3]	RW	1'h1	usb_phy_ponrst 0: To power on reset the USB transceiver. 1: No instruction
[2]	RW	1'h0	usb_sus_cfg 0 : To suspend the USB transceiver of USB 2.0 HS controller. 1 : In host mode, when bit[1] of <i>Power Control Register (Address: 0x2020,0001)</i> is set, USB controller and USB transceiver will be suspended. In client mode, bit[1] of <i>Power Control Register (Address: 0x2020,0001)</i> indicates current status of USB controller. - When bit[1] of <i>Power Control Register (Address: 0x2020,0001)</i> is cleared, it indicates that the USB controller and USB transceiver are not suspended. - When bit[1] of <i>Power Control Register (Address: 0x2020,0001)</i> is set, it indicates that the USB controller and USB transceiver are suspended.
[1]	RW	1'h0	usb_phy_pll_en 1: To enable the PLL of USB PHY.

Bits	Attributes	Reset	Description
			0: To disable the PLL of USB PHY.
[0]	RW	1'h0	USB_PHY_rst 1: To reset the transceiver of USB 2.0 HS controller. 0: To not reset the transceiver of USB 2.0 HS controller.

● PLL1 Bandwidth Adjustment Configuration (Address: 0x0800,00DC)

Bits	Attributes	Reset	Description
[31:20]	RW	12'hE	pll1_bwadj_cfg[11:0] PLL1 output frequency setting parameter4 In most cases, PLL1_BWADJ_CFG = (PLL1_CLKF_CFG+1)/2-1.
[19:0]	RO	20'h18	Reserved

● PLL2/PLL3 Bandwidth Adjustment Configuration (Address: 0x0800,00E0)

Bits	Attributes	Reset	Description
[31:24]	RO	8'h0	Reserved
[23:12]	RW	12'hE	pll3_bwadj_cfg[11:0] PLL3 output frequency setting parameter4 In most cases, PLL3_BWADJ_CFG = (PLL3_CLKF_CFG+1)/2-1.
[11:0]	RW	12'h13	pll2_bwadj_cfg[11:0] PLL2 output frequency setting parameter4 In most cases, PLL2_BWADJ_CFG = (PLL2_CLKF_CFG+1)/2-1.

● ISP/NPU/ENC VCLK Clock Source Configuration (Address: 0x0800,011C)

Bits	Attributes	Reset	Description
[31:24]	RO	8'h0	Reserved
[23]	RW	1'h1	enc_vclk_rst_cfg 1: Reset the module driven by enc_vclk 0: Not reset the module driven by enc_vclk
[22]	RW	1'h0	enc_src_clk_en_cfg 1: Enable enc_vclk 0: Disable enc_vclk
[21:19]	RW	3'h2	enc_src_clk_sel_cfg[2:0] ENC_VCLK clock source selection 3'h0: PLL1 Divide Clock 3'h1: PLL2 Divide Clock 3'h2: PLL3 Divide Clock 3'h4: PLL1 Output Clock 3'h5: PLL2 Output Clock 3'h6: PLL3 Output Clock Others: No clock source is in use.
[18:16]	RW	3'h1	enc_src_clk_div_num_cfg[2:0] enc_vclk=pll_clk/(enc_src_clk_div_num_cfg+1) enc_vclk frequency is 60MHz by default.
[15]	RW	1'h1	npu_vclk_rst_cfg 1: Reset the module driven by npu_vclk 0: Not reset the module driven by npu_vclk

Bits	Attributes	Reset	Description
[14]	RW	1'h0	npu_src_clk_en_cfg 1: Enable npu_vclk 0: Disable npu_vclk
[13:11]	RW	3'h0	npu_src_clk_sel_cfg[2:0] NPU_VCLK clock source selection 3'h0: PLL1 Divide Clock 3'h1: PLL2 Divide Clock 3'h2: PLL3 Divide Clock 3'h4: PLL1 Output Clock 3'h5: PLL2 Output Clock 3'h6: PLL3 Output Clock Others: No clock source is in use.
[10:8]	RW	3'h1	npu_src_clk_div_num_cfg[2:0] $\text{npu_vclk} = \text{pll_clk} / (\text{npu_src_clk_div_num_cfg} + 1)$ npu_vclk frequency is 60MHz by default.
[7]	RW	1'h1	isp_vclk_rst_cfg 1: Reset the module driven by isp_vclk 0: Not reset the module driven by isp_vclk
[6]	RW	1'h0	isp_src_clk_en_cfg 1: Enable isp_vclk 0: Disable isp_vclk
[5:3]	RW	3'h2	isp_src_clk_sel_cfg[2:0] ISP_VCLK clock source selection 3'h0: PLL1 Divide Clock 3'h1: PLL2 Divide Clock 3'h2: PLL3 Divide Clock 3'h4: PLL1 Output Clock 3'h5: PLL2 Output Clock 3'h6: PLL3 Output Clock Others: No clock source is in use.
[2:0]	RW	3'h1	isp_src_clk_div_num_cfg[2:0] $\text{isp_vclk} = \text{pll_clk} / (\text{isp_src_clk_div_num_cfg} + 1)$ isp_vclk frequency is 60MHz by default.

4.2.6 Programmer's Guide

4.2.6.1 PLLs Configuration Restrictions

AK3918AV100 series processors support three PLLs. Programmers should pay attention to the restrictions in the following chapters according to the requirements of each PLL and the implementation of the clock tree.

4.2.6.1.1 Restrictions of PLL1 and Related Clock Tree

- PLL parameters setting requirements:
 $500\text{MHz} \leq 24 * \text{NF} / \text{NR} \leq 2.5\text{GHz}$,
 In PLL1's working mode 1(CPU2X MEM1X): $\text{pll_clk} \leq 900\text{MHz}$;
 In PLL1's working mode 2(CPU1X MEM1X): $\text{pll_clk} \leq 444\text{MHz}$;
- It is strongly recommended that the setting parameter NR of PLL1 be set to 1, which can reduce the jitter of the PLL output clock, thereby improving the stability of DDR2.

- 3) Mode switching and frequency changing cannot be configured simultaneously.
- 4) The clock tree of PLL1 support 2 working modes. Before switching between mode1 and mode2, please ensure that working frequencies of each clock do not exceed the maximum allowed by the new mode.

4.2.6.1.2 Restrictions of PLL2 and Related Clock Tree

PLL2 parameters setting requirements

$500\text{MHz} \leq 24 * \text{NF} / \text{NR} \leq 2.5\text{GHz}$,

$\text{pll_clk} \leq 800\text{MHz}$, the maximum value depends on the MODE and VDD.

4.2.6.1.3 Restrictions of PLL3 and Related Clock Tree

PLL3 parameters setting requirements

$500\text{MHz} \leq 24 * \text{NF} / \text{NR} \leq 2.5\text{GHz}$,

$\text{pll_clk} \leq 800\text{MHz}$, the maximum value depends on the MODE and VDD.

4.2.6.2 PLLs Configuration

Programmers should operate according to the following steps.

4.2.6.2.1 Adjusting PLL1

The steps of adjusting PLL1 are described as follows:

- 1) Read the value of `cpu_freq_adj_cfg` and wait until it becomes 0.
- 2) Set the frequency parameters of the PLL1, including `cpu_pll_m_cfg`, `cpu_pll_n_cfg` and `cpu_pll_od_cfg`.
- 3) Set `cpu_freq_adj_cfg` to 1.

4.2.6.2.2 Adjusting PLL2

The steps of adjusting PLL2 are described as follows:

- 1) Read the value of `core_freq_adj_cfg` and wait until it becomes 0.
- 2) Set the frequency parameters of the PLL2, including `core_pll_m_cfg`, `core_pll_n_cfg` and `core_pll_od_cfg`.
- 3) Set `core_freq_adj_cfg` to 1.

4.2.6.2.3 Adjusting PLL3

The steps of adjusting PLL3 are described as follows:

- 1) Read the value of `audio_pll_freq_adj_cfg` and wait until it becomes 0.
- 2) Set the frequency parameters of the PLL3, including `audio_pll_ref_div_m_cfg` and `audio_pll_fb_div_n_cfg`.
- 3) Set `audio_pll_freq_adj_cfg` to 1.

4.2.6.3 Clock Divider Configuration

There are several clock dividers in the clock tree of the PLLs. Different types of dividers may have different characteristics and requirements.

No.	Clock Name	Divider Type	Calculation Formula
1	gclk	Special Even Divider	$1/(2(N+1))$
2	mac_opclk	Even Divider	$1/(N+1)$
3	dphy_cfgclk	Even Divider	$1/(N+1)$
4	spi0_hs_clk	Integer Divider	$1/(N+1)$

No.	Clock Name	Divider Type	Calculation Formula
5	sdadc_hscclk	Integer Divider	$1/(N+1)$
6	sddac_hscclk	Integer Divider	$1/(N+1)$
7	sdadc_clk	Integer Divider	$1/(N+1)$
8	sddac_clk	Integer Divider	$1/(N+1)$
9	sar_adc_clk	Integer Divider	$1/(N+1)$
10	isp_vclk	MS Integer Divider	$1/(N+1)$
11	npu_vclk	MS Integer Divider	$1/(N+1)$
12	enc_vclk	MS Integer Divider	$1/(N+1)$

4.2.6.3.1 Adjusting Clock Divider Frequency

The calculation formula for the integer, even and multiple source (MS) divider output clock frequency is the same, which is $OUT_CLK = REF_CLK / (N+1)$. But even divider requires that the division parameter N must be an odd number. Multiple source divider is finished by hardware automatically, whose status cannot be read by software.

- For those clocks with even divider, the configuration steps are similar. Taking mac_opclk as an example, the steps of adjusting mac_opclk are as follows:

- 1) Disable the module driven by mac_opclk(Ethernet MAC).
- 2) Enable the output clock when mac_opclk_div_vld_cfg(bit[9] of [CSI_SCLK/MAC_OPCLK Clock Control \(Address: 0x0800,0018\)](#) is 0 by setting mac_opclk_en_cfg(bit[8] of the same register) to be 1.
- 3) Change the divider coefficient mac_opclk_div_num_cfg(bits[5:0] of this register).
- 4) Set mac_opclk_div_vld_cfg(bit[9] of this register) to be 1 and wait until it is cleared.
- 5) Enable the module driven by mac_opclk(Ethernet MAC).

- For those clocks with integer divider, the configuration steps are similar. Taking sdadc_clk as an example, the steps of adjusting sdadc_clk are as follows:

- 1) Disable the module driven by sdadc_clk(Sigma-Delta ADC).
- 2) Set sdadc_clk_en_cfg(bit[8] of [SD_ADC/SD_DAC High Speed Clock Control\(Address: 0x0800,0010\)](#) to be 0 to disable the output clock when sdadc_clk_div_vld_cfg(bit[9] of the same register) is 0.
- 3) Change the divider coefficient sdadc_clk_div_num_cfg(bits[7:0] of this register).
- 4) Set sdadc_clk_div_vld_cfg(bit[9] of the same register) to be 1 and wait until it is cleared.
- 5) Set sdadc_clk_en_cfg(bit[8] of this register) to be 1 to enable the output clock.
- 6) Enable the module driven by sdadc_clk(Sigma-Delta ADC).

- For those clocks with MS divider, the configuration steps are similar. Taking isp_vclk as an example, the steps of adjusting isp_vclk are as follows:

- 1) Disable the module driven by isp_vclk(ISP module).
- 2) Set isp_src_clk_sel_cfg(bits[5:3] of [ISP/NPU/ENC VCLK Clock Source Configuration \(Address: 0x0800,011C\)](#) accordingly. If the clock source to be selected needs to be adjusted by frequency divider, programmers should configure the divider coefficient isp_src_clk_div_num_cfg(bits[2:0] of the same register).
- 3) Set isp_src_clk_en_cfg(bit[6] of this register) to be 1 to enable the output clock.
- 4) If frequency of ISP is needed, programmers need clear isp_src_clk_en_cfg first, turn off isp_vclk after several clock cycles and then repeat step1) ~3).

4.2.6.3.2 Frequency of `sddac_hclk`

The method of adjusting the `sddac_hclk` is the same as described in previous chapter, while the clock has the following requirements according to different sampling rates.

Sampling Rate	Frequency Range of <code>sddac_hclk</code>	Recommended frequency of <code>sddac_hclk</code>
96KHz	≥66.7MHz	72MHz
48KHz	≥33.3MHz	36MHz
24KHz	≥16.6MHz	18MHz
16KHz	≥11.1MHz	12MHz
8KHz	≥5.6MHz	6MHz

NOTE

`sddac_hclk` is used for Sigma-Delta DAC Filter. To handle Sigma-Delta DAC Filter's input data, at least $320 + 180 \times 2 + 106 = 690$ system clock cycles are needed. For Example, if the system clock is 70MHz, for Sigma-Delta DAC Filter's input data rate of 96KHz, $70\text{MHz}/96\text{KHz} = 729$, there are 729 system clock cycles can be used to handle one input data for interpolator.

4.2.6.3.3 Adjusting `sar_adc_clk`

This clock has no `div_vld_cfg` signal for frequency adjustment. Programmers should follow the steps below to adjust the `sar_adc_clk`.

1. Disable SAR ADC module.
2. Set '`sar_adc_clk_en_cfg`' to be 0 to disable the output clock
3. Change the divider coefficient '`sar_adc_clk_div_num_cfg`'
4. Set '`sar_adc_clk_en_cfg`' to be 1 to enable the output clock
5. Enable the module driven by this output clock

4.3 Chip ID

AK3918AV100 series processors are identified by its chip ID register, indicating its product number.

- Chip ID (Address: 0x0800,0000)

Bits	Attribute	Reset	Description
[31:8]	RO	0x535335	Chip ID The Chip ID is fixed as 0x535335.
[7:0]	RO	-	reserved NOTE: The reset value depends on the configuration.

4.4 Bootstrap Module

4.4.1 Overview

AK3018AV100 processor supports normal boot and secure boot to meet different application needs.

4.4.2 Features

Features of bootstrap module are as follows:

- Supports normal boot & secure boot:
 - SPI NOR Flash Boot
 - SPI NAND Flash Boot
 - eMMC/MMC/SD Boot

- USB Mass Storage Boot
- UART Boot

4.4.3 Block Diagram

Once powered on, AK3018AV100 processor will enter normal boot mode or secure boot mode based on security level. In normal boot mode, the bootstrap reads the value of GPIO[46]. When GPIO[46] is 1, the chip will boot from SPI NOR Flash/SPI NAND Flash. When GPIO[46] is 0, it will boot from eMMC/MMC/SD. When both SPI boot and eMMC/MMC/SD boot fail, the bootstrap will first try UART boot and then USB boot. The boot flow is shown in the following chart. For more details about security level and secure boot flow, please refer to *Secure Boot User Guide* (《安全启动使用指南》).

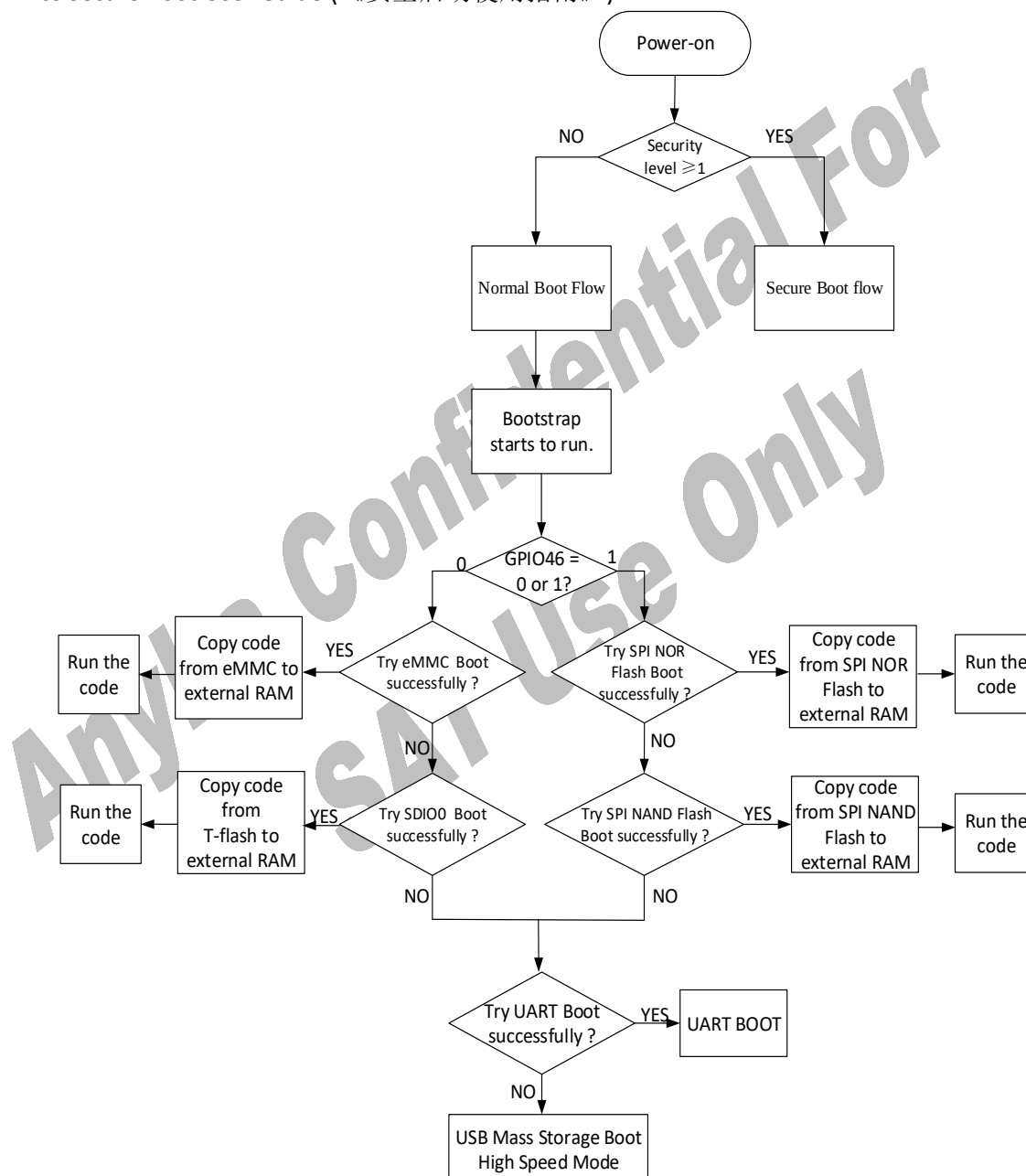


Figure 4-13 AK3918AV100 Series Processors Boot Flow

4.4.4 Register Definition

- RTC Control 1 and Boot Mode Flag (Address: 0x0800,0054)

Bits	Attributes	Reset	Description
[31:24]	RW	8'h0	Display current boot mode 8'h0: SPI0 - SPI NOR Flash BOOT 8'h1: SPI0 - SPI NAND Flash BOOT 8'h2: USB MASS STORAGE BOOT 8'h3: UART BOOT 8'h4: eMMC/MMC/SD BOOT Others: Reserved
[23:14]	RO	10'h0	Reserved
[13:0]	RO	14'h0	rtc_rcv_data The data to be read from RTC module.

4.4.5 Programmer's Guide

4.4.5.1 SPI NOR Flash Boot

Consecutive-read command format should be: 0x03 + Address Period 0 + [Address Period 1] + Address Period 2 + Address Period 3. (The content in square brackets is optional).

Configuration of Serial Flash is as follows:

Address	Value	Description
0x00 ~ 0x03	-	Reserved
0x04 ~ 0x0B	ANYKA322	Password: ANYKA322
0x0C ~ 0x0F	0x00~0xFFFFFFFF	Read data length (LSB first, unit: byte) NOTE: Programmers can firstly set any value they need to compile programs, then revise the value according to the length of BIN file, and compile the programs again.
0x10 ~ 0x13	0x00~0x7FFF	clk_div: SPI0 clock divider spi0 port clk = PLL2/(clk_div+1)/2
0x14 ~ 0x17	-	Reserved
0x18 ~ 0x1B	0x0/0x1/0x2	SPI0 read mode selection 0x0: Standard SPI (1-wire) 0x1: Dual SPI (2-wire) 0x2: Quad SPI (4-wire) NOTE: Before selecting one of these modes, please confirm that SPI NOR Flash supports this mode.
0x1C ~ 0x1F	0x00000006	Use the external RAM for SPI BOOT system. The unique value is 0x00000006.
0x20 ~ 0x23	user code copy address	Users can configure this address value so that the boot system can recognize the address to which the code should be copied.
0x24 ~ 0x27	0x0/0x1	Data transmission mode 0: CPU transmission mode 1: DMA transmission mode

Address	Value	Description
0x28 ~ 0x2B	-	Reserved
0x2C ~ 0x1FF	external RAM initialization sequence	The boot system will copy the user code from SPI NOR Flash to external RAM, so the external RAM must be initialized first. The initializing sequence is stored in these addresses. Unit: us NOTE: 1) The boot system considers the first word (0x2C~0x2F) as the register address, the second word (0x30~0x33) as the register value, the third word as the register address, the fourth word as the register value, and so on. The boot system will read all these data and set the value of the corresponding register in order. 2) When the boot system reads the value 0x88888888, it means the external RAM configuration is completed. 3) If the boot system reads the register address 0x66668888, it means the external RAM configuration will be delayed for a certain time configured by register value.

4.4.5.2 SPI NAND Flash Boot

Configuration of SPI NAND Flash is as follows:

Address	Value	Description
0x00 ~ 0x03	-	Reserved
0x04 ~ 0x0B	ANYKA322	Password: ANYKA322
0x0C ~ 0x0F	0x0	Must be 0x0 for SPI NAND Flash
0x10 ~ 0x13	0x00 ~ 0x7FFF	clk_div: SPI0 clock divider spi0 port clk = PLL2/(clk_div+1)/2
0x14	0x1 / 0x4 / 0x8	Page data size = value × 512B 0x1: Page data size = 1*512B 0x4: Page data size = 4*512B 0x8: Page data size = 8*512B
0x15 ~ 0x16	-	Reserved
0x17	0x00 ~ 0xFF	Corresponding to SPI NAND Cache Read Front dummy count eg. 0x00: no dummy 0x01: add 1*8 clock dummy

Address	Value	Description
		0x02: add 2*8 clock dummy 0xFF: add 255*8 clock dummy
0x18 ~ 0x1B	0x0 / 0x1 / 0x2	SPI0 read mode selection 0x0: Standard SPI(1-wire) 0x1: Dual SPI(2-wire) 0x2: Quad SPI(4-wire)
0x1C ~ 0x1F	0x0 ~ 0xFFFFFFFF	Corresponding to SPI NAND Flash spec parameter TRD. Unit: us
0x20 ~ 0x23	user code copy address	The user could configure this address value so that the boot system can recognize the address to which the code should be copied.
0x24 ~ 0x27	0x0/0x1	Data transmission mode: 0: CPU transmission mode 1: DMA transmission mode.
0x28 ~ 0x2B	0x01 ~ 0xFFFFFFFF	Pages count that will be read from SPI NAND Flash.
0x2C ~ 0x33	-	Reserved
0x34 ~ 0x1FF	external RAM initialization sequence	Boot system will copy the user code from SPI NAND Flash to external RAM, so the external RAM must be initialized first. The initializing sequence is stored in these addresses. Unit: us NOTE: 1) The boot system considers the first word (0x34~0x37) as the register address, the second word (0x38~0x3B) as the register value, the third word as the register address, the fourth word as the register value, and so on. The boot system will read all these data and set the value of the corresponding register in order. 2) When the boot system reads the value 0x88888888, it means the external RAM configuration is completed. 3) If the boot system reads the register address 0x66668888, it means the external RAM configuration will be delayed for a certain time configured by register value.

4.4.5.3 eMMC/MMC/SD Boot

Configuration of eMMC/MMC/SD is as follows:

Address	Value	Description
0x00 ~ 0x03	-	Reserved

Address	Value	Description
0x04 ~ 0x0B	ANYKA322	Password: ANYKA322
0x0C	-	Reserved
0x0D	0x00 ~ 0xFF	The eMMC/MMC/SD controller's clock divider corresponds to the value of bits[15:0] in Clock Control Register (Add: 0x2010, X004)
0x0E	0x1/0x4/0x8	The data bus width value of eMMC/MMC/SD. NOTE: 1) MMC interface supports 1/4/8 data line bus width 2) SDIO0 interface supports 1/4 data line bus width
0x0F	0x0/0x1/0x2	Data transmission mode: 0x0: Use CPU single block read mode to copy data from eMMC/MMC/SD to the external RAM. 0x1: Use DMA single block read mode to copy data from eMMC/MMC/SD to the external RAM. 0x2: Use DMA multi block read mode to copy data from eMMC/MMC/SD to the external RAM.
0x10 ~ 0x13	0x0 ~ 0xFFFFFFFF	The Data Timer Register value of eMMC/MMC/SD controller corresponds to the value of bits[31:0] in Data Timer Register (Add: 0x2010, X024)
0x14 ~ 0x17	0x0 ~ 0xFFFFFFFF	The block index with which the system will begin reading data from eMMC/MMC/SD. e.g. When this field is 8, it means the boot system will read data from eMMC/MMC/SD Flash card, beginning from the eighth block.
0x18 ~ 0x1B	0x0 ~ 0xFFFFFFFF	The number of blocks that will be read from eMMC/MMC/SD.
0x1C ~ 0x1F	0x00000006	Use the external RAM for eMMC/MMC/SD to boot the system. The unique value is 0x00000006.
0x20 ~ 0x23	user code copy address	The user can configure this address value so that the boot system can recognize the address to which the code should be copied.
0x24 ~ 0x1FF	external RAM initialization sequence	The boot system will copy the user code from eMMC/MMC/SD to external RAM, so the external RAM must be initialized first. The initializing sequence is stored in these addresses.

Address	Value	Description
		Unit: us NOTE: 1) The boot system considers the first word (0x24~0x27) as the register address, the second word (0x28~0x2B) as the register value, the third word as the register address, the fourth word as the register value, and so on. The boot system will read all these data and set the value of the corresponding register in order. 2) When the boot system reads the value 0x88888888, it means the external RAM configuration is completed. 3) If the boot system reads the register address 0x66668888, it means the external RAM configuration will be delayed for a certain time configured by register value.

4.5 CPU Core

4.5.1 Overview

The CPU core is a microprocessor named ARM926EJ-S from ARM Limited. The ARM926EJ-S processor is targeted at multi-tasking applications where full memory management, high performance, low die size, and low power are all important. The ARM926EJ-S processor supports the 32-bit ARM and 16-bit Thumb instruction sets, enabling the user to trade off between high performance and high code density. The ARM926EJ-S processor includes features for efficient execution of Java byte codes, providing Java performance similar to JIT, but without the associated code overhead. The ARM926EJ-S processor supports the ARM debug block diagram and includes logic to assist in both hardware and software debug.

4.5.2 Features

The ARM926EJ-S processor has a Harvard cached block diagram and provides a complete high-performance processor subsystem, including:

- An ARM926EJ-S integer core
- A Memory Management Unit (MMU)
- Separate instruction and data AMBA AHB bus interfaces

For more details on ARM926EJ-S, please refer to the technical manuals of ARM Limited.

4.6 Interrupt Module

4.6.1 Overview

The interrupt module collects interrupt requests from the core & analog module and provides an interface to the CPU.

4.6.2 Features

Features of interrupt module are as follows:

- Supports IRQ and FIQ interrupt
- Supports interrupt mask/not mask function
- Supports multiple interrupt sources

4.6.3 Architecture

CPU core responds to the interrupts according to priority and mask, when the CPU receives multiple interrupts, it should acknowledge the interrupts are masked or not and then decide which interrupt gets the higher priority the CPU will respond firstly.

4.6.3.1 Interrupt Priority

The interrupt module supports both fast and normal interrupts, when multiple interrupt requests are sent at the same time, the fast interrupts get the higher priority.

4.6.3.2 Interrupt Mask

The interrupt module can mask or unmask every interrupt source of normal and fast interrupts with specific bit of mask registers. Only the interrupt requests from unmasked interrupt source are responded. The core keeps on sending the interrupt request until it is informed by software that its request has been responded. The interrupt request can be generated from various sources which are listed in the following table.

Table 4-1 IRQ/FIQ Sources

IRQ/FIQ Sources	Interrupt Name
Peripherals	UART[2:0]
	TWI[2:0]
	SPI[2:0]
	Ethernet MAC
	MMC/SD/SDIO Controller[2:0]
	USB DMA/CPU
	GPIO
Memory	L2 Buffer[1:0]
	Memory Controller
Multimedia	Image Capture
	CIS[1:0] (Including the CIS error interrupt)
	Video Processor
	Encryption
	NPU
System	System Control
Analog	Sigma-Delta ADC Controller
	Sigma-Delta DAC Controller

4.6.4 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Register Address	Register Name
1	0x0800,0024	<i>IRQ Enable Configuration</i>
2	0x0800,0028	<i>FIQ Enable Configuration</i>

Register No.	Register Address	Register Name
3	0x0800,002C	TOP_INT Interrupt Mask
4	0x0800,0030	TOP_INT Interrupt Status
5	0x0800,004C	Interrupt Status

4.6.5 Register Definition

- IRQ Enable Configuration (Address: 0x0800,0024)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	Reserved
[30]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request when an error occur in CSI. 1: To unmask normal interrupt request when an error occur in CSI.
[29]	RO	1'h0	Reserved
[28]	RW	1'h0	irq_en_cfg 0: To mask normal CSI interrupt request. 1: To unmask normal CSI interrupt request.
[27]	RO	1'h0	Reserved
[26]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from NPU. 1: To unmask normal interrupt request from NPU.
[25]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from L2 Buffer1. 1: To unmask normal interrupt request from L2 Buffer1.
[24]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from UART2. 1: To unmask normal interrupt request from UART2.
[23]	RW	1'h0	irq_en_cfg 0: To mask normal SD plugged-in or plugged-off interrupt request. 1: To unmask normal SD plugged-in or plugged-off interrupt request.
[22]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from MMC/SD/SDIO Controller 2. 1: To unmask normal interrupt request from MMC/SD/SDIO Controller 2.
[21]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from TWI2. 1: To unmask normal interrupt request from TWI2.
[20]	RW	1'h0	irq_en_cfg 0: To mask normal TWI1 Interrupt request. 1: To unmask normal TWI1 Interrupt request.
[19]	RW	1'h0	irq_en_cfg 0: To mask normal USB DMA interrupt request. 1: To unmask normal USB DMA interrupt request.

Bits	Attributes	Reset	Description
[18]	RW	1'h0	irq_en_cfg 0: To mask normal USB CPU interrupt request. 1: To unmask normal USB CPU interrupt request.
[17]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from Encryption. 1: To unmask normal interrupt request from Encryption.
[16]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from Ethernet MAC. 1: To unmask normal interrupt request from Ethernet MAC.
[15]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from GPIO. 1: To unmask normal interrupt request from GPIO.
[14]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from SPI2. 1: To unmask normal interrupt request from SPI2.
[13]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from TWIO. 1: To unmask normal interrupt request from TWIO.
[12]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from L2 Buffer0. 1: To unmask normal interrupt request from L2 Buffer0.
[11]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from UART1. 1: To unmask normal interrupt request from UART1.
[10]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from UART0. 1: To unmask normal interrupt request from UART0.
[9]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from SPI1. 1: To unmask normal interrupt request from SPI1.
[8]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from SPI0. 1: To unmask normal interrupt request from SPI0.
[7]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from Sigma-Delta ADC Controller. 1: To unmask normal interrupt request from Sigma-Delta ADC Controller.
[6]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from Sigma-Delta DAC Controller. 1: To unmask normal interrupt request from Sigma-Delta DAC Controller.
[5]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from MMC/SD/SDIO Controller 1. 1: To unmask normal interrupt request from MMC/SD/SDIO Controller 1

Bits	Attributes	Reset	Description
[4]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from MMC/SD/SDIO Controller 0. 1: To unmask normal interrupt request from MMC/SD/SDIO Controller 0.
[3]	RW	1'h0	irq_en_cfg 0: To mask normal interrupt request from system control module. 1: To unmask normal interrupt request from system control module.
[2]	RW	1'h0	irq_en_cfg 1: To unmask normal interrupt request from Video processor. 0: To mask normal interrupt request from Video processor.
[1]	RW	1'h0	irq_en_cfg 1: To unmask normal interrupt request from Image capture. 0: To mask normal interrupt request from Image capture.
[0]	RW	1'h0	irq_en_cfg 1: To unmask normal interrupt request from memory controller. 0: To mask normal interrupt request from memory controller.

● FIQ Enable Configuration (Address: 0x0800,0028)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	Reserved
[30]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request when an error occur in CSI. 1: To unmask fast interrupt request when an error occur in CSI.
[29]	RO	1'h0	Reserved
[28]	RW	1'h0	fiq_en_cfg 0: To mask fast CSI interrupt request. 1: To unmask fast CSI interrupt request.
[27]	RO	1'h0	Reserved
[26]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from NPU. 1: To unmask fast interrupt request from NPU.
[25]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from L2 Buffer1. 1: To unmask fast interrupt request from L2 Buffer1.
[24]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from UART2. 1: To unmask fast interrupt request from UART2.
[23]	RW	1'h0	fiq_en_cfg 0: To mask fast SD plugged-in or plugged-off interrupt request.

Bits	Attributes	Reset	Description
			1: To unmask fast SD plugged-in or plugged-off interrupt request.
[22]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from MMC/SD/SDIO controller 2. 1: To unmask fast interrupt request from MMC/SD/SDIO controller 2.
[21]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from TWI2. 1: To unmask fast interrupt request from TWI2.
[20]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from TWI1. 1: To unmask fast interrupt request from TWI1.
[19]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from USB DMA. 1: To unmask fast interrupt request from USB DMA.
[18]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from USB CPU. 1: To unmask fast interrupt request from USB CPU.
[17]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from Encryption. 1: To unmask fast interrupt request from Encryption.
[16]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from Ethernet MAC. 1: To unmask fast interrupt request from Ethernet MAC.
[15]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from GPIO. 1: To unmask fast interrupt request from GPIO.
[14]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from SPI2. 1: To unmask fast interrupt request from SPI2.
[13]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from TWI0. 1: To unmask fast interrupt request from TWI0.
[12]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from L2 Buffer0. 1: To unmask fast interrupt request from L2 Buffer0.
[11]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from UART1. 1: To unmask fast interrupt request from UART1.
[10]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from UART0. 1: To unmask fast interrupt request from UART0.
[9]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from SPI1. 1: To unmask fast interrupt request from SPI1.
[8]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from SPI0.

Bits	Attributes	Reset	Description
			1: To unmask fast interrupt request from SPI0.
[7]	RW	1'h0	fiq_en_cfg Sigma-Delta ADC Controller Interrupt 0: To mask fast interrupt request from Sigma-Delta ADC Controller. 1: To unmask fast interrupt request from Sigma-Delta ADC Controller.
[6]	RW	1'h0	fiq_en_cfg Sigma-Delta DAC Controller Interrupt 0: To mask fast interrupt request from Sigma-Delta DAC Controller. 1: To unmask fast interrupt request from Sigma-Delta DAC Controller.
[5]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from MMC/SD/SDIO controller 1. 1: To unmask fast interrupt request from MMC/SD/SDIO controller 1.
[4]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from MMC/SD/SDIO controller 0. 1: To unmask fast interrupt request from MMC/SD/SDIO controller 0.
[3]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from system control module. 1: To unmask fast interrupt request from system control module.
[2]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from Video processor. 1: To unmask fast interrupt request from Video processor.
[1]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from Image Capture. 1: To unmask fast interrupt request from Image Capture.
[0]	RW	1'h0	fiq_en_cfg 0: To mask fast interrupt request from Memory Controller. 1: To unmask fast interrupt request from Memory Controller.

● TOP_INT Interrupt Mask (Address: 0x0800,002C)

Bits	Attributes	Reset	Description
[31:17]	RO	15'h0	Reserved
[16]	RW	1'h0	Tsensor_int_en 0: To disable Tsensor interrupt. 1: To enable Tsensor interrupt.
[15]	RW	1'h0	Timer5_int_en 0: To disable Timer5 interrupt. 1: To enable Timer5 interrupt.
[14]	RW	1'h0	Timer6_int_en

Bits	Attributes	Reset	Description
			0: To disable Timer6 interrupt. 1: To enable Timer6 interrupt.
[13]	RW	1'h0	Timer7_int_en 0: To disable Timer7 interrupt. 1: To enable Timer7 interrupt.
[12]	RW	1'h0	Timer8_int_en 0: To disable Timer8 interrupt. 1: To enable Timer8 interrupt.
[11]	RW	1'h0	Timer9_int_en 0: To disable Timer9 interrupt. 1: To enable Timer9 interrupt.
[10]	RW	1'h0	rtc_watchdog_int_en 0: To disable RTC watchdog timer interrupt. 1: To enable RTC watchdog timer interrupt.
[9]	RW	1'h0	rtc_period_timer_int_en 0: To disable RTC period timer interrupt. 1: To enable RTC period timer interrupt.
[8]	RW	1'h0	rtc_alarm_int_en 0: To disable RTC alarm interrupt. 1: To enable RTC alarm interrupt. NOTE: If an alarm has been set, an interrupt to the CPU will be generated when the time is matched with the alarm time. This kind of interrupt is called alarm interrupt.
[7]	RW	1'h0	rtc_rdy_int_en 0: To disable RTC ready interrupt. 1: To enable RTC ready interrupt. NOTE: When the RTC module is ready to communicate with main system, an interrupt will be generated. This kind of interrupt is called RTC ready interrupt.
[6]	RW	1'h0	wakeup_int_en 0: To disable wakeup trigger interrupt. 1: To enable wakeup trigger interrupt.
[5]	RW	1'h0	Timer0_int_en 0: To disable Timer0 interrupt. 1: To enable Timer0 interrupt.
[4]	RW	1'h0	Timer1_int_en 0: To disable Timer1 interrupt. 1: To enable Timer1 interrupt.
[3]	RW	1'h0	Timer2_int_en 0: To disable Timer2 interrupt. 1: To enable Timer2 interrupt.
[2]	RW	1'h0	Timer3_int_en 0: To disable Timer3 interrupt. 1: To enable Timer3 interrupt.
[1]	RW	1'h0	Timer4_int_en 0: To disable Timer4 interrupt. 1: To enable Timer4 interrupt.
[0]	RW	1'h0	sar_adc_int_en

Bits	Attributes	Reset	Description
			0: To disable SAR ADC sample interrupt. 1: To enable SAR ADC sample interrupt.

● TOP_INT Interrupt Status (Address: 0x0800,0030)

Bits	Attributes	Reset	Description
[31:17]	RO	15'h0	Reserved
[16]	RO	1'h0	Tsensor_int 0: No Tsensor interrupt has been generated. 1: A Tsensor interrupt has been generated.
[15]	RO	1'h0	Timer5_int 0: NoTimer5 interrupt has been generated. 1: A Timer5 interrupt has been generated.
[14]	RO	1'h0	Timer6_int 0: No Timer6 interrupt has been generated. 1: A Timer6 interrupt has been generated.
[13]	RO	1'h0	Timer7_int 0: No Timer7 interrupt has been generated. 1: A Timer7 interrupt has been generated.
[12]	RO	1'h0	Timer8_int 0: No Timer8 interrupt has been generated. 1: A Timer8 interrupt has been generated.
[11]	RO	1'h0	Timer9_int 0: No Timer9 interrupt has been generated. 1: A Timer9 interrupt has been generated.
[10]	RO	1'h0	rtc_watchdog_int 0: No RTC watchdog timer interrupt has been generated. 1: A RTC watchdog timer interrupt has been generated.
[9]	RO	1'h0	rtc_period_timer_int 0: No RTC period timer interrupt has been generated. 1: A RTC period timer interrupt has been generated.
[8]	RO	1'h0	rtc_alarm_int 0: No RTC alarm interrupt has been generated. 1: A RTC alarm interrupt has been generated. NOTE: If an alarm is set, an interrupt to the CPU will occur when the time is matched with the alarm time. This kind of interrupt is called alarm interrupt.
[7]	RO	1'h0	rtc_rdy_int 0: No RTC ready interrupt has been generated. 1: A RTC ready interrupt has been generated. NOTE: When the RTC module is ready to communicate with main system, an interrupt will be generated. This kind of interrupt is called RTC ready interrupt.
[6]	RO	1'h0	wakeup_int 0: No Wakeup trigger interrupt has been generated. 1: A Wakeup trigger interrupt has been generated.
[5]	RO	1'h0	Timer0_int 0: No Timer0 interrupt has been generated.

Bits	Attributes	Reset	Description
			1: A Timer0 interrupt has been generated.
[4]	RO	1'h0	Timer1_int 0: No Timer1 interrupt has been generated. 1: A Timer1 interrupt has been generated.
[3]	RO	1'h0	Timer2_int 0: No Timer2 interrupt has been generated. 1: A Timer2 interrupt has been generated.
[2]	RO	1'h0	Timer3_int 0: No Timer3 interrupt has been generated. 1: A Timer3 interrupt has been generated.
[1]	RO	1'h0	Timer4_int 0: No Timer4 interrupt has been generated. 1: A Timer4 interrupt has been generated.
[0]	RO	1'h0	sar_adc_int 0: No SAR ADC sampling interrupt has been generated. 1: A SAR ADC sampling interrupt has been generated.

● Interrupt Status (Address: 0x0800,004C)

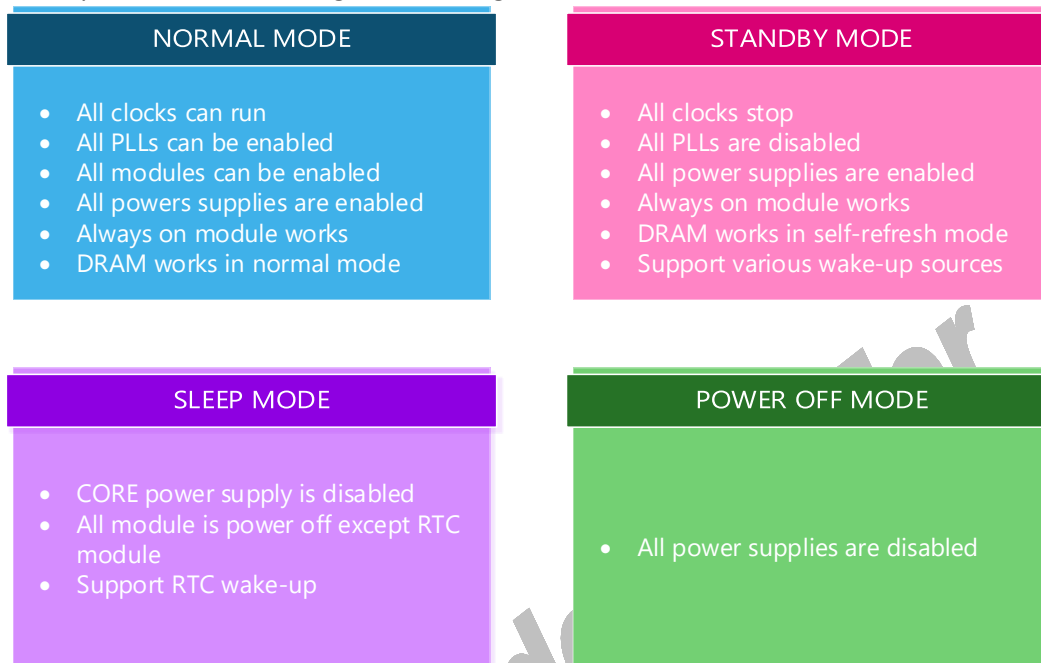
Bits	Attributes	Reset	Description
[31]	RO	1'h0	Reserved
[30]	RO	1'h0	0: An interrupt has not generated when an error occur in CSI. 1: An interrupt has generated when an error occur in CSI.
[29]	RO	1'h0	Reserved
[28]	RO	1'h0	0: CSI has not generated an interrupt. 1: CSI has generated an interrupt.
[27]	RO	1'h0	Reserved
[26]	RO	1'h0	0: NPU has not generated an interrupt. 1: NPU has generated an interrupt.
[25]	RO	1'h0	0: L2-buffer1 has not generated an interrupt. 1: L2-buffer1 has generated an interrupt.
[24]	RO	1'h0	0: UART2 has not generated an interrupt. 1: UART2 has generated an interrupt.
[23]	RO	1'h0	0: SD plugged-in or plugged-off has not generated an interrupt. 1: SD plugged-in or plugged-off has generated an interrupt.
[22]	RO	1'h0	0: MMC/SD/SDIO controller 2 has not generated an interrupt. 1: MMC/SD/SDIO controller 2 has generated an interrupt.
[21]	RO	1'h0	0: TWI2 has not generated an interrupt. 1: TWI2 has generated an interrupt.
[20]	RO	1'h0	0: TWI1 has not generated an interrupt. 1: TWI1 has generated an interrupt.
[19]	RO	1'h0	0: The USB DMA interrupt from USB 2.0 HS controller has not been generated. 1: The USB DMA interrupt from USB 2.0 HS controller has been generated.
[18]	RO	1'h0	0: The USB CPU interruption from USB 2.0 HS controller has not been generated.

Bits	Attributes	Reset	Description
			1: The USB CPU interruption from USB 2.0 HS controller has been generated.
[17]	RO	1'h0	0: Ecryption module has not generated an interrupt. 1: Ecryption module has generated an interrupt.
[16]	RO	1'h0	0: Ethernet MAC has not generated an interrupt. 1: Ethernet MAC has generated an interrupt.
[15]	RO	1'h0	0: GPIO has not generated an interrupt. 1: GPIO has generated an interrupt.
[14]	RO	1'h0	0: SPI2 has not generated an interrupt. 1: SPI2 has generated an interrupt.
[13]	RO	1'h0	0: TWI0 has not generated an interrupt. 1: TWI0 has generated an interrupt.
[12]	RO	1'h0	0: L2-buffer0 has not generated an interrupt. 1: L2-buffer0 has generated an interrupt.
[11]	RO	1'h0	0: UART1 has not generated an interrupt. 1: UART1 has generated an interrupt.
[10]	RO	1'h0	0: UART0 has not generated an interrupt. 1: UART0 has generated an interrupt.
[9]	RO	1'h0	0: SPI1 has not generated an interrupt. 1: SPI1 has generated an interrupt.
[8]	RO	1'h0	0: SPI0 has not generated an interrupt. 1: SPI0 has generated an interrupt.
[7]	RO	1'h0	0: Sigma-delta ADC Controller has not generated an interrupt. 1: Sigma-delta ADC Controller has generated an interrupt.
[6]	RO	1'h0	0: Sigma-delta DAC Controller has not generated an interrupt. 1: Sigma-delta DAC Controller has generated an interrupt.
[5]	RO	1'h0	0: MMC/SD/SDIO controller 1 has not generated an interrupt. 1: MMC/SD/SDIO controller 1 Controller has generated an interrupt.
[4]	RO	1'h0	0: MMC/SD/SDIO controller 0 has not generated an interrupt. 1: MMC/SD/SDIO controller 0 has generated an interrupt.
[3]	RO	1'h0	0: System control module has not generated an interrupt. 1: System control module has generated an interrupt. NOTE: If any of the bits in TOP_INT interrupt status (Address: 0x0800,0030) is 1, the top configuration interrupt will be generated.
[2]	RO	1'h0	0: Video processor has not generated an interrupt. 1: Video processor has generated an interrupt.
[1]	RO	1'h0	0: Image Capture module has not generated an interrupt. 1: Image Capture module has generated an interrupt.
[0]	RO	1'h0	0: Memory Controller has not generated an interrupt. 1: Memory Controller has generated an interrupt.

4.7 Working Modes

4.7.1 Overview

AK3918AV100 series processors implement a simple and powerful power control mechanism, which provides the following four working modes:



4.7.2 Wake-up Control

4.7.2.1 Overview

This module is used to wake up the AK3918AV100 series processors from standby/sleep mode when the wake-up signal is valid and the wake-up function of specific wake up source is enabled.

4.7.2.2 Features

Wake-up control module provides the following functions:

- For standby mode
 - Supports 28 GPIO wake-up ports
 - Supports other wake-up sources including RTC, AIN0, AIN1, USB_DM and USB_DP
- For sleep mode
 - Supports wake-up by powering on the external PMU
- Supports wake-up enabling and disabling function control
- Supports GPIO wake-up polarity selection

4.7.3 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the register name for the detailed description of a register.**

The following registers are used to define related parameters of the Wake-up GPIOs and Other Wake-up Source.

Register No.	Register Address	Register Name
1	0x0800,0034	<i>Other Wakeup Source/ USB Reference Control</i>
2	0x0800,0038	<i>Other Wakeup Source Triggered Status Register</i>
3	0x0800,003C	<i>Wake-up GPIO Polarity Selection Register</i>
4	0x0800,0040	<i>Wake-up GPIO Triggered Status Clear Register</i>
5	0x0800,0044	<i>Wake-up GPIO Function Enable Register</i>
6	0x0800,0048	<i>Wake-up GPIO Triggered Status Register</i>
7	0x0800,023C	<i>Wake-up IO Ports Polarity Selection Register</i>
8	0x0800,0240	<i>Wake-up IO Ports Function Enable Register</i>
9	0x0800,0244	<i>Wake-up IO Ports Triggered Status Clear Register</i>
10	0x0800,0248	<i>Wake-up IO Ports Triggered Status Register</i>

4.7.4 Register Definition

- Other Wakeup Source/ USB Reference Control Register (Address: 0x0800,0034)

This register enables/disables wakeup function of wake-up Other Source. Only when the wakeup function of corresponding wake-up Other Source is enabled, the wake-up Other Source can wake up the AK3916EV300 from standby mode. When the specific bit is set, the wakeup function of corresponding wake-up Other Source is enabled; when the specific bit is cleared, the wakeup function of corresponding wake-up Other Source is disabled.

Bits	Attributes	Reset	Description
[31:15]	RO	17'h0	Reserved
[14]	RW	1'h0	usb_dp_wu_en 0: Disabled 1: Enabled
[13]	RW	1'h0	rtc_timer_int_wu_en 0: To disable RTC timer wakeup when the chip is in standby mode. 1: To enable RTC timer wakeup when the chip is in standby mode.
[12]	RW	1'h0	rtc_int_wu_en 0: To disable RTC interrupt wakeup when the chip is in standby mode. 1: To enable RTC interrupt wakeup when the chip is in standby mode.
[11]	RW	1'h0	AIN0 wakeup enable 0: Disabled 1: Enabled
[10]	RO	1'h0	AIN1 wakeup enable 0: Disabled 1: Enabled
[9]	RW	1'h1	usb_dp_status_clr 1: To clear USB Host/Slave wakeup status.
[8]	RW	1'h1	rtc_timer_int_status_clr 1: To clear RTC timer interrupt status.
[7]	RW	1'h1	rtc_int_status_clr 1: Clear RTC interrupt triggered status.
[6]	RO	1'h1	ain0_wk_trig_status_clr

Bits	Attributes	Reset	Description
			1: Clear AIN0 wakeup triggered status
[5]	RO	1'h1	ain1_wk_status_clr 1: Clear AIN1 wakeup triggered status
[4:2]	RO	3'h0	Reserved
[1]	RW	1'h0	ana_trig_polar_cfg[1] 0: AIN0 is low level triggered. 1: AIN0 is high level triggered.
[0]	RO	1'h0	ana_trig_polar_cfg[0] 0: AIN1 is low level triggered. 1: AIN1 is high level triggered.

● Other Wakeup Source Triggered Status Register (Address: 0x0800,0038)

This register indicates current wake-up signal. When the specific bit is set, corresponding other wake-up sources are the wake-up signals.

Bits	Attributes	Reset	Description
[31:5]	RO	27'h0	Reserved
[4]	RO	1'h0	wk_int_sta USB wakeup interrupt status.
[3]	RO	1'h0	wk_int_sta RTC Timer wakeup interrupt status in standby mode.
[2]	RO	1'h0	wk_int_sta RTC wakeup interrupt status in standby mode.
[1]	RO	1'h0	wk_int_sta AIN0 wakeup interrupt status.
[0]	RO	1'h0	wk_int_sta AIN1 wakeup interrupt status.

● Wake-up GPIO Polarity Selection Register (Address: 0x0800,003C)

This register is used to select the wakeup GPIO polarity. When the specific bit is set to 0, the wake-up GPIO is high level triggered; when the specific bit is set to 1, the wake-up signal is low level triggered.

Bits	Attributes	Reset	Description
[31:30]	RO	2'h0	Reserved
[29]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[52] is high level triggered. 1: The wakeup GPIO[52] is low level triggered.
[28]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[51] is high level triggered. 1: The wakeup GPIO[51] is low level triggered.
[27]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[50] is high level triggered. 1: The wakeup GPIO[50] is low level triggered.
[26]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[49] is high level triggered. 1: The wakeup GPIO[49] is low level triggered.
[25]	RW	1'h0	trig_polar_cfg

Bits	Attributes	Reset	Description
			0: The wakeup GPIO[48] is high level triggered. 1: The wakeup GPIO[48] is low level triggered.
[24]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[45] is high level triggered. 1: The wakeup GPIO[45] is low level triggered.
[23]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[44] is high level triggered. 1: The wakeup GPIO[44] is low level triggered.
[22]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[42] is high level triggered. 1: The wakeup GPIO[42] is low level triggered.
[21:18]	RO	4'h0	Reserved
[17]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[37] is high level triggered. 1: The wakeup GPIO[37] is low level triggered.
[16:14]	RO	3'h0	Reserved
[13]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[27] is high level triggered. 1: The wakeup GPIO[27] is low level triggered.
[12]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[26] is high level triggered. 1: The wakeup GPIO[26] is low level triggered.
[11]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[25] is high level triggered. 1: The wakeup GPIO[25] is low level triggered.
[10]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[24] is high level triggered. 1: The wakeup GPIO[24] is low level triggered.
[9:6]	RO	4'h0	Reserved
[5]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[11] is high level triggered. 1: The wakeup GPIO[11] is low level triggered.
[4]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[10] is high level triggered. 1: The wakeup GPIO[10] is low level triggered.
[3]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[3] is high level triggered. 1: The wakeup GPIO[3] is low level triggered.
[2]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[2] is high level triggered. 1: The wakeup GPIO[2] is low level triggered.
[1]	RW	1'h0	trig_polar_cfg 0: The wakeup AIN1 is high level triggered. 1: The wakeup AIN1 is low level triggered.
[0]	RW	1'h0	trig_polar_cfg 0: The wakeup AIN0 is high level triggered. 1: The wakeup AIN0 is low level triggered.

● Wake-up GPIO Triggered Status Clear Register (Address: 0x0800,0040)

This register is used to clear wake-up GPIO status. When the specific bit is set, corresponding wake-up GPIO status bit is cleared. This register is cleared by being written 1s into corresponding bits.

This register must be cleared before entering standby mode.

Bits	Attributes	Reset	Description
[31:30]	RO	2'h3	Reserved
[29]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[52].
[28]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[51].
[27]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[50].
[26]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[49].
[25]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[48].
[24]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[45].
[23]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[44].
[22]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[42].
[21:18]	RO	4'hF	Reserved
[17]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[37].
[16:14]	RO	3'h7	Reserved
[13]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[27].
[12]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[26].
[11]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[25].
[10]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[24].
[9:6]	RO	4'hF	Reserved
[5]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[11].
[4]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[10].
[3]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[3].
[2]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[2].
[1]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of AIN1.
[0]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of AIN0.

● Wake-up GPIO Function Enable Register (Address: 0x0800,0044)

This register enables/disables wakeup function of wake-up GPIOs. Only when the wakeup function of corresponding wake-up GPIO is enabled, the wake-up GPIO can wake up the AK3918AV100 series processors from standby mode.

Bits	Attributes	Reset	Description
[31:30]	RO	2'h0	Reserved
[29]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[52] is disabled. 1: The wakeup function of GPIO[52] is enabled.
[28]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[51] is disabled. 1: The wakeup function of GPIO[51] is enabled.
[27]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[50] is disabled. 1: The wakeup function of GPIO[50] is enabled.
[26]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[49] is disabled. 1: The wakeup function of GPIO[49] is enabled.
[25]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[48] is disabled. 1: The wakeup function of GPIO[48] is enabled.
[24]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[45] is disabled. 1: The wakeup function of GPIO[45] is enabled.
[23]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[44] is disabled. 1: The wakeup function of GPIO[44] is enabled.
[22]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[42] is disabled. 1: The wakeup function of GPIO[42] is enabled.
[21:18]	RO	3'h0	Reserved
[17]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[37] is disabled. 1: The wakeup function of GPIO[37] is enabled.
[16:14]	RO	3'h0	Reserved
[13]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[27] is disabled. 1: The wakeup function of GPIO[27] is enabled.
[12]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[26] is disabled. 1: The wakeup function of GPIO[26] is enabled.
[11]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[25] is disabled. 1: The wakeup function of GPIO[25] is enabled.
[10]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[24] is disabled. 1: The wakeup function of GPIO[24] is enabled.
[9:6]	RO	4'h0	Reserved
[5]	RW	1'h0	trig_en_cfg

Bits	Attributes	Reset	Description
			0: The wakeup function of GPIO[11] is disabled. 1: The wakeup function of GPIO[11] is enabled.
[4]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[10] is disabled. 1: The wakeup function of GPIO[10] is enabled.
[3]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[3] is disabled. 1: The wakeup function of GPIO[3] is enabled.
[2]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[2] is disabled. 1: The wakeup function of GPIO[2] is enabled.
[1]	RW	1'h0	trig_en_cfg 0: The wakeup function of AIN1 is disabled. 1: The wakeup function of AIN1 is enabled.
[0]	RW	1'h0	trig_en_cfg 0: The wakeup function of AIN0 is disabled. 1: The wakeup function of AIN0 is enabled.

● Wake-up GPIO Triggered Status Register (Address: 0x0800,0048)

This register indicates current wake-up signal. When the specific bit is set to 1, corresponding wake-up GPIO is the wake-up signal.

This register must be cleared before entering standby mode.

Bits	Attributes	Reset	Description
[31:30]	RO	2'h0	Reserved
[29]	RO	1'h0	rig_sta 1: The wakeup GPIO[52] has been triggered. 0: The wakeup GPIO[52] has not been triggered.
[28]	RO	1'h0	rig_sta 1: The wakeup GPIO[51] has been triggered. 0: The wakeup GPIO[51] has not been triggered.
[27]	RO	1'h0	rig_sta 1: The wakeup GPIO[50] has been triggered. 0: The wakeup GPIO[50] has not been triggered.
[26]	RO	1'h0	rig_sta 1: The wakeup GPIO[49] has been triggered. 0: The wakeup GPIO[49] has not been triggered.
[25]	RO	1'h0	rig_sta 1: The wakeup GPIO[48] has been triggered. 0: The wakeup GPIO[48] has not been triggered.
[24]	RO	1'h0	trig_sta 1: The wakeup GPIO[45] has been triggered. 0: The wakeup GPIO[45] has not been triggered.
[23]	RO	1'h0	trig_sta 1: The wakeup GPIO[44] has been triggered. 0: The wakeup GPIO[44] has not been triggered.
[22]	RO	1'h0	trig_sta 1: The wakeup GPIO[42] has been triggered.

Bits	Attributes	Reset	Description
			0: The wakeup GPIO[42] has not been triggered.
[21:18]	RO	4'h0	Reserved
[17]	RO	1'h0	trig_sta 1: The wakeup GPIO[37] has been triggered. 0: The wakeup GPIO[37] has not been triggered.
[16:14]	RO	3'h0	Reserved
[13]	RO	1'h0	trig_sta 1: The wakeup GPIO[27] has been triggered. 0: The wakeup GPIO[27] has not been triggered.
[12]	RO	1'h0	trig_sta 1: The wakeup GPIO[26] has been triggered. 0: The wakeup GPIO[26] has not been triggered.
[11]	RO	1'h0	trig_sta 1: The wakeup GPIO[25] has been triggered. 0: The wakeup GPIO[25] has not been triggered.
[10]	RO	1'h0	trig_sta 1: The wakeup GPIO[24] has been triggered. 0: The wakeup GPIO[24] has not been triggered.
[9:6]	RO	4'h0	Reserved
[5]	RO	1'h0	trig_sta 1: The wakeup GPIO[11] has been triggered. 0: The wakeup GPIO[11] has not been triggered.
[4]	RO	1'h0	trig_sta 1: The wakeup GPIO[10] has been triggered. 0: The wakeup GPIO[10] has not been triggered.
[3]	RO	1'h0	trig_sta 1: The wakeup GPIO[3] has been triggered. 0: The wakeup GPIO[3] has not been triggered.
[2]	RO	1'h0	trig_sta 1: The wakeup GPIO[2] has been triggered. 0: The wakeup GPIO[2] has not been triggered.
[1]	RO	1'h0	trig_sta 1: The wakeup AIN1 has been triggered. 0: The wakeup AIN1 has not been triggered.
[0]	RO	1'h0	trig_sta 1: The wakeup AIN0 has been triggered. 0: The wakeup AIN0 has not been triggered.

● Wake-up IO Ports Polarity Selection Register (Address: 0x0800,023C)

This register is used to select the wakeup polarity of IO ports. When the specific bit is set to 0, the wake-up IO port is high level triggered; when the specific bit is set to 1, the wake-up IO port is low level triggered.

Bits	Attributes	Reset	Description
[31:15]	RO	17'h0	Reserved
[14]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[69] is high level triggered. 1: The wakeup GPIO[69] is low level triggered.

Bits	Attributes	Reset	Description
[13]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[68] is high level triggered. 1: The wakeup GPIO[68] is low level triggered.
[12]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[67] is high level triggered. 1: The wakeup GPIO[67] is low level triggered.
[11]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[66] is high level triggered. 1: The wakeup GPIO[66] is low level triggered.
[10]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[65] is high level triggered. 1: The wakeup GPIO[65] is low level triggered.
[9:4]	RO	6'h0	Reserved
[3]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[58] is high level triggered. 1: The wakeup GPIO[58] is low level triggered.
[2]	RO	1'h0	Reserved
[1]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[56] is high level triggered. 1: The wakeup GPIO[56] is low level triggered.
[0]	RW	1'h0	trig_polar_cfg 0: The wakeup GPIO[55] is high level triggered. 1: The wakeup GPIO[55] is low level triggered.

● Wake-up IO Ports Function Enable Register (Address: 0x0800,0240)

This register enables/disables wakeup function of wake-up IO ports. Only when the wakeup function of corresponding wake-up IO port is enabled, the wake-up IO port can wake up the AK3918AV100 series processors from standby mode.

Bits	Attributes	Reset	Description
[31:15]	RO	17'h0	Reserved
[14]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[69] is disabled. 1: The wakeup function of GPIO[69] is enabled.
[13]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[68] is disabled. 1: The wakeup function of GPIO[68] is enabled.
[12]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[67] is disabled. 1: The wakeup function of GPIO[67] is enabled.
[11]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[66] is disabled. 1: The wakeup function of GPIO[66] is enabled.
[10]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[65] is disabled. 1: The wakeup function of GPIO[65] is enabled.
[9:4]	RO	6'h0	Reserved
[3]	RW	1'h0	trig_en_cfg

Bits	Attributes	Reset	Description
			0: The wakeup function of GPIO[58] is disabled. 1: The wakeup function of GPIO[58] is enabled.
[2]	RO	1'h0	Reserved
[1]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[56] is disabled. 1: The wakeup function of GPIO[56] is enabled.
[0]	RW	1'h0	trig_en_cfg 0: The wakeup function of GPIO[55] is disabled. 1: The wakeup function of GPIO[55] is enabled.

● Wake-up IO Ports Triggered Status Clear Register (Address: 0x0800,0244)

This register is used to clear wake-up IO ports status. When the specific bit is set, corresponding wake-up IO port status bit is cleared. This register is cleared by being written 1s into corresponding bits.

This register must be cleared before entering standby mode.

Bits	Attributes	Reset	Description
[31:15]	RO	17'h1FFFF	Reserved
[14]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[69].
[13]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[68].
[12]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[67].
[11]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[66].
[10]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[65].
[9:4]	RO	6'h3F	Reserved
[3]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[58].
[2]	RO	1'h1	Reserved
[1]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[56].
[0]	RW	1'h1	trig_sta_clr 1: To clear the wakeup status of GPIO[55].

● Wake-up IO Ports Triggered Status Register (Address: 0x0800,0248)

This register indicates current wake-up signal. When the specific bit is set to 1, corresponding wake-up IO port is the wake-up signal.

This register must be cleared before entering standby mode.

Bits	Attributes	Reset	Description
[31:15]	RO	17'h0	Reserved
[14]	RO	1'h0	trig_sta 1: The wakeup GPIO[69] has been triggered.
[13]	RO	1'h0	trig_sta 1: The wakeup GPIO[68] has been triggered.

Bits	Attributes	Reset	Description
[12]	RO	1'h0	trig_sta 1: The wakeup GPIO[67] has been triggered.
[11]	RO	1'h0	trig_sta 1: The wakeup GPIO[66] has been triggered.
[10]	RO	1'h0	trig_sta 1: The wakeup GPIO[65] has been triggered.
[9:4]	RO	6'h0	Reserved
[3]	RO	1'h0	trig_sta 1: The wakeup GPIO[58] has been triggered.
[2]	RO	1'h0	Reserved
[1]	RO	1'h0	trig_sta 1: The wakeup GPIO[56] has been triggered.
[0]	RO	1'h0	trig_sta 1: The wakeup GPIO[55] has been triggered.

4.7.5 Programmer's Guide

4.7.5.1 Entering Standby Mode

4.7.5.1.1 Entering Standby Mode by Wake-up GPIO

Programmers should operate according to the following steps:

1. Power off all application programs, and copy the program of entering standby mode to on-chip RAM;
2. Disable the cache, MMU and write buffer of CPU core;
3. Close sar_adc_clk, sddac_clk, sdadc_clk, sddac_hclk, sdadc_hclk, OPCLK/RMII, usb_phy_clk and 12MHz crystal;
4. Set DRAM to enter self-refresh mode;
5. Set bit[31] of *RAM IO Control Register (Address: 0x2100,0078)* to 1. In this way, DDR DQ/DQS/DQSn PADS are forced to enter idle mode in order to save power consumption;
6. Set corresponding wakeup ports to be used as GPIOs
7. Set the wakeup port(s) to input state;
8. Clear Wake-up Status Register (to set *Wake-up GPIO Trigger Status Clear Register (Address: 0x0800, 0040)* to 0xFFFF, FFFF; and make sure that *Wake-up GPIO Trigger Status Register (Address: 0x0800,0048)* is 0x0000,0000); then to clear *Wake-up GPIO Trigger Status Clear Register (Address: 0x0800, 0040)* to 0x0000,0000;
9. Set the polarity of wake-up GPIOs (*Wake-up GPIO Polarity Select Register (Address: 0x0800,003C)*);
10. Unmask corresponding wake-up GPIOs (*Wake-up GPIO Function Enable Register (Address: 0x0800,0044)*);
11. Write 1 to address 0x2180,0000 and the system will enter standby mode.

4.7.5.1.2 Entering Standby Mode by Other Sources

Programmers should operate according to the following steps:

1. Power off all application programs, and copy the program of entering standby mode to on-chip RAM;
2. Disable the cache, MMU and write buffer of CPU core;
3. Close sar_adc_clk, sddac_clk, sdadc_clk, sddac_hclk, sdadc_hclk, mac0_opclk, mac1_opclk, USB_CLK and crystal oscillator;
4. Set DRAM to enter self-refresh mode;

5. Set bit[31] of *RAM IO Control Register (Address: 0x2100,0078)* to 1. In this way, DDR DQ/DQS/DQSn PADs are forced to enter idle mode in order to save power consumption;
6. If USB Host/Slave DP is used to wake up system, set bit[14] of *USB Operate Mode and I2S Mode Control Register (Address: 0x0800,0058)*; If RTC is used to wake up system in standby mode, set corresponding RTC time and interrupt first;
7. Clear Other Source Wake-up Status Register (to set bit[9:5] of *Other Wakeup Source/USB Reference Control Register (Address: 0x0800,0034)*) to 0x1F; and make sure that *Other Wakeup Source Triggered Status Register (Address: 0x0800,0038)* is 0x0000,0000; then clear bit[9:5] of *Other Wakeup Source/USB Reference Control Register (Address: 0x0800,0034)* to 0x0;
8. Set the polarity of other source (bit[1:0]) of *Other Wakeup Source/USB Reference Control Register (Address: 0x0800,0034)*;
9. Unmask corresponding other source (bit[14:10]) of *Other Wakeup Source/USB Reference Control Register (Address: 0x0800,0034)*;
10. Write 1 to address 0x2180,0000 and the system will enter standby mode.

4.7.5.2 Exiting Standby Mode

AK3918AV100 series processors can exit the standby mode by the following two ways:

- One of the wake-up GPIOs is asserted or deasserted;
- An alarm occurs.

When AK3918AV100 series processors are woken up, software should take the following steps to enter normal working mode:

1. Read *Wake-up GPIO Trigger Status Register (Address: 0x0800,0048)* or *Other Wakeup Source Triggered Status Register (Address: 0x0800,0038)* and go to wakeup sub-program;
2. Set bit[31] of *RAM IO Control Register (Address: 0x2100,0078)* to 0. In this way, DDR DQ/DQS/DQSn PADs can enter normal mode;
3. Set the DRAM to exit self-refresh mode;
4. Implement wakeup sub-program.

4.7.5.3 Entering Sleep Mode

Programmers should configure bit[0] of *RTC Control 0 (Address: 0x0800,0050)* as 1. More details on how to write into RTC registers, please refer to Chapter 4.9.4.

4.7.5.4 Exiting Sleep Mode

AK3918AV100 series processors can exit the sleep mode by powering on the external PMU.

4.7.5.5 Entering Power off mode

Users can directly power off the supply. Programmers should check corresponding status bits to confirm that the operations have been implemented successfully.

4.8 Timer/PWM

4.8.1 Overview

The Timer module mainly implements timing and counting functions. AK3918AV100 series processors include 10 general purpose timers (Timer0~9), each of which has its own 32-bit down-counter.

4.8.2 Features

The PWM/Timer controller has the following features:

- 5 pairs of PWM/Timer units(Timer0~PWM0,, Timer4~PWM4)
- 16-bit high-level pulse control and 16-bit low-level pulse control in PWM mode
- Minimum duty cycle: 1/65536 (on for 1/65536, and off for the other 65535/65536)
- Maximum duty cycle: 65535/65536 (on for 65535/65536, and off for the 1/65536)
- Supports PWM frequency ranging from 92Hz to 6MHz (frequency pre-divider disabled)
- Current value of Timer counter could be read back by host processor
- Timer counters are down counters
- A clock pre-divider is provided to each PWM/Timer
- Supports three kinds of working modes: PWM Mode, Auto-reload Timer Mode and One-shot Timer Mode

4.8.3 Functional Description

These timers can work in three modes, PWM Mode (Timer0~4), Auto-reload Timer Mode, and one-shot Timer Mode. When timer counts down to 0, a level signal of interrupt is generated.

4.8.3.1 Working Mode

4.8.3.1.1 PWM Mode

When the register of 32-bit timer is enabled, the rising edge is symbolled as starting up to load the number of clock cycle under low level. Timer is decremented by 1 from the predefined value when pre-divider counting equals to pre-divider value under enabled condition, decreasing progressively until Timer's predefined value equals to 0. When decrementing to 0, the number of clock cycle under high level is loaded for counting and restart to count, so the work flow begins again. In PWM Mode, the duty cycle and frequency of PWM are configurable.

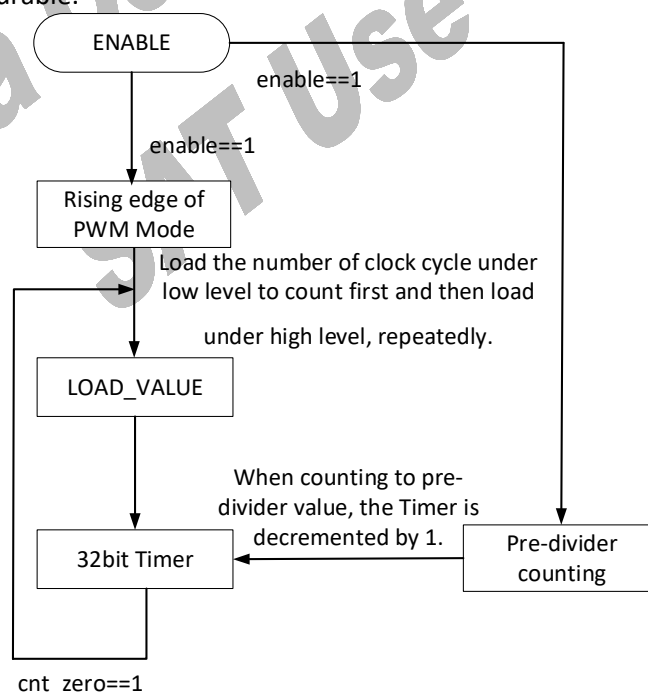


Figure 4-14 Timer work-flow of PWM Mode

4.8.3.1.2 Auto-reload Timer Mode

In Auto-reload Timer Mode, the timer loads predefined value, and an interrupt is generated when it down counts to zero. The Auto-reload Timer keeps counting unless it is disabled.

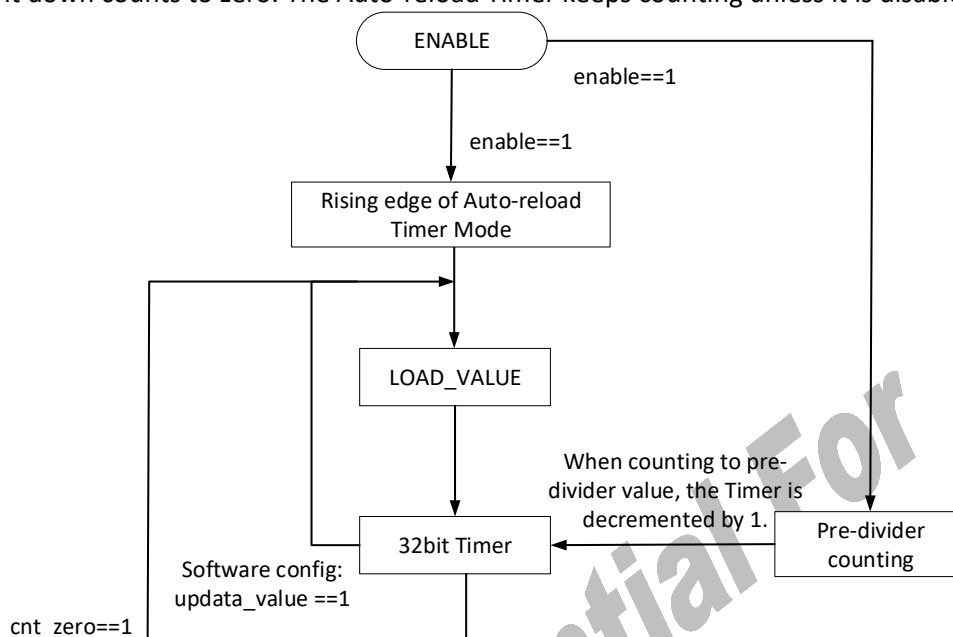


Figure 4-15 Timer work-flow of Auto-reload Timer Mode

4.8.3.1.3 One-shot Timer Mode

In One-shot Timer Mode, the timer will stop as it reaches zero and count only once. It will start to recount until another One-shot Timer Mode begins or it is converted into Auto-reload Timer Mode.

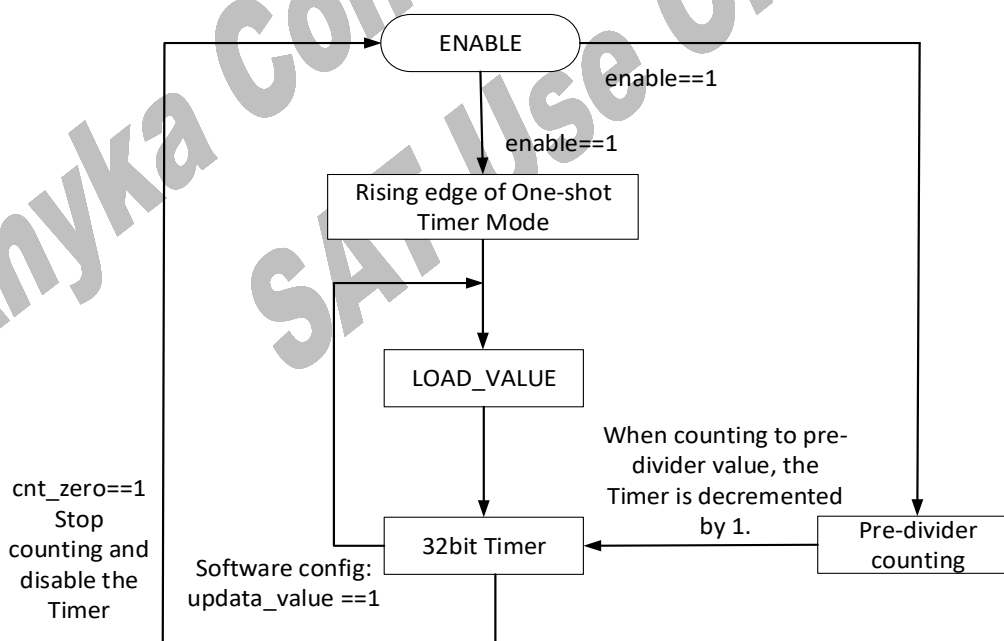


Figure 4-16 Timer work-flow of One-shot Timer Mode

4.8.3.2 Predefined Value Setting

In One-shot mode, the predefined value is set at the beginning of the whole counting procedure.

In Auto-reload Timer Mode, the predefined value is set when the cycle symbol is 1 which means bit[29] of the [PWM/Timer0 Control Register 1 \(Address: 0x0800,00B8\)](#) is 1 and the write signal is high.

4.8.4 Pulse Width Modulation

Pulse Width Modulation (PWM) is a method of digitally encoding analog signal levels. Through the use of high-resolution counters, the duty cycle of the square wave is modulated to encode the level of a specific analog signal. The PWM signal is still digital, because at any given moment, the full-scale DC power supply is either completely (ON) or completely absent (OFF). So the output voltage adjustment is achieved by adjusting the high and low level frequencies.

PWM frequency refers to the number of times the signal goes from high level to low level to high level in 1 second (one cycle), that is, how many cycles of PWM in one second.

PWM duty cycle: in a pulse period, the ratio of the duration of the positive pulse to the total period of the pulse, that is, the ratio of the cycle time occupied by the high level to the entire cycle time (t:T).

4.8.5 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Timer 0~4 support both timer mode and PWM mode and each timer is controlled by 2 registers, the definitions of these two registers for each PWM/Timer are the same. Timer 5~9 only support timer mode and are controlled by two registers with the same definitions. In the following section, only the definitions of registers for PWM0/Timer0 and Timer5 will be described.

Register No.	Register Address	Register Name and Description
1	0x0800,00B4	PWM/Timer0 Control Register 0
2	0x0800,00B8	PWM/Timer0 Control Register 1
3	0x0800,00BC	PWM/Timer1 control register 0 Refer to PWM/Timer0 Control Register 0 for details.
4	0x0800,00C0	PWM/Timer1 control register 1 Refer to PWM/Timer0 Control Register 1 for details.
5	0x0800,00C4	PWM/Timer2 control register 0 Refer to PWM/Timer0 Control Register 0 for details.
6	0x0800,00C8	PWM/Timer2 control register 1 Refer to PWM/Timer0 Control Register 1 for details.
7	0x0800,00CC	PWM/Timer3 control register 0 Refer to PWM/Timer0 Control Register 0 for details.

Register No.	Register Address	Register Name and Description
8	0x0800,00D0	PWM/Timer3 control register 1 Refer to PWM/Timer0 Control Register 1 for details.
9	0x0800,00D4	PWM/Timer4 control register 0 Refer to PWM/Timer0 Control Register 0 for details.
10	0x0800,00D8	PWM/Timer4 control register 1 Refer to PWM/Timer0 Control Register 1 for details.
11	0x0800,00F4	Timer5 control register 0
12	0x0800,00F8	Timer5 control register 1
12	0x0800,00FC	Timer6 control register 0 Refer to Timer5 control register 0 for details.
13	0x0800,0100	Timer6 control register 1 Refer to Timer5 control register 0 for details.
14	0x0800,0104	Timer7 control register 0 Refer to Timer5 control register 0 for details.
15	0x0800,0108	Timer7 control register 1 Refer to Timer5 control register 1 for details.
16	0x0800,010C	Timer8 control register 0 Refer to Timer5 control register 0 for details.
17	0x0800,0110	Timer8 control register 1 Refer to Timer5 control register 1 for details.
18	0x0800,0114	Timer9 control register 0 Refer to Timer5 control register 0 for details.
19	0x0800,0118	Timer9 control register 1 Refer to Timer5 control register 1 for details.

4.8.6 Register Definition

- PWM/Timer0 Control Register 0 (Address: 0x0800,00B4)

Bits	Attributes	Reset	Description
[31:16]	RW	16'h0	high level/timer cnt In PWM mode: PWM high pulse cycle number PWM output frequency: $(12M / (clk_pre_div+1)) / ((high_level+1)+(low_level+1))$ In Timer mode: Timer counter length[31:16]
[15:0]	RW	16'h0	low level/timer cnt In PWM mode: PWM low pulse cycle number PWM output frequency: $(12M / (clk_pre_div+1)) / ((high_level+1)+(low_level+1))$ In Timer mode: Timer counter length[15:0]

- PWM/Timer0 Control Register 1 (Address: 0x0800,00B8)

Bits	Attributes	Reset	Description
------	------------	-------	-------------

Bits	Attributes	Reset	Description
[31]	RW	1'h0	Reserved
[30]	RW	1'h0	timeout_clr 1: To clear timeout pulse.
[29]	RW	1'h0	feed_timer 1: To feed timer0.
[28]	RW/SC	1'h0	pwm/timer_en 0: To disable PWM/Timer0. 1: To enable PWM/Timer0. NOTE: 1. This bit is self-cleared in One Shot Timer mode. 2. All state, except timeout pulse, could be cleared after clearing this bit. It is recommended to clear this bit before changing the PWM/Timer0 configurations.
[27]	RW	1'h0	timeout_sta 1: Timeout pulse has been generated.
[26]	RW	1'h0	timer_read_sel 0: The predefined counter value of Control Register 0 of PWM/Timer0 is read back when reading Control Register 0 of PWM/Timer0. 1: The current counter value is read back when reading Control Register 0 of PWM/Timer0.
[25:24]	RW	2'h0	work_mode PWM/Timer0 working modes selection. 2'h0: PWM/Timer0 works as an Auto-reload Timer. 2'h1: PWM/Timer0 works as a One Shot Timer. 2'h2: PWM/Timer0 works as a PWM. 2'h3: Reserved.
[23:16]	RW	8'h0	clk_pre_div Clock pre-divider. The clock of PWM/Timer0 is divided by (clk_pre_div + 1).
[15:0]	RW	16'h0	Reserved

● Timer5 Control Register 0 (Address: 0x0800,00F4)

Bits	Attributes	Reset	Description
[31:16]	RW	16'h0	Timer5 counter length[31:16]
[15:0]	RW	16'h0	Timer5 counter length[15:0]

● Timer5 Control Register 1 (Address: 0x0800,00F8)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	Reserved
[30]	RW	1'h0	timeout_clr 1: To clear timeout pulse.
[29]	RW	1'h0	feed_timer 1: To feed Timer5.
[28]	RW/SC	1'h0	timer_en 0: To disable Timer5. 1: To enable Timer5.

Bits	Attributes	Reset	Description
			NOTE: 1) This bit is self-cleared in One Shot Timer mode. 2) All state, except timeout pulse, could be cleared after clearing this bit. It is recommended to clear this bit before changing Timer5 configurations.
[27]	RW	1'h0	timeout_sta 1: Timeout pulse has been generated.
[26]	RW	1'h0	timer_read_sel 0: The value of Control Register 0 of Timer5 is read back when reading Control Register 0 of Timer5. 1: The current counter value is read back when reading Control Register 1 of Timer5.
[25:24]	RW	2'h0	work_mode Timer5 working modes selection. 0: Timer5 works as an Auto-reload Timer. 1: Timer5 works as a One Shot Timer. Others: Reserved.
[23:16]	RW	8'h0	clk_pre_div Clock pre-divider. The clock of Timer5 is divided by (clk_pre_div + 1).
[15:0]	RW	16'h0	Reserved

4.8.7 Programmer's Guide

- Write 0, 1, 2 to bit[25:24] of *PWM/Timer0 Control Register 1 (Address: 0x0800,00B8)* means PWM/Timer0 works in Auto-reload Timer mode, One Shot Timer mode or PWM mode respectively.
- Set 0 to bit[26] of *PWM/Timer0 Control Register 1 (Address: 0x0800,00B8)* to read back the predefined value of Control Register 0 of PWM/Timer0; Set 1 to read back the current counter value.
- Write 1 to bit[30] of *PWM/Timer0 Control Register 1 (Address: 0x0800,00B8)* to clear timeout interrupt.

4.9 RTC

4.9.1 Overview

The RTC module serves as a real-time clock with calendar function, which sources from an external 32.768 KHz crystal or the internal 32 KHz RC oscillator. Except RTC, all the modules will be powered off when AK3918AV100 series processor is in sleep/power down mode. While the system is powered off, this module may be driven by the backup battery.

4.9.2 Features

This module mainly provides the following functions:

- Supports real-time clock
- Alarms or wake-up signals in sleep mode
- Interrupts in normal operating mode
- A watchdog timer
- An RTC timer
- Watchdog event interrupt output

4.9.3 Block Diagram

Power supply of RTC is shown in the following figure.

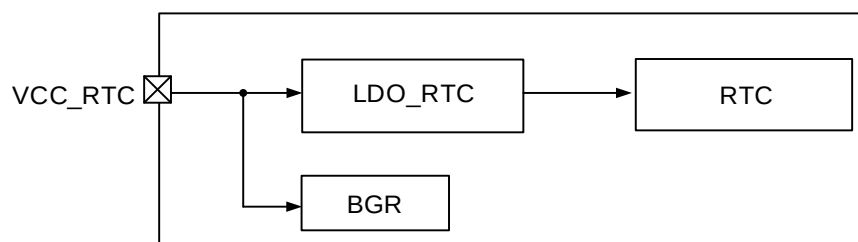


Figure 4-17 Power Supply of RTC

4.9.3.1 On-chip LDO_RTC

On-chip LDO_RTC is always on and it provides power rail for RTC module, and the output voltage of on-chip LDO_RTC is configurable.

4.9.3.2 BGR

An on-chip Band Gap Reference (BGR) is integrated with a default value of 1.5V, which is used to provide reference voltage for AK3918AV100 series processors during normal mode.

4.9.4 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Register Address	Register Name and Description
1	0x0800,0050	<i>RTC Control 0</i>
2	0x0800,0054	<i>Refer to Section 3.6.4 RTC Control 1 and Boot Mode Flag</i>
3	0x0	<i>Real Time Register 0</i>
4	0x1	<i>Real Time Register 1</i>
5	0x2	<i>Real Time Register 2</i>
6	0x3	<i>Alarm Time Register 0</i>
7	0x4	<i>Alarm Time Register 1</i>
8	0x5	<i>Alarm Time Register 2</i>
9	0x6	<i>Watchdog Timer/RTC Timer Configuration Register 0</i>
10	0x7	<i>Watchdog Timer/RTC Timer Configuration Register 1</i>
11	0x8	<i>RTC 32k Adjust Register 0</i>
12	0x9	<i>RTC 32k Adjust Register 1</i>
13	0xA	<i>RTC Setting Register</i>
14	0xB	<i>RTC Signals to Analog Configuration Register</i>

4.9.5 Register Definition

● RTC Control 0 (Address: 0x0800,0050)

Bits	Attributes	Reset	Description
[31:26]	RO	6'h0	Reserved
[25]	RW	1'h0	rtc_wr_enable, RTC register write/read enable 1: Enabled 0: Disabled NOTE: This bit should be configured as 0 before the system enters power down mode.
[24]	RW	1'h0	rtc_enable 1: RTC is enabled. 0: RTC is disabled.
[23]	RO	1'h0	Reserved
[22]	RW	1'h0	synchronous bit 0: No instruction. 1: Start data transfer.
[21:19]	RW	3'h0	Protection bits These bits must be set to 0x02.
[18]	RW	1'h0	Select data transfer direction selection 0: Write data/instruction into RTC module. 1: Read data from RTC module.
[17:14]	RW	4'h0	RTC internal register address selection. 4'h0: Real Time Register 0 4'h1: Real Time Register 1 4'h2: Real Time Register 2 4'h3: Alarm Time Register 0 4'h4: Alarm Time Register 1 4'h5: Alarm Time Register 2 4'h6: RTC Timer Configuration Register 4'h7: Watchdog Timer 4'h8: RTC 32k Adjustment Register 0(low 12-bit calibration value and 32KHz clock source selection) 4'h9: RTC 32k Adjustment Register 1(high 12-bit calibration value and calibration enable control) 4'hA: RTC Setting Register 4'hB: RTC signals to analog 4'hC~4'hF: Reserved
[13:0]	RW	14'h0	The data to be written to RTC module.

● Real Time Register 0 (Address: 0x0)

Bits	Attributes	Reset	Description
[13:12]	RO	2'h0	Reserved
[11:6]	RW	6'h0	min_cnt Current minute. (ranging from 0 to 59)
[5:0]	RW	6'h0	sec_cnt Current second. (ranging from 0 to 59)

● Real Time Register 1 (Address: 0x1)

Bits	Attributes	Reset	Description
[13]	RO	1'h0	Reserved
[12:10]	RW	3'h1	day_week_cnt The day order in a week. (ranging from 1 to 7)
[9:5]	RW	5'h1	day_mon th_cnt The day order in a month. (ranging from 1 to 31) NOTE: The total days in a month depends on the specific month and whether it is a leap year or not.
[4:0]	RW	5'h0	hour_cnt Current hour. (ranging from 0 to 23)

● Real Time Register 2 (Address: 0x2)

Bits	Attributes	Reset	Description
[13:11]	RO	3'h0	Reserved
[10:4]	RW	7'h0	year_cnt Current year. (ranging from 0 to 127) The base year like 2000 should be configured by software. Then if the base year is 2000, years ranging from 2000 to 2127 can be recorded.
[3:0]	RW	4'h1	month_cnt The month order in a year. (ranging from 1 to 12)

● Alarm Time Register 0 (Address: 0x3)

Bits	Attributes	Reset	Description
[13]	RW	1'h0	low_time_mask_en 1: Enable low time mask.
[12]	RO	1'h0	Reserved
[11:0]	RW	12'hFFF	low time mask low_time_mask[11:6]: minute low_time_mask[5:0]: second

● Alarm Time Register 1 (Address: 0x4)

Bits	Attributes	Reset	Description
[13]	RW	1'h0	mid_time_mask_en 1: Enable mid time mask.
[12:10]	RO	3'h0	Reserved
[9:0]	RW	10'h3FF	mid_time_mask mid_time_mask[9:5]: day mid_time_mask[4:0]: hour

● Alarm Time Register 2 (Address: 0x5)

Bits	Attributes	Reset	Description
[13]	RW	1'h0	high_time_mask_en 1: Enable high time mask

Bits	Attributes	Reset	Description
			0: Disable high time mask
[12:11]	RO	2'h0	Reserved
[10:0]	RW	11'h7F F	High time mask. low_time_mask[10:4]: year low_time_mask[3:0]: month

● Watchdog Timer/RTC Timer Configuration Register 0 (Address: 0x6)

Bits	Attributes	Reset	Description
[13]	RW	1'h0	rtc_timer_mask_en 1: Enable RTC timer. NOTE: Only when this bit is enabled, the RTC timer interrupt can be generated.
[12:0]	RW	13'h1F FF	rtc_timer_mask This bit is used to predefine the interval between two cyclical interrupts or two cyclical wakeup of the chip. NOTE: When rtc_timer_mask_en is enabled, timers within RTC will start counting. Once timers reach the predefined value, a RTC timer interrupt will be generated.(unit: (1/1024s)

● Watchdog Timer/RTC Timer Configuration Register 1 (Address: 0x7)

The watchdog timer protects system by providing a method of escaping from unexpected events or programming errors. Once activated, the timer must be serviced by software on a periodic basis. The watchdog timer resets the RTC module by sending a signal to the pin WAKEUP.

Bits	Attributes	Reset	Description
[13]	RW	1'h0	rtc_watchdog_mask_en 1: To enable watchdog timer. NOTE: 1) Only when this bit is set, the watchdog signal may be generated. 2) If programmers need to change the configuration of rtc_watchdog_mask, it is recommended to clear this bit first, then write new values to rtc_watchdog_mask and write 1 to this bit simultaneously.
[12:0]	RW	13'h1F FF	rtc_watchdog_mask This bit is used to predefine the time needed for watchdog signals to be produced after watchdog timers in RTC module are enabled. NOTE: When rtc_watchdog_mask_en is enabled, timers within RTC will start counting. Once timers reach the predefined value, a watchdog signal will be generated.(unit: (1/1024s)

● RTC 32KHz Adjustment Register 0 (Address: 0x8)

Bits	Attributes	Reset	Description
[13]	RW	1'h1	osc_sel 32KHz RTC OSC Source Selection 0: To use external 32K XTAL. 1: To use internal RC OSC.
[12:0]	RO	13'h0	osc_adj RC oscillation adjustment value. NOTE: Low 13 bits, together with high 13 bits of <i>RTC 32KHz Adjustment Register 0 (Address: 0x8)</i> , are the number of 24MHz clock cycles contained in 8192 RC oscillation cycles, which decides the fractional divider.

● RTC 32KHz Adjustment Register 1 (Address: 0x9)

Bits	Attributes	Reset	Description
[13]	RW	1'h0	rtc_adj_en The time adjustment will be started when its value is set to 1. After time adjustment, this bit will be automatically cleared.
[12:0]	RO	13'h0	osc_adj RC oscillation adjustment value. NOTE: Low 13 bits, together with high 13 bits of <i>RTC 32KHz Adjustment Register 1 (Address: 0x9)</i> , are the number of 24MHz clock cycles contained in 8192 RC oscillation cycles, which decides the fractional divider.

● RTC Setting Register (Address: 0xA)

Bits	Attributes	Reset	Description
[13]	RO	1'h0	Reserved
[12]	RO,W1C	1'h0	rtc_watchdog_int_sta_clr 1: Clear RTC Watchdog interrupt status. NOTE: This bit will be cleared automatically after being set.
[11]	RW	1'h0	rtc_watchdog_inner_req 1: Request the watchdog signal to pass through the reset circuit on the control chip. NOTE: This bit will be cleared automatically after being set.
[10]	RW	1'h0	real_time_rd_wait_sel wait time selection signal when real time is read 0: read operation started, real time value latched, waiting 0~7ms. 1: read operation started, real time value latched, waiting 0~500ms.
[9:7]	RO	3'h0	Reserved
[6]	RW	1'h0	rtc_watchdog_fed Within the time set by bits[12:0] of <i>Watchdog Time/RTC</i>

Bits	Attributes	Reset	Description
			<i>Timer Configuration Register (Address 0x7)</i> , this bit should be set to 1 to feed the watchdog timer. NOTE: This bit will be cleared automatically after being set.
[5]	RO	1'h0	Reserved
[4]	RW	1'h0	real_time_read_req 1: To request RTC module to read the value of real timer circuit into Real Time Register 0, 1 & 2. NOTE: 1) This bit will be cleared automatically after being set. 2) Software can get the real time value by following steps: - Set this bit to 1; - Read the value of Real Time Register 0, 1 & 2 sequentially. 3) This bit and bit[3] of this register can not be set to 1 at the same time.
[3]	RW	1'h0	real_time_write_req 1: To request RTC module to write values of Real Time Register 0, 1 & 2 into the real timer circuit. NOTE: 1) This bit will be cleared automatically after being set. 2) Software can set a time as the real time by following steps: - Write the time accordingly into Real Time Register 0, 1 & 2; - Set this bit to 1; - Read this bit to check whether it becomes 0 or not. - It should be noted that there should be a minimum interval of 30us between two operations of setting real time. 3) This bit and bit[4] of this register can not be set to 1 at the same time.
[2]	RO	1'h0	Reserved
[1]	RO,W1C	1'h0	rtc_timer_int_sta_clr 1: Clear RTC timer interrupt status. NOTE: This bit will be cleared automatically after being set.
[0]	RO, W1C	1'h0	timer_int_sta_clr 1: Clear alarm interrupt status. NOTE: This bit will be cleared automatically after being set.

● RTC Signals to Analog Configuration Register (Address: 0xB)

Bits	Attributes	Reset	Description
[13]	RO	1'h0	Reserved

Bits	Attributes	Reset	Description
[12:11]	RW	2'h1	UV_SEN[1:0] The chip reset(high to low level) threshold and the corresponding reset release(low to high level) threshold 2'h0: 2.79V and 3.0V 2'h1: 2.65V and 2.86V 2'h2: 2.55V and 2.78V 2'h3: 2.46V and 2.68V
[10]	RW	1'h0	efuse_valid efuse ready flag 0: Not ready 1: Ready
[9:5]	RW	5'h0	Calibrated reference voltage
[4]	RW	1'h0	pd_rc32k 32KHz RC oscillator power on control bit 0: Power on 1: Power down
[3]	RW	1'h0	ldo_rtc_sel[3] LDO11_RTC output voltage selection Dangling bit NOTE: This bit can be configured as any values.
[2:1]	RW	2'h1	ldo_rtc_sel[2:1] LDO11_RTC output voltage selection 2'h0: 0.95V 2'h1: 1.1V 2'h2: 1.15V 2'h3: 1.2V
[0]	RW	1'h0	en_cry32kd 32KHz oscillation enable control 1: Enable 0: Disable

4.9.6 Programmers' Guide

4.9.6.1 Writing into/Reading from RTC Register

Follow steps below to write into RTC registers:

1. Configure bit[24] of *RTC Control 0 (Address: 0x0800,0050)* as 1 to enable RTC module.
2. Configure bit[25] of *RTC Control 0 (Address: 0x0800,0050)* as 1 to enable RTC write/read register.
3. Wait for bit[7] of *TOP_INT Interrupt Status (Address: 0x0800,0030)* to become 1;
4. Set bit[22], bits[21:19] and bit[18] of *RTC Control 0 (Address: 0x0800,0050)* to 1'h1, 4'h2 and 1'h0 respectively. Write the address of RTC internal register that you want to configure into bits[17:14] of *RTC Control 0 (Address: 0x0800,0050)* ;
5. Write the instruction or data into bits[13:0] of *RTC Control 0 (Address: 0x0800,0050)* (The definition of bits[13:0] is mapped with that of the register addressed at where you write into bits[17:14] just now.
6. Wait for 100 us;
7. Wait for bit[7] of *TOP_INT Interrupt Status (Address: 0x0800,0030)* to become 1;

Follow the steps below to read from the RTC registers:

1. Configure bit[24] of *RTC Control 0 (Address: 0x0800,0050)* as 1 to enable RTC module.
2. Configure bit[25] of *RTC Control 0 (Address: 0x0800,0050)* as 1 to enable RTC write/read register.
3. Wait for bit[7] of *TOP_INT Interrupt Status (Address: 0x0800,0030)* to become 1;
4. Set bit[22], bits[21:19] and bit[18] of *RTC Control 0 (Address: 0x0800,0050)* to 1'h1,4'h2 and 1'h1 respectively. Write the address of RTC internal register that you want to configure into bits[17:14] of *RTC Control 0 (Address: 0x0800,0050)* ;
5. Wait for 100 us;
6. Wait for bit[7] of *TOP_INT Interrupt Status (Address: 0x0800,0030)* to become 1
7. Read bits[13:0] of *RTC Control 1 and Boot Mode Flag (Address: 0x0800,0054)* (The definition of bits[13:0] is mapped with that of the register addressed at where you write into bits[17:14] of *RTC Control 0 (Address: 0x0800,0050)* just now.

4.9.6.2 Setting Time

Follow the steps below to set time:

1. Write the time needed to be set into *Real Time Register 0 (Address: 0x0)*, *Real Time Register 1 (Address: 0x1)*, and *Real Time Register 2 (Address: 0x2)*.
2. Set bit[3] of *RTC Setting Register (Address: 0xA)* to 1.
3. Poll bit[3] of *RTC Setting Register (Address: 0xA)* until it becomes 0.

NOTE

The interval between two consecutive time settings should not be shorter than 40us.

4.9.6.3 Reading Time

Follow the steps below to read time:

1. Write 1 to the bit[4] of *RTC Setting Register (Address: 0xA)* .
2. Read the value of bits[11:0] of *Real Time Register 0 (Address: 0x0)*, bits[9:0] of *Real Time Register 1 (Address: 0x1)*, bits[10:0] of *Real Time Register 2 (Address: 0x2)* in order.

4.9.6.4 Power-on of the Processor

Programmers should operate according to the following steps:

- Powering on the AK3918AV100 series processors by software

Set bit[8] of *RTC Setting Register (Address: 0xA)* to 1 to power on the processor.

- Powering off the AK3918AV100 series processors by software

1. Set bit[0] of *Watchdog Control (Address: 0x0800,00E8)* to 0 to disable the watchdog.
2. Set bit[9] or bit[7] of *RTC Setting Register (Address: 0xA)* to 1 to power off the processor.

4.9.6.5 On-chip Watchdog Reset

Programmers should operate according to the following steps:

1. Set bit[11] to 1 of the *RTC Setting Register (Address: 0xA)* to 1 to enable on-chip watchdog reset function.
2. Write the watchdog timer period to *Watchdog Timer/RTC Timer Configuration Register 0(Address: 0x6)*; set bit[13] to 1 in *Watchdog Timer/RTC Timer Configuration Register 0(Address: 0x6)* to enable the RTC timer.
3. If the program runs normally, set bit[6] of the *RTC Setting Register (Address: 0xA)* to 1 to feed the watchdog within the watchdog timer period; If program run-away happens, the system will reset automatically.

4.9.6.6 Clearing RTC Interrupt Status

Programmers should operate according to the following steps:

1. Write 1 to bit[0] of *RTC Setting Register (Address: 0xA)* to clear RTC alarm interrupt status.
2. Write 1 to bit[1] of *RTC Setting Register (Address: 0xA)* to clear RTC timer interrupt status.
3. Write 1 to bit[12] of *RTC Setting Register (Address: 0xA)* to clear RTC watchdog interrupt status.

4.9.6.7 Selecting RTC OSC Source

Programmers should operate according to the following steps:

Set bit[13] of *RTC 32k Adjust Register 0 (Address: 0x8)* to 0 to use external 32K XTAL as RTC OSC Source; set this bit to 1 to use internal RC OSC as RTC OSC Source.

4.9.6.8 Enabling RTC 32KHz Adjustment

If internal RC OSC is used to generate 32KHz, time adjustment should be intermittently implemented to assure the precision of RTC clock. Programmers should operate according to the following steps:

1. Set bit[13] of *RTC 32k Adjust Register 1 (Address: 0x9)* to 1 to enable time adjustment.
2. Wait for bit[7] of *TOP_INT Interrupt Status (Address: 0x0800,0030)* to become 1, or poll bit[13] of *RTC 32k Adjust Register 1 (Address: 0x9)* until it becomes 0, which marks the end of adjustment.

4.10 Tsensor

4.10.1 Overview

Tsensor module is mainly integrated to monitor chip temperature. When current temperature of the chip reaches one of the threshold values(configurable by software), an interrupt will be generated so that programmer can reconfigure the chip accordingly.

4.10.2 Features

The features of Tsenor controller are as follows:

- Tsensor works at 12MHz clock domain
- Support one-time/cyclic temperature monitoring
- Support threshold interrupt and sample completion interrupt function
- Temperature detection range: -40℃ ~ 120℃

4.10.3 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Register Address	Register Name
1	0x0800,024C	<i>Tsensor Control Register</i>
2	0x0800,0250	<i>Tsensor Interrupt Register</i>
3	0x0800,0254	<i>Tsensor Temperature Record Register0</i>
4	0x0800,0258	<i>Tsensor Temperature Record Register1</i>
5	0x0800,025C	<i>Tsensor Temperature Record Register2</i>
6	0x0800,0260	<i>Tsensor Temperature Record Register3</i>

4.10.4 Register Definition

- Tsensor Control Register (Address: 0x0800,024C)

Bits	Attributes	Reset	Description
[31]	SC	1'h0	Tsensor_en Tsensor enable control 0: Disabled 1: Enabled
[30]	RW	1'h0	Tsensor sample mode Tsensor sample mode selection 0: one-time mode 1: cyclic mode
[29:22]	RO	8'h0	Reserved
[21:20]	RW	2'h0	Tsensor_time_sel Selection of the interval time between the starting of Tsenor and data collection. 2h'0: 200us 2h'1: 400us 2h'2: 800us 2h'3: 1600us
[19:10]	RW	10'h0	Tsensor_high_threshold The high threshold NOTE: For the relationship between the value of these bits and temperature, please refer to the NOTE in chapter 4.11.5 .
[9:0]	RW	10'h0	Tsensor_low_threshold The low threshold NOTE: For the relationship between the value of these bits and temperature, please refer to the NOTE in chapter 4.11.5 .

- Tsensor Interrupt Register (Address: 0x0800,0250)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	Tsensor_end_int_en Data sampling completion triggered interrupt enable 0: Disabled 1: Enabled
[30]	RW	1'h0	Tsensor_low_int_en Low temperature interrupt enable 0: Disabled 1: Enabled
[29]	RW	1'h0	Tsensor_high_int_en High temperature interrupt enable 0: Disabled 1: Enabled
[28:27]	RO	2'h0	Reserved
[26:4]	RW	23'h0	Tsensor_monitor_period The period for cyclic Tsensor temperature monitoring(12MHz)

Bits	Attributes	Reset	Description
			NOTE: Tsensor_monitor_period ≥ 2ms
[3]	RO	1'h0	Reserved
[2]	RC	1'h0	Tsensor_high_int High temperature interrupt When the temperature is higher than the high temperature threshold, a high temperature interrupt will be generated. 0: A high temperature interrupt is not generated. 1: A high temperature interrupt is generated.
[1]	RC	1'h0	Tsensor_low_int Low temperature interrupt When the temperature is lower than the low temperature threshold, a low temperature interrupt will be generated. 0: A low temperature interrupt is not generated. 1: A low temperature interrupt is generated.
[0]	RC	1'h0	Tsensor_end_int The end of data collection interrupt After finishing the data sampling, a data-sampled completion interrupt will be generated. 0: A data-sampled completion interrupt is not generated. 1: A data-sampled completion interrupt is generated.

When a data-sampled completion interrupt is generated, the detected result will be written into Tsensor Temperature Record Register x. In one-time mode, this result will be stored directly in *Tsensor Temperature Record Register0 (Address: 0x0800,0254)*, while in cyclic mode, the 8 sampling results will be written into *Tsensor Temperature Record Register0~3 (Address: 0x0800,0254~0x0800,0260)*.

- Tsensor Temperature Record Register0 (Address: 0x0800,0254)

Bits	Attributes	Reset	Description
[31:26]	RO	6'h0	Reserved
[25:16]	RW	10'h0	Tsensor_result1 Temperature detection result1
[15:10]	RO	6'h0	Reserved
[9:0]	RW	10'h0	Tsensor_result0 Temperature detection result0

- Tsensor Temperature Record Register1 (Address: 0x0800,0258)

Bits	Attributes	Reset	Description
[31:26]	RO	6'h0	Reserved
[25:16]	RW	10'h0	Tsensor_result3 Temperature detection result3
[15:10]	RO	6'h0	Reserved
[9:0]	RW	10'h0	Tsensor_result2 Temperature detection result2

● Tsensor Temperature Record Register2 (Address: 0x0800,025C)

Bits	Attributes	Reset	Description
[31:26]	RO	6'h0	Reserved
[25:16]	RW	10'h0	Tsensor_result5 Temperature detection result5
[15:10]	RO	6'h0	Reserved
[9:0]	RW	10'h0	Tsensor_result4 Temperature detection result4

● Tsensor Temperature Record Register3 (Address: 0x0800,0260)

Bits	Attributes	Reset	Description
[31:26]	RO	6'h0	Reserved
[25:16]	RW	10'h0	Tsensor_result7 Temperature detection result7
[15:10]	RO	6'h0	Reserved
[9:0]	RW	10'h0	Tsensor_result6 Temperature detection result6

4.10.5 Programmers' Guide

As mentioned above, Tsenor supports one-time mode and cyclic mode. Configuration steps for different working modes are given below.

4.10.5.1 One-time Mode Configuration

Programmers should follow the steps below to configure one-time mode:

1. Select the one-time mode by setting bit[30] in *Tsensor Control Register(Address: 0x0800,024C)* to 0 and configure bits[19:10] and bits[9:0] to set the threshold of the temperature and bits[21:20] to select the interval time.
2. Set bit[31], bit[30], bit[29] to 1 in *Tsensor Interrupt Register (Add: 0x0800,0250)* to enable the Data sampling completion triggered interrupt, High temperature interrupt and low temperature interrupt.
3. Set bit[31] in *Tsensor Control Register(Address: 0x0800,024C)* to 1 to start Tsensor.
4. Read bit[2], bit[1] and bit[0] in *Tsensor Control Register(Address: 0x0800,024C)* for the interrupt status.
5. Read the value of bits[9:0] in *Tsensor Interrupt Register (Add: 0x0800,0250)* when an interrupt triggered by data sampling completion signal is generated.
6. Calculate the corresponding value of Tsensor based on the formula listed below^{see note} and the acquired result.

4.10.5.2 Cyclic Mode Configuration

Programmers should follow the steps below to configure cyclic mode:

1. Select the cyclic mode by setting bit[30] in *Tsensor Control Register(Address: 0x0800,024C)* to 1 and configure the bits[19:10] and bits[9:0] to set the threshold of the temperature and bits[21:20] to select the interval time.
2. Set bit[31], bit[30] and bit[29] in *Tsensor Interrupt Register (Add: 0x0800,0250)* to 1 to enable the data sampling completion triggered interrupt, high temperature interrupt and low temperature interrupt, and configure bits[26:4] to set cyclic period accordingly.
3. Set bit[31] in *Tsensor Control Register(Address: 0x0800,024C)* to 1 to enable Tsensor.

4. Detect the bit[2], bit[1] and bit[0] in *Tsensor Interrupt Register (Add: 0x0800,0250)* for the interrupt status.
5. Read the values of bits[9:0] and bit[25:16] in *Tsensor Temperature Record Registers (Address: 0x0800,0254~ 0x0800,0260)* to gain the values of result0-7 when an interrupt triggered by data sampling completion signal is generated.
6. Calculate the corresponding value of Tsensor based on the formula listed below^{see note} and the acquired result.

NOTE

In theory, PWM duty cycle should be 19% in 120°C and 87.85% in -40°C. The value of TSEN_DUTY_CYC is the result of PWM duty cycle multiplied by 1023 and there is a linear relationship between temperature and TSEN_DUTY_CYC. TSEN_DUTY_CYC is a binary number in *Tsensor Temperature Record Registers (Address: 0x0800,0254~ 0x0800,0260)*, programmers should convert it to its corresponding decimal number with the following equation.

$$TSEN_DUTY_CYC = \frac{1024 * (19.0\% - 87.8\%)}{[120 - (-40)]} (T - 164)$$

Then the chip temperature can be calculated with the equation given below:

$$T = 164 - \frac{TSEN_DUTY_CYC * 160}{1024 * (87.8\% - 19.0\%)}$$

5 Memory

5.1 Memory Map

5.1.1 System Memory Map

The system address bus is 32-bit, and the address space is mainly allocated to the following parts.

- Memory type, including boot ROM, SRAM and DRAM
- Configuration register address space, mainly including the top layer and the sub-modules

Reserved	0xFFFF_FFFF
External DRAM	0xA000_0000
Reserved	0x8000_0000
On-chip SRAM (L2 Buffer 1)	0x6000_0000
Reserved	0x5800_0000
On-chip SRAM (L2 Buffer 0)	0x5000_0000
Reserved	0x4800_0000
Module Config Registers	0x3000_0000
Reserved	0x2000_0000
Top Config Registers	0x1000_0000
Reserved	0x0800_0000
Reserved	0x0000_0000

Figure 5-1 Memory Map

5.1.2 Sub-modules Memory Map

The figure below shows the register base address of each sub-module in the system.

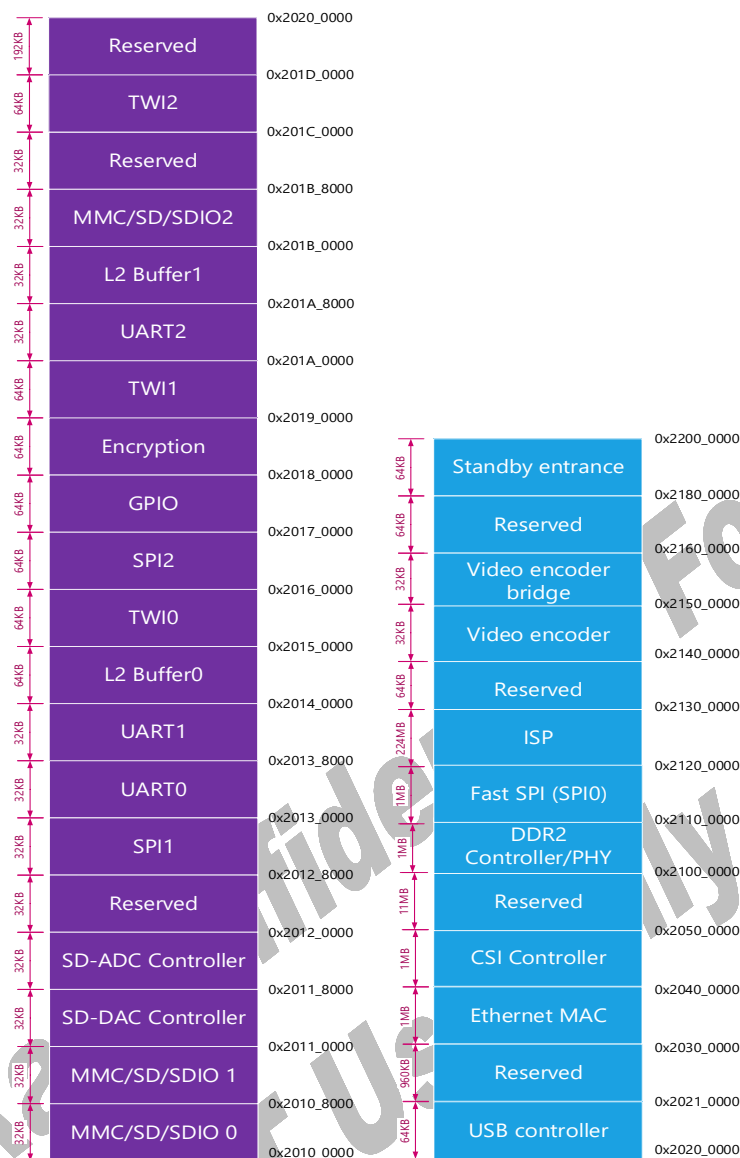


Figure 5-2 Module's Register Address Space

5.2 L2 Memory

L2 memory is an on-chip SRAM, which is considered as a bridge to connect external memory and peripherals. In order to satisfy the demands of high performance, 2 buffers are integrated for data transmission.

5.2.1 L2 Buffer0

5.2.1.1 Overview

L2 buffer0 contains 1344 x 32bits memory for data transfer between two different interfaces or between external memory and a certain interface.

5.2.1.2 Features

Features of L2 buffer0 are as follows:

- 16 blocks

- 32-bit data width
- Fixed priority
- Supports data transfer to/from external memory by DMA mode
- Supports data transfer between two blocks
- Supports direct access of CPU by AHB bus
- Used as system status and command cache when the processor is booted or in standby mode

5.2.1.3 Block Diagram

Block diagram of L2 buffer0 is shown in the following figure.

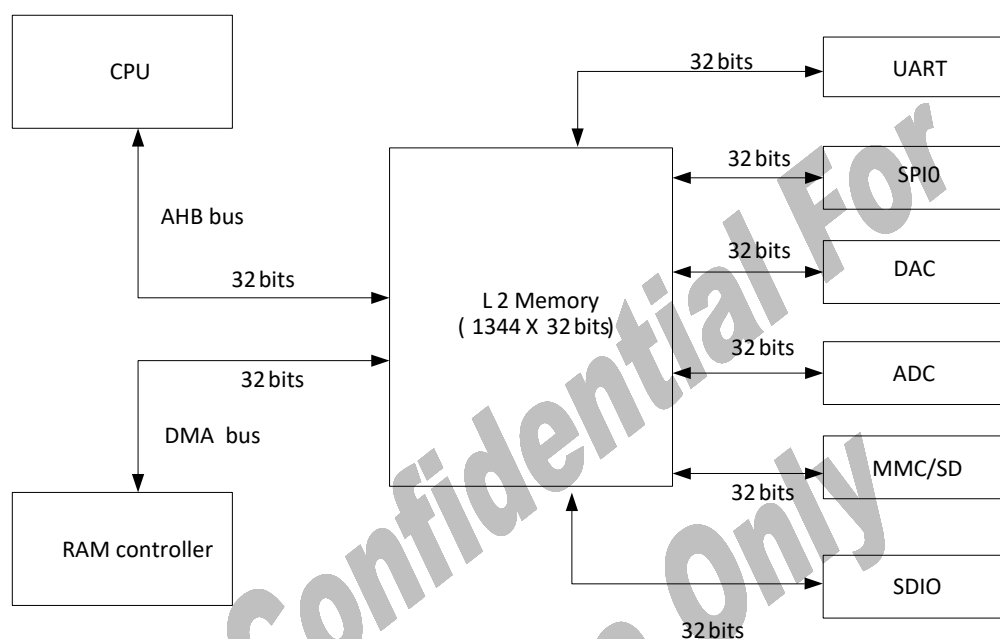


Figure 5-3 L2 Memory Block Diagram

5.2.1.4 Block Assignment

L2 memory is divided into 17 blocks, as shown in the following table.

Table 5-1 Block Assignment

START ADDRESS	END ADDRESS	SIZE	ASSIGNMENT
0x00	0x7F	512 Bytes	Buf 0
0x80	0xFF	512 Bytes	Buf 1
0x100	0x17F	512 Bytes	Buf 2
0x180	0x1FF	512 Bytes	Buf 3
0x200	0x27F	512 Bytes	Buf 4
0x280	0x2FF	512 Bytes	Buf 5
0x300	0x38F	512 Bytes	Buf 6
0x380	0x3FF	512 Bytes	Buf 7
0x400	0x41F	128 Bytes	UART0 TX Buf (Buf 8)
0x420	0x43F	128 Bytes	UART0 RX Buf (Buf 9)
0x440	0x45F	128 Bytes	UART1 TX Buf (Buf 10)
0x460	0x47F	128 Bytes	UART1 RX Buf (Buf 11)
0x480	0x49F	128 Bytes	UART2 TX buf(Buf 12)
0x4A0	0x4BF	128 Bytes	UART2 RX buf(Buf 13)

START ADDRESS	END ADDRESS	SIZE	ASSIGNMENT
0x4C0	0x4DF	128 Bytes	Reserved
0x4E0	0x4FF	128 Bytes	Reserved
0x500	0x53F	256 Bytes	USB

NOTE

Buf0, Buf1, Buf2, ..., Buf7 are not assigned to given interfaces. Programmers can assign these 8 blocks in accordance with particular applications.

5.2.1.5 Priority

L2 memory employs a fixed priority arbitrator. If multiple interfaces send requests at the same time, request from the interface with higher priority is responded first.

Table 5-2 Priority of L2 Memory (from highest to lowest)

PRIORITY	SOURCE
1	DMA controller
2	SPI0 (SPI0 interface)
3	UART (UART0, UART1 and UART2)
4	MMC/SD/SDIO, DAC or ADC
5	CPU
6	Data transfer between two interfaces

NOTE

If the requests are from the interfaces with same priority, such as UART0 and UART1, L2 buffer 0 responds to the requests on different time slots.

5.2.1.6 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Register Address	Register Name and Description
1	0x2014,0000 ~ 0x2014,003C	<i>DMA Address Information Registers of Buffer 0~15</i>
2	0x2014,0040 ~ 0x2014,007C	<i>DMA Operation Times Configuration Register of Buffer 0~15</i>
3	0x2014,0080	<i>DMA Request Register</i>
4	0x2014,0084	<i>Fraction DMA Address Information Register</i>
5	0x2014,0088	<i>Buffer0 ~ Buffer 7 Configuration Register</i>
6	0x2014,008C	<i>CPU-Controlled Buffer and Buffer8 ~ Buffer15 Configuration Register</i>
7	0x2014,0090	<i>Buffer Assignment Register 0</i>
8	0x2014,0094	<i>Buffer Assignment Register 1</i>
9	0x2014,0098	<i>Configuration Register of LDMA</i>
10	0x2014,009C	<i>Interrupt Enable Register</i>
11	0x2014,00A0	<i>Buffer Status Register 0</i>
12	0x2014,00A8	<i>Buffer Status Register 1</i>

NOTE

When those bits that can be cleared automatically need to be written into again after being read, programmers should set these bits as 0 unless the particular functions are required.

5.2.1.7 Register Definition

DMA Address Information Registers of Buffer 0 ~ Buffer 15 (addressed from 0x2014,0000 to 0x2014,003C) are used to define the start address of external RAM from which the data will be stored/sent.

Register 0x2014,0000 corresponds with the information of Buffer 0;

Register 0x2014,0004 corresponds to Buffer 1;

Register 0x2014,0008 corresponds to Buffer 2;

...,

Register 0x2014,003C corresponds to Buffer 15.

These 16 registers share the same register definitions. So the following section will only define one register.

● DMA Address Information Registers of Buffer 0~15 (Address:0x2014,0000~0x2014,003C)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	addr Address of external RAM from which the data will be stored/sent.

DMA Operation Times Configuration Register of Buffer 0 ~ Buffer 15 (addressed from 0x2014,0040 to 0x2014,007C) are used to define the DMA operation times to be implemented and to record those having been finished.

Register 0x2014,0040 corresponds to Buffer 0;

Register 0x2014,0044 corresponds to Buffer 1;

Register 0x2014,0048 corresponds to Buffer 2;

...,

Register 0x2014,007C, corresponds to Buffer 15.

These 16 registers share similar register definitions. So the following section will only define one register.

- DMA Operation Times Configuration Register of Buffer 0~15 (Address: 0x2014,0040~0x2014,007C)

Bits	Attributes	Reset	Description
[31:24]	RO	8'h0	cnt_current_h These bits record times of completed DMA operations. It should be noted that these bits are read only. NOTE: 1) When the data to be transferred is less than 8KB, times of completed DMA operations equal CNT_current. 2) When the data to be transferred is equal to or more than 8KB, times of completed DMA operations can be calculated from the equation: $CNT_current_H * CNT_cfg$.
[23:16]	RW	8'h0	cnt_cfg_h Extended bits of CNT_cfg. NOTE: 1) When the data to be transferred is less than 4KB, CNT_cfg_H must be 0. 2) When the data to be transferred is equal to or more than 4KB, • For buffer0 to buffer7, CNT_cfg must be $(N * 8)$, $(N \geq 1, N * 8 < 256)$ Then the data to be transferred will be $((CNT_cfg_H + 1) * N * 8 * 64)$ bytes • For buffer8 to buffer15, CNT_cfg must be $(N * 2)$, $(N \geq 1, N * 2 < 256)$ Then the data to be transferred will be $((CNT_cfg_H + 1) * N * 2 * 64)$ bytes 3) When CNT_cfg_H is valid, the DMA operation times of $(1 * CNT_cfg)$ is called a small loop, the DMA operation times of $(CNT_cfg_H * CNT_cfg)$ is called a big loop.
[15:8]	RO	8'h0	cnt_current These bits record the times of DMA operation having been finished. It should be noted that these bits are read only.
[7:0]	RW	8'h0	cnt_cfg Total times of DMA operations to be implemented. NOTE: Programmers should write the expected times of DMA operations into these bits. For example, 1024 bytes are expected to be transferred. Programmers should write 0x010000 into these bits.

Bits	Attributes	Reset	Description
			($\frac{1024bytes}{64bytes} = 16 = 010000B$, DMA controller transfers 64 bytes data in a DMA operation.)

● DMA Request Register (Address: 0x2014,0080)

This register is used to configure DMA request signal of each buffer and to set relative parameters of fraction DMA operation.

The DMA controller transfers 64-byte data in a DMA operation. If the number of data to be transferred is less than 64 bytes, fraction DMA operation is utilized.

Assuming that total number of data to be transferred is M,

$$M = 64 \times A + N$$

Then A is normal DMA operation times, and N is the data quantity fraction DMA operation needs to process.

Bits	Attributes	Reset	Description
[31]	RW/SC	1'h0	buf7_dma_req Buffer7 DMA request signal. When this bit is set by CPU, the Buffer7 DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[30]	RW/SC	1'h0	buf6_dma_req Buffer6 DMA request signal. When this bit is set by CPU, the Buffer6 DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[29]	RW/SC	1'h0	buf5_dma_req Buffer5 DMA request signal. When this bit is set by CPU, the Buffer5 DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[28]	RW/SC	1'h0	buf4_dma_req Buffer4 DMA request signal. When this bit is set by CPU, the Buffer4 DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[27]	RW/SC	1'h0	buf3_dma_req Buffer3 DMA request signal. When this bit is set by CPU, the Buffer3 DMA operation is enabled. This bit will be cleared automatically after the

Bits	Attributes	Reset	Description
			requested DMA operation has been finished successfully.
[26]	RW/SC	1'h0	buf2_dma_req Buffer2 DMA request signal. When this bit is set by CPU, the Buffer2 DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[25]	RW/SC	1'h0	buf1_dma_req Buffer1 DMA request signal. When this bit is set by CPU, the Buffer1 DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[24]	RW/SC	1'h0	buf0_dma_req Buffer0 DMA request signal. When this bit is set by CPU, the Buffer0 DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[23:22]	RO	2'h0	Reserved
[21]	RW/SC	1'h0	UART2_rx_buf_req UART2 RX buffer DMA request signal. When this bit is set by CPU, the UART2 RX buffer DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[20]	RW/SC	1'h0	UART2_tx_buf_req UART2 TX buffer DMA request signal. When this bit is set by CPU, the UART2 TX buffer DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[19]	RW/SC	1'h0	UART1_rx_buf_req UART1 RX buffer DMA request signal. When this bit is set by CPU, the UART1 RX buffer DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[18]	RW/SC	1'h0	UART1_tx_buf_req UART1 TX buffer DMA request signal. When this bit is set by CPU, the UART1 TX buffer

Bits	Attributes	Reset	Description
			DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[17]	RW/SC	1'h0	UART0_rx_buf_req UART0 RX buffer DMA request signal. When this bit is set by CPU, the UART0 RX buffer DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[16]	RW/SC	1'h0	UART0_tx_buf_req UART0 TX buffer DMA request signal. When this bit is set by CPU, the UART0 TX buffer DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[15:10]	RW	6'h0	frac_dma_len Quantity of data to be transferred in next fraction DMA operation. Number of data to be transferred = Frac_DMA_len + 1 NOTE: When to read data from external RAM, the number of data to be transferred in a Fraction DMA operation should be an even number. When to send data to external RAM, the number of data to be transferred in a Fraction DMA operation can be an odd or even number.
[9]	RW	1'h0	frac_dma_req Fraction DMA request signal. When this bit is set by CPU, the fraction DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[8]	RW	1'h0	frac_dma_dir 0: To read data from L2 memory in fraction DMA mode. 1: To write data into L2 memory in fraction DMA mode.
[7:1]	RW	7'h0	frac_dma_l2_add The address of L2 memory starting from which the data will be stored/sent in next fraction DMA operation. NOTE: L2 memory is divided into several 64-byte regions. The first 64-byte space is called Region 0; the second, Region 1; ... Programmers should write

Bits	Attributes	Reset	Description
			into these bits the address (Region X) from which the data will be stored/sent in next fraction DMA operation.
[0]	RW	1'h0	dma_en 0: To disable DMA function of L2 memory. 1: To enable DMA function of L2 memory.

● Fraction DMA Address Information Register (Address: 0x2014,0084)

The DMA controller transfers 64-byte data in a DMA operation. If the number of data to be transferred is less than 64 bytes, fraction DMA operation can be utilized. This working mode is called "fraction DMA mode".

This register is used to define the address of external RAM starting from which the data will be stored/sent, and to configure the status flag of two kinds of operations.

Bits	Attributes	Reset	Description
[31:30]	RW	2'h0	frac_dma_addr_h High address of external RAM from which the data will be stored/sent. The real address: (Frac_DMA_ADDR_H <<28) + Frac_DMA_ADDR_L
[29]	RW	1'h0	ahb_flag_en 0: The buffer status bit is not changed after the data transfer by AHB bus has been finished. 1: The buffer status bit may be changed after the data transfer by AHB bus has been finished.
[28]	RW	1'h0	ldma_flag_en 0: The buffer status bit is not changed after the LDMA operation has been finished. 1: The buffer status bit may be changed after the LDMA operation has been finished. NOTE: 1) The data transfer between two buffers of L2 memory is called Local DMA operation and abbreviated as LDMA operation. 2) Please configure Configuration Register of LDMA (Address: 0x2014, 0098) for other setting of the LDMA operation.
[27:0]	RW	28'h0	frac_dma_addr_l Low address of external RAM from which the data will be stored/sent. The real address: (Frac_DMA_ADDR_H <<28) + Frac_DMA_ADDR_L

● Buffer0 ~ Buffer 7 Configuration Register (Address: 0x2014,0088)

The register is used to set Buffer 0 ~ Buffer 7.

Bits	Attributes	Reset	Description
[31]	RW	1'h0	buffer7_flag_clr 1: To clear buffer7 status flag.

Bits	Attributes	Reset	Description
			NOTE: This bit is cleared automatically.
[30]	RW	1'h0	buffer6_flag_clr 1: To clear buffer6 status flag. NOTE: This bit is cleared automatically.
[29]	RW	1'h0	buffer5_flag_clr 1: To clear buffer5 status flag. NOTE: This bit is cleared automatically.
[28]	RW	1'h0	buffer4_flag_clr 1: To clear buffer4 status flag. NOTE: This bit is cleared automatically.
[27]	RW	1'h0	buffer3_flag_clr 1: To clear buffer3 status flag. NOTE: This bit is cleared automatically.
[26]	RW	1'h0	buffer2_flag_clr 1: To clear buffer2 status flag. NOTE: This bit is cleared automatically.
[25]	RW	1'h0	buffer1_flag_clr 1: To clear buffer1 status flag. NOTE: This bit is cleared automatically.
[24]	RW	1'h0	buffer0_flag_clr 1: To clear buffer0 status flag. NOTE: This bit is cleared automatically.
[23]	RW	1'h0	buffer7_vld 1: To enable buffer7.
[22]	RW	1'h0	buffer6_vld 1: To enable buffer6.
[21]	RW	1'h0	buffer5_vld 1: To enable buffer5.
[20]	RW	1'h0	buffer4_vld 1: To enable buffer4.
[19]	RW	1'h0	buffer3_vld 1: To enable buffer3.
[18]	RW	1'h0	buffer2_vld 1: To enable buffer2.
[17]	RW	1'h0	buffer1_vld 1: To enable buffer1.
[16]	RW	1'h0	buffer0_vld 1: To enable buffer0.
[15]	RW	1'h0	buffer7_dir 0: Read operation between buffer7 and RAM controller by DMA bus. 1: Write operation between buffer7 and RAM controller by DMA bus.
[14]	RW	1'h0	buffer6_dir 0: Read operation between buffer6 and RAM controller by DMA bus. 1: Write operation between buffer6 and RAM

Bits	Attributes	Reset	Description
			controller by DMA bus.
[13]	RW	1'h0	Buffer5_dir 0: Read operation between buffer5 and RAM controller by DMA bus. 1: Write operation between buffer5 and RAM controller by DMA bus.
[12]	RW	1'h0	buffer4_dir 0: Read operation between buffer4 and RAM controller by DMA bus. 1: Write operation between buffer4 and RAM controller by DMA bus.
[11]	RW	1'h0	buffer3_dir 0: Read operation between buffer3 and RAM controller by DMA bus. 1: Write operation between buffer3 and RAM controller by DMA bus.
[10]	RW	1'h0	buffer2_dir 0: Read operation between buffer2 and RAM controller by DMA bus. 1: Write operation between buffer2 and RAM controller by DMA bus.
[9]	RW	1'h0	buffer1_dir 0: Read operation between buffer1 and RAM controller by DMA bus. 1: Write operation between buffer1 and RAM controller by DMA bus.
[8]	RW	1'h0	buffer0_dir 0: Read operation between buffer0 and RAM controller by DMA bus. 1: Write operation between buffer0 and RAM controller by DMA bus.
[7]	RW	1'h0	buffer7_dma 0: To disable data transfer between buffer7 and RAM controller by DMA bus. 1: To enable data transfer between buffer7 and RAM controller by DMA bus.
[6]	RW	1'h0	buffer6_dma 0: To disable data transfer between buffer6 and RAM controller by DMA bus. 1: To enable data transfer between buffer6 and RAM controller by DMA bus.
[5]	RW	1'h0	buffer5_dma 0: To disable data transfer between buffer5 and RAM controller by DMA bus. 1: To enable data transfer between buffer5 and RAM controller by DMA bus.
[4]	RW	1'h0	buffer4_dma 0: To disable data transfer between buffer4 and

Bits	Attributes	Reset	Description
			RAM controller by DMA bus. 1: To enable data transfer between buffer4 and RAM controller by DMA bus.
[3]	RW	1'h0	buffer3_dma 0: To disable data transfer between buffer3 and RAM controller by DMA bus. 1: To enable data transfer between buffer3 and RAM controller by DMA bus.
[2]	RW	1'h0	buffer2_dma 0: To disable data transfer between buffer2 and RAM controller by DMA bus. 1: To enable data transfer between buffer2 and RAM controller by DMA bus.
[1]	RW	1'h0	buffer1_dma 0: To disable data transfer between buffer1 and RAM controller by DMA bus. 1: To enable data transfer between buffer1 and RAM controller by DMA bus.
[0]	RW	1'h0	buffer0_dma 0: To disable data transfer between buffer0 and RAM controller by DMA bus. 1: To enable data transfer between buffer0 and RAM controller by DMA bus.

- CPU-Controlled Buffer & Buffer8 ~ Buffer15 Configuration Register (Address:0x2014,008C)

The register is used to configure CPU-controlled buffer and Buffer8~Buffer15.

Bits	Attributes	Reset	Description
[31]	RO	1'h0	Reserved
[30]	RW	1'h0	uart2_dma_vld 0: To disable data transfer between UART2 TX/RX Buf (Buf12/Buf13) and RAM controller by DMA bus. 1: To enable data transfer between UART2 TX/RX Buf (Buf12/Buf13) and RAM controller by DMA bus.
[29]	RW	1'h0	uart1_dma_vld 0: To disable data transfer between UART1 TX/RX Buf (Buf10/Buf11) and RAM controller by DMA bus. 1: To enable data transfer between UART1 TX/RX Buf (Buf10/Buf11) and RAM controller by DMA bus.
[28]	RW	1'h0	uart0_dma_vld 0: To disable data transfer between UART0 TX/RX Buf (Buf8/Buf9) and RAM controller by DMA bus. 1: To enable data transfer between UART0 TX/RX Buf (Buf8/Buf9) and RAM controller by DMA bus.
[27:22]	RO	6'h0	Reserved
[21]	RW	1'h0	uart2_rx_clr 1: To clear UART2 RX Buf status flag.

Bits	Attributes	Reset	Description
			NOTE: This bit is cleared automatically.
[20]	RW	1'h0	uart2_tx_clr 1: To clear UART2 TX Buf status flag. NOTE: This bit is cleared automatically.
[19]	RW	1'h0	uart1_rx_clr 1: To clear UART1 RX Buf status flag. NOTE: This bit is cleared automatically.
[18]	RW	1'h0	uart1_tx_clr 1: To clear UART1 TX Buf status flag. NOTE: This bit is cleared automatically.
[17]	RW	1'h0	uart0_rx_clr 1: To clear UART0 RX Buf status flag. NOTE: This bit is cleared automatically.
[16]	RW	1'h0	uart0_tx_clr 1: To clear UART0 TX Buf status flag. NOTE: This bit is cleared automatically.
[15:8]	RO	8'h0	Reserved
[7:4]	WO	4'h0	cpu_buf_val Current number of data contained in the CPU-controlled buffer. Number of data = CPU_buf_val×64 (unit: byte) NOTE: 0 ≤ CPU_buf_val ≤ 8. The value is loaded into the corresponding status bits of Buffer Status Register 0(Address: 0x2014,00A0) . For instance, that Buffer0's status can be controlled by CPU is wanted. Therefore, you should set bit[3] and bits[2:0] of this register as 1B and 000B respectively, and write the data number you want into these bits. Then programmers can check bits[3:0] of Buffer Status Register 0(Address: 0x2014,00A0) to confirm whether the configuration is valid or not.
[3]	WO	1'h0	cpu_buf_sel_en 0: To disable CPU-controlled buffer function(the setting of bits[7:4] and bits[2:0] is invalid). 1: To enable CPU-controlled buffer function(the setting of bits[7:4] and bits[2:0] is valid).
[2:0]	WO	3'h0	cpu_buf_sel To select the buffer whose status can be controlled by CPU. 3'h0: buffer0 3'h1: buffer1 3'h2: buffer2 3'h3: buffer3 3'h4: buffer4 3'h5: buffer5 3'h6: buffer6 3'h7: buffer7

Buffer Assignment Register 0 & 1 assign the dynamic buffers (Buf0 ~ Buf 7) to interfaces.

● Buffer Assignment Register 0 (Address: 0x2014,0090)

Bits	Attributes	Reset	Description
[31:30]	RO	2'h0	Reserved
[29:27]	RW	3'h0	dac_add Buffer address for DAC control module. 0 corresponds with Buffer 0; 1, Buffer 1; 2, Buffer 2; ..., 7, Buffer 7.
[26:24]	RW	3'h0	SPIOr_add Buffer address for RX buffer of SPI0 Controller. 0 corresponds with Buffer 0; 1, Buffer 1; 2, Buffer 2; ..., 7, Buffer 7.
[23:21]	RW	3'h0	SPI0w_add Buffer address for TX buffer of SPI0 Controller. 0 corresponds with Buffer 0; 1, Buffer 1; 2, Buffer 2; ..., 7, Buffer 7.
[20:18]	RO	3'h0	MCI2_add Buffer address for MCI2 Interface. 0 corresponds with Buffer 0; 1, Buffer 1; 2, Buffer 2; ..., 7, Buffer 7.
[17:15]	RW	3'h0	MCI1_add Buffer address for MCI1 Interface. 0 corresponds with Buffer 0; 1, Buffer 1; 2, Buffer 2; ..., 7, Buffer 7.
[14:12]	RW	3'h0	MCI0_add Buffer address for MCI0 Interface. 0 corresponds with Buffer 0; 1, Buffer 1; 2, Buffer 2; ..., 7, Buffer 7.
[11:0]	RO	12'h0	Reserved

● Buffer Assignment Register 1 (Address: 0x2014,0094)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	mem_gate 1: To gate control the enabling signal of SRAM. NOTE: When this bit is set, power consumption is reduced in standby mode.
[30:15]	RO	16'h0	Reserved
[14:12]	RW	3'h0	sdadc_add Buffer address for ADC. 0 corresponds with Buffer 0; 1, Buffer 1; 2, Buffer 2; ..., 7, Buffer 7.
[11:0]	RO	12'h0	Reserved

● Configuration Register of LDMA (Address: 0x2014,0098)

This register is used to configure the LDMA parameters.

Programmers should notice the following items:

- 1) The data transfer between two buffers of L2 memory is called Local DMA operation and abbreviated as LDMA operation.

- 2) That the status bit is changed or not after the LDMA operation is decided by bit[28] of [Fraction DMA Address Information Register \(Address: 0x2014,0084\)](#).

Bits	Attributes	Reset	Description
[31]	RW	1'h0	ldma_vld 1: To enable data transfer between two kinds of interfaces. NOTE: When the data transfer has been finished successfully, this bit is set to 0 automatically.
[30:14]	RW	17'h0	cnt Number of data to be transferred (unit: 4bytes)
[13:7]	RW	7'h0	d_add L2 memory address in which the data will be stored. (unit: 64bytes)
[6:0]	RW	7'h0	s_add L2 memory address from which the data will be transmitted. (unit: 64bytes)

● Interrupt Enable Register (Address: 0x2014,009C)

This register enables/disables the buffer interrupts.

For bits[16:0] of this register, If CNT_cfg_H is valid, the DMA operation requested by corresponding Buffer refers to the DMA operation of the big loop. If CNT_cfg_H is invalid, the DMA operation requested by corresponding Buffer refers to the DMA operation defined by CNT_cfg. For details of CNT_cfg_H, big loop and CNT_cfg, please refer to bits[23:16] and bits[7:0] of [DMA Operation Times Configuration Register of Buffer0 \(Address: 0x2014,0040\)](#).

Bits	Attributes	Reset	Description
[31:20]	RO	12'h0	Reserved
[19]	RW	1'h0	DMA_loop_int_en 0: To disable small loop interrupt. 1: To enable small loop interrupt. NOTE: When the DMA operation of a small loop is finished, an interrupt is generated. This kind of interrupt is called small loop interrupt. For the details of small loop, please refer to bits[23:16] of DMA Operation Times Configuration Register of Buffer0 (Address: 0x2014, 0040) .
[18]	RW	1'h0	Reserved
[17]	RW	1'h0	ldma_vld 0: To disable ldma_vld interrupt. 1: To enable ldma_vld interrupt. NOTE: To enable data transfer between two buffers of L2 memory is called ldma_vld. When ldma_vld interrupt is enabled, an interrupt is generated when a LDMA operation has finished.
[16]	RW	1'h0	buf7_dma_req 0: To disable buf7_dma_req interrupt. 1: To enable buf7_dma_req interrupt. NOTE: The Buf7_dma_req interrupt is generated

Bits	Attributes	Reset	Description
			when the DMA operation requested by Buffer 7 has finished.
[15]	RW	1'h0	buf6_dma_req 0: To disable buf6_dma_req interrupt. 1: To enable buf6_dma_req interrupt. NOTE: The Buf6_dma_req interrupt is generated when the DMA operation requested by Buffer 6 has finished.
[14]	RW	1'h0	buf5_dma_req 0: To disable buf5_dma_req interrupt. 1: To enable buf5_dma_req interrupt. NOTE: The Buf5_dma_req interrupt is generated when the DMA operation requested by Buffer 5 has finished.
[13]	RW	1'h0	buf4_dma_req 0: To disable buf4_dma_req interrupt. 1: To enable buf4_dma_req interrupt. NOTE: The Buf4_dma_req interrupt is generated when the DMA operation requested by Buffer 4 has finished.
[12]	RW	1'h0	buf3_dma_req 0: To disable buf3_dma_req interrupt. 1: To enable buf3_dma_req interrupt. NOTE: The Buf3_dma_req interrupt is generated when the DMA operation requested by Buffer 3 has finished.
[11]	RW	1'h0	buf2_dma_req 0: To disable buf2_dma_req interrupt. 1: To enable buf2_dma_req interrupt. NOTE: The Buf2_dma_req interrupt is generated when the DMA operation requested by Buffer 2 has finished.
[10]	RW	1'h0	buf1_dma_req 0: To disable buf1_dma_req interrupt. 1: To enable buf1_dma_req interrupt. NOTE: The Buf1_dma_req interrupt is generated when the DMA operation requested by Buffer 1 has finished.
[9]	RW	1'h0	buf0_dma_req 0: To disable buf0_dma_req interrupt. 1: To enable buf0_dma_req interrupt. NOTE: The Buf0_dma_req interrupt is generated when the DMA operation requested by Buffer 0 has finished.
[8:7]	RO	2'h0	Reserved
[6]	RW	1'h0	uart2w_dma_req 0: To disable uart2w_dma_req interrupt. 1: To enable uart2w_dma_req interrupt.

Bits	Attributes	Reset	Description
			NOTE: The uart2w_dma_req interrupt is generated when the DMA operation requested by UART2 RX buffer has finished.
[5]	RW	1'h0	uart2r_dma_req 0: To disable uart2r_dma_req interrupt. 1: To enable uart2r_dma_req interrupt. NOTE: The uart2r_dma_req interrupt is generated when the DMA operation requested by UART2 TX buffer has finished.
[4]	RW	1'h0	uart1w_dma_req 0: To disable uart1w_dma_req interrupt. 1: To enable uart1w_dma_req interrupt. NOTE: The uart1w_dma_req interrupt is generated when the DMA operation requested by UART1 RX buffer has finished.
[3]	RW	1'h0	uart1r_dma_req 0: To disable uart1r_dma_req interrupt. 1: To enable uart1r_dma_req interrupt. NOTE: The uart1r_dma_req interrupt is generated when the DMA operation requested by UART1 TX buffer has finished.
[2]	RW	1'h0	uart0w_dma_req 0: To disable uart0w_dma_req interrupt. 1: To enable uart0w_dma_req interrupt. NOTE: The uart0w_dma_req interrupt is generated when the DMA operation requested by UART0 RX buffer has finished.
[1]	RW	1'h0	uart0r_dma_req 0: To disable uart0r_dma_req interrupt. 1: To enable uart0r_dma_req interrupt. NOTE: The uart0r_dma_req interrupt is generated when the DMA operation requested by UART0 TX buffer has finished.
[0]	RW	1'h0	frac_dma_req_cfg 0: To disable frac_dma_req interrupt. 1: To enable frac_dma_req interrupt. NOTE: The frac_dma_req interrupt is generated when the fraction DMA operation has finished.

Buffer Status Register 0&1 indicate current status (data quantity) of each buffer.

- Buffer Status Register 0 (Address: 0x2014,00A0)

Bits	Attributes	Reset	Description
[31:28]	RO	4'h0	buf7_cnt Total quantity of data contained in Buffer7. The number of data = Buf7_sta x 64 (unit: byte) 0 <= Buf7_sta <= 8
[27:24]	RO	4'h0	buf6_cnt

Bits	Attributes	Reset	Description
			Total quantity of data contained in Buffer6. The number of data = Buf6_sta x 64 (unit: byte) 0 ≤ Buf6_sta ≤ 8
[23:20]	RO	4'h0	buf5_cnt Total quantity of data contained in Buffer5. The number of data = Buf5_sta x 64 (unit: byte) 0 ≤ Buf5_sta ≤ 8
[19:16]	RO	4'h0	buf4_cnt Total quantity of data contained in Buffer4. The number of data = Buf4_sta x 64 (unit: byte) 0 ≤ Buf4_sta ≤ 8
[15:12]	RO	4'h0	buf3_cnt Total quantity of data contained in Buffer3. The number of data = Buf3_sta x 64 (unit: byte) 0 ≤ Buf3_sta ≤ 8
[11:8]	RO	4'h0	buf2_cnt Total quantity of data contained in Buffer2. The number of data = Buf2_sta x 64 (unit: byte) 0 ≤ Buf2_sta ≤ 8
[7:4]	RO	4'h0	buf1_cnt Total quantity of data contained in Buffer1. The number of data = Buf1_sta x 64 (unit: byte) 0 ≤ Buf1_sta ≤ 8
[3:0]	RO	4'h0	buf0_cnt Total quantity of data contained in Buffer0. The number of data = Buf0_sta x 64 (unit: byte) 0 ≤ Buf0_sta ≤ 8

● Buffer Status Register 1 (Address: 0x2014,00A8)

Bits[31:16] of this register are valid only when cnt_cfg_h is valid. When the DMA operation of a small loop is finished, an interrupt is generated. This kind of interrupt is called small loop interrupt. For details of small loop and cnt_cfg_h, please refer to bits[23:16] of [DMA Operation Times Configuration Register of Buffer0 \(Address: 0x2014,0040\)](#)

Bits	Attributes	Reset	Description
[31:30]	RO	2'h0	Reserved
[29]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of UART2 RX Buffer is not generated. 1: A small loop interrupt of UART2 RX Buffer is generated.
[28]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of UART2 TX Buffer is not generated. 1: A small loop interrupt of UART2 TX Buffer is generated.
[27]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of UART1 RX Buffer is not

Bits	Attributes	Reset	Description
			generated. 1: A small loop interrupt of UART1 RX Buffer is generated.
[26]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of UART1 TX Buffer is not generated. 1: A small loop interrupt of UART1 TX Buffer is generated.
[25]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of UART0 RX Buffer is not generated. 1: A small loop interrupt of UART0 RX Buffer is generated.
[24]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of UART0 TX Buffer is not generated. 1: A small loop interrupt of UART0 TX Buffer is generated.
[23]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 7 is not generated. 1: A small loop interrupt of Buffer 7 is generated.
[22]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 6 is not generated. 1: A small loop interrupt of Buffer 6 is generated.
[21]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 5 is not generated. 1: A small loop interrupt of Buffer 5 is generated.
[20]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 4 is not generated. 1: A small loop interrupt of Buffer 4 is generated.
[19]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 3 is not generated. 1: A small loop interrupt of Buffer 3 is generated.
[18]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 2 is not generated. 1: A small loop interrupt of Buffer 2 is generated.
[17]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 1 is not generated. 1: A small loop interrupt of Buffer 1 is generated.
[16]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 0 is not

Bits	Attributes	Reset	Description
			generated. 1: A small loop interrupt of Buffer 0 is generated.
[15:12]	RO	4'h0	Reserved
[11:10]	RO	2'h0	uart2_rx_cnt Total quantity of data contained in UART2 RX Buf (Buf 13). The number of data = uart2_rx_cnt x 64 (unit: byte) 0 <= uart2_rx_cnt <= 2
[9:8]	RO	2'h0	uart2_tx_cnt Total quantity of data contained in UART2 TX Buf (Buf 12). The number of data = uart2_tx_cnt x 64 (unit: byte) 0 <= uart2_tx_cnt <= 2
[7:6]	RO	2'h0	uart1_rx_cnt Total quantity of data contained in UART1 RX Buf (Buf 11). The number of data = uart1_rx_cnt x 64 (unit: byte) 0 <= uart1_rx_cnt <= 2
[5:4]	RO	2'h0	uart1_tx_cnt Total quantity of data contained in UART1 TX Buf (Buf 10). The number of data = uart1_tx_cnt x 64 (unit: byte) 0 <= uart1_tx_cnt <= 2
[3:2]	RO	2'h0	uart0_rx_cnt Total quantity of data contained in UART0 RX Buf (Buf 9). The number of data = uart0_rx_cnt x 64 (unit: byte) 0 <= uart0_rx_cnt <= 2
[1:0]	RO	2'h0	uart0_tx_cnt Total quantity of data contained in UART0 TX Buf (Buf 8). The number of data = uart0_tx_cnt x 64 (unit: byte) 0 <= uart0_tx_cnt <= 2

5.2.2 L2 Buffer1

5.2.2.1 Overview

L2 buffer1 contains 1088 x 32bits memory for data transfer between two different interfaces, or between external memory and a certain interface.

5.2.2.2 Features

Features of L2 buffer1 are as follows:

- 9 blocks
- 32-bit data width
- Fixed priority
- Supports data transfer to/from external memory by DMA mode
- Supports data transfer between two blocks
- Supports direct access of CPU by AHB bus

- Used as system status and command cache when the processor is booted or in standby mode

5.2.2.3 Block Diagram

Block diagram of L2 buffer1 is shown in the following figure.

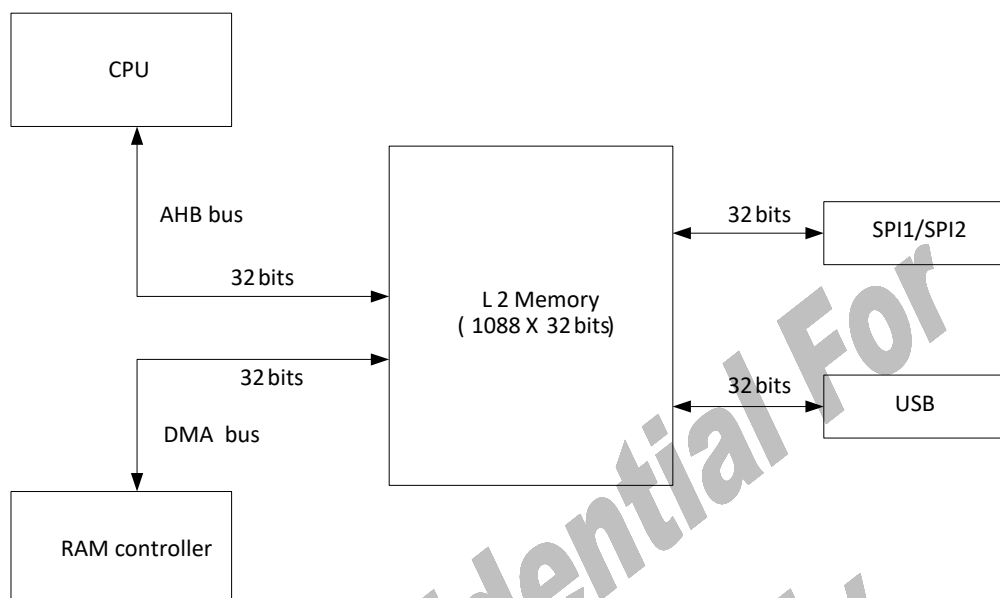


Figure 5-4 L2 Memory Block Diagram

5.2.2.4 Block Assignment

L2 memory is divided into 9 blocks, as shown in the following table.

Table 5-3 Block Assignment

START ADDRESS	END ADDRESS	SIZE	ASSIGNMENT
0x00	0x7F	512 Bytes	Buf0
0x80	0xFF	512 Bytes	Buf1
0x100	0x17F	512 Bytes	Buf2
0x180	0x1FF	512 Bytes	Buf3
0x200	0x27F	512 Bytes	Buf4
0x280	0x2FF	512 Bytes	Buf5
0x300	0x37F	512 Bytes	Buf6
0x380	0x3FF	512 Bytes	Buf7
0x400	0x43F	256 Bytes	USB

NOTE

Buf0, Buf1, Buf2, ..., Buf7 are not assigned to given interfaces. Programmers can assign these 8 blocks in accordance with particular applications.

5.2.2.5 Priority

L2 memory employs a fixed priority arbitrator. If multiple interfaces send requests at the same time, request from the interface with higher priority is responded first.

Table 5-4 Priority of L2 Memory (from highest to lowest)

PRIORITY	SOURCE
1	DMA controller
2	SPI (SPI1/SPI2)
3	CPU
4	USB
5	Data transfer between two interfaces

NOTE

If the requests are from the interfaces with same priority, such as SPI1 and SPI2, L2 buffer1 responds to the requests on different time slots.

5.2.2.6 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Register Address	Register Name and Description
1	0x201A,8000 ~ 0x201A,801C	<i>DMA Address Information Registers of Buffer 0~7</i>
2	0x201A,8040 ~ 0x201A,805C	<i>DMA Operation Times Configuration Register of Buffer 0~7</i>
3	0x201A,8080	<i>DMA Request Register</i>
4	0x201A,8084	<i>Fraction DMA Address Information Register</i>
5	0x201A,8088	<i>Buffer0 ~ Buffer 7 Configuration Register</i>
6	0x201A,808C	<i>CPU-Controlled Buffer Configuration Register</i>
7	0x201A,8090	<i>Buffer Assignment Register 0</i>
8	0x201A,8094	<i>Buffer Assignment Register 1</i>
9	0x201A,8098	<i>Configuration Register of LDMA</i>
10	0x201A,809C	<i>Interrupt Enable Register</i>
11	0x201A,80A0	<i>Buffer Status Register 0</i>
12	0x201A,80A8	<i>Buffer Status Register 1</i>

NOTE

When those bits that can be cleared automatically need to be written into again after being read, programmers should set these bits as 0 unless the particular functions are required.

5.2.2.7 Register Definition

DMA Address Information Registers of Buffer 0 ~ Buffer 7 (addressed from 0x201A,8000 to 0x201A,801C) are used to define the address of external RAM starting from which the data will be stored/sent.

Register 0x201A,8000 corresponds with the information of Buffer 0;

Register 0x201A,8004 corresponds to Buffer 1;

Register 0x201A,8008 corresponds to Buffer 2;

...,

Register 0x201A,801C corresponds to Buffer 7.

These 16 registers share similar register definitions. So the following section will only define one register.

● DMA Address Information Registers of Buffer 0~7 (Address: 0x201A,8000~0x201A,801C)

Bits	Attributes	Reset	Description
[31:0]	RW	-	addr Address of external RAM from which the data will be stored/sent.

DMA Operation Times Configuration Register of Buffer 0 ~ Buffer 7 (addressed from 0x201A,8040 to 0x201A,805C) are used to define the DMA operation times to be implemented and to record those having been finished.

Register 0x201A,8040 corresponds to Buffer 0;

Register 0x201A,8044 corresponds to Buffer 1;

Register 0x201A,8048 corresponds to Buffer 2;

...,

Register 0x201A,805C, corresponds to Buffer 7.

These 16 registers share similar register definitions. So the following section will only define one register.

● DMA Operation Times Configuration Registers of Buffer 0~7 (Address: 0x201A,8040~0x201A,805C)

Bits	Attributes	Reset	Description
[31:24]	RO	-	cnt_current_h These bits record times of completed DMA operations. It should be noted that these bits are read only. NOTE: 1) When the data to be transferred is less than 8KB, times of completed DMA operations equal CNT_current. 2) When the data to be transferred is equal to or more than 8KB, times of completed DMA operations can be calculated from the equation: CNT_current_H * CNT_cfg.
[23:16]	RW	-	cnt_cfg_h Extended bits of CNT_cfg. NOTE: 1) When the data to be transferred is less than 4KB, CNT_cfg_H must be 0. 2) When the data to be transferred is equal to or more than 4KB, For buffer0 to buffer7, CNT_cfg must be $(N * 8)$, $(N \geq 1, N * 8 < 256)$ Then the data to be transferred will be $((CNT_cfg_H + 1) * N * 8 * 64)$ bytes

Bits	Attributes	Reset	Description
			- For buffer8 to buffer15, CNT_cfg must be $(N * 2)$, ($N \geq 1$, $N * 2 < 256$) Then the data to be transferred will be $((CNT_cfg_H + 1) * N * 2 * 64)$ bytes 3) When CNT_cfg_H is valid, the DMA operation times of $(1 * CNT_cfg)$ is called a small loop, the DMA operation times of $(CNT_cfg_H * CNT_cfg)$ is called a big loop.
[15:8]	RO	-	cnt_current These bits record the times of DMA operation having been finished. It should be noted that these bits are read only.
[7:0]	RW	-	cnt_cfg Total times of DMA operations to be implemented. NOTE: Programmers should write the expected times of DMA operations into these bits. For example, 1024 bytes are expected to be transferred. Programmers should write 0x010000 into these bits. $(\frac{1024bytes}{64bytes} = 16 = 010000B)$, DMA controller transfers 64 bytes data in a DMA operation.)

● DMA Request Register(Address: 0x201A,8080)

This register is used to configure DMA request signal of each buffer and to set relative parameters of fraction DMA operation.

The DMA controller transfers 64-byte data in a DMA operation. If the number of data to be transferred is less than 64 bytes, fraction DMA operation is utilized.

Assuming that total number of data to be transferred is M,

$$M = 64 \times A + N$$

Then A is normal DMA operation times, and N is the data quantity fraction DMA operation needs to process.

Bits	Attributes	Reset	Description
[31]	RW/SC	1'h0	buf7_dma_req Buffer7 DMA request signal. When this bit is set by CPU, the Buffer7 DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[30]	RW/SC	1'h0	buf6_dma_req Buffer6 DMA request signal. When this bit is set by CPU, the Buffer6 DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[29]	RW/SC	1'h0	buf5_dma_req

Bits	Attributes	Reset	Description
			Buffer5 DMA request signal. When this bit is set by CPU, the Buffer5 DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[28]	RW/SC	1'h0	buf4_dma_req Buffer4 DMA request signal. When this bit is set by CPU, the Buffer4 DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[27]	RW/SC	1'h0	buf3_dma_req Buffer3 DMA request signal. When this bit is set by CPU, the Buffer3 DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[26]	RW/SC	1'h0	buf2_dma_req Buffer2 DMA request signal. When this bit is set by CPU, the Buffer2 DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[25]	RW/SC	1'h0	buf1_dma_req Buffer1 DMA request signal. When this bit is set by CPU, the Buffer1 DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[24]	RW/SC	1'h0	buf0_dma_req Buffer0 DMA request signal. When this bit is set by CPU, the Buffer0 DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[23:16]	RO	8'h0	Reserved
[15:10]	RW	6'h0	frac_dma_len Quantity of data to be transferred in next fraction DMA operation. Number of data to be transferred = Frac_DMA_len + 1 NOTE: When to read data from external RAM, the

Bits	Attributes	Reset	Description
			number of data to be transferred in a Fraction DMA operation should be an even number. When to send data to external RAM, the number of data to be transferred in a Fraction DMA operation can be an odd or even number.
[9]	RW	1'h0	frac_dma_req Fraction DMA request signal. When this bit is set by CPU, the fraction DMA operation is enabled. This bit will be cleared automatically after the requested DMA operation has been finished successfully.
[8]	RW	1'h0	frac_dma_dir 0: To read data from L2 memory in fraction DMA mode. 1: To write data into L2 memory in fraction DMA mode.
[7:1]	RW	7'h0	frac_dma_l2_add The address of L2 memory starting from which the data will be stored/sent in next fraction DMA operation. NOTE: L2 memory is divided into several 64-byte regions. The first 64-byte space is called Region 0; the second, Region 1; ... Programmers should write into these bits the address (Region X) from which the data will be stored/sent in next fraction DMA operation.
[0]	RW	1'h0	dma_en 0: To disable DMA function of L2 memory. 1: To enable DMA function of L2 memory.

● Fraction DMA Address Information Register (Address: 0x201A,8084)

The DMA controller transfers 64-byte data in a DMA operation. If the number of data to be transferred is less than 64 bytes, fraction DMA operation can be utilized. This working mode is called "fraction DMA mode".

This register is used to define the address of external RAM starting from which the data will be stored/sent, and to configure the status flag of two kinds of operations.

Bits	Attributes	Reset	Description
[31:30]	RW	2'h0	frac_dma_addr_h High address of external RAM from which the data will be stored/sent. The real address: (Frac_DMA_ADDR_H <<28) + Frac_DMA_ADDR_L
[29]	RW	1'h0	ahb_flag_en 0: The buffer status bit is not changed after the data transfer by AHB bus has been finished. 1: The buffer status bit may be changed after the

Bits	Attributes	Reset	Description
			data transfer by AHB bus has been finished.
[28]	RW	1'h0	ldma_flag_en 0: The buffer status bit is not changed after the LDMA operation has been finished. 1: The buffer status bit may be changed after the LDMA operation has been finished. NOTE: The data transfer between two buffers of L2 memory is called Local DMA operation and abbreviated as LDMA operation. Please configure Configuration Register of LDMA (Address: 0x201A, 8098) for other setting of the LDMA operation.
[27:0]	RW	28'h0	frac_dma_addr_l Low address of external RAM from which the data will be stored/sent. The real address: (Frac_DMA_ADDR_H <<28) + Frac_DMA_ADDR_L

● Buffer0 ~ Buffer 7 Configuration Register (Address: 0x201A,8088)

The register is used to set Buffer 0 ~ Buffer 7.

Bits	Attributes	Reset	Description
[31]	RW	1'h0	buffer7_flag_clr 1: To clear buffer7 status flag. NOTE: This bit is cleared automatically.
[30]	RW	1'h0	buffer6_flag_clr 1: To clear buffer6 status flag. NOTE: This bit is cleared automatically.
[29]	RW	1'h0	buffer5_flag_clr 1: To clear buffer5 status flag. NOTE: This bit is cleared automatically.
[28]	RW	1'h0	buffer4_flag_clr 1: To clear buffer4 status flag. NOTE: This bit is cleared automatically.
[27]	RW	1'h0	buffer3_flag_clr 1: To clear buffer3 status flag. NOTE: This bit is cleared automatically.
[26]	RW	1'h0	buffer2_flag_clr 1: To clear buffer2 status flag. NOTE: This bit is cleared automatically.
[25]	RW	1'h0	buffer1_flag_clr 1: To clear buffer1 status flag. NOTE: This bit is cleared automatically.
[24]	RW	1'h0	buffer0_flag_clr 1: To clear buffer0 status flag. NOTE: This bit is cleared automatically.
[23]	RW	1'h0	buffer7_vld

Bits	Attributes	Reset	Description
			1: To enable buffer7.
[22]	RW	1'h0	buffer6_vld 1: To enable buffer6.
[21]	RW	1'h0	buffer5_vld 1: To enable buffer5.
[20]	RW	1'h0	buffer4_vld 1: To enable buffer4.
[19]	RW	1'h0	buffer3_vld 1: To enable buffer3.
[18]	RW	1'h0	buffer2_vld 1: To enable buffer2.
[17]	RW	1'h0	buffer1_vld 1: To enable buffer1.
[16]	RW	1'h0	buffer0_vld 1: To enable buffer0.
[15]	RW	1'h0	buffer7_dir 0: Read operation between buffer7 and RAM controller by DMA bus. 1: Write operation between buffer7 and RAM controller by DMA bus.
[14]	RW	1'h0	buffer6_dir 0: Read operation between buffer6 and RAM controller by DMA bus. 1: Write operation between buffer6 and RAM controller by DMA bus.
[13]	RW	1'h0	Buffer5_dir 0: Read operation between buffer5 and RAM controller by DMA bus. 1: Write operation between buffer5 and RAM controller by DMA bus.
[12]	RW	1'h0	buffer4_dir 0: Read operation between buffer4 and RAM controller by DMA bus. 1: Write operation between buffer4 and RAM controller by DMA bus.
[11]	RW	1'h0	buffer3_dir 0: Read operation between buffer3 and RAM controller by DMA bus. 1: Write operation between buffer3 and RAM controller by DMA bus.
[10]	RW	1'h0	buffer2_dir 0: Read operation between buffer2 and RAM controller by DMA bus. 1: Write operation between buffer2 and RAM controller by DMA bus.
[9]	RW	1'h0	buffer1_dir 0: Read operation between buffer1 and RAM controller by DMA bus.

Bits	Attributes	Reset	Description
			1: Write operation between buffer1 and RAM controller by DMA bus.
[8]	RW	1'h0	buffer0_dir 0: Read operation between buffer0 and RAM controller by DMA bus. 1: Write operation between buffer0 and RAM controller by DMA bus.
[7]	RW	1'h0	buffer7_dma 0: To disable data transfer between buffer7 and RAM controller by DMA bus. 1: To enable data transfer between buffer7 and RAM controller by DMA bus.
[6]	RW	1'h0	buffer6_dma 0: To disable data transfer between buffer6 and RAM controller by DMA bus. 1: To enable data transfer between buffer6 and RAM controller by DMA bus.
[5]	RW	1'h0	buffer5_dma 0: To disable data transfer between buffer5 and RAM controller by DMA bus. 1: To enable data transfer between buffer5 and RAM controller by DMA bus.
[4]	RW	1'h0	buffer4_dma 0: To disable data transfer between buffer4 and RAM controller by DMA bus. 1: To enable data transfer between buffer4 and RAM controller by DMA bus.
[3]	RW	1'h0	buffer3_dma 0: To disable data transfer between buffer3 and RAM controller by DMA bus. 1: To enable data transfer between buffer3 and RAM controller by DMA bus.
[2]	RW	1'h0	buffer2_dma 0: To disable data transfer between buffer2 and RAM controller by DMA bus. 1: To enable data transfer between buffer2 and RAM controller by DMA bus.
[1]	RW	1'h0	buffer1_dma 0: To disable data transfer between buffer1 and RAM controller by DMA bus. 1: To enable data transfer between buffer1 and RAM controller by DMA bus.
[0]	RW	1'h0	buffer0_dma 0: To disable data transfer between buffer0 and RAM controller by DMA bus. 1: To enable data transfer between buffer0 and RAM controller by DMA bus.

● CPU-Controlled Buffer Configuration Register (Address: 0x201A,808C)

The register is used to configure CPU-controlled buffer.

Bits	Attributes	Reset	Description
[31:8]	RO	24'h0	Reserved
[7:4]	WO	4'h0	cpu_buf_val Current number of data contained in the CPU-controlled buffer. Number of data = CPU_buf_val×64 (unit: byte) NOTE: 0 ≤ CPU_buf_val ≤ 8. The value is loaded into the corresponding status bits of Buffer Status Register 0 (Address: 0x201A,80A0). For instance, that Buffer0's status can be controlled by CPU is wanted. Therefore, you should set bit[3] and bits[2:0] of this register as 1B and 000B respectively, and write the data number you want into these bits. Then programmers can check bits[3:0] of Buffer Status Register 0 (Address: 0x201A,80A0) to confirm whether the configuration is valid or not.
[3]	WO	1'h0	cpu_buf_sel_en 0: To disable CPU-controlled buffer function(the setting of bits[7:4] and bits[2:0] is invalid). 1: To enable CPU-controlled buffer function(the setting of bits[7:4] and bits[2:0] is valid).
[2:0]	WO	3'h0	cpu_buf_sel To select the buffer whose status can be controlled by CPU. 3'h0: buffer0 3'h1: buffer1 3'h2: buffer2 3'h3: buffer3 3'h4: buffer4 3'h5: buffer5 3'h6: buffer6 3'h7: buffer7

Buffer Assignment Register 0&1 assign the dynamic buffers (Buf0 ~ Buf 7) to interfaces.

● Buffer Assignment Register 0 (Address: 0x201A,8090)

Bits	Attributes	Reset	Description
[31:27]	RO	5'h0	Reserved
[26:24]	RW	3'h0	SPI1r_add Buffer address for read buffer of SPI1 Controller. 0 corresponds with Buffer 0; 1, Buffer 1; 2, Buffer 2; ..., 7, Buffer 7.
[23:21]	RW	3'h0	SPI1w_add Buffer address for write buffer of SPI1 Controller. 0 corresponds with Buffer 0; 1, Buffer 1; 2, Buffer 2; ..., 7, Buffer 7.
[20:9]	RO	12'h0	Reserved

Bits	Attributes	Reset	Description
[8:6]	RW	3'h0	USB_endpoint3_add Buffer address for USB endpoint3
[5:3]	RW	3'h0	USB_endpoint2_add Buffer address for USB endpoint2
[2:0]	RW	3'h0	USB_endpoint1_add Buffer address for USB endpoint1

● Buffer Assignment Register 1(Address: 0x201A,8094)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	mem_gate 1: To gate control the enabling signal of SRAM. NOTE: When this bit is set, power consumption is reduced in standby mode.
[30:27]	RO	4'h0	Reserved
[26:24]	RW	3'h0	USB_endpoint0_add Buffer address for USB endpoint0
[23:21]	RW	3'h0	USB_endpoint6_add Buffer address for USB endpoint6
[20:18]	RW	3'h0	USB_endpoint5_add Buffer address for USB endpoint5
[17:15]	RW	3'h0	USB_endpoint4_add Buffer address for USB endpoint4
[14:6]	RO	9'h0	Reserved
[5:3]	RW	3'h0	SPI2r_add Buffer address for read buffer of SPI2 Controller. 0 corresponds with Buffer 0; 1, Buffer 1; 2, Buffer 2; ..., 7, Buffer 7.
[2:0]	RW	3'h0	SPI2w_add Buffer address for write buffer of SPI2 Controller. 0 corresponds with Buffer 0; 1, Buffer 1; 2, Buffer 2; ...,7, Buffer 7.

● Configuration Register of LDMA (Address: 0x201A,8098)

This register is used to configure the LDMA parameters.

Programmers should notice the following items:

- 1) The data transfer between two buffers of L2 memory is called Local DMA operation and abbreviated as LDMA operation.
- 2) That the status bit is changed or not after the LDMA operation is decided by bit[28] of [Fraction DMA Address Information Register \(Address: 0x201A,8084\)](#).

Bits	Attributes	Reset	Description
[31]	RW	1'h0	ldma_vld 1: To enable data transfer between two different interfaces. NOTE: When the data transfer has been finished successfully, this bit is set to 0 automatically.
[30:14]	RW	17'h0	cnt Number of data to be transferred (unit: 4bytes)

Bits	Attributes	Reset	Description
[13:7]	RW	7'h0	d_add L2 memory address in which the data will be stored. (unit: 64bytes)
[6:0]	RW	7'h0	s_add L2 memory address from which the data will be transmitted. (unit: 64bytes)

● Interrupt Enable Register (Address: 0x201A,809C)

This register enables/disables the buffer interrupts.

For bits[16:0] of this register, If CNT_cfg_H is valid, the DMA operation requested by corresponding Buffer refers to the DMA operation of the big loop. If CNT_cfg_H is invalid, the DMA operation requested by corresponding Buffer refers to the DMA operation defined by CNT_cfg. For details of CNT_cfg_H, big loop and CNT_cfg, please refer to bits[23:16] and bits[7:0] of [DMA Operation Times Configuration Register of Buffer0 \(Address: 0x201A,8040\)](#).

Bits	Attributes	Reset	Description
[31:20]	RO	12'h0	Reserved
[19]	RW	1'h0	DMA_loop_int_en 0: To disable small loop interrupt. 1: To enable small loop interrupt. NOTE: When the DMA operation of a small loop is finished, an interrupt is generated. This kind of interrupt is called small loop interrupt. For the details of small loop, please refer to bits[23:16] of DMA Operation Times Configuration Register of Buffer0 (Address: 0x201A,8040) .
[18]	RO	1'h0	Reserved
[17]	RW	1'h0	ldma_vld 0: To disable ldma_vld interrupt. 1: To enable ldma_vld interrupt. NOTE: To enable data transfer between two buffers of L2 memory is called ldma_vld. When ldma_vld interrupt is enabled, an interrupt is generated when a LDMA operation has finished.
[16]	RW	1'h0	buf7_dma_req 0: To disable buf7_dma_req interrupt. 1: To enable buf7_dma_req interrupt. NOTE: The Buf7_dma_req interrupt is generated when the DMA operation requested by Buffer 7 has finished.
[15]	RW	1'h0	buf6_dma_req 0: To disable buf6_dma_req interrupt. 1: To enable buf6_dma_req interrupt. NOTE: The Buf6_dma_req interrupt is generated when the DMA operation requested by Buffer 6 has finished.
[14]	RW	1'h0	buf5_dma_req

Bits	Attributes	Reset	Description
			0: To disable buf5_dma_req interrupt. 1: To enable buf5_dma_req interrupt. NOTE: The Buf5_dma_req interrupt is generated when the DMA operation requested by Buffer 5 has finished.
[13]	RW	1'h0	buf4_dma_req 0: To disable buf4_dma_req interrupt. 1: To enable buf4_dma_req interrupt. NOTE: The Buf4_dma_req interrupt is generated when the DMA operation requested by Buffer 4 has finished.
[12]	RW	1'h0	buf3_dma_req 0: To disable buf3_dma_req interrupt. 1: To enable buf3_dma_req interrupt. NOTE: The Buf3_dma_req interrupt is generated when the DMA operation requested by Buffer 3 has finished.
[11]	RW	1'h0	buf2_dma_req 0: To disable buf2_dma_req interrupt. 1: To enable buf2_dma_req interrupt. NOTE: The Buf2_dma_req interrupt is generated when the DMA operation requested by Buffer 2 has finished.
[10]	RW	1'h0	buf1_dma_req 0: To disable buf1_dma_req interrupt. 1: To enable buf1_dma_req interrupt. NOTE: The Buf1_dma_req interrupt is generated when the DMA operation requested by Buffer 1 has finished.
[9]	RW	1'h0	buf0_dma_req 0: To disable buf0_dma_req interrupt. 1: To enable buf0_dma_req interrupt. NOTE: The Buf0_dma_req interrupt is generated when the DMA operation requested by Buffer 0 has finished.
[8:1]	RO	8'h0	Reserved
[0]	RW	1'h0	frac_dma_req_cfg 0: To disable frac_dma_req interrupt. 1: To enable frac_dma_req interrupt. NOTE: The frac_dma_req interrupt is generated when the fraction DMA operation has finished.

● Buffer Status Register 0 (Address: 0x201A,80A0)

Bits	Attributes	Reset	Description
[31:28]	RO	4'h0	buf7_cnt Total quantity of data contained in Buffer7. The number of data = Buf7_sta x 64 (unit: byte)

Bits	Attributes	Reset	Description
			0 ≤ Buf7_sta ≤ 8
[27:24]	RO	4'h0	buf6_cnt Total quantity of data contained in Buffer6. The number of data = Buf6_sta x 64 (unit: byte) 0 ≤ Buf6_sta ≤ 8
[23:20]	RO	4'h0	buf5_cnt Total quantity of data contained in Buffer5. The number of data = Buf5_sta x 64 (unit: byte) 0 ≤ Buf5_sta ≤ 8
[19:16]	RO	4'h0	buf4_cnt Total quantity of data contained in Buffer4. The number of data = Buf4_sta x 64 (unit: byte) 0 ≤ Buf4_sta ≤ 8
[15:12]	RO	4'h0	buf3_cnt Total quantity of data contained in Buffer3. The number of data = Buf3_sta x 64 (unit: byte) 0 ≤ Buf3_sta ≤ 8
[11:8]	RO	4'h0	buf2_cnt Total quantity of data contained in Buffer2. The number of data = Buf2_sta x 64 (unit: byte) 0 ≤ Buf2_sta ≤ 8
[7:4]	RO	4'h0	buf1_cnt Total quantity of data contained in Buffer1. The number of data = Buf1_sta x 64 (unit: byte) 0 ≤ Buf1_sta ≤ 8
[3:0]	RO	4'h0	buf0_cnt Total quantity of data contained in Buffer0. The number of data = Buf0_sta x 64 (unit: byte) 0 ≤ Buf0_sta ≤ 8

● Buffer Status Register 1 (Address: 0x201A,80A8)

Bits[31:16] of this register are valid only when cnt_cfg_h is valid. When the DMA operation of a small loop is finished, an interrupt is generated. This kind of interrupt is called small loop interrupt. For details of small loop and cnt_cfg_h, please refer to bits[23:16] of [DMA Operation Times Configuration Register of Buffer0 \(Address: 0x201A,8040\)](#)

Bits	Attributes	Reset	Description
[31]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 7 is not generated. 1: A small loop interrupt of Buffer 7 is generated.
[30]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 6 is not generated. 1: A small loop interrupt of Buffer 6 is generated.
[29]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 5 is not generated.

Bits	Attributes	Reset	Description
			1: A small loop interrupt of Buffer 5 is generated.
[28]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 4 is not generated. 1: A small loop interrupt of Buffer 4 is generated.
[27]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 3 is not generated. 1: A small loop interrupt of Buffer 3 is generated.
[26]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 2 is not generated. 1: A small loop interrupt of Buffer 2 is generated.
[25]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 1 is not generated. 1: A small loop interrupt of Buffer 1 is generated.
[24]	RO	1'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 0 is not generated. 1: A small loop interrupt of Buffer 0 is generated.
[23:0]	RO	24'h0	small_loop_int_sta 0: The small loop interrupt of Buffer 7 is not generated. 1: A small loop interrupt of Buffer 7 is generated.

5.2.3 Programmer's Guide

As shown in Figure 5-5, A and B are two kinds of interfaces. With the help of L2 memory, Interface A can communicate with CPU or other devices in three ways: 1) between Interface A and Interface B; 2) between Interface A and external RAM by DMA bus; 3) between interface A and ARM by AHB bus.

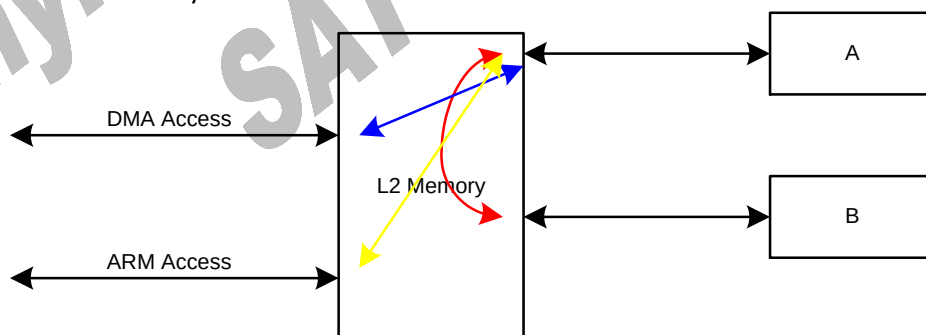


Figure 5-5 Working Mode of L2 Memory

5.2.3.1.1 Data Transfer between Interface A and External RAM

Normal working Mode

1. Interface A sends a DMA operation request to begin the data transfer from external RAM to Interface A.

2. To check whether a specific buffer has been assigned to Interface A or not. If yes, to wait until current operation has been finished; if not, to assign an idle buffer. See NOTE 1 for explanation.
3. To enable the buffer and buffer DMA operation function, and set the DMA operation direction.
4. To enable the buffer interrupt function
5. To configure the source address and the number of data to be transferred. See NOTE 2 for explanation.
6. To enable the DMA operation or Fraction DMA operation.
7. To transfer data from the buffer to Interface A.
8. To release the buffer.

NOTE

- 1) In normal operation, every time a data transfer operation has been finished, the buffer of L2 memory assigned to the specific interface is released. If the buffers of L2 memory are settled to be assigned to given interfaces, L2 driver should manage the buffer assignment, and tell the system that the new request follows current operation.
- 2) If number of data to be transferred is less than 64 bytes, Fraction DMA mode is recommended. And programmers should configure the fraction DMA operation parameters instead.
- 3) Only when the buffer flag is "1" (i.e. buffer is full), the data in buffer can be read; and only when the buffer flag is "0" (i.e. buffer is empty), data can be written into the buffer.
- 4) The buffer transmits 4 bytes data in a data transfer operation.
- 5) The buffer flag is set automatically when the last address of the buffer is accessed just now. If the number of data received is too few to fill the buffer, the buffer flag should be set manually to inform the receiver that data is ready.
- 6) If there is data left in the buffer after the reading operation, and the buffer is still needed in next operation, the buffer flag can also be cleared manually.

Ping-Pang Mode

If large number of data is needed to be transferred, and more than one buffer is in idle state, the Ping-Pang mode can be utilized.

Buffer X is assigned to Interface A

Interface A requests to read data from Buffer X

Buffer X downloads data from external RAM

When Buffer X is full of data downloaded from external RAM, the request from Interface A can be responded

Buffer Y is assigned to external RAM and downloads data from external RAM

When Buffer X is empty and Buffer Y is full, Buffer Y is assigned to Interface A and Buffer X is assigned to external RAM

When Buffer Y is empty and Buffer X is full, Buffer X is assigned to Interface A and Buffer Y is assigned to external RAM

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5.2.3.1.2 Data Transfer between Interface A and Interface B

Same Buffer Operation

If data transfer between Interface A and Interface B is needed, the same buffer (Buffer 0 ~ Buffer 7) can be assigned to both interfaces. When the buffer is full of the data received from Interface A, Interface B can read the data instantly. If the number of received data is

less than the size of buffer, buffer flag should be set manually so that Interface B can read data from the buffer. Similarly, the buffer flag should be cleared manually when Interface B has finished the reading operation.

NOTE

Prior to data transmission, Interface A driver and Interface B driver should define the direction of data transfer between the buffer and Interface A, between the buffer and Interface B properly.

LDMA Mode

If one of the two interfaces (or both interfaces) is (are) settled to use a specific buffer (i.e. the same buffer cannot be assigned to two interfaces), LDMA mode can be utilized.

LDMA mode, the abbreviation of local DMA mode, is a fast working mode of data transfer from one buffer to the other. [Configuration Register of LDMA \(Address: 0x2014,0098\)](#)/[Configuration Register of LDMA \(Address: 0x201A,8098\)](#) is provided to configure the necessary parameters.

A typical LDMA operation is as follows:

1. To assign a buffer to Interface A, and transfer data from Interface A to the buffer
2. To assign a buffer to Interface B
3. To configure [Configuration Register of LDMA \(Address: 0x2014,0098\)](#)/[Configuration Register of LDMA \(Address: 0x201A,8098\)](#)
4. To transfer data from the buffer to Interface B

5.2.3.1.3 Buffer Management

Every buffer of L2 memory is divided into several blocks, each of which can store 64-byte data. If every block is full of data, then the buffer is in FULL status (data can be read; but no more data can be written into). If no data is stored in any block, then the buffer is in EMPTY status (data can be written into, but no data can be read); if data is not stored in all but some blocks, data can be read and new data can be written into. One time data transmission transfers one block (64 bytes) data. When the quantity of data left is less than one block (64 bytes), buffer status should be controlled by software. That is, you should set buffer status (full or empty) manually.

5.2.3.1.4 UART Buffers

As described in above sections, 4 buffers are settled to be assigned to the three UART ports. Each one of these 8 buffers can transfer the data in one direction only. The TX buffers get data by DMA bus, CPU mode or LDMA mode, and then send to a UART port; and the RX buffers get data from a UART port, and then send to external RAM by DMA bus, CPU by AHB bus, or other interface by LDMA mode.

TX buffers

To enable corresponding UART data transfer function and set the number of data to be transferred

UARTn requests to get 4 bytes data from corresponding TX buffer

UARTn sends the 4 bytes data

UARTn requests to get another 4 bytes data from corresponding TX buffer

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NOTE

- 1) A TX buffer is divided into two blocks, each of which contains 16x4 bytes data.
- 2) UARTn can request to get data from TX buffer, only when one block of the TX buffer is full which is indicated by that Address 15 is accessed just now. That is, if CPU writes the data into Address 15 of the TX buffer directly, for example, 0x40F, the block is seen as full.
- 3) If two blocks are full, the data cannot be downloaded from external RAM by DMA bus until the buffer flag is cleared.
- 4) There are two ways to clear the buffer flag: one is to wait until the data in buffer is sent out generally; the other is to clear the buffer flag by *CPU-Controlled Buffer and Buffer8 ~ Buffer15 Configuration Register (Address: 0x2014,008C)* or *CPU-Controlled Buffer Configuration Register(Address: 0x201A,808C)*.

RX buffers

To enable corresponding UART data transfer function.

UARTn sends a request to write data into RX buffer when 4 bytes data has been received.

4 bytes data is sent to RX buffer

UARTn receives new data.

.....

NOTE

- 1) A RX buffer is divided into two blocks, each of which contains 16x4 bytes data.
- 2) UARTn can request to write data into RX buffer, only when one block of the RX buffer is empty which is indicated by that Address 15 is accessed just now. That is, if CPU reads the data from Address 15 of the RX buffer directly, for example, 0x42F, the block is seen as empty.
- 3) If two blocks are full, UARTn cannot write any data into the RX buffer until the buffer flag is cleared.
- 5) There are two ways to clear the buffer flag: one is to wait until the data in buffer is sent out generally; the other is to clear the buffer flag by *CPU-Controlled Buffer and Buffer8 ~ Buffer15 Configuration Register (Address: 0x2014,008C)*, *CPU-Controlled Buffer Configuration Register(Address: 0x201A,808C)*.

5.2.3.1.5 USB Buffers

There are five kinds of requests from USB 2.0 HS Host/Slave Controller to L2 memory, namely, ISO, bulking, bulkout, USB Control, and USB Interrupt, all of which are taken as requests from interface engaged in the assignment of dynamic buffers. No matter what the buffer flag is, these five kinds of requests are responded within two GCLK cycles. Therefore, L2 memory has no mechanism to make sure the data in the buffer is correct. It should be USB driver's responsibility to ensure the correctness. Furthermore, when USB Control and USB Interrupt send request to L2 memory, Fraction DMA mode and CPU mode are recommended, because the number of data to be transferred is always equal to or less than 64 bytes.

5.3 RAM Controller

5.3.1 Overview

The RAM controller is the bridge logic that connects an internal unit to the stacked DDR2 SDRAM.

5.3.2 Features

Features of RAM controller are as follows:

- Supports 16-bit data bus width
- Supports DDR2 SDRAM only
- Up to 450MHz DDR2 clock
- One ODT control pin for DDR2
- One chip selected only
- Differential DQS only
- Sequential burst type only
- 4-bit burst length only
- Configurable CAS latency (2, 2.5, 3, 4, 5, 6 and 7, among which DDR2 only supports 2, 3, 4, 5 and 6)
- No Additive Latency
- The clock frequency ratio between PHY/Controller_Core (high speed clock) and System Bus can be 2:1 or 1:1
- The frequency of high speed clock is the same as memory interface clock

5.3.3 Block Diagram

The RAM controller provides memory services to eight AHB channels. Only one of the eight channels can be activated at a time. If multiple channels require memory services at the same time, only the channel with the highest priority will be active. The priority of these eight channels is configurable.

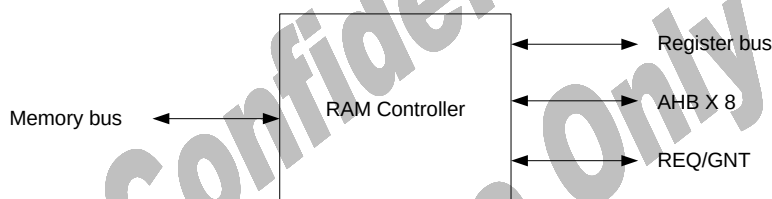


Figure 5-6 RAM Controller Unit Block Diagram

Table 5-5 Definition of AHB Channels

CHANNEL	INITIATOR	CHANNEL	INITIATOR
1	ISP/Pre-processor DMA	2	NPU
3	Video	4	ARM
5	MAC/Encryption	6	L2 DMA
7	Reserved	8	Reserved

5.3.4 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Register Address	Register Name
1	0x2100,0000	<i>Memory Controller Configuration Register 0</i>

Register No.	Register Address	Register Name
2	0x2100,0004	Memory Controller Configuration Register 1
3	0x2100,0008	Memory Controller Configuration Register 2
4	0x2100,000C	Memory Controller Configuration Register 3
5	0x2100,0010	CPU Command Register
6	0x2100,0014	AHB Time Slice Configuration Register
7	0x2100,0018	AHB Priority Configuration Register 0
8	0x2100,001C	AHB Priority Configuration Register 1
9	0x2100,0020	AHB Time Slice Release Configuration Register
10	0x2100,0024	Read Timing Calibration Configuration Register
11	0x2100,0028	Delay Line Control Register 0
12	0x2100,002C	Delay Line Control Register 1
13	0x2100,0030	Delay Line Control Register 2
14	0x2100,0034	Delay Line Control Register 3
15	0x2100,0038	Delay Line Control Register 4
16	0x2100,004C	Delay Line Control Register 5
17	0x2100,0050	Delay Line Control Register 6
18	0x2100,0054	Delay Line Control Enable Register 0
19	0x2100,0058	Delay Line Control Enable Register 1
20	0x2100,0060	Fail Status Register 0
21	0x2100,0064	Fail Status Register 1
22	0x2100,0068	Fail Status Register 2
23	0x2100,006C	Fail Status Register 3
24	0x2100,0070	Fail Status Register 4
25	0x2100,0078	RAM IO Control Register
26	0x2100,007C	AXI DMA Buffer Configuration Register 0
27	0x2100,0080	AXI DMA Buffer Configuration Register 1
28	0x2100,0084	Calibration Register
29	0x2100,0088	Success Status Register 0
30	0x2100,008C	Success Status Register 1
31	0x2100,0090	Success Status Register 2
32	0x2100,0094	Success Status Register 3
33	0x2100,00AC	I-Cache Hit Counting Register
34	0x2100,00B0	I-Cache Miss Counting Register
35	0x2100,00B4	D-Cache Hit Counting Register
36	0x2100,00B8	D-Cache Miss Counting Register
37	0x2100,00BC	L1 Cache Hit/Miss Counting Control Register

5.3.5 Register Definition

Memory Controller Configuration Register 0, 1, 2 & 3 configure RAM controller and configurable parameters.

- Memory Controller Configuration Register 0 (Address: 0x2100,0000)

Bits	Attributes	Reset	Description
[31:24]	RO	8'h0	Reserved
[23]	RW	1'h1	CpuMemClkGateBypassCFG 0: Enable memorys clock gate for power saving

Bits	Attributes	Reset	Description
			1: Bypass memorys clock gate
[22]	RW	1'h0	DmsMemClkGateBypassCFG 0: Enable memorys clock gate for power saving 1: Bypass memorys clock gate
[21]	RW	1'h0	BankAddrLenCFG1, Select memory bank address width. {BankAddrLenCFG1,BankAddrLenCFG0} 1'h0: Bank address is 2 bits (4 bank) 1'h1: Bank address is 3 bits (8 bank) 1'h2: Bank address is 0 bits (1 bank) 1'h3: Bank address is 1 bits (2 bank) NOTE: When Bank address is 0/1 bit, ColAddrLenCFG[2:0] is allowed to be set 1/2/3 and RowAddrLenCFG[2:0] is allowed to be set 3. But the true row address length of SDRAM can be less than 14 bits.
[20]	RW	1'h1	AHBReadBufDepthSel To decide how many words will be read into inner buffer before CPU reads DRAM. 0: 4 words 1: 8 words
[19:18]	RW	2'h0	CmdDepthSel Command FIFO depth selection 2'h0: 32 command. 2'h1: 16 command. 2'h2: 8 command. 2'h3: 8 command.
[17]	RW	1'h0	ODTWriteCon Software or hardware selection to control ODT of Anyka chips when DDR2 is read. 0: To select software. 1: To select hardware (opened only when DDR2 is written).
[16]	RW	1'h0	ODTReadCon Software or hardware selection to control ODT of Anyka chips when DDR2 is read. 0: To select software. 1: To select hardware (opened only when DDR2 is read).
[15:13]	RW	3'h3	RowAddrLenCFG Memory Row address width selection. 3'h0: The Row address width is 11-bit. 3'h1: The Row address width is 12-bit. 3'h2: The Row address width is 13-bit. Others: Reserved.
[12:10]	RW	3'h3	ColAddrLenCFG Memory Column address width selection. 3'h1: The Column address is 8-bit.

Bits	Attributes	Reset	Description
			3'h2: The Column address is 9-bit. 3'h3: The Column address is 10-bit. 3'h4: The Column address is 11-bit. Others: Reserved. NOTE: For SDRAM, the Column address more than 10 bits is not supported.
[9]	RW	1'h1	BankAddrLenCFG0, Select memory bank address width. {BankAddrLenCFG1,BankAddrLenCFG0} 1'h0: Bank address is 2 bits (4 bank) 1'h1: Bank address is 3 bits (8 bank) 1'h2: Bank address is 0 bits (1 bank) 1'h3: Bank address is 1 bits (2 bank) NOTE: When Bank address is 0/1 bit, ColAddrLenCFG[2:0] is allowed to be set as 1/2/3 and RowAddrLenCFG[2:0] is allowed to be set as 3. But the true row address length of SDRAM can be less than 14 bits.
[8:6]	RW	3'h2	WriteLatencyCFG This bit is used to configure the delay between the Write command and the first write data. The Unit is MEM CLK cycle. 3'h0: The Write latency is 0. 3'h1: The Write latency is 1. 3'h2: The Write latency is 2. 3'h3: The Write latency is 3. 3'h4: The Write latency is 4. 3'h5: The Write latency is 5. 3'h6: The Write latency is 6. Others: Reserved.
[5:3]	RW	3'h2	ReadLatencyCFG This bit is used to configure the delay between a Read command and the first read data. The Unit is MEM CLK cycle. 3'h0: The Read latency is 2. 3'h1: The Read latency is 2.5. 3'h2: The Read latency is 3. 3'h3: The Read latency is 4. 3'h4: The Read latency is 5. 3'h5: The Read latency is 6. 3'h6: The Read latency is 7. Others: Reserved.
[2]	RO	1'h0	Reserved
[1:0]	RW	2'h0	DDRSelCFG Select memory type 0: DDR2 Others: Reserved

NOTE

The value of *Memory Controller Configuration Register 0 (Address: 0x2100,0000)* should be changed when DRAM controller is in idle state, if programmers change the value of this register when DRAM controller is in its normal running state, unpredictable errors can be induced to hardware.

● Memory Controller Configuration Register 1 (Address: 0x2100,0004)

Bits	Attributes	Reset	Description
[31:30]	RW	2'h0	PVREF_CFG
[29:24]	RW	6'hF	tRC_CFG Memory timing parameter tRC configuration. tRC: the delay between two Active commands for the same bank, or delay between the Active command and the next Active command. $tRC = (tRC_cfg + 1)$ high speed CLK cycle.
[23:21]	RW	3'h5	tRRD_CFG Memory timing parameter tRRD configuration. tRRD: the delay between two Active commands for different Banks, at least 2 MEM CLK cycles $tRRD = (tRRD_cfg + 1)$ high speed CLK cycle
[20:16]	RW	5'hE	tRAS_CFG Memory timing parameter tRAS configuration. tRAS: the delay between the Active command and the next Precharge command $tRAS = (tRAS_cfg + 1)$ high speed CLK cycle
[15:12]	RW	4'h5	tRP_CFG Memory timing parameter tRP configuration tRP: the delay between the Precharge command and the next Active/Refresh command $tRP = (tRP_cfg + 1)$ high speed CLK cycle
[11:8]	RW	4'h3	tRCD_CFG Memory timing parameter tRCD configuration tRCD: the delay between the Active command and the next Read/Write command. $tRCD = (tRCD_cfg + 1)$ high speed CLK cycle
[7:0]	RW	8'h20	tRFC_CFG Memory timing parameter tRFC configuration tRFC: the delay between the Refresh command and the next Active command or subsequent Refresh command $tRFC = (tRFC_cfg + 1)$ high speed CLK cycle

● Memory Controller Configuration Register 2 (Address: 0x2100,0008)

Bits	Attributes	Reset	Description
[31:26]	RO	6'h0	Reserved
[25:18]	RW	8'h5	tDQSQ_tQHS_CFG The configuration of the sum of DDR/DDR2 timing parameters tDQSQ, tQHS and tDQSCK (the skew

Bits	Attributes	Reset	Description
			between DQS and its associated DQ). This timing parameter is used only by calibration. $t_{DQSQ} + t_{QHS} + 2 * t_{DQSCK} = (t_{DQSQ_tQHS_cfg} * 45) \text{ ps.}$ NOTE: 1) 45ps is the step of delay line. It is the simulation result. The actual value of a chip is TBD (to be determined). 2) The value of these bits must be less than 67 to ensure that $(t_{DQSQ} + t_{QHS} + 2 * t_{DQSCK})$ does not exceed a half cycle of high speed CLK.
[17:12]	RW	6'hF	tRTW_CFG DDR2/DDR/SDRAM timing parameter tRTW configuration tRTW: the delay between the Read command and the next Write command $t_{RTW} = (t_{RTW_cfg} + 1) \text{ high speed CLK cycle}$
[11:8]	RW	4'h5	tRTP_CFG DDR2 timing parameter tRTP configuration tRTP: the delay between the Read command and the next Precharge command $t_{RTP} = (t_{RTP_cfg} + 1) \text{ high speed CLK cycle}$
[7:4]	RW	4'h5	tWTR_CFG DDR2/DDR timing parameter tWTR configuration tWTR: the delay between the last data input and the next Read command $t_{WTR} = (t_{WTR_cfg} + 1) \text{ high speed CLK cycle}$
[3:0]	RW	4'h5	tWR_CFG DRAM timing parameter tWR configuration tWR: the delay between the last data input and the next Precharge command $t_{WR} = (t_{WR_cfg} + 1) \text{ high speed CLK cycle}$

● Memory Controller Configuration Register 3 (Address: 0x2100,000C)

Bits	Attributes	Reset	Description
[31:18]	RO	14'h0	Reserved
[17]	RO	1'h0	FIFOReadDataFull 1: Read Data FIFO is full.
[16]	RO	1'h1	FIFOReadDataEmpty 1: Read Data FIFO is empty.
[15]	RO	1'h0	FIFOCMDFull 1: Command FIFO is full.
[14]	RO	1'h1	FIFOCMDEmpty 1: Command FIFO is empty.
[13:1]	RW	13'h618	tREFI_CFG DRAM timing parameter tREFI configuration tREFI: the average periodic refresh interval.

Bits	Attributes	Reset	Description
			tREFI = (tREFI_cfg + 1) high speed CLK cycle.
[0]	RW	1'h0	AutoRefEnableCFG Auto-refresh enable bit 0: Disable Auto-Refresh. 1: Enable Auto-Refresh.

● CPU Command Register (Address: 0x2100,0010)

This register holds the memory command output by CPU.

Bits	Attributes	Reset	Description
[31:29]	RO	3'h0	Reserved
[28]	RW	1'h0	ODTSetModeCFG Configuring the mode of setting the value of ODT_cfg to DRAM_ODT. 0: To set the new value of ODT_cfg to DRAM_ODT immediately. 1: To set the new value of ODT_cfg to DRAM_ODT after Cpu_cmd_req is set as 1.
[27]	RW	1'h0	ODTCFG Memory command output by CPU This bit corresponds to pin DRAM_ODT.
[26]	RW	1'h0	CKESetModeCFG Configuring the mode of setting the value of CKE_cfg to DRAM_CKE 0: Set the new value of CKE_cfg to DRAM_CKE immediately. 1: Set the new value of CKE_cfg to DRAM_CKE after Cpu_cmd_req is set as 1.
[25]	RW	1'h0	CKECFG Memory command output by CPU This bit corresponds to pin DRAM_CKE.
[24]	RO	1'h0	Reserved
[23]	RW	1'h0	CPUCommandReq 0: Disable the transmission of CPU command. 1: Enable the transmission of CPU command. NOTE: This bit will be automatically cleared when the transmission of CPU command is finished.
[22]	RW	1'h1	CPUWEnCFG Memory command output by CPU This bit corresponds to pin #DRAM_WE.
[21]	RW	1'h1	CPUCASnCFG Memory command output by CPU This bit corresponds to pin #DRAM_CAS.
[20]	RW	1'h1	CPURASnCFG Memory command output by CPU This bit corresponds to pin #DRAM_RAS.
[19]	RW	1'h1	CPUCSnCFG Memory command output by CPU

Bits	Attributes	Reset	Description
			This bit corresponds to pin #DRAM_CS.
[18]	RW	1'h0	CPUBACFG Memory command output by CPU This bit corresponds to pin DRAM_BANK[2].
[17]	RW	1'h0	CPUBACFG Memory command output by CPU This bit corresponds to pin DRAM_BANK[1].
[16]	RW	1'h0	CPUBACFG Memory command output by CPU This bit corresponds to pin DRAM_BANK[0].
[15]	RO	1'h0	Reserved
[14:0]	RW	15'h0	CPUADDRCFG Memory command output by CPU These bits correspond to MADDR[15:0].

● AHB Time Slice Configuration Register (Address: 0x2100,0014)

Bits[2:0] configures the time slice of AHB channel gets when core module accesses external DRAM. Among them, AHB channel1 is ISP/Pre-Processor DMA; AHB channel2 is NPU; AHB channel3 is Video Processor; AHB channel4 is ARM; AHB channel5 is MAC Controller/Encryption module; AHB channel6 is L2 DMA. AHB channel7/AHB channel8 is reserved.

Bits	Attributes	Reset	Description
[31:26]	RO	3'h0	AHBChannel8TimeSliceCFG. Configure the numbers of time slice when AHB channel 8 wins the arbitration, which ranges from 1~8. Time piece number = AHBChannel8TimeSliceCFG +1
[28:26]	RW	3'h0	AHBChannel7TimeSliceCFG. Configure the numbers of time slice when AHB channel 7 wins the arbitration, which ranges from 1~8. Time piece number = AHBChannel7TimeSliceCFG +1
[25:23]	RW	3'h0	AHBChannel6TimeSliceCFG. Configure the numbers of time slice when AHB channel 6 wins the arbitration, which ranges from 1~8. Time piece number = AHBChannel6TimeSliceCFG +1
[22:20]	RW	3'h0	AHBChannel5TimeSliceCFG. Configure the numbers of time slice when AHB channel 5 wins the arbitration, which ranges from 1~8. Time piece number = AHBChannel5TimeSliceCFG +1
[19:17]	RW	3'h0	AHBChannel4TimeSliceCFG. Configure the numbers of time slice when AHB channel 4 wins the arbitration, which ranges from 1~8. Time piece number = AHBChannel4TimeSliceCFG +1

Bits	Attributes	Reset	Description
[16:12]	RW	5'h3	AHBChannel3TimeSliceCFG. Configure the numbers of time slice when AHB channel 3 wins the arbitration, which ranges from 1~32. Time slice number = AHBChannel3TimeSliceCFG + 1
[11:9]	RW	3'h2	AHBChannel2TimeSliceCFG. Configure the numbers of time slice when AHB channel 2 wins the arbitration, which ranges from 1~8. Time slice number = AHBChannel2TimeSliceCFG + 1
[8:4]	RW	5'h3	AHBChannel1TimeSliceCFG. Configure the numbers of time slice when AHB channel 1 wins the arbitration, which ranges from 1~32. Time slice number = AHBChannel1TimeSliceCFG + 1
[3]	RW	1'h1	AHBChannelTimeSliceStartCFG. Configure the time slice Timer starting point when AHB channel wins arbitration. Unit is DRAM clock cycle 0: Start point is 0 1: Random. Compatible with previous design.
[2:0]	RW	3'h1	AHBChannelTimeCFG Configure time slice of each AHB channel accessing DRAM. The Unit is MEM CLK. 3'h0: The time slice is 4. 3'h1: The time slice is 8. 3'h2: The time slice is 12. 3'h3: The time slice is 16. 3'h4: The time slice is 20. 3'h5: The time slice is 24. 3'h6: The time slice is 28. 3'h7: The time slice is 32. NOTE: The external DRAM can only serve on one GCLK module at a time, so the GCLKs get the same time slice when accessing DRAM. The setting of time slice is according to DRAM throughput rate. If higher throughput rate is available (like 32-bit DDR/DDR2), the time slice should be set shorter; If lower throughput rate is available (like 16-bit SDRAM), the time slice should be set longer.

The RAM controller provides memory services to eight AHB channels. Only one of the eight channels can be set active at a time. If multiple channels require memory services at the same time, only the channel with the highest priority will be set active. The default priority of these eight channels is: AHB1 (the highest), AHB2, AHB3, AHB4, AHB5, AHB6, AHB7, AHB8 (the lowest).

● AHB Priority Configuration Register 0 (Address: 0x2100,0018)

Bits	Attributes	Reset	Description
[31]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 4 0: The priority of AHB channel 8 is lower than that of AHB channel 4. 1: The default priority
[30]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 4 0: The priority of AHB channel 7 is lower than that of AHB channel 4. 1: The default priority
[29]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 4 0: The priority of AHB channel 6 is lower than that of AHB channel 4. 1: The default priority
[28]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 4 0: The priority of AHB channel 5 is lower than that of AHB channel 4. 1: The default priority
[27]	RW	1'h1	AHBStaticPriorityCFG No instruction.
[26]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 4 0: The priority of AHB channel 3 is lower than that of AHB channel 4. 1: The default priority
[25]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 4 0: The priority of AHB channel 2 is lower than that of AHB channel 4. 1: The default priority
[24]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 4 0: The priority of AHB channel 1 is lower than that of AHB channel 4. 1: The default priority
[23]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 3 0: The priority of AHB channel 8 is lower than that of AHB channel 3. 1: The default priority
[22]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 3 0: The priority of AHB channel 7 is lower than that of AHB channel 3. 1: The default priority

Bits	Attributes	Reset	Description
[21]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 3 0: The priority of AHB channel 6 is lower than that of AHB channel 3. 1: The default priority
[20]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 3 0: The priority of AHB channel 5 is lower than that of AHB channel 3. 1: The default priority
[19]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 3 0: The priority of AHB channel 4 is lower than that of AHB channel 3. 1: The default priority
[18]	RW	1'h1	AHBStaticPriorityCFG No instruction.
[17]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 3 0: The priority of AHB channel 2 is lower than that of AHB channel 3. 1: The default priority
[16]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 3 0: The priority of AHB channel 1 is lower than that of AHB channel 3. 1: The default priority
[15]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 2 0: The priority of AHB channel 8 is lower than that of AHB channel 2. 1: The default priority
[14]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 2 0: The priority of AHB channel 7 is lower than that of AHB channel 2. 1: The default priority
[13]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 2 0: The priority of AHB channel 6 is lower than that of AHB channel 2. 1: The default priority
[12]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 2 0: The priority of AHB channel 5 is lower than that of AHB channel 2. 1: The default priority
[11]	RW	1'h1	AHBStaticPriorityCFG

Bits	Attributes	Reset	Description
			AHB static priority control of AHB channel 2 0: The priority of AHB channel 4 is lower than that of AHB channel 2. 1: The default priority
[10]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 2 0: The priority of AHB channel 3 is lower than that of AHB channel 2. 1: The default priority
[9]	RW	1'h1	AHBStaticPriorityCFG No instruction.
[8]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 2 0: The priority of AHB channel 1 is lower than that of AHB channel 2. 1: The default priority
[7]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 1 0: The priority of AHB channel 8 is lower than that of AHB channel 1. 1: The default priority
[6]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 1 0: The priority of AHB channel 7 is lower than that of AHB channel 1. 1: The default priority
[5]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 1 0: The priority of AHB channel 6 is lower than that of AHB channel 1. 1: The default priority
[4]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 1 0: The priority of AHB channel 5 is lower than that of AHB channel 1. 1: The default priority
[3]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 1 0: The priority of AHB channel 4 is lower than that of AHB channel 1. 1: The default priority
[2]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 1 0: The priority of AHB channel 3 is lower than that of AHB channel 1. 1: The default priority
[1]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 1

Bits	Attributes	Reset	Description
			0: The priority of AHB channel 2 is lower than that of AHB channel 1. 1: The default priority
[0]	RW	1'h1	AHBStaticPriorityCFG No instruction.

● AHB Priority Configuration Register 1 (Address: 0x2100,001C)

Bits	Attributes	Reset	Description
[31]	RW	1'h1	AHBStaticPriorityCFG No instruction.
[30]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 8 0: The priority of AHB channel 7 is lower than that of AHB channel 8. 1: The default priority
[29]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 8 0: The priority of AHB channel 6 is lower than that of AHB channel 8. 1: The default priority
[28]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 8 0: The priority of AHB channel 5 is lower than that of AHB channel 8. 1: The default priority
[27]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 8 0: The priority of AHB channel 4 is lower than that of AHB channel 8. 1: The default priority
[26]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 8 0: The priority of AHB channel 3 is lower than that of AHB channel 8. 1: The default priority
[25]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 8 0: The priority of AHB channel 2 is lower than that of AHB channel 8. 1: The default priority
[24]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 8 0: The priority of AHB channel 1 is lower than that of AHB channel 8. 1: The default priority
[23]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 7

Bits	Attributes	Reset	Description
			0: The priority of AHB channel 8 is lower than that of AHB channel 7. 1: The default priority
[22]	RW	1'h1	AHBStaticPriorityCFG No instruction.
[21]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 7 0: The priority of AHB channel 6 is lower than that of AHB channel 7. 1: The default priority
[20]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 7 0: The priority of AHB channel 5 is lower than that of AHB channel 7. 1: The default priority
[19]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 7 0: The priority of AHB channel 4 is lower than that of AHB channel 7. 1: The default priority
[18]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 7 0: The priority of AHB channel 3 is lower than that of AHB channel 7. 1: The default priority
[17]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 7 0: The priority of AHB channel 2 is lower than that of AHB channel 7. 1: The default priority
[16]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 7 0: The priority of AHB channel 1 is lower than that of AHB channel 7. 1: The default priority
[15]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 6 0: The priority of AHB channel 8 is lower than that of AHB channel 6. 1: The default priority
[14]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 6 0: The priority of AHB channel 7 is lower than that of AHB channel 6. 1: The default priority
[13]	RW	1'h1	AHBStaticPriorityCFG No instruction.
[12]	RW	1'h1	AHBStaticPriorityCFG

Bits	Attributes	Reset	Description
			AHB static priority control of AHB channel 6 0: The priority of AHB channel 5 is lower than that of AHB channel 6. 1: The default priority
[11]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 6 0: The priority of AHB channel 4 is lower than that of AHB channel 6. 1: The default priority
[10]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 6 0: The priority of AHB channel 3 is lower than that of AHB channel 6. 1: The default priority
[9]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 6 0: The priority of AHB channel 2 is lower than that of AHB channel 6. 1: The default priority
[8]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 6 0: The priority of AHB channel 1 is lower than that of AHB channel 6. 1: The default priority
[7]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 5 0: The priority of AHB channel 8 is lower than that of AHB channel 5. 1: The default priority
[6]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 5 0: The priority of AHB channel 7 is lower than that of AHB channel 5. 1: The default priority
[5]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 5 0: The priority of AHB channel 6 is lower than that of AHB channel 5.
[4]	RW	1'h1	AHBStaticPriorityCFG No instruction.
[3]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 5 0: The priority of AHB channel 4 is lower than that of AHB channel 5. 1: The default priority
[2]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 5 0: The priority of AHB channel 3 is lower than that of

Bits	Attributes	Reset	Description
			AHB channel 5. 1: The default priority
[1]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 5 0: The priority of AHB channel 2 is lower than that of AHB channel 5. 1: The default priority
[0]	RW	1'h1	AHBStaticPriorityCFG AHB static priority control of AHB channel 5 0: The priority of AHB channel 1 is lower than that of AHB channel 5. 1: The default priority

● AHB Time Slice Release Configuration Register (Address: 0x2100,0020)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	AHBChannelTimeSliceFreeReleaseEnableCFG This bit is used to enable the deassertion of time slice/arbiter function when timeout occurs in AHB channels. 0: Disabled 1: Enabled
[30]	RW	1'h0	AHBChannelTimeSlicePriorityReleaseSelectCFG This bit is used to configure the time point when operations of AHB channels with higher priority disrupt those with lower priority. 0: Within one time slice of the current AHB channel. CPU will respond to AHB channels with higher priority immediately. Waiting time can be obviously shortened especially when the time slice is configured as a relatively higher values. 1: In the last cycle of one time slice of the current AHB channel.
[29:24]	RO	6'h0	Reserved
[23:16]	RW	8'h0	AHBChannelTimeSlicePriorityReleaseCFG This bit is used to enable AHB channels with higher priority to disrupt the current operations of AHB channels with lower priority and to deassert the current time slice signal and start a new round of arbitration. 0: Disabled 1: Enabled NOTE: Bits[23:16] correspond to the eight AHB channels.
[15:14]	RW	2'h3	AHBChannel8TimeSliceFreeReleaseCFG This bit is used to configure that in which cycle the current time slice and the arbiter will be deasserted when there is no read/write request from AHB

Bits	Attributes	Reset	Description
			channel8 in a row after AHB channel8 is set active. It can be made from the following equation: AHBChannel8TimeSliceFreeReleaseCFG+1
[13:12]	RW	2'h3	AHBChannel7TimeSliceFreeReleaseCFG This bit is used to configure that in which cycle the current time slice and the arbiter will be deasserted when there is no read/write request from AHB channel7 in a row after AHB channel7 is set active. It can be made from the following equation: AHBChannel7TimeSliceFreeReleaseCFG+1
[11:10]	RW	2'h3	AHBChannel6TimeSliceFreeReleaseCFG This bit is used to configure that in which cycle the current time slice and the arbiter will be deasserted when there is no read/write request from AHB channel6 in a row after AHB channel6 is set active. It can be made from the following equation: AHBChannel6TimeSliceFreeReleaseCFG+1
[9:8]	RW	2'h3	AHBChannel5TimeSliceFreeReleaseCFG This bit is used to configure that in which cycle the current time slice and the arbiter will be deasserted when there is no read/write request from AHB channel5 in a row after AHB channel5 is set active. It can be made from the following equation: AHBChannel5TimeSliceFreeReleaseCFG+1
[7:6]	RW	2'h1	AHBChannel4TimeSliceFreeReleaseCFG This bit is used to configure that in which cycle the current time slice and the arbiter will be deasserted when there is no read/write request from AHB channel4 in a row after AHB channel4 is set active. It can be made from the following equation: AHBChannel4TimeSliceFreeReleaseCFG+1
[5:4]	RW	2'h1	AHBChannel3TimeSliceFreeReleaseCFG This bit is used to configure that in which cycle the current time slice and the arbiter will be deasserted when there is no read/write request from AHB channel3 in a row after AHB channel3 is set active. It can be made from the following equation: AHBChannel3TimeSliceFreeReleaseCFG+1
[3:2]	RW	2'h1	AHBChannel2TimeSliceFreeReleaseCFG This bit is used to configure that in which cycle the current time slice and the arbiter will be deasserted when there is no read/write request from AHB channel2 in a row after AHB channel2 is set active. It can be made from the following equation: AHBChannel2TimeSliceFreeReleaseCFG+1
[1:0]	RW	2'h1	AHBChannel1TimeSliceFreeReleaseCFG This bit is used to configure that in which cycle the current time slice and the arbiter will be deasserted

Bits	Attributes	Reset	Description
			when there is no read/write request from AHB channel1 in a row after AHB channel1 is set active. It can be made from the following equation: AHBChannel1TimeSliceFreeReleaseCFG+1

● Read Timing Calibration Configuration Register (Address: 0x2100,0024)

This register configures the necessary parameters of Read Timing Colibration and indicates the calibration status.

Bits	Attributes	Reset	Description
[31:10]	RW	12'h0	ReadTimingCalibStartAddress Read Timing Calibration start address During the calibration, memory area from start address (Cal_start_add << 0x7) to end address ((Cal_start_add << 0x7) + 0x7F) is used by Read Timing Calibration. This memory area is 128 bytes in total. NOTE: 1) During the calibration, the content in this memory area will be changed. Programmer should assure that the content in this memory area is unuseful before calibration. 2) The Read Timing Calibration start address configured here is a relative address. And the absolute start address will be (0x8000,0000 + (Cal_start_add << 0x7)).
[9]	RW	1'h0	ReadTimingCalibResultMapEn By setting this bit after the read timing calibration is finished, hardware will map the delay control information to Delay Line Control Register 0 (Address: 0x2100,0028) ~ Delay Line Control Enable Register 1 (Address: 0x2100,0058).
[8]	RW	1'h0	DqsDelayAddInitialDir When normal read timing calibration flow is started, software can adjust the DQS DCDL in the first stage of calibration 0: Accelerate mode: DQS DDEs = MDLU/16 + DqsDelayAddInitial[7:0] Non - accelerate mode: DQS DDEs = MDLU/8 + DqsDelayAddInitial[7:0] 1: Accelerate mode: DQS DDEs =MDLU/16 - DqsDelayAddInitial[7:0] Non - accelerate mode: DQS DDEs = MDLU/8 - DqsDelayAddInitial[7:0] NOTE: 1) Only the value of SDLU DDE is configured, not including the value of ADE. 2) For DqsDelayAddInitial[7:0] please refer to bits[7:0] in Calibration Register .

Bits	Attributes	Reset	Description
[7]	RW	1'h0	ReadTimingCalibAccelerate When normal read timing calibration flow is started and this bit is set, hardware will accelerate the first stage of DQS/DQ calibration by setting DQS DCDLs directly 0: No need to accelerate 1: DQS DCDL = DQS DDEs = MDLU/16 +/- DqsDelayAddInitial[7:0] NOTE: 1) Only the value of SDLU DDE is configured, not including the value of ADE. 2) For DqsDelayAddInitial[7:0] please refer to bits[7:0] in Calibration Register.
[6]	RO	1'h0	ReadTimingSoftCalibDone 0: The software read timing calibration is not finished or not started. 1: The software read timing calibration is finished.
[5]	RW	1'h0	ReadTimingSoftCalibStart To start the software read timing calibration flow. When this bit is set, programmers can calibrate the read timing by configuring the delay line control word directly. 0: No instruction 1: To start software read timing calibration flow. NOTE: Only when this bit is set, the Delay Line Control Register 0 ~ 6 (Address: 0x2100,0028 ~ 0x2100,0038, 0x2010,004C, 0x2100,0050) and Delay Line Control Enable Register 0 ~ 1 (Address: 0x2100,0054 – 0x2100,0058) are valid.
[4]	RO	1'h0	ReadTimingFastCalibDone 0: The fast read timing calibration is not finished or not started. 1: The fast read timing calibration is finished.
[3]	RW	1'h0	ReadTimingFastCalibStart If the RAM controller working clock is no more than 125MHz, this mode is used to calibrate read timing. 0: No instruction 1: To start fast read timing calibration flow.
[2]	RO	1'h0	ReadTimingCalibFail 0: The normal read timing calibration is successful. 1: The normal read timing calibration failed.
[1]	RO	1'h0	ReadTimingCalibDone 0: The normal read timing calibration is not finished or not started. 1: The normal read timing calibration is finished.
[0]	RW	1'h0	ReadTimingCalibStart Start normal read timing calibration flow If the RAM controller working clock is more than

Bits	Attributes	Reset	Description
			125MHz, this mode is used to calibrate read timing. 0: No instruction. 1: Start normal read timing calibration flow.

Delay Line Control Register 0 ~ 6 (Address: 0x2100, 0028 ~ 0x2100, 0050) controls delays of DQSs, DQs, and etc.

● Delay Line Control Register 0 (Address: 0x2100,0028)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15:8]	RW	8'h0	Dqs1DelayControl Control words for delay line of DQS[1]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.
[7:0]	RW	8'h0	Dqs0DelayControl Control words for delay line of DQS[0]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.

● Delay Line Control Register 1 (Address: 0x2100,002C)

Bits	Attributes	Reset	Description
[31:24]	RW	8'h0	Dq3DelayControl Control words for delay line of DQ [3]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.
[23:16]	RW	8'h0	Dq2DelayControl Control words for delay line of DQ [2]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.
[15:8]	RW	8'h0	Dq1DelayControl Control words for delay line of DQ [1]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not

Bits	Attributes	Reset	Description
			including the value of ADE.
[7:0]	RW	8'h0	Dq0DelayControl Control words for delay line of DQ[0]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.

● Delay Line Control Register 2 (Address: 0x2100,0030)

Bits	Attributes	Reset	Description
[31:24]	RW	8'h0	Dq7DelayControl Control words for delay line of DQ [7]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.
[23:16]	RW	8'h0	Dq6DelayControl Control words for delay line of DQ [6]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.
[15:8]	RW	8'h0	Dq5DelayControl Control words for delay line of DQ [5]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.
[7:0]	RW	8'h0	Dq4DelayControl Control words for delay line of DQ [4]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.

● Delay Line Control Register 3 (Address: 0x2100,0034)

Bits	Attributes	Reset	Description
[31:24]	RW	8'h0	Dq11DelayControl Control words for delay line of DQ [11]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD.

Bits	Attributes	Reset	Description
			2) Only the value of SDLU DDE is configured, not including the value of ADE.
[23:16]	RW	8'h0	Dq10DelayControl Control words for delay line of DQ [10]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.
[15:8]	RW	8'h0	Dq9DelayControl Control words for delay line of DQ [9]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.
[7:0]	RW	8'h0	Dq8DelayControl Control words for delay line of DQ [8]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.

● Delay Line Control Register 4 (Address: 0x2100,0038)

Bits	Attributes	Reset	Description
[31:24]	RW	8'h0	Dq15DelayControl Control words for delay line of DQ[15]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.
[23:16]	RW	8'h0	Dq14DelayControl Control words for delay line of DQ[14]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.
[15:8]	RW	8'h0	Dq13DelayControl Control words for delay line of DQ[13]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.

Bits	Attributes	Reset	Description
[7:0]	RW	8'h0	Dq12DelayControl Control words for delay line of DQ[12]. (Unit: 45ps) NOTE: 1) The unit of 45ps is a simulation result. The actual value of a chip is TBD. 2) Only the value of SDLU DDE is configured, not including the value of ADE.

● Delay Line Control Register 5 (Address: 0x2100,004C)

Bits	Attributes	Reset	Description
[31:15]	RO	16'h0	Reserved
[14:12]	RW	3'h0	ReadDataValidWindowDelayControl Control words for delay cycle of read data valid window. (Unit: MEM CLK cycle)
[11:10]	RO	2'h0	Reserved
[9]	RW	1'h0	DQSGateMUXControl Control words for MUX of DQS[1], the MUX is used to select different registered Read Valid Window signal for gate DQS. 0: Read Valid Window signal is registered by positive-edge clock. 1: Read Valid Window signal is registered first by positive-edge clock, then by negative-edge clock.
[8]	RW	1'h0	DQSGateMUXControl Control words for MUX of DQS[0], the MUX is used to select different registered Read Valid Window signal for gate DQS. 0: Read Valid Window signal is registered by positive-edge clock. 1: Read Valid Window signal is registered first by positive-edge clock, then by negative-edge clock.
[7:6]	RO	2'h0	Reserved
[5]	RW	1'h0	DQReadMUX2Control Control words for MUXs of DQ[15:8] during Read, the MUX is used to select different delayed DQ to controller. 0: Front stage DQ value will be sent to controller. 1: Front stage DQ value will be registered and then sent to controller.
[4]	RW	1'h0	DQReadMUX2Control Control words for MUXs of DQ[7:0] during Read, the MUX is used to select different delayed DQ to controller. 0: Front stage DQ value will be sent to controller. 1: Front stage DQ value will be registered and then sent to controller.
[3:2]	RO	2'h0	Reserved

Bits	Attributes	Reset	Description
[1]	RW	1'h1	DQReadMUXControl Control words for MUXs of DQ[15:8] during Read, the MUX is used to select different registered DQ. 0: DQ is registered by positive-edge clock. 1: DQ is registered first by neative-edge clock, then by positive-edge clock.
[0]	RW	1'h1	DQReadMUXControl Control words for MUXs of DQ[7:0] during Read, the MUX is used to select different registered DQ. 0: DQ is registered by positive-edge clock. 1: DQ is registered first by neative-edge clock, then by positive-edge clock.

● Delay Line Control Register 6 (Address: 0x2100,0050)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15:8]	RW	8'h0	Dqs1GateDelayControl Control words for delay line of DQS[1] Gate circuit. (Unit: 45ps) NOTE: The unit of 45ps is a simulation result. The actual value of a chip is TBD.
[7:0]	RW	8'h0	Dqs0GateDelayControl Control words for delay line of DQS[0] Gate circuit. (Unit: 45ps) NOTE: The unit of 45ps is a simulation result. The actual value of a chip is TBD.

Delay Line Control Enable Register 0 ~ 1 enable/disable the delay line control defined in Delay Line Control Registers.

● Delay Line Control Enable Register 0 (Address: 0x2100,0054)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[15] Delay control[7:0]. 1: To enable DQ[15] Delay control[7:0].
[14]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[14] Delay control[7:0]. 1: To enable DQ[14] Delay control[7:0].
[13]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[13] Delay control[7:0]. 1: To enable DQ[13] Delay control[7:0].
[12]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[12] Delay control[7:0]. 1: To enable DQ[12] Delay control[7:0].
[11]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[11] Delay control[7:0]. 1: To enable DQ[11] Delay control[7:0].

Bits	Attributes	Reset	Description
[10]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[10] Delay control[7:0]. 1: To enable DQ[10] Delay control[7:0].
[9]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[9] Delay control[7:0]. 1: To enable DQ[9] Delay control[7:0].
[8]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[8] Delay control[7:0]. 1: To enable DQ[8] Delay control[7:0].
[7]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[7] Delay control[7:0]. 1: To enable DQ[7] Delay control[7:0].
[6]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[6] Delay control[7:0]. 1: To enable DQ[6] Delay control[7:0].
[5]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[5] Delay control[7:0]. 1: To enable DQ[5] Delay control[7:0].
[4]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[4] Delay control[7:0]. 1: To enable DQ[4] Delay control[7:0].
[3]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[3] Delay control[7:0]. 1: To enable DQ[3] Delay control[7:0].
[2]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[2] Delay control[7:0]. 1: To enable DQ[2] Delay control[7:0].
[1]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[1] Delay control[7:0]. 1: To enable DQ[1] Delay control[7:0].
[0]	RW	1'h0	DqDelayControlEnable 0: To disable DQ[0] Delay control[7:0]. 1: To enable DQ[0] Delay control[7:0].

● Delay Line Control Enable Register 1 (Address: 0x2100,0058)

Bits	Attributes	Reset	Description
[31:12]	RO	20'h0	Reserved
[11]	RW	1'h0	ReadDataValidWindowDelayControlEnable 0: To disable Read Data Valid Window Delay control[3:0]. 1: To enable Read Data Valid Window Delay control[3:0].
[10]	RW	1'h0	DQSGateMUXControlEnable 0: To disable DQS Gate MUX control[3:0]. 1: To enable DQS Gate MUX control[3:0].
[9]	RW	1'h0	DQReadMUXControlEnable 0: To disable MDATA Read MUX2 control[3:0].

Bits	Attributes	Reset	Description
			1: To enable MDATA Read MUX2 control[3:0].
[8]	RW	1'h0	DQReadMUXControlEnable 0: To disable MDATA Read MUX1 control[3:0]. 1: To enable MDATA Read MUX1 control[3:0].
[7:6]	RO	2'h0	Reserved
[5]	RW	1'h0	DqsGateDelayControlEnable 0: To disable DQS[1] Gate Delay control[7:0]. 1: To enable DQS[1] Gate Delay control[7:0].
[4]	RW	1'h0	DqsGateDelayControlEnable 0: To disable DQS[0] Gate Delay control[7:0]. 1: To enable DQS[0] Gate Delay control[7:0].
[3:2]	RO	2'h0	Reserved
[1]	RW	1'h0	DqsDelayControlEnable 0: To disable DQS[1] Delay control[7:0]. 1: To enable DQS[1] Delay control[7:0].
[0]	RW	1'h0	DqsDelayControlEnable 0: To disable DQS[0] Delay control[7:0]. 1: To enable DQS[0] Delay control[7:0].

● Fail Status Register 0 (Address: 0x2100,0060)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15:0]	RO	16'h0	FirstEdgeNotFound_Stage1 Fail status during normal calibration stage 1.

● Fail Status Register 1 (Address: 0x2100,0064)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15:0]	RO	16'h0	SecondEdgeNotFound_Stage1 Fail status during normal calibration stage 1.

● Fail Status Register 2 (Address: 0x2100,0068)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15:0]	RO	16'h0	BitTimeTooSmall_Stage1 Fail status during normal calibration stage 1.

● Fail Status Register 3 (Address: 0x2100,006C)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15:0]	RO	16'h0	SecondEdgeFoundFirstEdgeTooSmall_Stage1 Fail status during normal calibration stage 1.

● Fail Status Register 4 (Address: 0x2100,0070)

Bits	Attributes	Reset	Description
[31:28]	RO	4'h0	ReadDataCorrectMUX1DelayTooLarge_Stage4 Fail status during normal calibration stage 4
[27:24]	RO	4'h0	ReadDataErrorMUX0DelayExpired_Stage4 Fail status during normal calibration stage 4
[23:20]	RO	4'h0	ReadDataCorrectMUX0DelayIsZero_Stage4 Fail status during normal calibration stage 4
[19:16]	RO	4'h0	ReadDataErrorMUX1DelayExpired_Stage4 Fail status during normal calibration stage
[15:12]	RO	4'h0	DifferenceOfDelayTooLarge_Stage3 Fail status during normal calibration stage 3
[11:8]	RO	4'h0	ReadDataErrorDelayExpired_Stage3 Fail status during normal calibration stage 3
[7:0]	RO	8'h0	Reserved

● RAM IO Control Register (Address: 0x2100,0078)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	DDR DQ/DQS/DQSn PADS idle mode config This bit is configured to select working modes of DDR DQ/DQS/DQSn PADS. 1: DDR DQ/DQS/DQSn PADS are forced to enter idle mode to save power consumption, and in this case DDR cannot be read or written 0: DDR DQ/DQS/DQSn PADS work in normal mode NOTE: This bit must be set to 0 before using DDR interface.
[30:18]	RO	13'h0	Reserved
[17:15]	RW	3'h0	SSELcon_PAD_CFG[2:0] Control 'DS' ports of PAD except DQ, DM, DQS, CK and CKn. 'DS' ports control driving strength of PAD.
[14:12]	RW	3'h0	SSELclk_PAD_CFG[2:0] Control 'DS' ports of CK and CKn PAD. 'DS' ports control driving strength of PAD.
[11:9]	RW	3'h0	TSELdqs_PAD_CFG[2:0] Control 'TSEL' ports of DQS[1:0] PAD. 'TSEL' ports control ODT value of PAD.
[8:6]	RW	3'h0	SSELdqs_PAD_CFG[2:0] Control 'DS' ports of DQS[1:0] PAD. 'DS' ports control driving strength of PAD.
[5:3]	RW	3'h0	TSELdat_PAD_CFG[2:0] Control 'TSEL' ports of DQ[15:0] and DM[1:0] PAD. 'TSEL' ports control ODT value of PAD.
[2:0]	RW	3'h0	SSELdat_PAD_CFG Control 'DS' ports of DQ[15:0] and DM[1:0] PAD. 'DS' ports control driving strength of PAD.

● AXI DMA Buffer Configuration Register 0 (Address: 0x2100,007C)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	reg_wchn_aging_en AXI DMA Buffer write channel timeout enable bit 0: Disabled 1: Enabled
[30:16]	RW	15'h0	reg_wchn_aging This bit is used to set the timeout initial value of write channel. The smaller the initial value is, the higher priority of the write channel will be.
[15]	RW	1'h0	reg_rchn_aging_en AXI DMA Buffer read channel timeout enable bit 0: Disabled 1: Enabled
[14:0]	RW	15'h0	reg_rchn_aging This bit is used to set the timeout initial value of read channel. The smaller the initial value is, the higher priority of the write channel will be.

● AXI DMA Buffer Configuration Register 1 (Address: 0x2100,0080)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	reg_wchn1_max_bursts_en Continuous Max Bursts arbitration function configuration of Write Address Channel of AXI DMA Buffer Channel 1(NPU) 1: Enabled 0: Disabled
[30:24]	RW	7'h8	reg_wchn1_max_bursts These bits are used to set the times for which Write Address Channel of AXI DMA Buffer Channel 1(NPU) can be responded continuously.
[23]	RW	1'h1	reg_rchn1_max_bursts_en Continuous Max Bursts arbitration function configuration of Read Address Channel of AXI DMA Buffer Channel 1(NPU) 0: Disabled 1: Enabled
[22:16]	RW	7'h10	reg_rchn1_max_bursts These bits are used to set the times for which Read Address Channel of AXI DMA Buffer Channel 1(NPU) can be responded continuously.
[15]	RW	1'h1	reg_wchn2_max_bursts_en Continuous Max Bursts arbitration function configuration of Write Address Channel of AXI DMA Buffer Channel 2(Video Encoder) 1: Enabled 0: Disabled
[14:8]	RW	7'h10	reg_wchn2_max_bursts

Bits	Attributes	Reset	Description
			These bits are used to set the times for which Write Address Channel of AXI DMA Buffer Channel 2(Video Encoder) can be responded continuously.
[7]	RW	1'h1	reg_rchn2_max_bursts_en Continuous Max Bursts arbitration function configuration of Read Address Channel of AXI DMA Buffer Channel 2(Video Encoder) 0: Disabled 1: Enabled
[6:0]	RW	7'h20	reg_rchn2_max_bursts These bits are used to set the times for which Read Address Channel of AXI DMA Buffer Channel 2(Video Encoder) can be responded continuously.

● Calibration Register (Address: 0x2100,0084)

Bits	Attributes	Reset	Description
[31]	RW	1'h1	reg_wchn0_max_bursts_en Continuous Max Bursts arbitration function configuration of Write Address Channel of AXI DMA Buffer Channel 0(ISP) 1: Enabled 0: Disabled
[30:24]	RW	7'h8	reg_wchn0_max_bursts These bits are used to set the times for which Write Address Channel of AXI DMA Buffer Channel 0(ISP) can be responded continuously.
[23]	RW	1'h1	reg_rchn0_max_bursts_en Continuous Max Bursts arbitration function configuration of Read Address Channel of AXI DMA Buffer Channel 0(ISP) 0: Disabled 1: Enabled
[22:16]	RW	7'h10	reg_rchn0_max_bursts These bits are used to set the times for which Read Address Channel of AXI DMA Buffer Channel 0(ISP) can be responded continuously.
[15:8]	RW	8'hC	MinBitCycle_Stage1(Unit :45ps) The minimum bit cycle time used in the first stage of calibration. NOTE: It is just the simulation result and the actual value of chip is TBD.
[7:0]	RW	0x0	DqsDelayAddInitial(Unit :45ps) The added initial delay of DQS[3:0] for normal calibration flow. This is used when normal calibration failed during stage 1. Actually added initial delay of DQS[3:0] is (DqsDelayAddInitial[7:0] * 45ps).

Bits	Attributes	Reset	Description
			NOTE: 1) It is just the simulation result and the actual value of chip is TBD. 2) Result of the equation (DqsDelayAddInitial[7:0] +/- DLLDelayInfo[7:0]) cannot exceed 255.

● Success Status Register 0 (Address: 0x2100,0088)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15:0]	RO	16'h0	Success_OneEdgeFoundInitialValueValid_Stage1

● Success Status Register 1 (Address: 0x2100,008C)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15:0]	RO	16'h0	Success_OneEdgeFoundInitialValueNotValid_Stage1

● Success Status Register 2 (Address: 0x2100,0090)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15:0]	RO	16'h0	Success_TwoEdgeFoundInitialValueValid_Stage1

● Success Status Register 3 (Address: 0x2100,0094)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15:0]	RO	16'h0	Success_TwoEdgeFoundInitialValueNotValid_Stage1

● I-Cache Hit Counting Register (Address: 0x2100,00AC)

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	cpu_l1icache_hit_times This field is used to count the number of I-Cache hit at this stage after CPU L1 Cache hit/miss counting is enabled.

● I-Cache Miss Counting Register (Address: 0x2100,00B0)

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	cpu_l1icache_miss_times This field is used to count the number of I-Cache miss at this stage after CPU L1 Cache hit/miss counting is enabled.

● D-Cache Hit Counting Register (Address: 0x2100,00B4)

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	cpu_l1dcache_hit_times This field is used to count the number of D-Cache hit

Bits	Attributes	Reset	Description
			at this stage after CPU L1 Cache hit/miss counting is enabled.

- D-Cache Miss Counting Register (Address: 0x2100,00B8)

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	cpu_l1dcache_miss_times This field is used to count the number of D-Cache miss at this stage after CPU L1 Cache hit/miss counting is enabled.

- L1 Cache Hit/Miss Counting Control Register (Address: 0x2100,00BC)

Bits	Attributes	Reset	Description
[31:1]	RO	31'h0	Reserved
[0]	RW	1'h0	cpu_l1cache_static_clear CPU L1 Cache hit/miss counting enable bit 0 : Enabled 1 : Disabled

5.3.6 Programmers' Guide

When the system powers on or being reset, programmers should calibrate DDR2/DDR according to the following steps.

- 1) Enable the working clock of memory controller;
- 2) Configure all corresponding timing, I/O ports registers of DDR2 according to its working clock;
- 3) Deassert MDLU reset signal and wait until MDLU is latched;
- 4) Initialize DDR2/DDR;
- 5) Start Normal Calibration and wait until the operation is finished;
- 6) If the compensation function of SDLU needs to be enabled, please configure the threshold value of SDLU compensation circuit.
- 7) Enable Auto-Refreshing function;
- 8) The DDR2/DDR works normally.

NOTE

Those steps mentioned above should be implemented in L2 memory.

5.4 Fast SPI (SPI0)

5.4.1 Overview

AK3918AV100 series processors support one Fast SPI Controller (SPI0) for SPI NOR Flash or SPI NAND Flash.

5.4.2 Features

Features of SPI0 controller are as follows:

- Compatible with SPI Protocol, Version 2.11
- Works in master mode only
- Supports half-duplex transmission
- Supports polling mode and interrupt mode
- Supports CPU mode and DMA mode

- Supports Transmit Garbage Data Mode (TGDM) and Auto Reject Received Data Mode (ARRM)
- Supports Standard SPI, Dual SPI, and Quad SPI
- Supports two timing modes: mode 0 (CPOL = 0, CPHA = 0) and mode 3 (CPOL = 1, CPHA = 1)

5.4.3 Functional Description

5.4.3.1 Block Diagram

SPI0 controller of AK3918AV100 series processors work at up to 100MHz clock frequency. Figure 5-7 indicates the block diagram of SPI0 controller.

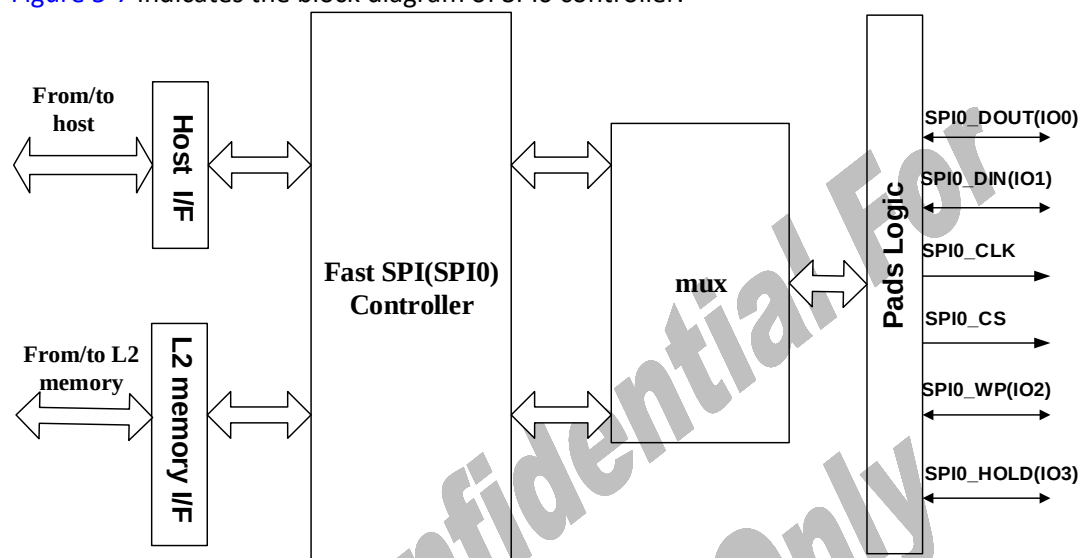


Figure 5-7 Block Diagram of SPI0 Controller

5.4.3.2 SPI Operations

SPI0 controller allows various Flash operations via satisfying the specific flash timing requirements, including read data, fast read, dual output fast read, dual I/O fast read, quad output fast read, quad I/O fast read and quad I/O word fast read. Timing diagrams of all SPI operations are illustrated as follows.

5.4.3.2.1 Read Data

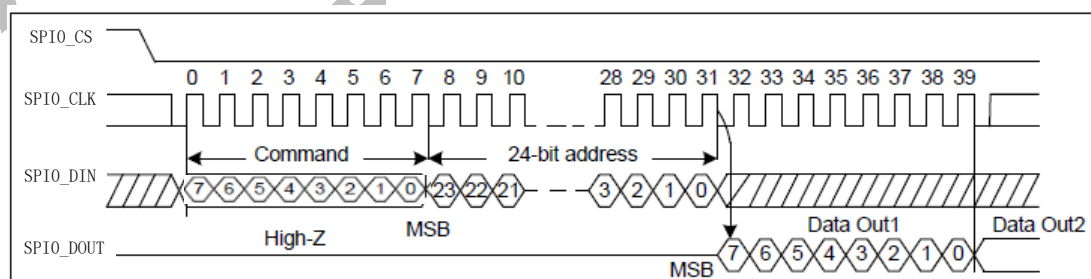


Figure 5-8 SPI0 Read data operation timing diagram

5.4.3.2.2 Fast Read

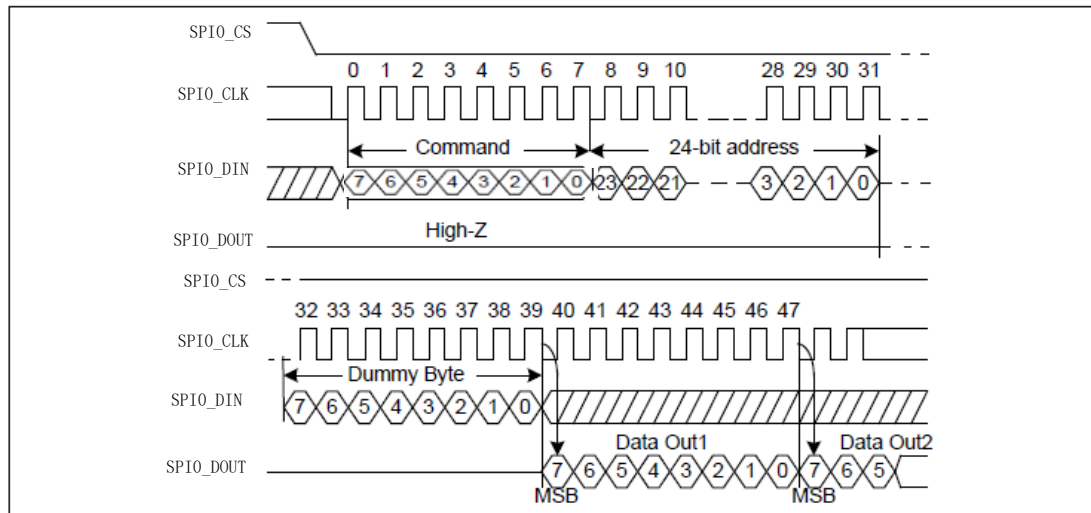


Figure 5-9 SPI0 Fast read operation timing diagram

5.4.3.2.3 Dual Output Fast Read

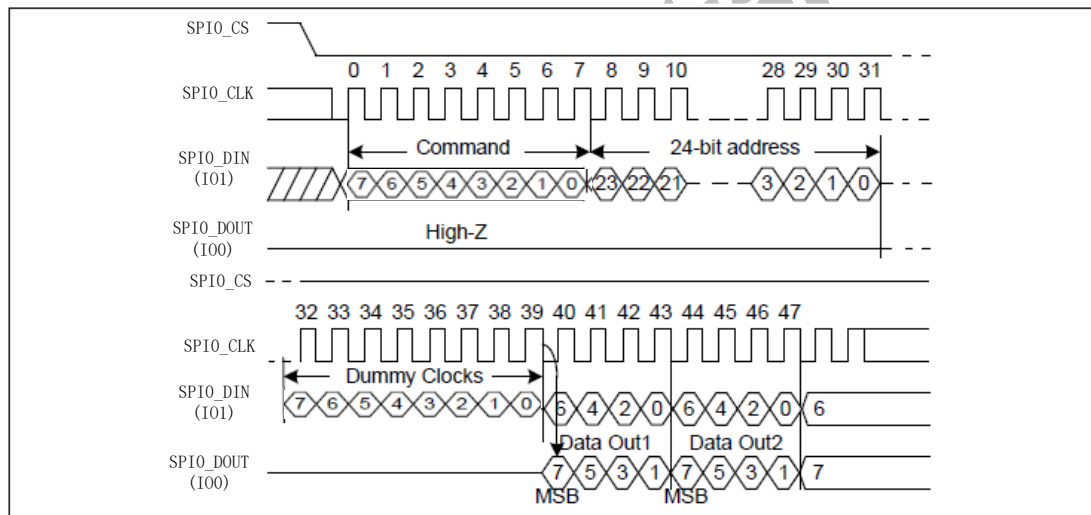


Figure 5-10 SPI0 Dual output fast read operation timing diagram

5.4.3.2.4 Dual I/O Fast Read

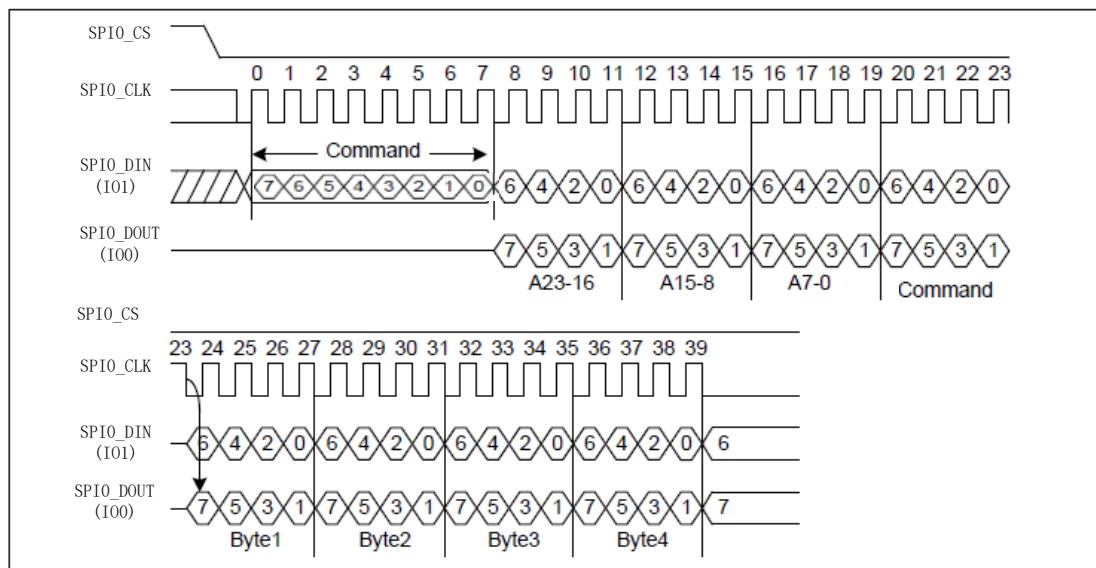


Figure 5-11 SPI0 Dual I/O fast read operation timing diagram

5.4.3.2.5 Quad Output Fast Read

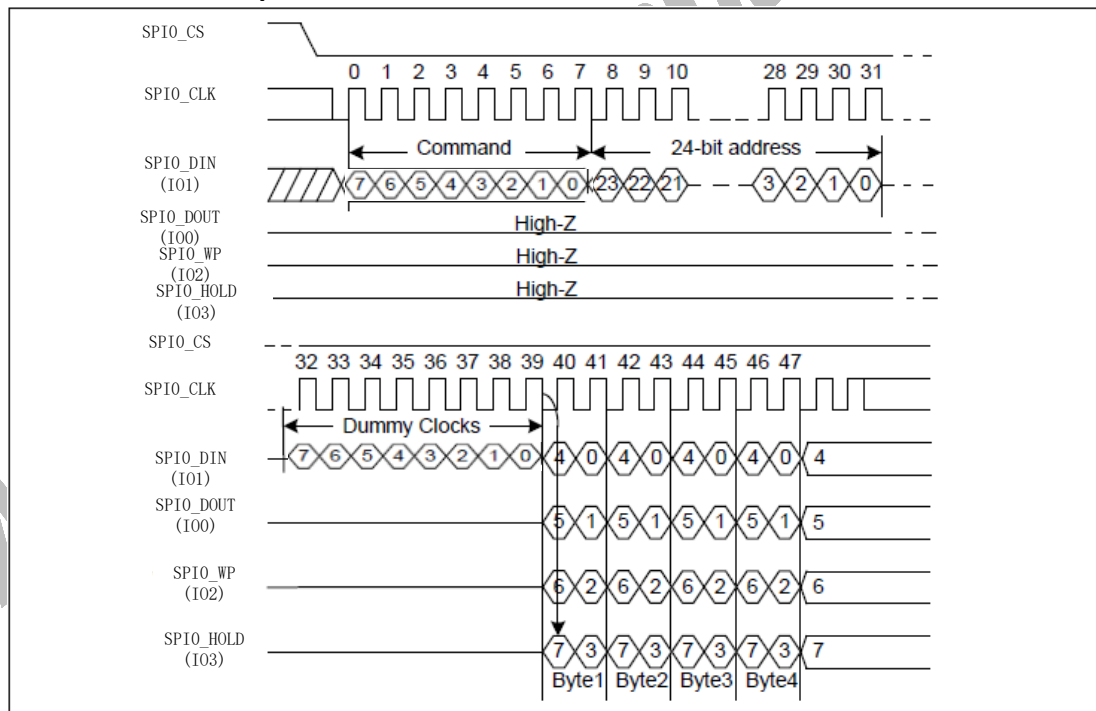


Figure 5-12 SPI0 Quad output fast read operation timing diagram

5.4.3.2.6 Quad I/O Fast Read

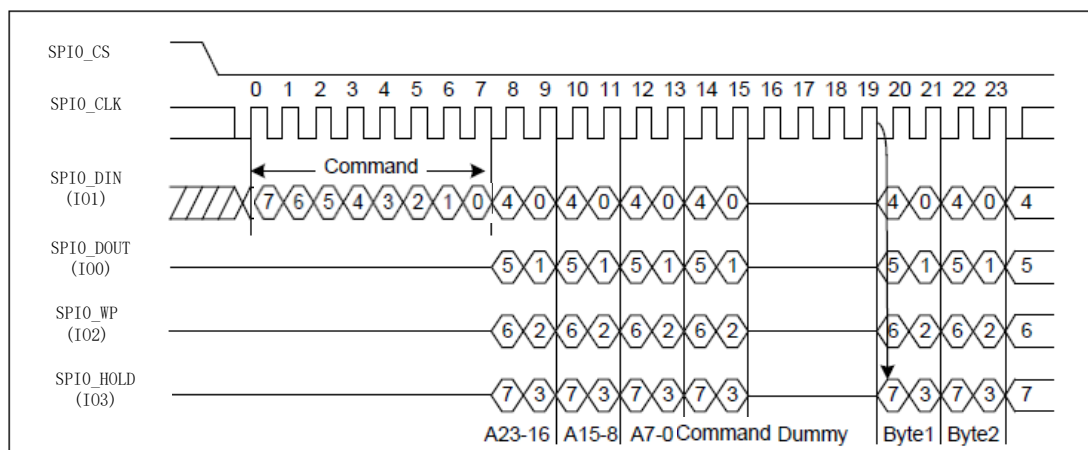


Figure 5-13 SPI0 Quad I/O fast read operation timing diagram

5.4.3.2.7 Quad I/O Word Fast Read

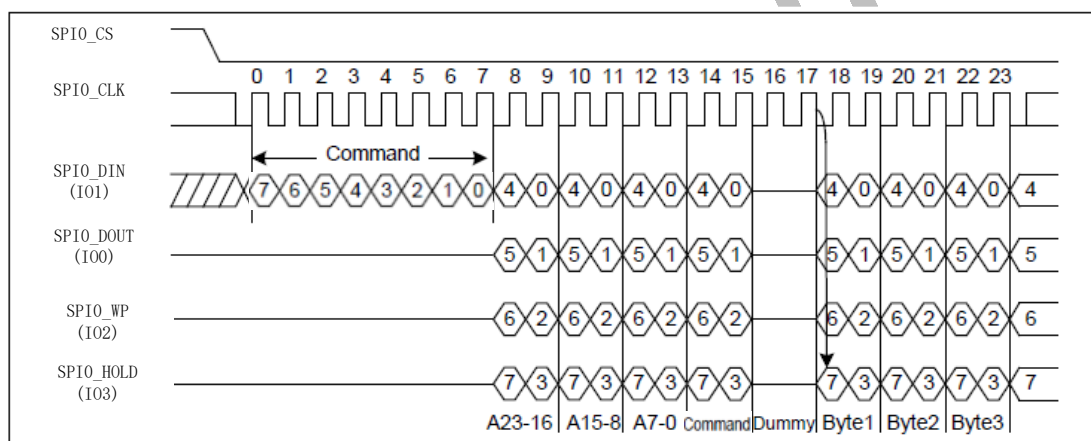
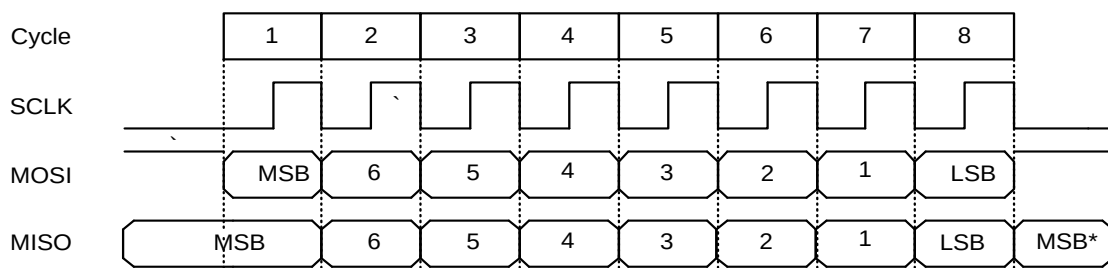


Figure 5-14 SPI0 Quad I/O word fast read operation timing diagram

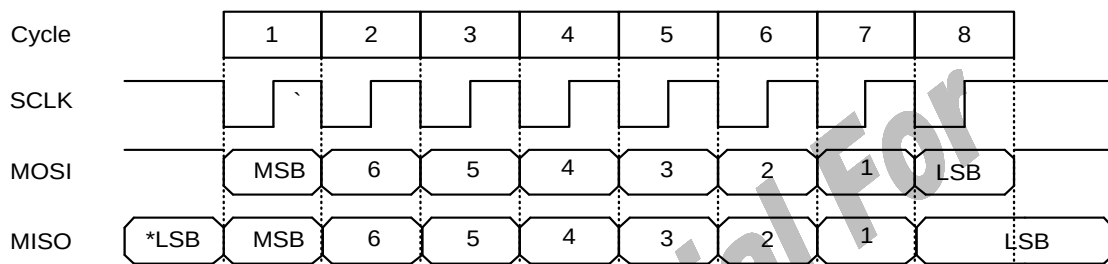
5.4.3.3 Transmission Format

During a SPI0 transmission, data are transmitted and received simultaneously. Data transmission formats between SPI devices are determined by CPOL (clock polarity) and CPHA (clock phase). CPOL defines the idle level of the SPI0 clock signal and CPHA determines at which toggling edge data sampling and data transmitting take place. CPOL and CPHA should be identical for SPI devices in communication. SPI0 interface of AK3918AV100 series processors support mode0 and mode3 transmission formats. The following figure illustrates the SPI-standard defined timing diagram of Mode 0 & Mode 3.



CPOL=0,CPHA=0(Format A)

* MSB of the character just received



CPOL=1,CPHA=1(Format B)

* LSB of the previously transmitted character

Notes:

MOSI: data output in master mode; data input in slave mode

MISO: data input in master mode; data output in slave mode

Figure 5-15 Timing Diagram of Mode 0 & Mode 3 of SPI0

5.4.3.4 Interface Timings

Timings of SPI0 interface are shown in the following figures.

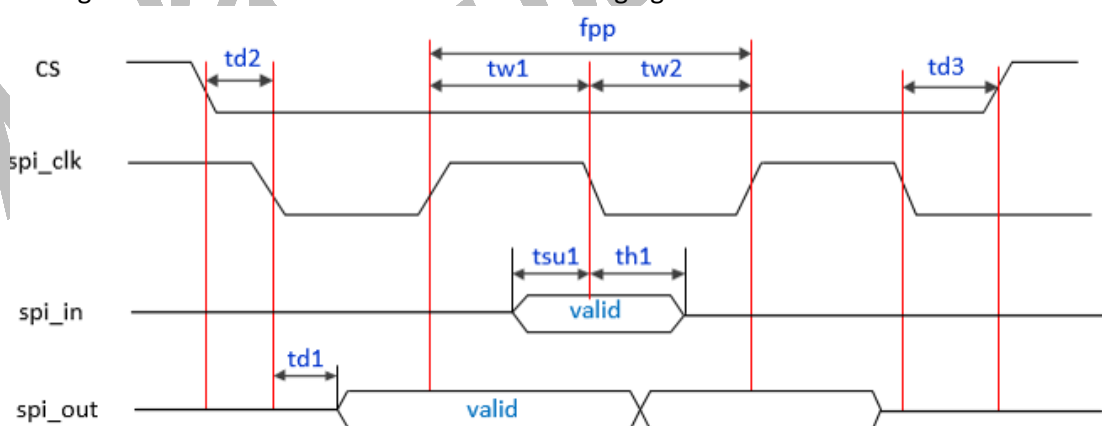


Figure 5-16 Timing of SPI0 master interface (Mode 3)

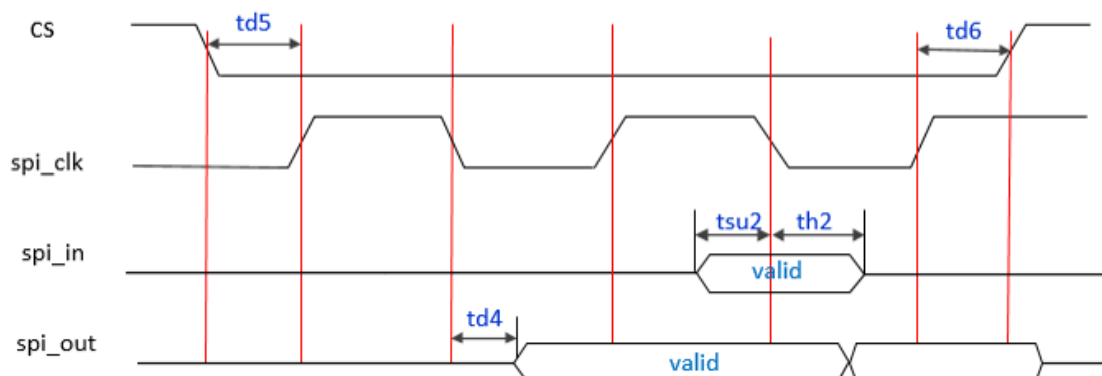


Figure 5-17 Timing of SPI0 master interface (Mode 0)

Parameters of SPI0 master interface timing are shown in the following table.

Table 5-6 Timing Parameters of SPI0 Master interface

Parameter	Symbol	Min	Max	Unit
Clock frequency	fpp	0	100	MHz
Clock high time	tw1	5	-	ns
Clock low time	tw2	5	-	ns
Output delay time	td1	0.3	1.8	ns
Delay time (CS falling edge to first SPI0_CLK rising or falling edge)	td2	1/fpp	-	ns
Delay time (CS rising edge to first SPI0_CLK rising or falling edge)	td3	1/2fpp	-	ns
Input set up time	tsu1	2.3	-	ns
Input hold time	th1	0	-	ns
Input set up time	tsu2	2.3	-	ns
Input hold time	th2	0	-	ns
Output delay time	td4	0.3	1.8	ns
Delay time (CS falling edge to first SPI0_CLK rising or falling edge)	td5	1/2fpp	-	ns
Delay time (CS rising edge to first SPI0_CLK rising or falling edge)	td6	1/fpp	-	ns

5.4.3.5 Data Transmission inside the Chip

SPI0 supports two data transmission modes, the process of which can be explained as follows.

● DMA mode

TX: SRAM → DMA → TX buffer of SPI0 interface → TX data line

RX: RX data line → RX buffer of SPI0 interface → DMA → SRAM

● CPU mode

TX operation: CPU writes data into TX buffer of SPI0 Interface → TX data line

RX operation: RX data line → CPU reads data from RX buffer of SPI0 Interface

5.4.3.6 Chip Boot

AK3918AV100 series processors support chip boot function via reading the boot code stored in the external Flash memory connected to SPI0 controller. For more details, please refer to section 4.4.

5.4.4 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Register Address	Register Name
1	0x2110,0000	<i>SPI0 Configuration Register</i>
2	0x2110,0004	<i>SPI0 Status Register</i>
3	0x 2110,0008	<i>SPI0 Interrupt Enable Register</i>
4	0x2110,000C	<i>SPI0 Data Count Register</i>
5	0x2110,0010	<i>SPI0 External TX Buffer Register</i>
6	0x2110,0014	<i>SPI0 External RX Buffer Register</i>
7	0x2110,0018	<i>SPI0 Output Data Register</i>
8	0x2110,001C	<i>SPI0 Input Data Register</i>
9	0x2110,0020	<i>SPI0 Control Register</i>
10	0x2110,0024	<i>SPI0 DMA/CPU Operation Register</i>

5.4.5 Register Definition

- SPI0 Configuration Register (Address: 0x2110,0000)

Bits	Attributes	Reset	Description
[31:18]	RO	14'h0	Reserved
[17:16]	RW	2'h0	master_data_mode 1/2/4-wire data selection for master mode 2'h0: 1-wire data mode 2'h1: 2-wire data mode 2'h2: 4-wire data mode 2'h3: 1-wire data mode NOTE: For slave mode, these bits must be 2'h0.
[15:7]	RO	9'h0	Reserved
[6]	RW	1'h0	spi0_en SPI0 enable control 0: Disabled 1: Enabled
[5]	RW	1'h0	cs 0: No instruction 1: To force #CS signal to be low. NOTE: Normally, the #CS signal is HIGH after a transmission of 8 bits data. If this bit is set, the #CS signal is LOW until the all the data transmission has been finished.
[4]	RO	1'h0	Reserved

Bits	Attributes	Reset	Description
[3]	RW	1'h0	cpha Initial phase of clock (CPHA) selection 0: Format A 1: Format B
[2]	RW	1'h0	cpol Active high or active low clock (CPOL) selection. 0: Active high 1: Active low
[1]	RW	1'h0	arm ARRM mode enable control 0: Disabled 1: Enabled NOTE: Once enabled, the SPI0 interface transmits and receives data simultaneously. If only data transmission is required, the ARRM (Auto Reject Received Data) mode can be enabled to discard the received data.
[0]	RW	1'h0	tgdm TGDM mode enable control 0: Disabled 1: Enabled NOTE: Once TGDM mode is enabled, the SPI0 interface transmits and receives data simultaneously. If only data reception is required, the TGDM (Transmit Garbage Data Mode) mode can be enabled to transmit dummy data on transmit line.

● SPI0 Status Register (Address: 0x2110,0004)

This register indicates current status of SPI0 controller, such as TX/RX buffer status, data transmission operation status, and etc.

Bits	Attributes	Reset	Description
[31:12]	RO	20'h0	Reserved
[11]	RO	1'h0	cache_in_progress 0: Data transmission operation of the whole cache mode is finished. 1: Data transmission operation of the whole cache mode is in progress.
[10]	RO	1'h0	cache_trans_finish 0: The whole cache transmission has not been finished. 1: The whole cache transmission has been finished.
[9]	RO	1'h0	mst_in_progress 0: The data transmission operation is finished in master mode. 1: The data transmission operation is in progress in master mode.
[8]	RO	1'h0	mst_trans_finish 0: The transmission operation has not been finished.

Bits	Attributes	Reset	Description
			1: The transmission operation has been finished.
[7]	RO	1'h0	rx_buf_overrun 1: RX buffer is written into even when it is full.
[6]	RO	1'h0	rx_buf_hfull 1: RX buffer of SPI0 interface is half full.
[5]	RO	1'h0	rx_buf_full 1: RX buffer of SPI0 interface is full.
[4]	RO	1'h1	rx_buf_empty 1: RX buffer of SPI0 interface is empty.
[3]	RO	1'h0	tx_buf_underrun 1: TX buffer is read even when it is empty.
[2]	RO	1'h1	tx_buf_hempty 1: TX buffer of SPI0 interface is half empty.
[1]	RO	1'h0	tx_buf_full 1: TX buffer of SPI0 interface is full.
[0]	RO	1'h1	tx_buf_empty 1: TX buffer of SPI0 interface is empty.

NOTE

This register will be cleared after being read.

- SPI0 Interrupt Enable Register (Address: 0x2110,0008)

The SPI0 controller supports various interrupts, each of which can be enabled or disabled independently by a specific bit of this register.

Bits	Attributes	Reset	Description
[31:12]	RO	20'h0	Reserved
[11]	RW	1'h0	cache_in_pro_int 0: To disable cache_in_pro interrupt 1: To enable cache_in_pro interrupt NOTE: the cache_in_pro interrupt is generated when the data transmission operation of the whole cache is in progress.
[10]	RW	1'h0	cache_fini_int 0: To disable cache_fini interrupt 1: To enable cache_fini interrupt NOTE: the cache_fini interrupt is generated when the transmission operation of the cache has been finished.
[9]	RW	1'h0	m_in_pro_int 0: To disable M_in_pro interrupt 1: To enable M_in_pro interrupt NOTE: The M_in_pro interrupt is generated when the data transmission operation is in progress in master mode.
[8]	RW	1'h0	trans_fini_int 0: To disable trans_fini interrupt 1: To enable trans_fini interrupt

Bits	Attributes	Reset	Description
			NOTE: The trans_fini interrupt is generated when the transmission operation has been finished.
[7]	RW	1'h0	rx_buf_ovrrun_int 0: To disable RX_buf_ovrrun interrupt 1: To enable RX_buf_ovrrun interrupt NOTE: The RX_buf_ovrrun interrupt is generated if the RX buffer is written into when it is full.
[6]	RW	1'h0	rx_buf_hfull_int 0: To disable RX_buf_hfull interrupt 1: To enable RX_buf_hfull interrupt NOTE: The RX_buf_hfull interrupt is generated when the RX buffer of SPI0 Interface is half full.
[5]	RW	1'h0	rx_buf_full_int 0: To disable RX_buf_full interrupt 1: To enable RX_buf_full interrupt NOTE: The RX_buf_full interrupt is generated when the RX buffer of SPI0 Interface is full.
[4]	RW	1'h0	rx_buf_emp_int 0: To disable RX_buf_emp interrupt 1: To enable RX_buf_emp interrupt NOTE: The RX_buf_emp interrupt is generated when the RX buffer of SPI0 Interface is empty.
[3]	RW	1'h0	tx_buf_underrun_int 0: To disable TX_buf_underrun interrupt 1: To enable TX_buf_underrun interrupt NOTE: The TX_buf_underrun interrupt is generated if the TX buffer is read when it is empty.
[2]	RW	1'h0	tx_buf_hemp_int 0: To disable TX_buf_hemp interrupt 1: To enable TX_buf_hemp interrupt NOTE: The TX_buf_hemp interrupt is generated when the TX buffer of SPI0 Interface is half empty.
[1]	RW	1'h0	tx_buf_full_int 0: To disable TX_buf_full interrupt 1: To enable TX_buf_full interrupt NOTE: The TX_buf_full interrupt is generated when the TX buffer of SPI0 Interface is full.
[0]	RW	1'h0	tx_buf_emp_int 0: To disable TX_buf_emp interrupt 1: To enable TX_buf_emp interrupt NOTE: The TX_buf_emp interrupt is generated when the TX buffer of SPI0 Interface is empty.

● SPI0 Data Count Register (Address: 0x2110,000C)

This register defines the total number of data to be sent or having been received in CPU/DMA mode.

Bits	Attributes	Reset	Description
------	------------	-------	-------------

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15:0]	RW	16'h0	data_cnt Unit: byte To transmit data: total number of data to be sent To receive data: total number of data to be received NOTE: The value of these bits is decreased by step length of "1" if one byte data has been sent/received when the SPI0 Interface works in the first three modes; and the value of these bits is increased by step length of "1" if one byte of data has been received when the SPI0 Interface works in the fourth mode.

● SPI0 External TX Buffer Register (Address: 0x2110,0010)

Bits	Attributes	Reset	Description
[31:17]	RW	15'h0	buf_len The size of the TX buffer (in L2 memory) assigned to SPI0 Interface
[16]	RW	1'h0	dma_en 0: To access external TX buffer by CPU port 1: To access external TX buffer by DMA port (SPI0 interface is ready and the TX buffer in L2 memory can request a DMA operation.)
[15:8]	RO	8'h0	Reserved
[7:1]	RW	7'h0	saddr Start address of external buffer in L2 memory, from which the data will be sent.
[0]	RW	1'h0	buf_en 0: To disable the TX buffer in L2 memory (i.e. only the 8-byte TX buffer in SPI0 Interface is utilized.) 1: To enable the TX buffer in L2 memory.

● SPI0 External RX Buffer Register (Address: 0x2110,0014)

Bits	Attributes	Reset	Description
[31:17]	RW	15'h0	buf_len The size of the RX buffer (in L2 memory) assigned to SPI0 Interface
[16]	RW	1'h0	DMA_en 0: To access external RX buffer by CPU port 1: To access external RX buffer by DMA port (SPI0 interface is ready, and the TX buffer in L2 memory can request a DMA operation.)
[15:8]	RO	8'h0	Reserved
[7:1]	RW	7'h0	start_addr Starting address of external buffer in L2 memory, from which the data will be stored.

Bits	Attributes	Reset	Description
[0]	RW	1'h0	buf_en 0: To disable the RX buffer in L2 memory (i.e. only the 8-byte RX buffer in SPI0 Interface is utilized.) 1: To enable the RX buffer in L2 memory

● SPI0 Output Data Register (Address: 0x2110,0018)

Bits	Attributes	Reset	Description
[31:0]	WO	32'h0	output_data CPU writes 4 bytes data to be transferred into this register in a writing operation. NOTE: 1) As the TX buffer of the SPI0 Interface is 8-byte wide, the 4 bytes data CPU first writes into is stored in the lower 4 bytes of the buffer; next 4 bytes data is stored in the higher 4 bytes of the buffer; the third 4 bytes is stored in the lower 4 bytes of the buffer; and the fourth 4bytes is stored in the higher 5 bytes of the buffer; ... 2) High-order alignment is utilized. Bits[31:24] corresponds with Byte 3 and Byte 7; Bits[23:16], Byte 2 and Byte 6; Bits[15:8], Byte 1 and Byte 5; Bits[7:0], Byte 0 and Byte 4.

● SPI0 Input Data Register (Address: 0x2110,001C)

The Output Data Register holds the data CPU writes into, and the Input Data Register holds the data from RX data lines.

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	input data CPU reads 4-byte received data

● SPI0 Control Register (Address: 0x2110,0020)

Bits	Attributes	Reset	Description
[31:16]	RW	16'h0	cs_delay_cycles When data reading finished, the SCLK cycles of cs delay to high.
[15:13]	RW	3'h0	cs_high_cycles The SPI0 CLK cycles of cs remain high after a read operation; the real cs high cycles = (cs_high_cycles + 1).
[12:8]	RW	5'h0	dummy_cycles The SCLK cycles of dummy data
[7:0]	RW	8'h3	spiflash_rd_cmd The read command of SPI flash

● SPI0 DMA/CPU Operation Register (Address: 0x2110,0024)

Bits	Attributes	Reset	Description
[31:1]	RO	31'h0	Reserved

Bits	Attributes	Reset	Description
[0]	RW	1'h0	1: To send a DMA/CPU operation request to SPI0 controller. 0: No instruction. NOTE: This bit must be configured as 1.

5.4.6 Programmer's Guide

Before CPU or DMA transmission, programmers should write 1 to bit[0] of *SPI0 DMA/CPU Operation Register(Address: 0x2110,0024)*, then a request will be sent to SPI0 controller. Programmers should poll bit[0] of *SPI0 DMA/CPU Operation Register(Address: 0x2110,0024)* continuously, as long as this bit is read as 1. It means DMA or CPU request is responded and the SPI0 controller can enter DMA or CPU mode and configure the related registers.

5.4.6.1 CPU Mode

For MOSI data transmission mode, programmers should operate according to the following steps:

1. Write 1 to bit[1] of *SPI0 Configuration Register(Address: 0x2110,0000)*.
2. Write the number of data to be sent into *SPI0 Data Count Register (Address: 0x2110,000C)*.
3. Write the data to be sent into *SPI0 Output Data Register (Address: 0x2110,0018)*.
4. Check bit[8] of *SPI0 Status Register (Address: 0x2110, 0004)* of SPI0 master to judge whether the data sending is finished or not.

For MOSI data transmission mode, programmers should operate according to the following steps:

1. Write 1 to bit[1] of *SPI0 Configuration Register(Address: 0x2110,0000)*.
2. Write the number of data to be received into *SPI0 Data Count Register (Address: 0x2110,000C)*.
3. Check bit[8] of *SPI0 Status Register (Address: 0x2110,0004)* of SPI0 master to judge whether the data receiving is finished or not.
4. Read *SPI0 Input Data Register (Address: 0x2110,001C)*.

5.4.6.2 DMA Mode

For MOSI data transmission mode, programmers should operate according to the following steps:

1. Write 1 to bit[1] of *SPI0 Configuration Register(Address: 0x2110,0000)*.
2. Write the number of data to be sent into *SPI0 Data Count Register (Address: 0x2110,000C)*.
3. Configure *SPI0 External TX Buffer Register(Address: 0x2110,0010)* to sending the data.
4. Check bit[8] of *SPI0 Status Register (Address: 0x2110,0004)* of SPI0 master to judge whether the data sending is finished or not.

For MISO data transmission mode, programmers should operate according to the following steps:

1. Write 1 to bit[1] of *SPI0 Configuration Register(Address: 0x2110,0000)*.
2. Configure *SPI0 External RX Buffer Register(0x2110,0014)* to start data-receiving.

3. Write the number of data to be sent into *SPI0 Data Count Register (Address: 0x2110,000C)*.
4. Check bit[8] of *SPI0 Status Register (Address: 0x2110,0004)* of SPI0 master to judge whether the data sending is finished or not.

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6 Multimedia

AK3918AV100 series processors integrate image capture, ISP, pre-processor and video processor to provide high multimedia performance.

6.1 Image Capture

6.1.1 Overview

The image capture module acquires data and synchronous signals from the external image sensor, performs data format conversion, shrinks the data, and routes the data to memory.

6.1.2 Features

The features of image capture module are as follows:

- Supports two kinds of mode
 - Single CMOS Image Sensor with DVP interface
 - Single CMOS Image Sensor with MIPI interface
- Supports DVP interfaces
 - Compatible with BT.601/BT.656/BT.1120 standard
 - Supports Bayer RGB data, YUV422 data and JPEG compressed data
- Supports 1/2-lane MIPI interfaces with Bayer RAW data
 - Compliant with MIPI Alliance Standard for Camera Serial Interface 2(CSI-2)
 - Supports Bayer RAW data, YUV422 data, YUV420 data, JPEG compress data and other commonly-used data
 - Supports high speed and low power lane operation
 - Supports GPIO mode on MIPI ports when MIPI CSI-2 CMOS Image Sensor is not required.
- Input data width: 8-bit/10-bit
- Maximum input image size: 2880*2880
- Supports data sampling at both rising edge and falling edge
- Supports continuous frames sampling and single frame sampling
- Supports DMA upload stride function
- Supports DMA overflow detection
- Supports downscaling of YUV422 data
- Image cropping
- Data line swap function

6.1.3 Block Diagram

Image capture supports both MIPI and DVP modes. The following figure is a schematic diagram of the chip's image capture interface.

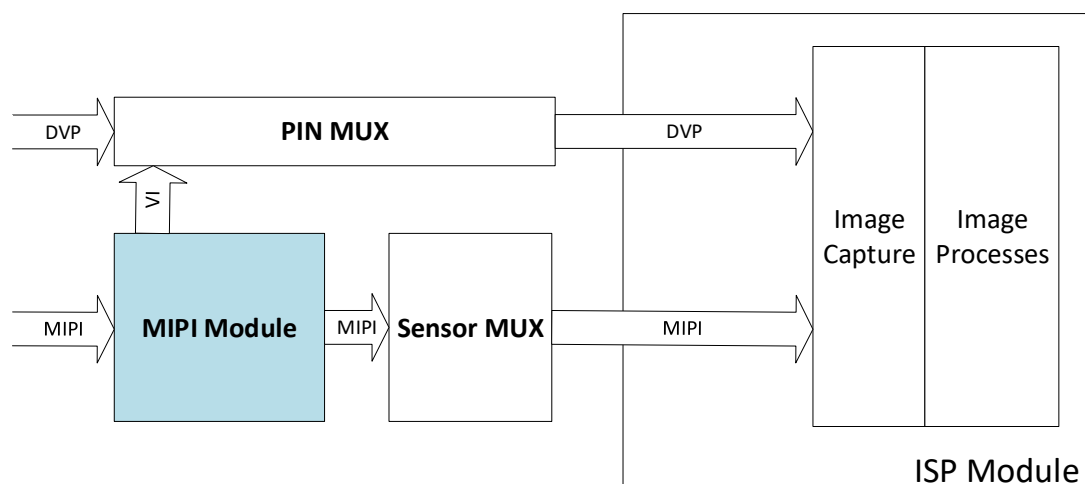


Figure 6-1 Block Diagram of Image Capture

NOTE

- 1) For the external CIS with DVP interface, data will be directly bypassed to the ISP module.
- 2) For the external CIS with MIPI interface, data will be received and passed by MIPI, then bypassed to ISP module.

6.1.4 Interface Timing

Timing diagrams of DVP interface and standard ITU-R BT.1120 format are shown in the following figures.

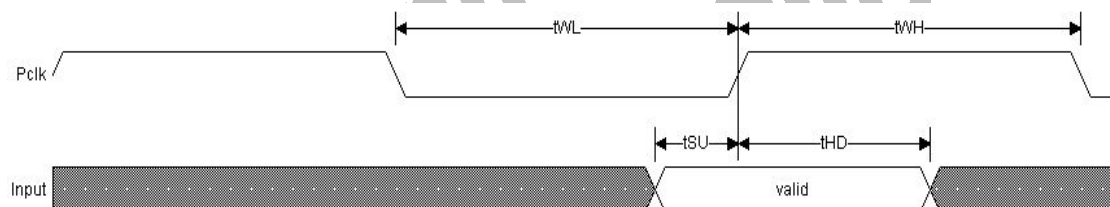


Figure 6-2 Timing Diagram of DVP Interface

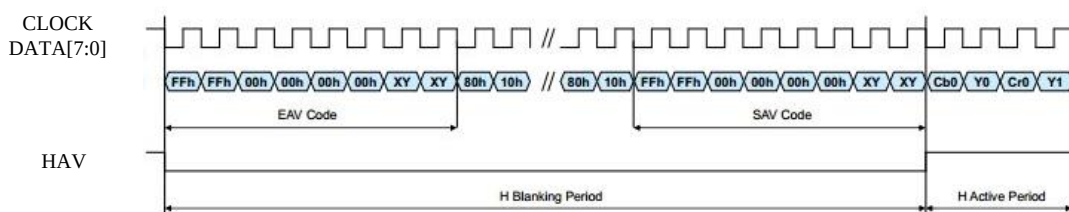


Figure 6-3 Timing Diagram of Standard ITU-R BT.1120 Format

Parameters of DVP interface timing interface are shown in the following table.

Table 6-1 Timing Parameters of DVP Interface

Parameter	Symbol	Min.	Max.	Unit
Input clock	Pclk	-	150	MHz
Clock low time	tWL	0.45	0.55	Tck

Parameter	Symbol	Min.	Max.	Unit
Clock high time	tWH	0.45	0.55	Tck
Referenced to CLK				
Input setup time	tSU	1.5	-	ns
Input hold time	tHD	1.5	-	ns

6.1.5 Registers Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Offset Address	Register Name and Description
1	0X0800,0218	<i>MIPI DPHY Power Control Register</i>
2	0X0800,0220	<i>MIPI DPHY Swap Selection Register</i>
3	0X0800,0224	<i>MIPI DPHY Reset and Control Register</i>
4	0X2040,0004	<i>Blcoks Enable Register</i>
5	0X2040,0008	<i>Lane Number and Dphy Selection Register</i>
6	0X2040,0050	<i>MIPI DPHY Lane Control Register</i>
7	0X2040,0100	<i>Stream Data Output Configuration Register</i>
8	0X2040,0104	<i>CSI Slave Controller Status Register</i>
9	0X2040,0108	<i>Data Type and Virtual Channel Configuration Register</i>
10	0X2040,010C	<i>Output Mode and FIFO Configuration Register</i>
11	0X2040,0110	<i>FIFO Fill Level Mod Configuration Register</i>
12	0x2120,0060	<i>DVP Interface Data Map Register 0</i>
13	0x2120,0064	<i>DVP Interface Data Map Register 1</i>

6.1.6 Register Definition

- MIPI DPHY Power Control Register (Address: 0X0800,0218)

Bits	Attributes	Reset	Description
[31:23]	RO	9'h0	Reserved
[22]	RW	1'h1	AON_POWER_READY_N Isolation signal that prevents in-rush currents. 1'h0: Power is ready. 1'h1: Power is not ready. NOTE: - Set Resetb(bit[30] of <i>MIPI DPHY Swap Selection Register (Address: 0X0800,0220)</i> to 1'h0 and this bit as 1'h1 to put MIPI controller into power isolation mode. - Before starting the power-up sequence of MIPI controller, this bit must be deasserted. And configuration of this module must be completed before Resetb(bit[30] of <i>MIPI DPHY Swap Selection Register (Address: 0X0800,0220)</i> is set to 1'h1.
[21:20]	RO	2'h0	Reserved
[19:12]	RW	8'h64	DPHY: RG_CFGCLK_1US_CNT[7:0] The input signal RG_CFGCLK_1US_CNT[7:0] should be

Bits	Attributes	Reset	Description
			changed based on the selected frequency. For more please refer to bit[9] in SAR_ADC/SPIO/ SD_DAC Clock Control (Address: 0X0800,000C) . RG_CFGCLK_1US_CNT[7:0]=1 us/(1/cfgclk_in frequency)
[11:0]	RO	12'h0	Reserved

● MIPI DPHY Swap Selection Register (Address: 0X0800,0220)

Bits	Attributes	Reset	Description
[31]	RO	1'h0	Reserved
[30]	RW	1'h0	Resetsb This signal is used to place the PHY into disabled mode. All analog blocks are in power-down mode and the digital logic is reset. NOTE: 1) Set Resetsb(bit[30] of MIPI DPHY Swap Selection Register (Address: 0X0800,0220) to 1'h0 and this bit as 1'h1 to put MIPI controller into power isolation mode. 2) Before starting the power-up sequence of MIPI controller, this bit must be deasserted. And configuration of this module must be completed before Resetsb(bit[30] of MIPI DPHY Swap Selection Register (Address: 0X0800,0220) is set to 1'h1.
[29:23]	RO	7'h0	Reserved
[22:20]	RW	3'h0	CFG_DPDN_SWAP[2:0] This field is used to enable DP/DN swap function. Setting the CFG_DPDN_SWAP[0] bit to 1 will swap the DP/DN of physical lane 0. Setting the CFG_DPDN_SWAP[1] bit to 1 will swap the DP/DN of physical lane 1, and so on.
[19:17]	RO	3'h5	Reserved
[16:14]	RW	3'h2	CFG_C0_SWAP_SEL[2:0] This field is used to configure Clock Lane 0 control source. For example, setting this signal to 3'h2 will source clock lane 0 signals from physical lane 2 (MIPI_RX_L2P/N).
[13:8]	RO	6'h1A	Reserved
[7:5]	RW	3'h1	CFG_D1_SWAP_SEL[2:0] This field is used to configure Data Lane 1 control source. For example, setting this signal to 3'h1 will source data lane 1 signals from physical lane 1 (MIPI_RX_L1P/N).
[4:2]	RW	3'h0	CFG_D0_SWAP_SEL[2:0] This field is used to configure Data Lane 0 control source. For example, setting this signal to 3'h0 will source data lane 0 signals from physical lane 0 (MIPI_RX_L0P/N). NOTE: 3) Only Data Lane 0 supports the LP Turnaround (BTA) function. 4) For the MP Test Ip_loopback mode, Data Lane 0 will be

Bits	Attributes	Reset	Description
			used to transmit LP test patterns. It is strongly recommended to set CFG_D0_SWAP_SEL (bits[4:2] of this register) to 3'h0. Otherwise, the lp_loopback mode can't work.
[1:0]	RO	2'h0	Reserved

● MIPI DPHY Reset and Control Register (Address: 0X0800,0224)

Bits	Attributes	Reset	Description
[31:24]	RO	8'h0	Reserved
[23:12]	RW	12'h0	DA_CDRX_L_SW_EN 0: MIPI_CSI sensor 1: GPIO/DVP pins NOTE: 1) Bit[23] is used to configure MIPI_RX_L5N, bit[22] corresponds to MIPI_RX_L5P, bit[21] to MIPI_RX_L4N, bit[20] to MIPI_RX_L4P.....and bit[12] corresponds to MIPI_RX_LOP. 2) When MIPI pins are used as GPIO/DVP function, this field should be set to 1 and the corresponding function should be enabled. For more details about the configuration precautions, please refer to Chapter 3 .
[11]	RW	1'h1	ppi_rx_word_reset_n_CSI_cfg
[10:0]	RO	11'h0	Reserved

● Blocks Enable Register (Address: 0X2040,0004)

This register is used to enable front and protocol blocks. Front and Protocol blocks will be held in soft reset until this register is set.

Bits	Attributes	Reset	Description
[31:1]	RO	31'h0	Reserved
[0]	RW	1'h0	Start Writing 1'b1 into this bit to enable the front and protocol blocks.

● Lane Number and Dphy Selection Register (Address: 0X2040,0008)

This register is used to configure the number of lanes being used, external DPHY selection and support for CSIRX v2.0. This register should be set prior to enabling the streams and must not be updated when the stream is running. In ASF configurations this register is parity protected.

Bits	Attributes	Reset	Description
[31:8]	RO	24'h0	Reserved
[7:4]	RW	4'h1	lane_nb The number of lanes 4'h1: 1 lane 4'h2: 2 lanes Others: reserved
[3]	RW	1'h0	lrte_epd_option_select

Bits	Attributes	Reset	Description
			LRTE EPD Option Selection 0: Option 1 1: Option 2
[2]	RW	1'h0	enable_lрте Enable LRTE support
[1]	RW	1'h0	enable_descrambling De-scrambling control. Implemented as per MIPI CSI v2.1 Specification
[0]	RW	1'h0	v3p0_support_enable Enable RAW24 support as per MIPI CSI v3.0 Specification

- MIPI DPHY Lane Control Register (Address: 0X2040,0050)

This register is used to enable and reset DPHY lane control for data and clock lanes. In ASF configurations, this register is parity protected.

Bits	Attributes	Reset	Description
[31:9]	RO	23'h0	Reserved
[8]	RW	1'h0	cl_enable DPHY clock lane enable 0: Disabled 1: Enabled
[7:2]	RO	6'h0	Reserved
[1]	RW	1'h0	dl1_enable DPHY data lane 1 enable 0: Disabled 1: Enabled
[0]	RW	1'h0	dl0_enable DPHY data lane 0 enable 0: Disabled 1: Enabled

- Stream Data Output Configuration Register (Address: 0X2040,0100)

This register is used to control CSIRX Stream Data output Datapath. Start and Stop commands are independent for each output with the exception of pixel outputs that can never be enabled together. If a pixel output is started while the other is already running, the start command will be ignored. If both pixel outputs are enabled in a single register access, then both start commands are ignored and no pixel output is started.

Bits	Attributes	Reset	Description
[31:2]	RO	30'h0	reserved
[1]	RW	1'h0	Stop Writing 1 in this register will cause csirx to stop streaming on the corresponding output at the end of the current frame. If the command is issued during frame blanking, then the data path will immediately stop streaming data on that output. stream_stop_irq is generated on completion of the stop operation.
[0]	RW	1'h0	start When 0 the stream is held in a reset condition. Writing 1 in

Bits	Attributes	Reset	Description
			this register enables the corresponding datapath output. It will start streaming data at the start of the next frame that complies with the output configuration. stream_status[31] running indicates when data stream is enabled.

● CSI Slave Controller Status Register (Address: 0X2040,0104)

This register is used to indicate the CSI Slave Controller Status.

Bits	Attributes	Reset	Description
[31]	RO	1'h0	Running The stream is enabled.
[30:9]	RO	22'h0	Reserved
[8]	RO	1'h0	ready_state Indicates the state of pixel_ready_if for this stream
[7:4]	RO	4'h1	stream_fsm Output to Stream FSM states: 4'h0: DEFAULT Should never hit this since FSM is one-hot 4'h1: STREAM_WAIT_CTRL_DATA Expecting control data next 4'h2: STREAM_CTRL Check contents of Ctrl packet and extract header information 4'h3: STREAM_DATA Pixel stream pixel data unpacking 4'h4: STREAM_CONV_PIX_NB 1st cycle delay for byte2pixel conversion 4'h5: STREAM_CONV_PIX_NB_EXT 2nd cycle delay for byte2pixel conversion 4'h6: STREAM_DATA_START Assert Hsync 4'h7: STREAM_DATA_END De-assert Hsync 4'h8: STREAM_FILL_WAIT Elastic Buffer cfg - wait until fill level is reached 4'h9: STREAM_STOP Stop at the end of Frame - used to set irq 4'hA: STREAM_WAIT_CRC

Bits	Attributes	Reset	Description
			Wait until CRC check has completed - Full Line cfg 4'hB: STREAM_PKT_DONE Packet complete - no error conditions after this point 4'hC: STREAM_BLANK Blank Packet – Initiate blanking handling
[3:2]	RO	2'h0	Reserved
[1:0]	RO	2'h0	protocol_fsm Input to Stream FSM states: 2'h0: PROT_IDLE 2'h1: PROT_WAIT_CTRL 2'h2: PROT_CTRL 2'h3: PROT_DATA

● Data Type and Virtual Channel Configuration Register (Address: 0X2040,0108)

This register is used to configure the data types and virtual channels to be processed and output by this stream. In ASF configurations this register is parity protected.

Bits	Attributes	Reset	Description
[31:16]	RW	16'h0	c_select Selection of Virtual Channels to be processed: Default '0' -> All Virtual Channels are processed vc_select0[16] : Virtual Channel Select 0 is processed vc_select1[17] : Virtual Channel Select 1 is processed vc_select2[18] : Virtual Channel Select 2 is processed vc_select3[19] : Virtual Channel Select 3 is processed vc_select4[20] : Virtual Channel Select 4 is processed vc_select5[21] : Virtual Channel Select 5 is processed vc_select6[22] : Virtual Channel Select 6 is processed vc_select7[23] : Virtual Channel Select 7 is processed vc_select8[24] : Virtual Channel Select 8 is processed vc_select9[25] : Virtual Channel Select 9 is processed vc_select10[26] : Virtual Channel Select 10 is processed vc_select11[27] : Virtual Channel Select 11 is processed vc_select12[28] : Virtual Channel Select 12 is processed vc_select13[29] : Virtual Channel Select 13 is processed vc_select14[30] : Virtual Channel Select 14 is processed vc_select15[31] : Virtual Channel Select 15 is processed
[15]	RW	1'h0	enable_dt1 Enable processing of dt1
[14]	RO	1'h0	Reserved
[13:8]	RW	6'h0	datatype_select1 Second data type format that this stream will process. Datatypes use the encoding as defined in the MIPI standard. Also see csirx_dt_constants.v.

Bits	Attributes	Reset	Description
[7]	RW	1'h0	enable_dt0 Enable processing of dt0
[6]	RO	1'h0	Reserved
[5:0]	RW	6'h0	datatype_select0 First data type format that this stream will process. Datatypes use the encoding as defined in the MIPI standard. Also see csirx_dt_constants.v.

● Output Mode and FIFO Configuration Register (Address: 0X2040,010C)

This register is used to configure the output mode and set up some Stream FIFO related settings. In ASF configurations this register is parity protected.

Bits	Attributes	Reset	Description
[31:16]	RW	16'h0	fifo_fill Set the FIFO_FILL_LEVEL which is used to hold data in the FIFO until this level is reached before allowing data to be pulled. NOTE: This setting is only used when fifo_mode is set for Large Buffer operation. MAXIMUM level is 85% of stream FIFO depth.
[15:12]	RW	4'h0	bpp_bypass Force unpacking of any Data type as selected RAW type. 4'h1: unpack 6 bits per pixel 4'h2: unpack 7 bits per pixel 4'h3: unpack 8 bits per pixel 4'h4: unpack 10 bits per pixel 4'h5: unpack 12 bits per pixel 4'h6: unpack 14 bits per pixel 4'h7: unpack 16 bits per pixel 4'h8: unpack 20 bits per pixel 4'h9: unpack 24 bits per pixel Others: No bypass
[11:10]	RO	2'h0	Reserved
[9:8]	RW	2'h0	fifo_mode Stream FIFO configuration, which must be set in accordance with FIFO sizing, flow control and the relationship between the link and pixel interface data rates. 2'h0: Full Line Buffer. Hold data in FIFO until CRC check completes. 2'h1: Large Buffer (Fill Level Controlled). Hold data in FIFO until FIFO_FILL_LEVEL is reached. Others: Short Buffer. When pixel output data rate can match link data rate, a small buffer can be used to accommodate CDC, pixel data packing, and data rate matching.
[7:6]	RO	2'h0	Reserved
[5:4]	RW	2'h0	num_pixels Number of pixels to output from the stream. Valid value is 1.

Bits	Attributes	Reset	Description
			The width of the pixel interface and the bits per pixel for the selected datatype will determine how many pixels can be output in a single cycle. 2'h0: 1 pixel per clock
[3:1]	RO	3'h0	Reserved
[0]	RW	1'h0	interface_mode Select the output configuration. 0: Pixel mode 1: Packed mode

● FIFO Fill Level Mod Configuration Register (Address: 0X2040,0110)

This register is used to configure the mode of fifo fill level monitor. In ASF configurations the read-write bits of this register are parity protected.

Bits	Attributes	Reset	Description
[31:18]	RO	14'h0	Reserved
[17:16]	RW	2'h0	Mode 2'h0: Fill level detection disabled 2'h1: Mode 1. In this mode, Monitor peak fill level until the stream is stopped. 2'h2: Mode 2. In this mode, Monitor peak fill level when the first FIFO read is made during a frame. 2'h3: Reserved
[15:10]	RO	6'h0	Reserved
[9:0]	RO	10'h0	Count Peak fill level of FIFO

● DVP Interface Data Map Register 0 (Address: 0X2120,0060)

Bits	Attributes	Reset	Description
[31:28]	RW	4'h7	isp_dvp_din7_map_cfg Indicate which DVP IOs is connected to CMOS Image sensor data line D7. 4'h0: CSI_DATA0 4'h1: CSI_DATA1 4'h2: CSI_DATA2 4'h3: CSI_DATA3 4'h4: CSI_DATA4 4'h5: CSI_DATA5 4'h6: CSI_DATA6 4'h7: CSI_DATA7 4'h8: CSI_DATA8 4'h9: CSI_DATA9 Others: not used
[27:24]	RW	4'h6	isp_dvp_din6_map_cfg Indicate which DVP IOs is connected to CMOS Image sensor data line D6. 4'h0: CSI_DATA0 4'h1: CSI_DATA1

Bits	Attributes	Reset	Description
			4'h2: CSI_DATA2 4'h3: CSI_DATA3 4'h4: CSI_DATA4 4'h5: CSI_DATA5 4'h6: CSI_DATA6 4'h7: CSI_DATA7 4'h8: CSI_DATA8 4'h9: CSI_DATA9 Others: not used
[23:20]	RW	4'h5	isp_dvp_din5_map_cfg Indicate which DVP IOs is connected to CMOS Image sensor data line D5. 4'h0: CSI_DATA0 4'h1: CSI_DATA1 4'h2: CSI_DATA2 4'h3: CSI_DATA3 4'h4: CSI_DATA4 4'h5: CSI_DATA5 4'h6: CSI_DATA6 4'h7: CSI_DATA7 4'h8: CSI_DATA8 4'h9: CSI_DATA9 Others: not used
[19:16]	RW	4'h4	isp_dvp_din4_map_cfg Indicate which DVP IOs is connected to CMOS Image sensor data line D4. 4'h0: CSI_DATA0 4'h1: CSI_DATA1 4'h2: CSI_DATA2 4'h3: CSI_DATA3 4'h4: CSI_DATA4 4'h5: CSI_DATA5 4'h6: CSI_DATA6 4'h7: CSI_DATA7 4'h8: CSI_DATA8 4'h9: CSI_DATA9 Others: not used
[15:12]	RW	4'h3	isp_dvp_din3_map_cfg Indicate which DVP IOs is connected to CMOS Image sensor data line D3. 4'h0: CSI_DATA0 4'h1: CSI_DATA1 4'h2: CSI_DATA2 4'h3: CSI_DATA3 4'h4: CSI_DATA4 4'h5: CSI_DATA5 4'h6: CSI_DATA6 4'h7: CSI_DATA7

Bits	Attributes	Reset	Description
			4'h8: CSI_DATA8 4'h9: CSI_DATA9 Others: not used
[11:8]	RW	4'h2	isp_dvp_din2_map_cfg Indicate which DVP IOs is connected to CMOS Image sensor data line D2. 4'h0: CSI_DATA0 4'h1: CSI_DATA1 4'h2: CSI_DATA2 4'h3: CSI_DATA3 4'h4: CSI_DATA4 4'h5: CSI_DATA5 4'h6: CSI_DATA6 4'h7: CSI_DATA7 4'h8: CSI_DATA8 4'h9: CSI_DATA9 Others: not used
[7:4]	RW	4'h1	isp_dvp_din1_map_cfg Indicate which DVP IOs is connected to CMOS Image sensor data line D1. 4'h0: CSI_DATA0 4'h1: CSI_DATA1 4'h2: CSI_DATA2 4'h3: CSI_DATA3 4'h4: CSI_DATA4 4'h5: CSI_DATA5 4'h6: CSI_DATA6 4'h7: CSI_DATA7 4'h8: CSI_DATA8 4'h9: CSI_DATA9 Others: not used
[3:0]	RW	4'h0	isp_dvp_din0_map_cfg Indicate which DVP IOs is connected to CMOS Image sensor data line D0. 4'h0: CSI_DATA0 4'h1: CSI_DATA1 4'h2: CSI_DATA2 4'h3: CSI_DATA3 4'h4: CSI_DATA4 4'h5: CSI_DATA5 4'h6: CSI_DATA6 4'h7: CSI_DATA7 4'h8: CSI_DATA8 4'h9: CSI_DATA9 Others: not used

● DVP Interface Data Map Register 1 (Address: 0X2120,0064)

Bits	Attributes	Reset	Description
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Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15:12]	RW	4'hB	isp_dvp_din11_map_cfg Indicate which DVP IOs is connected to CMOS Image sensor data line D11. 4'h0: CSI_DATA0 4'h1: CSI_DATA1 4'h2: CSI_DATA2 4'h3: CSI_DATA3 4'h4: CSI_DATA4 4'h5: CSI_DATA5 4'h6: CSI_DATA6 4'h7: CSI_DATA7 4'h8: CSI_DATA8 4'h9: CSI_DATA9 Others: not used
[11:8]	RW	4'Ha	isp_dvp_din10_map_cfg Indicate which DVP IOs is connected to CMOS Image sensor data line D10. 4'h0: CSI_DATA0 4'h1: CSI_DATA1 4'h2: CSI_DATA2 4'h3: CSI_DATA3 4'h4: CSI_DATA4 4'h5: CSI_DATA5 4'h6: CSI_DATA6 4'h7: CSI_DATA7 4'h8: CSI_DATA8 4'h9: CSI_DATA9 Others: not used
[7:4]	RW	4'h9	isp_dvp_din9_map_cfg Indicate which DVP IOs is connected to CMOS Image sensor data line D9. 4'h0: CSI_DATA0 4'h1: CSI_DATA1 4'h2: CSI_DATA2 4'h3: CSI_DATA3 4'h4: CSI_DATA4 4'h5: CSI_DATA5 4'h6: CSI_DATA6 4'h7: CSI_DATA7 4'h8: CSI_DATA8 4'h9: CSI_DATA9 Others: not used
[3:0]	RW	4'h8	isp_dvp_din8_map_cfg Indicate which DVP IOs is connected to CMOS Image sensor data line D8. 4'h0: CSI_DATA0 4'h1: CSI_DATA1

Bits	Attributes	Reset	Description
			4'h2: CSI_DATA2 4'h3: CSI_DATA3 4'h4: CSI_DATA4 4'h5: CSI_DATA5 4'h6: CSI_DATA6 4'h7: CSI_DATA7 4'h8: CSI_DATA8 4'h9: CSI_DATA9 Others: not used

6.1.7 Programmer's Guide

6.1.7.1 MIPI Configuration

Programmers should operate according to the following steps:

1. Configure bit[9], bit[8] and bits[7:4] in *SAR_ADC/SPI0/ SD_DAC Clock Control (Address: 0x0800,000C)* to set the clock of MIPI DPHY accordingly. It should be noted that dphy_cfgclk ranges from 80MHz to 120MHz.
2. Configure bits[19:12] in *MIPI DPHY Power Control Register (Address: 0X0800,0218)* based on the value of dphy_cfgclk.
3. Configure *MIPI DPHY Swap Selection Register (Address: 0X0800,0220)* accordingly and set bits[23:12] in *MIPI DPHY Reset and Control Register (Address: 0X0800,0224)* as 0 to enable MIPI_CSI sensor related ports.
4. Configure bit[22] in *MIPI DPHY Power Control Register (Address: 0X0800,0218)* as 0 to power on MIPI DPHY.
5. Configure bit[30] in *MIPI DPHY Swap Selection Register (Address: 0X0800,0220)* as 1 to reset MIPI DPHY.
6. Configure bit[10] and bit[11] in *MIPI DPHY Reset and Control Register (Address: 0X0800,0224)* and bit[16] and bit[19] in *Software Reset Control (Address: 0x0800,0020)* based on the application needs.
7. Configure *Lane Number and Dphy Selection Register (Address: 0X0008)*, *MIPI DPHY Lane Control Register (Address: 0X0050)*, *CSI Slave Controller Status Register (Address: 0x0104)*, *Data Type and Virtual Channel Configuration Register (Address: 0x0108)*, *Output Mode and FIFO Configuration Register (Address: 0x010C)* and *FIFO Fill Level Mod Configuration Register (Address: 0x0110)* based on the application needs. It should be noted that programmers should not change the default value of bits[9:8] in *Output Mode and FIFO Configuration Register (Address: 0x010C)* during the usage of this module.
8. Configure bit[0] of *Blcoks Enable Register (Address: 0X0004)* and that of *Stream Data Output Configuration Register (Address: 0X0100)* as 1 to start front and protocol blocks and enable the corresponding datapath output after the steps mentioned above have been completed.

6.1.7.2 DVP IOs Connection

Programmers can configure *DVP Interface Data Map Register 0 (Address: 0X2120,0060)* and *DVP Interface Data Map Register 1 (Address: 0X2120,0064)* to connect the DVP IOs to CMOS Image Sensor data line.

6.2 ISP

6.2.1 Overview

The ISP is integrated to improve image quality and to add new features.

6.2.2 Features

Features of ISP are as follows:

- Black level correction and digital gain of four independent channels
- Lens brightness correction
- Gamma correction
- Direction-based color interpolation algorithm
- Noise reduction/Blue fringing removal/Green balance
- White balance correction
- Color correction
- color space conversion (RGB to YUV, compatible with BT.601 and BT.709)
- YUV HDR compressing
- YUV saturation enhance
- 3D Noise Reduction/Motion detection/Data compression of the reference frame
- 2D Noise Reduction
- Sharpening/False color suppression
- Luma Transmit Improvement
- Image flipping and image mirroring

6.3 Pre-processor

6.3.1 Overview

The pre-processor of AK3918AV100 series processors support scaling down/up, OSD (timestamp, watermark), privacy mask and rectangle drawing.

The following figure shows the pre-processor block diagram.

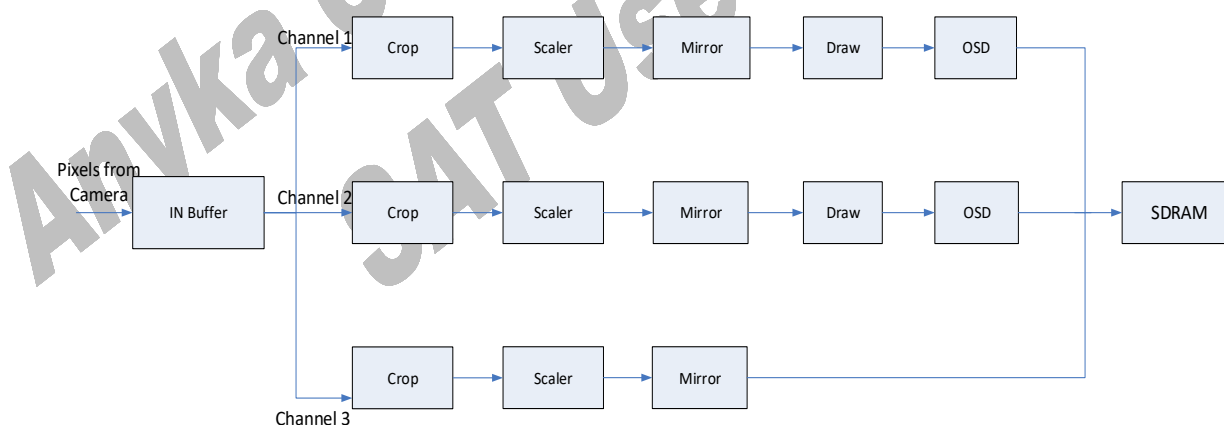


Figure 6-4 Pre-processor Block Diagram

6.3.2 Features

The features of pre-processor are as follows:

- 3 independent channels
- Supports mirroring and flipping of input image
- Input format: YUV420/YUV422/data stream

- Output format: YUV420/YUV420/YUV444/RGB888 packed/semi-planar/full planar
- Resolution and scaling factors
 - Maximum input 2880 x 2880 and minimum input 64 x 64
 - Channel 1: maximum output 3072 x 3072; scaling factor: [0.50, 2.00]
 - Channel 2: maximum output 1280x1024; scaling factor: [0.125, 1.00]
 - Channel 3: maximum output 640 x 480; scaling factor: [0.03125, 1.00]
- Each channel supports 4 frame/slice buffers and stride parameter settings.
- Channel 1 & channel 2 support OSD and rectangle Drawing

6.4 NPU

6.4.1 Overview

Based on CNN algorithm, Neural-network Processing Unit (NPU) is mainly integrated for object detection/tracking and face detection/recognition.

6.4.2 Features

The features of hardware accelerator are as follows:

- Completes one DNN algorithm processing ,including convolution operation, the corresponding PReLU function operation and pooling operation(which can disabled), in a single run
- Input image data format: supports signed char INT8 and unsigned char INT8
- Supports zero-padding on the top/bottom/left/right margin of the convolved image data when the convolution kernel is 2*2 or 3*3 DPI
- Supports zero-padding on the bottom/right margin of the pooled image data
- Supports general convolution mode/Depthwise separable convolution mode
- Supports 1*1/ 2*2/3*3 DPI convolution kernel
- Programmable pooling:
 - bypassing pooling/max-pooling/ mean-pooling
 - pooling layer: 2*2/3*3 DPI
- Programmable convolution stride
 - Convolution kernel = 1*1 DPI, convolution stride= 1 DPI
 - Convolution kernel = 2*2/3*3 DPI, convolution stride= 1/2 DPI
 - Pooling layer = 2*2/3*3 DPI, pooling stride= 1/2 DPI

6.5 Video Processor

6.5.1 Overview

The video processor of AK3918AV100 series processors provide hardware acceleration for the compression and decompression of videos and still pictures. It supports MJPEG/H.264/H.265 encoding.

6.5.2 Features

- H.265/H.264 encoding of multiple streams:
 - Supports output resolution up to 3072*2048
 - 2880*1620@15fps+640*480@15fps+320*240@15fps+2880*1620@1fps JPEG
 - 2560*1440@20fps+640*480@20fps+320*240@20fps+2560*1440@1fps JPEG
 - 2304*1296@25fps+640*480@25fps+320*240@25fps+2304*1296@1fps JPEG
 - 1920*1080@30fps+640*480@30fps+320*240@30fps+1920*1080@1fps JPEG

- H.265 encoder
 - Standard: MP Level 1~5
 - Bit rate: VBR, CBR
- H.264 encoder
 - Standard: BP level 1~5/MP Level 1~5/HP level 1~5
 - Bitrate: VBR, CBR
- JPEG encoder
 - Standard: baseline profile (DCT sequential)
 - supports resolution up to 16384*8192

6.6 Encryption

6.6.1 Overview

AES/DES/3DES Encryption module is mainly used to encrypt encoded video code stream. H264/MJPEG Encoding module reads ISP captured data which is saved in the memory, encodes it, and then saves the code stream in the memory; AES/DES/3DES module reads the code stream data by DMA mode, encrypts it and saves the cipher text in the memory in the same way; The cipher text by then can be transmitted to the server by MAC module or network. The figure below shows the data interaction among AES/DES/3DES Encryption module and other modules.

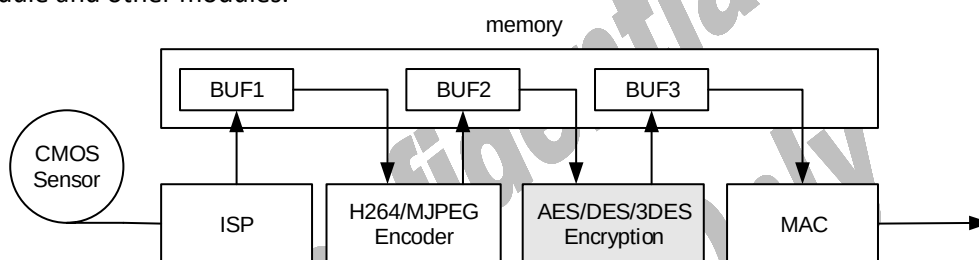


Figure 6-5 Encryption Module Interacting with Other Modules

6.6.2 Features

Features of AES/DES/3DES module are as follows:

- AES algorithm
 - Conforms to the FIPS-197 standard.
 - Supports 128-bit/192-bit/256-bit key length.
 - Supports working modes like ECB, CBC, 1/8/128-CFB, OFB, and CTR, all of which conforms to NIST Special Publication 800-38a standard.
- DES/3DES algorithm
 - Compatible with the FIPS46-3 standard.
 - Supports 64-bit key length.
 - 3DES supports 3 different keys mode and 2 different keys mode.
 - Supports working modes like ECB, CBC, 1/8/64-CFB and 1/8/64-OFB, all of which conforms to the FIPS-81 standard.
- ECB, CBC, CFB, OFB working modes support multi-group encryption as well as single group encryption in a single run.
- CTR working mode of AES only supports single group encryption and decryption in a single run.
- Supports operation suspension of the module by force
- Supports time-out interrupt

- Supports interrupt status inquiry, interrupt mask and interrupt clearing
- Supports byte/bit sequence adjustment to grouped input/output data
- Multiplexing multiple encryption channels at different time via software configuration

6.6.3 Block Diagram

The following figure shows the internal block diagram of AES/DES/3DES module.

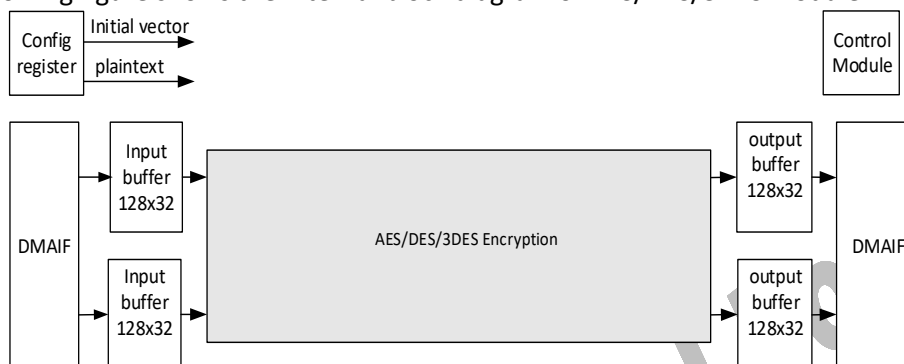


Figure 6-6 Encryption Module Block Diagram

6.6.4 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Register Address	Register Name
1	0x2018,0000	<i>Encryption Control Register</i>
2	0x2018,0004	<i>Interrupt Status Register</i>
3	0x2018,0008	<i>Time-out Period Register</i>
4	0x2018,000C	<i>Grouping Input Data Register 0</i>
5	0x2018,0010	<i>Grouping Input Data Register 1</i>
6	0x2018,0014	<i>Grouping Input Data Register 2</i>
7	0x2018,0018	<i>Grouping Input Data Register 3</i>
8	0x2018,001C	<i>Vector Input Register 0</i>
9	0x2018,0020	<i>Vector Input Register 1</i>
10	0x2018,0024	<i>Vector Input Register 2</i>
11	0x2018,0028	<i>Vector Input Register 3</i>
12	0x2018,002C	<i>Key Input Register 0</i>
13	0x2018,0030	<i>Key Input Register 1</i>
14	0x2018,0034	<i>Key Input Register 2</i>
15	0x2018,0038	<i>Key Input Register 3</i>
16	0x2018,003C	<i>Key Input Register 4</i>
17	0x2018,0040	<i>Key Input Register 5</i>
18	0x2018,0044	<i>Key Input Register 6</i>
19	0x2018,0048	<i>Key Input Register 7</i>
20	0x2018,004C	<i>Grouping Output Data Register 0</i>
21	0x2018,0050	<i>Grouping Output Data Register 1</i>

Register No.	Register Address	Register Name
22	0x2018,0054	Grouping Output Data Register 2
23	0x2018,0058	Grouping Output Data Register 3
24	0x2018,005C	Plaintext Starting Address Register
25	0x2018,0060	Data Length Register
26	0x2018,0064	Ciphertext Starting Address Register
27	0x2018,006C	Vector Output Register 0
28	0x2018,0070	Vector Output Register 1
29	0x2018,0074	Vector Output Register 2
30	0x2018,0078	Vector Output Register 3

6.6.5 Register Definition

- Encryption Control Register (Address: 0x2018,0000)

Bits	Attributes	Reset	Description
[31:23]	RO	9'h0	Reserved
[22]	RW	1'b0	iv_in_bit_sequence 0: Not to adjust bit sequence of vector input data. 1: To adjust bit sequence of vector input data.
[21]	RW	1'b0	key_bit_sequence 0: Not to adjust bit sequence of key. 1: To adjust bit sequence of key.
[20]	RW	1'b0	clk_gate_en 0: To close the internal clock gate. All functional module clocks will be opened. 1: To enable internal clock gate. Controlled by the clock gate, all functional module clocks which are not working will be closed.
[19]	RW	1'b0	output_byte_sequence 0: Not to adjust byte sequence of output grouping data. 1: To adjust byte sequence of output grouping data.
[18]	RW	1'b0	output_bit_sequence 0: Not to adjust bit sequence of output grouping data. 1: To adjust bit sequence of output grouping data.
[17]	RW	1'b0	input_byte_sequence 0: Not to adjust byte sequence of input grouping data. 1: To adjust byte sequence of input grouping data.
[16]	RW	1'b0	input_bit_sequence 0: Not to adjust bit sequence of input grouping data. 1: To adjust bit sequence of input grouping data.
[15]	RO	1'b0	Reserved
[14]	RW	1'b1	iv_mode_sel IV input mode selection 0: To choose the IV generated in last encryption operation. 1: To choose the IV configured by register. NOTE: When 0 is selected, it's Valid for CBC, CFB and

Bits	Attributes	Reset	Description
			OFB modes. It applies to the situation where encryption of a data packet is finished in separate operations and IV is set only once.
[13]	RW	1'b0	cipher_mode_sel Encryption mode selection. 0: Single group operation. 1: Multiple group operation.
[12:10]	RW	3'b0	algorithm_sel Encryption algorithm selection. 000: DES. 001: 3DES (3 keys). 010: 3DES (2 keys). 011: AES-128 (128-bit key length). 100: AES-192 (192-bit key length). 101: AES-256 (256-bit key length). Others: Reserved
[9:8]	RW	2'b0	width_sel Data bit width selection For DES/3DES algorithm: 2'h0: 64-bit 2'h1: 8-bit 2'h2: 1-bit 2'h3: 64-bit For AES algorithm: 2'h0: 128-bit 2'h1: 8-bit 2'h2: 1-bit 2'h3: 128-bit
[7:5]	RW	3'h0	operate mode Working mode selection. 3'h0: ECB. 3'h1: CBC. 3'h2: CFB. 3'h3: OFB. 3'h4: CTR. Others: Reserved
[4]	RW	1'b0	timeout_int_en 0: To mask time-out interrupt. 1: To enable time-out interrupt.
[3]	RW	1'b0	encrypt_int_en 0: To mask interrupt triggered when an encryption operation is completed. 1: To enable interrupt triggered when an encryption operation is completed.
[2]	RW	1'b0	operation_status Operation status. 0: No encryption operation is in progress. 1: An encryption operation is in progress.

Bits	Attributes	Reset	Description
[1]	RW	1'b0	encryption_stop 1: To forcefully terminate encryption operation. NOTE: This bit will be cleared automatically.
[0]	RW	1'b0	encryption_start 1: To start encryption operation. NOTE: This bit will be cleared automatically.

NOTE

- 1) The value of CLK_gate_en (bit[20] of [Encryption Control Register\(Address:0x2018,0000\)](#)) can only be changed before the encryption operation, not in the process of encryption.
- 2) When byte sequence adjustment is enabled, the sequence of 4 bytes in a 32-bit data will be reversed. That is to say, the original order {byte3, byte2, byte1, byte0} (byte3 in the higher position and byte0 in the lower position) will be changed as {byte0, byte1, byte2, byte3} (byte0 in the higher position and byte3 in the lower position). Byte sequence adjustment is allowed to be used only when the length of the data to be encrypted is integer multiple of 4. CFB_1bit, OFB_1bit, CFB_8bit and OFB_8bit in single grouping mode do not support byte sequence adjustment. Byte sequence adjustment applies to single grouping mode as well as multiple grouping modes.
- 3) When bit sequence adjustment is enabled, the 8-bit order in a byte will be reversed. In other words, the original bit sequence {bit7, bit6, bit5, bit4, bit3, bit2, bit1, bit0} (bit7 in the higher position and bit0 in the lower position) will be changed as {bit0, bit1, bit2, bit3, bit4, bit5, bit6, bit7}. Bit sequence adjustment is allowed to be used only when the length of the data to be encrypted is integer multiple of a byte. Bit sequence adjustment applies to single grouping mode as well as multiple grouping modes.
- 4) CTR working mode of AES supports only encryption operations of a single grouping data once a time and does not support that of multiple grouping data.
- 5) In multiple grouping modes, when software starts a forced stop command, if the DMA bus operation is working, Hardware will stop the module after DMA bus operation is finished. Otherwise, hardware will stop the module at once.

Interrupt Status Register (Address: 0x2018,0004)

This register displays time-out interrupt and encryption completed interrupt.

Bits	Attributes	Reset	Description
[31:2]	RO	30'h0	Reserved
[1]	RO	1'b0	Time-out status. 0: Not to generate time-out interrupt. 1: To generate time-out interrupt. NOTE: This bit will be cleared after being read.
[0]	RO	1'b0	encryption_done Encryption completed interrupt. 0: Encryption operation uncompleted. 1: Encryption operation completed. NOTE: This bit will be cleared after being read.

NOTE

- 1) The Interrupt Status Register is cleared by reading operation.
- 2) Encryption completed interrupt is generated when the encryption is completed normally or when encryption is forcefully stopped.

● Time-out Period Register (Address: 0x2018,0008)

This register is mainly used to set time-out period.

Once encryption is enabled, if encryption can not be completed when the counter exceeds the value set in the Time-out Period Register, a time-out interrupt will be generated.

Bits	Attributes	Reset	Description
[31:0]	RW	32'hFFFF,FFF F	time_out_val Time-out setting.

● Grouping Input Data Register 0 (Address: 0x2018,000C)

Bits	Attributes	Reset	Description
[31:0]	W	32'h0	data_in1 Grouping input data bit[31:0]

● Grouping Input Data Register 1 (Address: 0x2018,0010)

Bits	Attributes	Reset	Description
[31:0]	W	32'h0	data_in2 Grouping input data bit[63:32]

● Grouping Input Data Register 2 (Address: 0x2018,0014)

Bits	Attributes	Reset	Description
[31:0]	W	32'h0	data_in3 Grouping input data bit[95:64]

● Grouping Input Data Register 3 (Address: 0x2018,0018)

Bits	Attributes	Reset	Description
[31:0]	W	32'h0	data_in4 Grouping input data bit[127:96]

NOTE

Registers (Address: 0x2018,000C~ 0x2018,0018) are used to define grouping input data of the encryption module.

- 1) Configure this register when single grouping operation is chosen.
 - a. If 1-bit width is chosen, only 1 bit is valid, i.e. bit[0] of *Grouping Input Data Register 0* is valid.
 - b. If 8-bit width is chosen, the lower 8 bits are valid, i.e. bit[7:0] of *Grouping Input Data Register 0* are valid.
 - c. If 64-bit width is chosen, the lower 64 bits are valid, i.e. bit[31:0] of *Grouping Input Data Register 1* and bit[31:0] of *Grouping Input Data Register 0* are valid.
 - d. If 128-bit width is chosen, the lower 128 bits are valid, i.e. bit[31:0] of *Grouping Input Data Register 3*, bit[31:0] of *Grouping Input Data Register 2*, bit[31:0] of *Grouping Input Data Register 1* and bit[31:0] of *Grouping Input Data Register 0* are valid.
- 2) For AES encryption, the corresponding relation between *Grouping Input Data Registers* and the 128-bit grouping data (bit[0:127] of plaintext) to be encrypted are as follows: bit[127:0] of Grouping input data corresponds with bit[127:0] of plaintext. The same relation is reflected between memory data and the grouping data to be encrypted.

● Vector Input Register 0(Address: 0x2018,001C)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	iv_in1 Vector input data bit[31:0]. NOTE: Counter[31:0] when in CTR working mode.

● Vector Input Register 1(Address: 0x2018,0020)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	iv_in2 Vector input data bit[63:32]. NOTE: Counter[63:32] when in CTR working mode.

● Vector Input Register 2 (Address: 0x2018,0024)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	iv_in3 Vector input data bit[95:64]. NOTE: Counter[95:64] when in CTR working mode.

● Vector Input Register 3(Address: 0x2018,0028)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	iv_in4 Vector input data bit[127:96]. NOTE: Counter[127:96] when in CTR working mode.

NOTE

Registers (Address: 0x2018,001C~ 0x2018,0028) are used to define the vector input of the encryption module.

- 1) Configure this register, if vector configuration is required in a single grouping operation. In AES operation, IV_in[127:0] are valid data; in DES or 3DES operation, the lower 64 bits are valid, i.e. IV_in[63:0] are valid data.
- 2) Configure this register, if vector configuration is required in a multiple grouping operation. The first input vector of grouping operation is acquired from this register.
- 3) In CTR working mode, IV input is not required and the Vector Input Registers are used to configure the Counter.

● Key Input Register 0(Address: 0x2018,002C)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	key_in1 KEY input bit[31:0].

● Key Input Register 1 (Address: 0x2018,0030)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	key_in2 KEY input bit[63:32].

● Key Input Register 2(Address: 0x2018,0034)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	key_in3 KEY input bit[95:64].

● Key Input Register 3 (Address: 0x2018,0038)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	key_in4 KEY input bit[127:96].

● Key Input Register 4 (Address: 0x2018,003C)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	key_in5 KEY input bit[159:128].

● Key Input Register 5 (Address: 0x2018,0040)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	key_in6 KEY input bit[191:160].

● Key Input Register 6 (Address: 0x2018,0044)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	key_in7 KEY input bit[223:192].

● Key Input Register 7 (Address: 0x2018,0048)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	key_in8 KEY input bit[255:224].

NOTE

[Registers\(Address:0x2018,002C~0x2018,0048\)](#) are used to define key input of the encryption module.

- 1) For DES operation, the lower 64 bits are valid, i.e. Key[63:0] is valid data.
- 2) In 3DES operation:
 - a. If three keys are chosen, the lower 192 bits are valid, at this time:
Key[63:0] indicates the first key.
Key[127:64] indicates the second key.
Key[191:128] indicates the third key.
 - b. If two keys are chosen, the lower 128 bits are valid, at this time:
Key[63:0] indicates the first key.
Key[127:64] indicates the second key.
- 3) For AES operation:
 - a. If 128-bit key operation is chosen, the lower 128 bits are valid, i.e. Key[127:0] is valid data.
 - b. If 192-bit key operation is chosen, the lower 192 bits are valid, i.e. Key[191:0] is valid data.

c. If 256-bit key operation is chosen, all the 256 bits are valid data.

● Grouping Output Data Register 0(Address: 0x2018,004C)

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	data_out1 Grouping output data bit[31:0].

● Grouping Output Data Register 1 (Address: 0x2018,0050)

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	data_out2 Grouping output data bit[63:32].

● Grouping Output Data Register 2 (Address: 0x2018,0054)

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	data_out3 Grouping output data bit[95:64].

● Grouping Output Data Register 3 (Address: 0x2018,0058)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	data_out4 Grouping output data bit[127:96].

NOTE

[Registers\(Address:0x2018,004C~0x2018,0058\)](#) are used to define grouping output data of the encryption module. In single grouping operation, data read from this register is the result data of single grouping operation. In multiple grouping operations, the result data of grouping operation is saved in the memory, having no to do with this register.

- 1) If 1-bit width is chosen, only 1 bit is valid, i.e. bit[0] of [Grouping Output Data Register0](#).
 - 2) If 8-bit width is chosen, the lower 8 bits are valid, i.e.bit[7:0] of [Grouping Output Data Register0](#).
 - 3) If 64-bit width is chosen, the lower 64 bits are valid, i.e. bit[31:0] of [Grouping Output Data Register1](#) and bit[31:0] of [Grouping Output Data Register0](#) are valid data.
- If 128-bit width is chosen, the lower 128 bits are valid, i.e. bit[31:0] of [Grouping Output Data Register3](#), bit[31:0] of [Grouping Output Data Register2](#), bit[31:0] of [Grouping Output Data Register1](#) and bit[31:0] of [Grouping Output Data Register0](#) are valid data.

● Plaintext Starting Address Register (Address: 0x2018,005C)

Plaintext Starting Address Register defines the starting address of memory saving the grouping data to be encrypted. This register needs to be configured when multiple grouping operation is processed.

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	src_addr NOTE: 1) The starting address of the grouping data saved in the memory before encryption. 2) SRC_Addr is required to be the integer multiple of 4.

● Data Length Register (Address: 0x2018,0060)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	data_length Length(bytes) of grouping data to be encrypted.

NOTE

Data Length Register defines the length of the grouping data to be encrypted (a unit of length: byte). Configure this register when multiple grouping operation is required.

1) For AES operation

If either ECB, CBC, OFB or CTR mode is chosen, the Data_length is required to be integer multiple of 16.

If 128-CFB mode is chosen, the Data_length is required to be integer multiple of 16.

If 8-CFB mode is chosen, there is no requirement to Data_length.

If 1-CFB mode is chosen, there is no requirement to Data_length. The data in a byte is processed from the lower bits to the higher bits, i.e. from bit[0] to bit[7] in succession. If the data is less than a byte, it is required to fill in the higher position.

2) For DES/3DES operation

If ECB or CBC mode is chosen, the Data_length is required to be integer multiple of 8.

If 64-CFB or 64-OFB mode is chosen, the Data_length is required to be integer multiple of 8.

If 8-CFB or 8-OFB mode is chosen, there is no requirement to Data_length.

If 1-CFB or 1-OFB mode is chosen, there is no requirement to Data_length. The data in a byte is processed from the lower bits to the higher bits, i.e. from bit[0] to bit[7] in succession. If the data is less than a byte, it is required to fill in the higher position.

● Ciphertext Starting Address Register (Address: 0x2018,0064)

Ciphertext Starting Address Register defines the starting address of memory saving the grouping data after being encrypted. This register needs to be configured when multiple grouping operation is processed.

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	des_addr The starting address of the grouping data saved in the memory after encryption. NOTE: DES_Addr is required to be integer multiple of 4.

● Vector Output Register0 (Address: 0x2018,006C)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	iv_out0 Vector output data bit[31:0]

● Vector Output Register1 (Address: 0x2018,0070)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	iv_out1 Vector output data bit[63:32]

● Vector Output Register2 (Address: 0x2018,0074)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	iv_out2

Bits	Attributes	Reset	Description
			Vector output data bit[95:64]

- Vector Output Register3 (Address: 0x2018,0078)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	iv_out3 Vector output data bit[127:96]

NOTE

For [registers \(Address: 0x2018,006C ~ 0x2018,0078\)](#), programmers should note the following statements.

- 1) When multiple channels of data needs to be encrypted, the encryption module processes one channel of data at a time. Because the IV value of each channel is different, softwares needs to read and save the IV output data when switching among different channels.
- 2) In AES operation, values of iv_out[127:0] are valid; while in DES/3DES operation, only values of iv_out[63:0] are valid.

6.6.6 Programmer's Guide

6.6.6.1 Single Grouping Mode

Single grouping mode can only encrypt a single grouping data every time. Programmers should operate according to the following steps:

1. Check encryption status (bit[2] of [Encryption Control Register\(Address:0x2018,0000\)](#)).If this bit is cleared, enter step 2. If not, wait until it is cleared.
2. Configure Grouping Input Data Registers, Vector Input Registers and Key Input Registers.
3. Write 0 to bit[13] of [Encryption Control Register\(Address:0x2018,0000\)](#)and configure other corresponding items of Encryption Control Register.
4. Write 1 to bit[0] of [Encryption Control Register\(Address:0x2018,0000\)](#)to start encryption operation.
5. Wait until encryption operation is finished, which can be realized by interrupt mode or checking the encryption status (bit[2] of [Encryption Control Register \(Address:0x2018,0000\)](#)).
6. Read encryption result data from Grouping Output Data Registers.

6.6.6.2 Multiple Grouping Mode

Multiple grouping mode can continuously encrypt several grouping data in a single run. Programmers should operate according to the following steps:

1. Check encryption status (bit[2] of [Encryption Control Register\(Address:0x2018,0000\)](#)).If this bit is cleared, enter step 2. If not, wait until it is cleared.
2. Configure the start addresses of grouping data before being encrypted, the length of grouping data to be encrypted, and the start address of the grouping data after being encrypted.
3. Configure Vector Input Registers and Key Input Registers.
4. Write 1 to bit [13] of [Encryption Control Register\(Address:0x2018,0000\)](#) and configure other corresponding items of Encryption Control Register.
5. Write 1 to bit [0] of [Encryption Control Register\(Address:0x2018,0000\)](#) to start encryption operation.

6. Wait until encryption operation is finished, which can be realized by interrupt mode or checking the encryption status (bit[2] of [Encryption Control Register \(Address:0x2018,0000\)](#)).

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7 Analog

7.1 Overview

The analog control module includes all analog functions for the AK3918AV100 series processors. It mainly consists of:

- One reference voltage generator
- One 16-bit Sigma-Delta ADC
- One 22-bit Sigma-Delta DAC
- Headphone driver
- Microphone differential input
- One 12-bit SAR ADC
- Battery monitor

To make block diagram presentation easier, the digital filters associated with the Sigma-Delta ADC and Sigma-Delta DAC are included.

7.2 Reference Voltage Generator

The reference voltage generator provides reference VREF (1.5V), VCM2 (1.5V) and VCM3 (3.0V) for audio codec and 12-bit SAR ADC.

As illustrated in [Figure 7-1](#), after being regulated by BGR(1.5V), UVDET outputs POR signals for Brown-out detection(3.3V) of the chip and the reference buffer provides a 3.0V reference voltage for SAR ADC. SAR ADC can also acquire reference voltage from HPVDD(3.3V) based on the application needs. After being adjusted by the voltage divider, HPVDD(3.3V) generates a 1.5V reference voltage for VCM2 and VCM3, which thus provide 1.5V and 3.0V reference voltage for audio modules respectively.

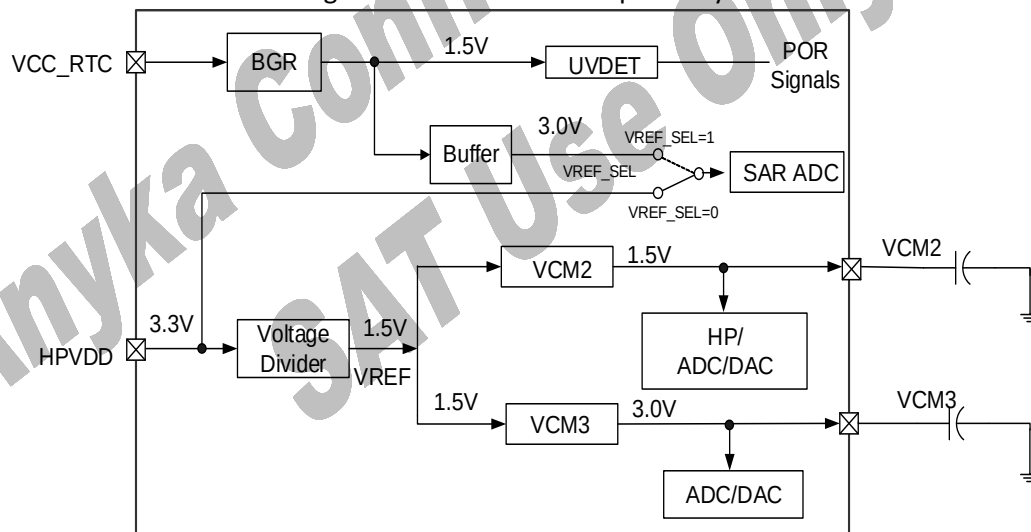


Figure 7-1 Block Diagram of Reference Voltage Generator

7.3 Audio Codec

Audio codec of AK3918AV100 series processors is a high-performance stereo playback/mono recording codec, integrated with headphone, line-in and microphone driver. The block diagram of audio codec of AK3918AV100 series processors is shown in the following figure.

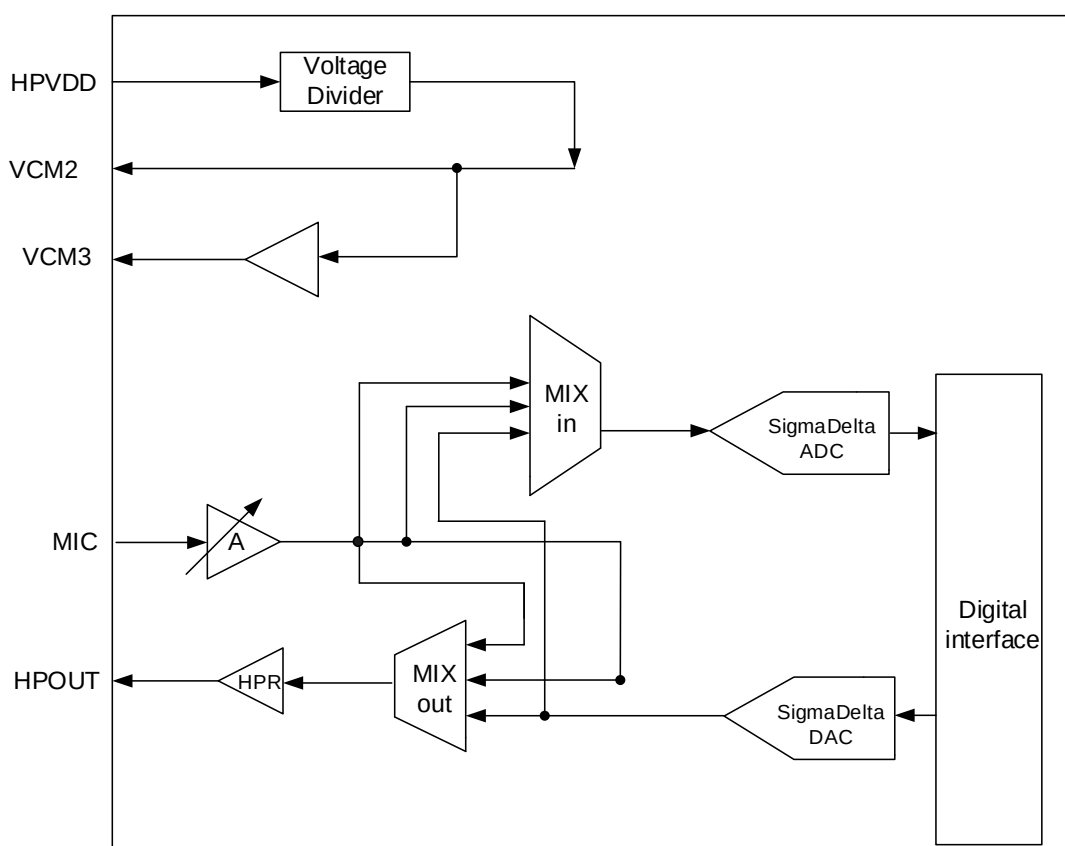


Figure 7-2 Block Diagram of Audio Codec

NOTE

HPVDD, VCM2, VCM3, MIC and HPOUT are external pins. Please refer to *AK3918AV100 Series Processors Specification* for details.

7.3.1 Sigma-Delta ADC

Aiming to provide high-performance voice and music mono recording, AK3918AV100 series processors include a 16-bit Sigma-Delta ADC. As illustrated in [Figure 7-2](#), ADC can accept signals from microphone interface or from the output of DAC, the latter of which is used for loopback test of Sigma-Delta ADC.

The high impedance and low capacitance microphone interface is compatible with a wide range of microphones with different dynamics and sensitivities. It contains a programmable gain amplifier (PGA), and a passive filter to prevent high frequencies from folding back into the audio band. VCM3 is used as the bias voltage for microphone. The microphone interface supports differential/single end input.

7.3.2 Sigma-Delta DAC

The AK3918AV100 series processors contain one 22-bit DAC, which accepts input either from CPU or L2 memory. All of these inputs can be switched off independently. Its output may be sent to headphone driver. The headphone driver is able to drive a 32Ω headphone. As it only supports AC output, a DC blocking capacitor is required in the output of HP.

7.4 SAR ADC

SAR ADC includes three channels, one for battery monitoring, and the other two for analog keypad input or general purpose analog-to-digital conversion. Please note that these three channels cannot be enabled simultaneously, which means that once a certain channel is selected, the other two will be powered down.

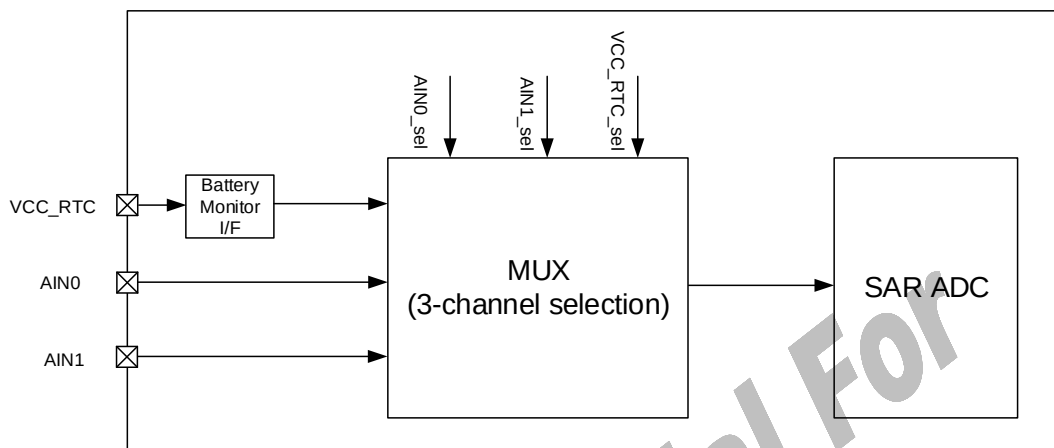


Figure 7-3 Block Diagram of SAR ADC

NOTE

- 1) VCC_RTC, AIN0 and AIN1 are external pins.
- 2) VCC_RTC_sel: select battery monitor input; AIN0_sel: select AIN0 signal; AIN1_sel: select AIN1 signal;
- 3) When VCC_RTC is used for battery monitoring, the voltage of the external batteries should be low than 3.7V.

7.4.1 Battery Monitoring

Battery voltage is monitored by VCC_RTC channel of SAR ADC. And the maximum input voltage of VCC_RTC is 3.63V.

7.4.2 AIN0 Channel and AIN1 Channels

AIN0 and AIN1 channels are used for analog keypad input or general analog-to-digital conversion. It should be noted that once AIN0 or AIN1 channel is enabled, VCC_RTC channel will be disabled. And the voltage of AIN0 and AIN1 ranges from 0V to 3.3V. If the input voltage to be sampled by AIN0 or AIN1 exceeds 3.3V, an external voltage divider is needed.

7.5 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Register Address	Register Name and Description
1	0x0800,009C	<i>Analog Control Register 0</i>
2	0x0800,00A0	<i>Analog Control Register 1</i>
3	0x0800,00A4	<i>Analog Control Register 2</i>
4	0x0800,005C	<i>SAR ADC Sample Control Register</i>
5	0x0800,0060	<i>SAR ADC Sample Rate Register</i>
6	0x0800,0064	<i>SAR ADC Sample Threshold Register</i>
7	0x0800,0068	<i>SAR ADC Sample Data Register</i>
8	0x0800,006C	<i>SAR ADC Sample Interrupt Status Register</i>
9	0x0800,0070	<i>DAC Fadeout Control Register</i>
10	0x2011,8000	<i>ADC Mode Configuration Register</i>
11	0x2011,8004	<i>ADC Data Receive Register</i>
12	0x2011,0000	<i>DAC Mode Configuration Register</i>
13	0x2011,0004	<i>I2S Configuration Register</i>
14	0x2011,0008	<i>CPU Data Register</i>

NOTE

The internal Sigma-Delta ADC only transmits the data of left channel.

7.6 Register Definition

Analog Control Register 0 ~ 2 are used to configure the analog control module.

- Analog Control Register 0 (Address: 0x0800,009C)

Bits	Attributes	Reset	Description
[31:29]	RW	3'b111	dischg_hp Electricity discharging control bits of the headphone driver connected off-chip AC coupling capacitor 3'bxx0: Electricity discharging function is disabled. 3'b001: A 30uA of electricity is discharged. 3'b011: A 90uA of electricity is discharged. 3'b101: A 120uA of electricity is discharged. 3'b111: A 180uA of electricity is discharged. NOTE: 'x' can be 0 or 1.
[28:25]	RW	4'hF	pd_hp_ctrl Fade-out rate of HP driver, active low. 4'h0: The fastest fade-out rate. ... 4'hE: The slowest fade-out rate. 4'hF: To disable PD_HP_ctrl. NOTE: The larger value this field is, the lower the fade-out rate will be.
[24]	RW	1'h1	pd2_hp 0: Power on PMOS part of headphone driver. 1: Power off PMOS part of headphone driver.
[23]	RW	1'h1	pd1_hp

Bits	Attributes	Reset	Description
			0: Power on NMOS part of headphone driver. 1: Power off NMOS part of headphone driver.
[22:18]	RW	5'b10000	hp_gain Headphone driver gain control. 5'b00000: 1 time (0dB) 5'b00001: 0.9 time (-0.9dB) 5'b0001x: 0.8 time (-1.9dB) 5'b001xx: 0.5 time (-6dB) 5'b01xxx: 0.2 time (-14dB) 5'b1xxxx: 0.1 time (-20dB) NOTE: 'x' can be 0 or 1.
[17]	RW	1'h1	pre_en1 De-pipa noise control bit in AC mode 0: Disable de-pipa noise control. 1: Enable de-pipa noise control. NOTE: When this bit is enabled, a 3K Ω resistor should be pulled down to GND.
[16:14]	RW	3'h0	hp_in Select the input signal source to headphone driver 3'h0: Headphone is mute. 3'h1: Microphone 3'h2: Line-in 3'h3: Mixed signals from line-in and Sigma-Delta DAC 3'h4: Output from DAC 3'h5: Mixed signals from microphone and Sigma-Delta DAC 3'h6: Mixed signals from microphone and Line-in 3'h7: Mixed signals from microphone, Line-in and Sigma-Delta DAC
[13]	RW	1'h1	rst_dac 0: Normal mode 1: Reset Sigma-Delta DAC output to middle voltage.
[12]	RW	1'h1	pd_ck 0: Power on Sigma-Delta DAC clock. 1: Power off Sigma-Delta DAC clock.
[11]	RW	1'h1	pd_op 0: Power on the integrator in Sigma-Delta DAC. 1: Power off the integrator in Sigma-Delta DAC.
[10:6]	RW	5'h1F	dis_chg_vcm2 Discharge VCM2(Audio_VREF) NOTE: Bit[6] must be set before discharging. 5'h0: Disable discharge 5'h1: 2.5uA 5'h2: 5uA 5'h3: 7.5uA 5'h4: 10uA 5'h5: 12.5uA 5'h6: 15uA

Bits	Attributes	Reset	Description
			5'h7: 17.5uA 5'h8: 20uA 5'h9: 22.5uA 5'hA: 25uA 5'hB: 27.5uA 5'hC: 30uA 5'hD: 32.5uA 5'hE: 35uA 5'hF: 37.5uA 5'h10: 40uA 5'h11: 42.5uA 5'h12: 45uA 5'h13: 47.5uA 5'h14: 50uA 5'h15: 52.5uA 5'h16: 55uA 5'h17: 57.5uA 5'h18: 60uA 5'h19: 62.5uA 5'h1A: 65uA 5'h1B: 67.5uA 5'h1C: 70uA 5'h1D: 72.5uA 5'h1E: 75uA 5'h1F: 77.5uA
[5]	RW	1'h0	en_vp2.5v 0: A AVCC1.8V voltage is generated for the efuse VP. 1: A AVCC2.5V voltage is generated for efuse VP.
[4]	RW	1'h0	pon_vp 0: Power down the EFUSE_VP. 1: Power on the EFUSE_VP.
[3]	RW	1'h1	AVCC33_SEL Only when this bit is set as 0, configuration of en_vp2.5v and pon_vp is valid.
[2]	RW	1'h1	pl_vcm3 0: Normal. 1: Pull down VCM3 with 2KΩ resistor to ground.
[1]	RW	1'h1	pd_vcm3 0: Power on VCM3. 1: Power off VCM3.
[0]	RW	1'h1	pd_bias 0: Power on the bias generator. 1: Power off the bias generator.

● Analog Control Register 1 (Address: 0x0800,00A0)

Bits	Attributes	Reset	Description
[31:29]	RW	3'h0	adc_in_sel

Bits	Attributes	Reset	Description
			Select the input signal source to the Sigma-delta ADC 3'h0: Sigma-delta ADC is mute. 3'h1: Output from Sigma-Delta DAC 3'h2: Microphone 3'h3: Mixed signals from Line-in and Sigma-Delta DAC 3'h4: Line-in 3'h5: Mixed signals from Line-in and Sigma-Delta DAC 3'h6: Mixed signal from microphone and Line-in 3'h7: Mixed signals from microphone, Line-in and Sigma-Delta DAC
[28]	RW	1'h0	limen 0: Disable limit. 1: Limit Sigma-Delta ADC input from VCM3 to 0.
[27]	RW	1'h1	pdr_s2d 0: Power on single-differential conversion in Sigma-Delta ADC. 1: Power off single-differential conversion in Sigma-Delta ADC.
[26]	RW	1'h1	pd_adc_core 0: Power on the Sigma-delta core in ADC. 1: Power off the Sigma-delta core in ADC.
[25:22]	RW	4'h6	linein_gain Linein gain control bits 4'h0: x 0.2 (-14dB) 4'h1: x 0.25 (-12dB) 4'h2: x 0.395 (-8dB) 4'h3: x 0.5 (-6dB) 4'h4: x 0.63 (-4dB) 4'h5: x 0.8 (-1.93dB) 4'h6: x 1 (0dB) 4'h7: x 1.195 (1.5dB) 4'h8: x 1.4 (3dB) 4'h9: x 1.7 (4.6dB) 4'hA: x 2 (6dB) 4'hB: x 2.4 (7.6dB) 4'hC: x 2.8 (9dB) 4'hD: x 3.35 (10.5dB) 4'hE: x 4 (12dB) 4'hF: x 4.75 (13.5dB)
[21]	RW	1'h1	pd_line 0: Power on Line-in. 1: Power off Line-in.
[20]	RW	1'h1	pd_micn 0: Power on differential MIC. 1: Power off differential MIC. NOTE: To power on differential MIC, bits [20:19] of this register should be set to 0.
[19]	RW	1'h1	pd_micp

Bits	Attributes	Reset	Description
			0: Power on mono MIC. 1: Power off mono MIC.
[18]	RW	1'h0	mic_gainbst Double MIC amplifier gain, active high. 0: MIC amplifier gain controlled by MIC_GAIN 1: MIC gain is doubled.
[17:15]	RW	3'h0	mic_gain MIC gain control bits (mono microphone input). When bit[18] is set as 0, MIC gain can be configured as follows: 3'h0: x 10(20dB) 3'h1: x 12.5(22dB) 3'h2: x 16(24dB) 3'h3: x 20(26dB) 3'h4: x 25(28dB) 3'h5: x 32(30dB) 3'h6: x 40(32dB) 3'h7: x 60(35.6dB) When bit[18] is set as 1, MIC gain can be configured as follows: 3'h0: x 10(26dB) 3'h1: x 12.5(28dB) 3'h2: x 16(30.1dB) 3'h3: x 20(32dB) 3'h4: x 25(34dB) 3'h5: x 32(36.1dB) 3'h6: x 40(38dB) 3'h7: x 60(41.6dB)
[14]	RO	1'h1	Reserved
[13:10]	RW	4'h1	antipop_en Audio VREF (1.5V) charge current control When bit[10] is 0, audio VREF with 200KΩ resistor is enabled to charge the audio_ref pin with a capacitance of 1uF. When bit[11] is 1, the audio VREF is 0.5V. When bit[12] is 1, the audio VREF is 0.2V. When bit[13] is 1, the audio VREF is 0.1V. NOTE: When audio VREF approaches 1.5V, programmers should set these bits to 0001.
[9]	RW	1'h0	ain1_wk_en Enable AIN1 wake up, active high.
[8]	RW	1'h0	ain0_wk_en Enable AIN0 wake up, active high.
[7]	RW	1'h0	en_batdiv Enable BAT voltage divider divided by 2, active high.
[6]	RW	1'h0	ain1_sel Select AIN1 channel as SARADC measurement input,

Bits	Attributes	Reset	Description
			active high.
[5]	RW	1'h0	ain0_sel Select AIN0 channel as SARADC measurement input, active high.
[4]	RW	1'h0	VCC_RTC_sel Select VCC_RTC channel as SARADC measurement input, active high.
[3]	RW	1'h1	pd_saradcref Power down SAR ADC reference voltage (3.0V), active high.
[2]	RW	1'h1	saradc_ibctrl SAR ADC bias current control 0: 5uA 1: 2.5uA
[1]	RW	1'h0	saradc_reset Reset SAR ADC analog output, active low.
[0]	RW	1'h1	pd_saradc Power down SAR ADC, active high.

● Analog Control Register 2 (Address: 0x0800,00A4)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	vref_test_en Enable the BGR output VREF test with the audio_ref pin, active high.
[30:28]	RO	3'h0	Reserved
[27]	RW	1'h1	VREF_SEL SAR ADC VREF selection 0: AVCC 3.3V is selected as VREF. 1: BGR 3.0V is selected as VREF.
[26]	RW	1'h0	micn_gpi_en Enable MICN pin to be multiplexed as GPI, active high
[25]	RW	1'h0	micp_gpi_en Enable MICP pin to be multiplexed as GPI, active high
[24]	RW	1'h0	adc_in_sel 1: Select Line-inL as the input to Sigma-delta ADC. 0: No instruction.
[23]	RW	1'h1	pd_mic_dts Power down MIC differential input to single output reference buffer, active high. NOTE: This bit can be set 0 to avoid the VCM2 poor driving capability.
[22]	RW	1'h1	pd_hpvcm Power down VMID buffer of headphone driver, active high.
[21:20]	RW	2'h3	cfg_ib10 Bias current control 2'h0: 10uA

Bits	Attributes	Reset	Description
			2'h1: 7.5uA 2'h2: 5 uA 2'h3: 2.5uA
[19:18]	RW	2'h3	cfg_ib9 Bias current control 2'h0: 10uA 2'h1: 7.5uA 2'h2: 5uA 2'h3: 2.5uA
[17:16]	RW	2'h3	cfg_ib8 Headphone bias current control 2'h0: 10uA 2'h1: 7.5uA 2'h2: 5uA 2'h3: 2.5uA
[15:14]	RW	2'h3	cfg_ib7 Sigma-Delta DAC core OPAMP bias current control 2'h0: 10μA 2'h1: 7.5μA 2'h2: 5μA 2'h3: 2.5μA
[13:12]	RW	2'h3	cfg_ib6 Sigma-Delta ADC second stage integrator's bias current control 2'h0: 10μA 2'h1: 7.5μA 2'h2: 5μA 2'h3: 2.5μA
[11:10]	RW	2'h3	cfg_ib5 Sigma-Delta ADC first stage integrator's bias current control 2'h0: 10μA 2'h1: 7.5μA 2'h2: 5μA 2'h3: 2.5μA
[9:8]	RW	2'h3	cfg_ib4 Sigma-Delta ADC single to differential buffer bias current 2'h0: 10μA 2'h1: 7.5μA 2'h2: 5μA 2'h3: 2.5μA
[7:6]	RW	2'h3	cfg_ib3 Microphone negative terminal buffer's bias current control 2'h0: 10μA 2'h1: 7.5μA 2'h2: 5μA 2'h3: 2.5μA

Bits	Attributes	Reset	Description
[5:4]	RW	2'h3	cfg_ib2 Microphone AMP single to differential buffer bias current control 2'h0: 10μA 2'h1: 7.5μA 2'h2: 5μA 2'h3: 2.5μA
[3:2]	RW	2'h3	cfg_ib1 Microphone positive terminal buffer's bias current control 2'h0: 10μA 2'h1: 7.5μA 2'h2: 5μA 2'h3: 2.5μA
[1:0]	RW	2'h3	cfg_ib0 Line-in buffer bias current control 2'h0: 10μA 2'h1: 7.5μA 2'h2: 5μA 2'h3: 2.5μA

● SAR ADC Sample Control Register (Address: 0x0800,005C)

Bits	Attributes	Reset	Description
[31:22]	RO	10'h0	Reserved
[21:14]	RW	8'h0	spl_wait Sample waiting time These bits are used to define the start moment of SOC, which will be invalid after two clocks (unit: the number of SAR_ADC_CLK cycles). NOTE: For single channel, 1) Spl_wait >= 2 2) Spl_hold >= (Spl_wait + 22) 3) Spl_cycle >= (Spl_hold+1) For multiple channels, 1) Spl_wait >= 2 2) Spl_hold >= (Spl_wait + 22) 3) Spl_cycle >= (Spl_hold+1), Spl_cycle >= (Spl_wait+27)
[13]	RW	1'h0	bat_threshold_polarity 0: An interrupt is generated when difference between two consecutive sampling values from VCC_RTC channel is less than the threshold. 1: An interrupt is generated when difference between two consecutive sampling values from VCC_RTC channel exceeds the threshold.
[12]	RW	1'h0	ain1_threshold_polarity 0: An interrupt is generated when difference between

Bits	Attributes	Reset	Description
			two consecutive sampling values from AIN1 channel is less than the threshold. 1: An interrupt is generated when difference between two consecutive sampling values from AIN1 channel is more than the threshold.
[11]	RW	1'h0	ain0_threshold_polarity 0: An interrupt is generated when difference between two consecutive sampling values from AIN0 channel is less than the threshold. 1: An interrupt is generated when difference between two consecutive sampling values from AIN0 channel is more than the threshold.
[10:8]	RW	3'h0	spl_num Sampling times Read the sampling data for N times and then calculate the average value. NOTE: N ranges from 1 to 7.
[7]	RW	1'h0	bat_spl_en 0: Disable sampling operation of VCC_RTC channel. 1: Enable sampling operation of VCC_RTC channel.
[6]	RW	1'h0	ain1_spl_en 0: Disable sampling operation of AIN1 channel. 1: Enable sampling operation of AIN1 channel.
[5]	RW	1'h0	ain0_spl_en 0: Disable sampling operation of AIN0 channel. 1: Enable sampling operation of AIN0 channel.
[4]	RW	1'h0	bat_threshold_int_en 0: Disable VCC_RTC interrupt. 1: Enable VCC_RTC interrupt.
[3]	RW	1'h0	ain1_threshold_int_en 0: Disable AIN1 interrupt. 1: Enable AIN1 interrupt.
[2]	RW	1'h0	ain0_threshold_int_en 0: Disable AIN0 interrupt. 1: Enable AIN0 interrupt.
[1]	RW	1'h0	spl_done_int_en 0: Disable interrupt after data-sampling is completed. 1: Enable interrupt after data-sampling is completed.
[0]	RW	1'h0	saradc_en 0: Disable data sampling. 1: Enable data sampling.

● SAR ADC Sample Rate Register (Address: 0x0800,0060)

Bits	Attributes	Reset	Description
[31:16]	RW	16'h0	spl_hold_cfg Channel selection hold time. (unit: cycles of SAR ADC_CLK)

Bits	Attributes	Reset	Description
			NOTE: For single channel, 1) Spl_wait >= 2 2) Spl_hold >= (Spl_wait + 22) 3) Spl_cycle >= (Spl_hold+1) For multiple channels, 1) Spl_wait >= 2 2) Spl_hold >= (Spl_wait + 22) 3) Spl_cycle >= (Spl_hold+1), Spl_cycle >= (Spl_wait+27)
[15:0]	RW	16'h0	spl_cyc_cfg Sampling cycle. SAR ADC_CLK cycle time needed for one sampling operation. NOTE: For single channel, 1) Spl_wait >= 2 2) Spl_hold >= (Spl_wait + 22) 3) Spl_cycle >= (Spl_hold+1) For multiple channels, 1) Spl_wait >= 2 2) Spl_hold >= (Spl_wait + 22) 3) Spl_cycle >= (Spl_hold+1), Spl_cycle >= (Spl_wait+27)

● SAR ADC Sample Threshold Register (Address: 0x0800,0064)

Bits	Attributes	Reset	Description
[31:24]	RO	8'h0	Reserved
[23:12]	RW	12'h0	ain1_threshold Threshold of analog key
[11:0]	RW	12'h0	ain0_threshold Threshold of analog key

● SAR ADC Sample Data Register (Address: 0x0800,0068)

Bits	Attributes	Reset	Description
[31:24]	RO	8'h0	Reserved
[23:12]	RO	12'h0	ain1_data_cur Sample data of analog key
[11:0]	RO	12'h0	ain0_data_cur Sample data of analog key

● SAR ADC Sample Interrupt Status Register (Address: 0x0800,006C)

Bits	Attributes	Reset	Description
[31:20]	RW	12'h0	bat_dat_cur Sampling data from VCC_RTC channel
[19:8]	RW	12'h0	VCC_RTC_threshold Threshold of battery voltage
[7:4]	RO	4'h0	Reserved
[3]	RW	1'h0	bat_threshold_int

Bits	Attributes	Reset	Description
			1: An interrupt is generated when the value of VCC_RTC exceeds the threshold. 0: No interrupt is generated when the value of VCC_RTC exceeds the threshold. NOTE: This bit can be cleared by writing 0.
[2]	RW	1'h0	ain1_threshold_int Analog key threshold interrupt status 1: An interrupt is generated when the value of AIN1 exceeds the threshold. 0: No interrupt is generated when the value of AIN1 exceeds the threshold. NOTE: This bit can be cleared by writing 0.
[1]	RW	1'h0	ain0_threshold_int Analog key threshold interrupt status 1: An interrupt is generated when the value of AINO exceeds the threshold. 0: No interrupt is generated when the value of AINO exceeds the threshold. NOTE: This bit can be cleared by writing 0.
[0]	RW	1'h0	spl_done_int Interrupt status when data sampling is finished. 1: An interrupt is generated after data-sampling is completed. 0: No interrupt is generated after data-sampling is completed. NOTE: This bit can be cleared by writing 0.

● DAC Fadeout Control Register (Address: 0x0800,0070)

This register configures Sigma-Delta DAC filter, Sigma-Delta DAC fade-out and over sample rate.

Bits	Attributes	Reset	Description			
[31]	RW	1'h0	48K_mode 0: To select 16K mode. 1: To select 48K mode.			
			Mode	adc_clk	Output data rate	Oversample rate ratio
			48K mode	12.288MHz	48KHz	256
			16K mode	8.192MHz	16KHz	512
[30]	RW	1'h0	dac_fade_out_done The Sigma-Delta DAC fade-out process is completed.			
[29:25]	RW	5'h0	dac_fade_out_factor Sigma-Delta DAC fade-out factor.			
[24]	RW	1'h0	dac_fade_out_en 0: Disable Sigma-Delta DAC fade-out control. 1: Enable Sigma-Delta DAC fade-out control.			
[23:4]	RO	20'h0	Reserved			
[3]	RW	1'h0	dac_filter_en			

Bits	Attributes	Reset	Description																																													
			0: Disable Sigma-Delta DAC filter. 1: Enable Sigma-Delta DAC filter.																																													
[2:0]	RW	3'h0	osr Sigma-Delta DAC sampling rate selection																																													
			<table><tr><th>OSR</th><th>bclk</th><th>lrclk</th><th>over-sampling rate</th><th>Interpolation ratio</th></tr><tr><td>3'h0</td><td>x/4</td><td>bclk/64</td><td>256</td><td>32</td></tr><tr><td>3'h1</td><td>x/4</td><td>bclk/68</td><td>272</td><td>34</td></tr><tr><td>3'h2</td><td>x/4</td><td>bclk/66</td><td>264</td><td>33</td></tr><tr><td>3'h3</td><td>x/4</td><td>bclk/62</td><td>248</td><td>31</td></tr><tr><td>3'h4</td><td>x/4</td><td>bclk/60</td><td>240</td><td>30</td></tr><tr><td>3'h5</td><td>x/2</td><td>bclk/68</td><td>136</td><td>68</td></tr><tr><td>3'h6</td><td>x/2</td><td>bclk/64</td><td>128</td><td>64</td></tr><tr><td>3'h7</td><td>x/2</td><td>bclk/60</td><td>120</td><td>60</td></tr></table>	OSR	bclk	lrclk	over-sampling rate	Interpolation ratio	3'h0	x/4	bclk/64	256	32	3'h1	x/4	bclk/68	272	34	3'h2	x/4	bclk/66	264	33	3'h3	x/4	bclk/62	248	31	3'h4	x/4	bclk/60	240	30	3'h5	x/2	bclk/68	136	68	3'h6	x/2	bclk/64	128	64	3'h7	x/2	bclk/60	120	60
			OSR	bclk	lrclk	over-sampling rate	Interpolation ratio																																									
			3'h0	x/4	bclk/64	256	32																																									
			3'h1	x/4	bclk/68	272	34																																									
			3'h2	x/4	bclk/66	264	33																																									
			3'h3	x/4	bclk/62	248	31																																									
			3'h4	x/4	bclk/60	240	30																																									
			3'h5	x/2	bclk/68	136	68																																									
			3'h6	x/2	bclk/64	128	64																																									
3'h7	x/2	bclk/60	120	60																																												
NOTE: The relationship between DAC CLK and OSR can be reflected with the following equator: DAC CLK = DAC Sample Frequency x OSR (over-sampling rate).																																																

● ADC Mode Configuration Register (Address: 0x2011,8000)

This register configures Sigma-Delta ADC, and it is applicable to both embedded/external Sigma-Delta ADCs.

Bits	Attributes	Reset	Description
[31:12]	RO	20'h0	Reserved
[11:8]	RW	4'h0	word_len_config Word length configuration bits Word length = word_len_config + 1 (unit: bit)
[7:5]	RO	3'h0	Reserved
[4]	RW	1'h0	i2s_receive_en 0: Internal Sigma-Delta ADC mode 1: I2S receive mode
[3]	RW	1'h0	channel_pol_sel Channel polarity selection 0: Receive the left channel data when LRCK is low. 1: Receive the left channel data when LRCK is high.
[2]	RW	1'h0	host_int_sta 0: An ARM read interrupt is not generated. 1: An ARM read interrupt is generated. NOTE: An ARM read interrupt is generated when sampling data have been read successfully (the ARM core reads sample data from Sigma-Delta ADC).
[1]	RW	1'h0	l2_en 0: No instruction. 1: Sigma-Delta ADC sends data to L2 memory.
[0]	RW	1'h0	adc_en Sigma-Delta ADC controller enable 0: No instruction.

Bits	Attributes	Reset	Description
			1: Enable Sigma-Delta ADC controller.

- ADC Data Receive Register (Address: 0x2011,8004)

This register holds the sample data from Sigma-Delta ADC or external Sigma-Delta ADC.

Bits	Attributes	Reset	Description
[31:16]	RW	16'h0	sample_data For internal Sigma-Delta ADC, The (2N + 1) slice of sample data received from Sigma-Delta ADC. NOTE: N is an integer. For external Sigma-Delta ADC: Sample data received from the right channel.
[15:0]	RW	16'h0	sample_data For internal Sigma-Delta ADC, The 2N slice of sample data received from Sigma-Delta ADC. NOTE: N is an integer. For external Sigma-Delta ADC, Sample data received from the left channel.

- DAC Mode Configuration Register (Address: 0x2011,0000)

The Sigma-Delta DAC can input signals from ARM or L2 memory. This register selects signal input sources of Sigma-Delta DAC, and enables the interrupt function, which is applicable to both embedded and external Sigma-Delta DAC.

Bits	Attributes	Reset	Description
[31:5]	RO	17'h0	Reserved
[4]	RW	1'h0	data_storage_format 0: Normal format 1: Memory saving format
[3]	RW	1'h0	arm_int Host load interrupt status. 0: An ARM interrupt is not generated. 1: An ARM interrupt is generated. NOTE: ARM interrupt is generated when sampled data have been transmitted successfully (Sigma-Delta DAC(s) accepts (accept) input from ARM.)
[2]	RW	1'h0	mute 0: No instruction 1: Send the same amount of data as that in last clock cycle
[1]	RW	1'h0	l2_en 0: No instruction. 1: Sigma-Delta DAC(s) accepts (accept) data from L2 memory.
[0]	RW	1'h0	dac_en Sigma-Delta DAC controller enable control 0: No instruction

Bits	Attributes	Reset	Description
			1: Enable Sigma-Delta DAC controller

- I2S Configuration Register (Address: 0x2011,0004)

This register defines the word length, selects channel polarity and reflects the current value of I2S_LRCLK.

Bits	Attributes	Reset	Description
[31:7]	RO	25'h0	Reserved
[6]	RO	1'h0	lrclk Current value of I2S_LRCLK
[5]	RW	1'h1	chn_pol Channel polarity selection bits 0: Send the data of left channel when I2S_LRCLK is low. 1: Send the data of left channel when I2S_LRCLK is high.
[4:0]	RW	5'h0	Word length

- CPU Data Register (Address: 0x2011,0008)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	Double-buffered transmit control bits

7.7 Programmer's Guide

7.7.1 Sigma-delta ADC Configuration

7.7.1.1 MIC Amplifier Gain Control

Configure bits[17:15], bit[19] and bit[20] in [Analog Control Register 1 \(Address: 0x0800,00A0\)](#) accordingly.

7.7.1.2 Line-in Amplifier Gain Control

Configure bit[19] and bits[25:22] in [Analog Control Register 1 \(Address: 0x0800,00A0\)](#) accordingly.

7.7.1.3 Sigma-delta ADC Working Modes Switching

Configure bits[31:20] in [Analog Control Register 1 \(Address: 0x0800,00A0\)](#) and bit[24] in [Analog Control Register 2 \(Address: 0x0800,00A4\)](#) accordingly.

7.7.2 Sigma-delta DAC Configuration

Headphone amplifier control can be realized by configuring bits[16:14] and bits[22:18] in [Analog Control Register 0 \(Address: 0x0800,009C\)](#).

7.7.3 SAR ADC Configuration

7.7.3.1 SAR ADC Working Modes Switching

Configure bit[4]/bit[5]/bit[6]/bit[7] in [Analog Control Register 1 \(Address: 0x0800,00A0\)](#) to switch between different working modes of SAR ADC.

7.7.3.2 Battery Monitoring

Programmers should follow the steps to measure the battery voltage:

1. Set the SAR ADC properly, such as working frequency, bit cycle, sample rate, and etc..
The value of sar_adc_clk ranges from 1MHz to 4MHz (4MHz is recommended).

- Wait at least 1 bit cycle.
- Enable battery monitor(bit[7] of *SAR ADC Sample Control Register (Address: 0x0800,005C)*).
- Enable SAR ADC (bit[0] of *SAR ADC Sample Control Register (Address: 0x0800,005C)*).
- Read the battery voltage from *SAR ADC Sample Interrupt Status Register (Address: 0x0800,006C)*.
- Read the sample data for N times at intervals of 1 millisecond's (D1, D2,..., DN). Average

$\frac{D1 + \dots + DN}{N}$

the read value with the equation: the average read value = $\frac{D1 + \dots + DN}{N}$. In AK3918AV100 series processors, it is suggested to read the sample data for 7 times before getting the average value, in which case, the average value = $\frac{D1 + D2 + D3 + D4 + D5 + D6 + D7}{7}$. Actual battery voltage can be calculated according to the equation: actual battery voltage = $\frac{average_value \times 2 \times VREF_SARADC}{2^{12}}$.

NOTE

To reduce power consumption, it is recommended to disable battery monitor function (bit[7] of *SAR ADC Sample Control Register (Address: 0x0800,005C)*) after taking the battery measurement.

7.7.3.3 AIN0/AIN1 channel

Programmers should follow the steps to get the actual AIN0/AIN1 voltage.

- Set the SAR ADC properly, such as working frequency, bit cycle, sample rate, and etc. through configuring the corresponding bits in these following registers: (*SAR_ADC/SPI0/ SD_DAC Clock Control (Address: 0x0800,000C)*, *SAR ADC Sample Rate Register (Address: 0x0800,0060)*, and *Analog Control Register 1 (Address: 0x0800,00A0)*).
- Wait at least 10ms;
- Select AIN0/AIN1 channel mode (bit[5]/bit[6] of *SAR ADC Sample Control Register (Address: 0x0800,005C)*);
- Enable SAR ADC (bit[0] of *SAR ADC Sample Control Register (Address: 0x0800,005C)*);
- Read AIN0/AIN1 data from *SAR ADC Sample Data Register (Address: 0x0800,0068)*;
- Actual AIN0/AIN1 voltage can be calculated according to the equation: Actual

$\frac{AIN_data \times VREF_SARADC}{2^{12}}$

AIN0/AIN1 voltage =

8 Peripherals

8.1 I2S

8.1.1 Overview

I2S is a popular 3-wire serial bus standard protocol for transmission of 2 channel (stereo) Pulse Code Modulation digital data between audio devices, where MSB of each audio sample is sent first.

8.1.2 Features

AK3918AV100 series processors support one I2S interface and its features are as follows:

- Support master and slave mode
- Support up to 32-bit data width
- Support 8KHz ~ 96KHz sampling rate
- Support full-duplex transmission

8.1.3 Functional Description

8.1.3.1 Block Diagram

As shown in the following figure, audio data can be received from the external ADC and sent to the memory or DAC controller to be processed through I2S interface. I2S interface is also able to gain the data from the memory or ADC controller and then transfer the data to the external DAC.

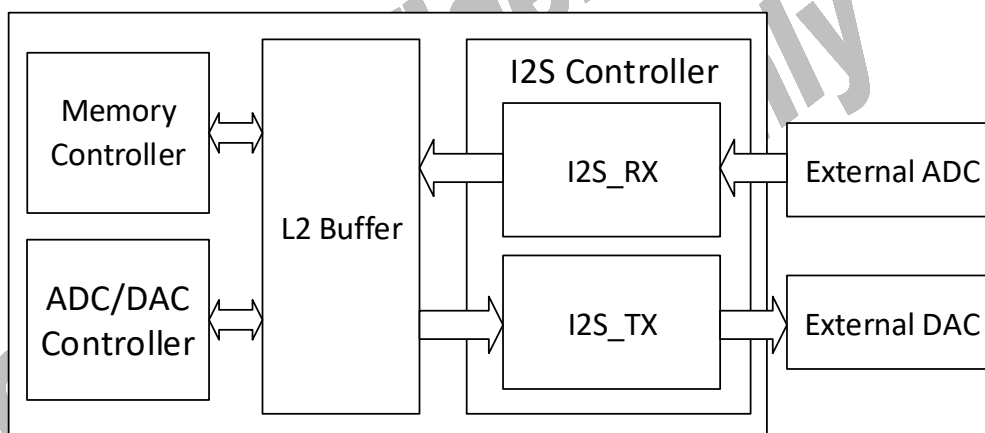


Figure 8-1 I2S module block diagram

8.1.3.2 Interface Timings

About the I2S timing, the BCLK and LRCLK are only generated by master device. The LRCLK indicates which channel is being transmitted, whose polarity can be configured. I2S interface also provides a programmable clock output (MCLK) signal for the external Sigma-Delta ADC/DAC. The timing diagrams of I2S interface is shown in the following figures.

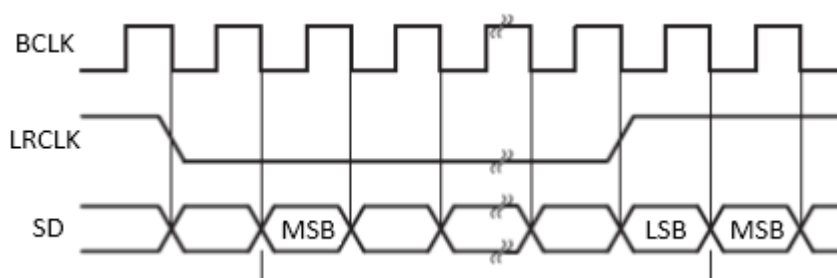


Figure 8-2 I2S interface timing

The following figures show the detail of the I2S interface timing parameter:

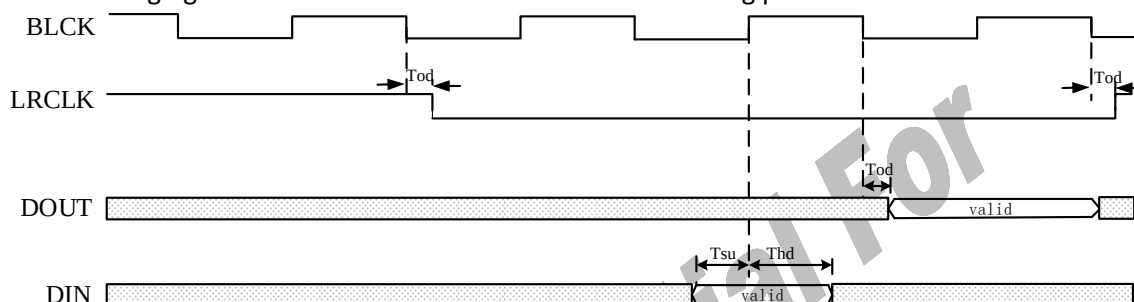


Figure 8-3 Timing of I2S master interface

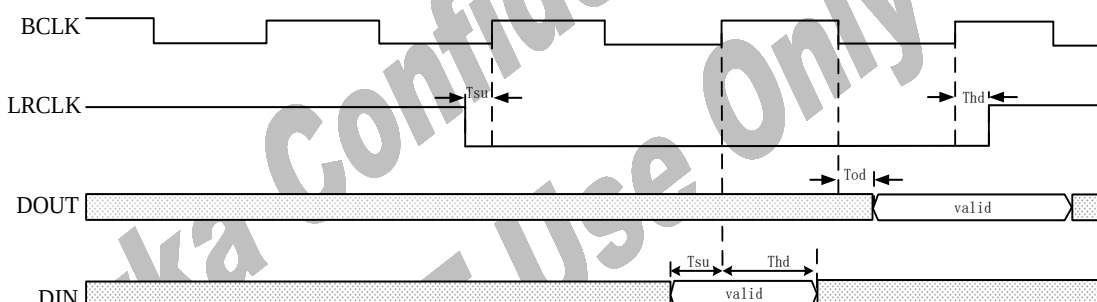


Figure 8-4 Timing of I2S slave interface

The following tables show the detail of the timing parameters:

Table 8-1 Timing Parameters of I2S Master DAC & I2S Master ADC

Parameter	Symbol	Min.	Max.	Unit
Input data setup time	Tsu	10	-	ns
Input data hold time	Thd	35	-	ns
BCLK to LRCLK output delay	Tod	0	3	ns
Output data delay	Tod	32	40	ns

Table 8-2 Timing Parameters of I2S Slave DAC

Parameter	Symbol	Min.	Max.	Unit
LRCK to BCLK input setup time	Tsu	5	-	ns
LRCK to BCLK input hold time	Thd	50	-	ns
Input data setup time	Tsu	-	-	ns
Input data hold time	Thd	-	-	ns

Parameter	Symbol	Min.	Max.	Unit
Output data delay	Tod	40	50	ns

Table 8-3 Timing Parameters of I2S Slave ADC

Parameter	Symbol	Min.	Max.	Unit
LRCK to BCLK input setup time	Tsu	5	-	ns
LRCK to BCLK input hold time	Thd	50	-	ns
Input data setup time	Tsu	5	-	ns
Input data hold time	Thd	42	-	ns
Output data delay	Tod	-	-	ns

8.1.3.3 Working Modes

According to the direction of the data transfer, the I2S controller can be described as transmitter and receiver. The working situation of the I2S interface shows as follow:

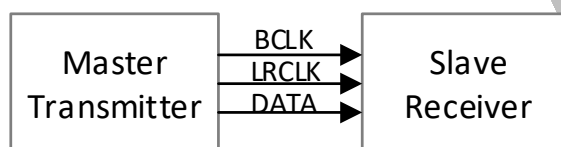


Figure 8-5 Master Transmitter & Slave Receiver mode

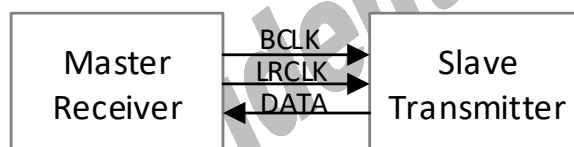


Figure 8-6 Master Receiver & Slave Transmitter mode

➤ Transmitter

When the controller works as a transmitter, it supports up to 32bits word length which can be configured by Sigma-Delta DAC controller. If the word length of external Sigma-Delta DAC is more than 32 bits, 0s are added to the end of LSBs.

➤ Receiver

The word length that controller can receive is up to 16bits and can be configured by Sigma-Delta ADC controller. If the word length of external Sigma-Delta ADC is more than 16 bits, the LSBs are ignored.

8.1.4 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Register Address	Register Name and Description
1	0x0800,0058	Refer to Section 3.2.5 USB Operate Mode and I2S Mode Control
2	0x2011,8000	Refer to Section 6.6 ADC Mode Configuration Register
3	0x2011,8004	Refer to Section 6.6 ADC Data Receive Register
4	0x2011,0000	Refer to Section 6.6 DAC Mode Configuration Register

5	0x2011,0004	Refer to Section 6.6 I2S Configuration Register
6	0x2011,0008	Refer to Section 6.6 CPU Data Register

NOTE

The internal Sigma-Delta ADC only transmits the data of left channel.

8.2 TWI

8.2.1 Overview

TWI bus is a two-wire bi-directional serial bus that provides a simple and efficient method of data exchange while minimizing the interconnection between devices.

8.2.2 Features

Features of TWI are as follows:

- Compatible with SCCB protocol
- Supports three TWI ports: TWI0, TWI1 and TWI2
- Supports master and slave mode
- Supports standard mode, fast mode, and fast mode plus

8.2.3 Functional Description

8.2.3.1 Working Mode

- 1) TWI supports master and slave mode.
 - In master mode, TWI controller generates the SCL signal for the slave device which works at slave mode as receiver or transmitter. It also manages the data transmission on the bus with the start or stop signal.
 - In slave mode, the TWI controller can be addressed by the external master devices on the bus and communicate with the master devices. In this mode, controller is not able to generate the SCL signal, it only receives SCL from the external master devices.
- 2) The TWI controller can work as a transmitter or receiver. When the R/W bit is 0, the controller is described as a transmitter to write data; when R/W is 1, the controller is considered as a receiver to read data from other devices.

8.2.3.2 Data Transfer Format

The possible transfer formats are described in the following section. The explanation of the data transfer symbol refers to the [Figure 8-7](#).

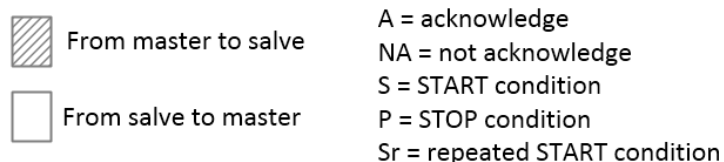


Figure 8-7 Explanation of the data symbol

8.2.3.2.1 Write Data Format

This format is used for data transmission from a master-transmitter to a slave-receiver. The transfer direction will not change and the slave receiver acknowledges each byte.

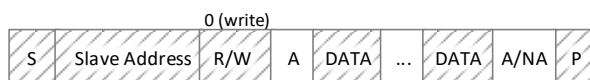


Figure 8-8 A master-transmitter addressing a slave receiver with a 7-bit address

8.2.3.2.2 Read Data Format

This format is used for master reading slave immediately after the first byte. At the moment of the first acknowledge, the master-transmitter becomes a master-receiver and the slave-receiver becomes a slave-transmitter. This first acknowledge is still generated by the slave and the master generates subsequent acknowledges. The STOP condition is generated by the master, which sends a not-acknowledge (NACK) just before the STOP condition.

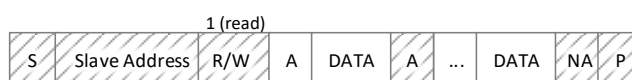


Figure 8-9 A master reading a slave immediately after the first byte

8.2.3.2.3 Combined Format

The combined format is used when the transfer direction is changed. The START signal and slave address will be repeated during the change of direction. If master receiver sends a repeated START condition, it sends a not acknowledge signal just before the repeated START condition.



Figure 8-10 A combined format

8.2.3.2.4 10-bit addressing

TWI also supports 10-bit addressing in master mode. This section describes the 10-bit address format when the controller works in master mode as transmitter or receiver.

8.2.3.2.4.1 Transmitter

In this situation, the transfer direction will not change. The first byte contains a 7-bit address and the R/W bit is 0. The first 7-bit address uses the format “11110XX”, of which the last two bits are the first two MSB of the whole 10-bit address and the second byte is the last eight LSB of the 10-bit address.

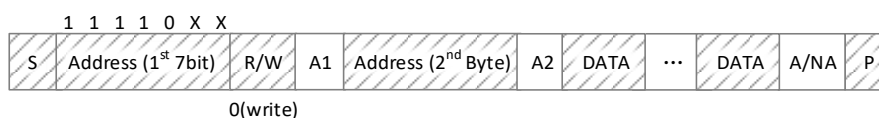


Figure 8-11 transmitter addresses receiver with a 10-bit address

8.2.3.2.4.2 Receiver

In this situation, the first byte and the second byte have the same format in this transmission situation. The transmitter will check if the 7-bit address following the Sr are the same as the

first 7-bit address and if the R/W bit is 1. When the target address is matched, the transmitter will give an acknowledge signal and then transmit the data to the master device.

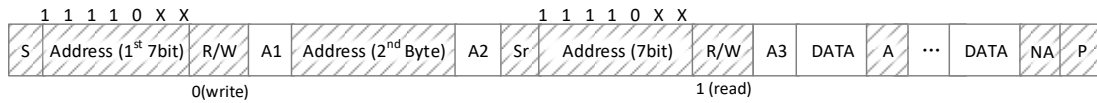


Figure 8-12 receiver addresses transmitter with a 10-bit address

8.2.4 Programmer's Guide

8.2.4.1 Master Mode

8.2.4.1.1 Write Operation

When TWI controller works in master mode as transmitter, programmers should operate according to the following steps:

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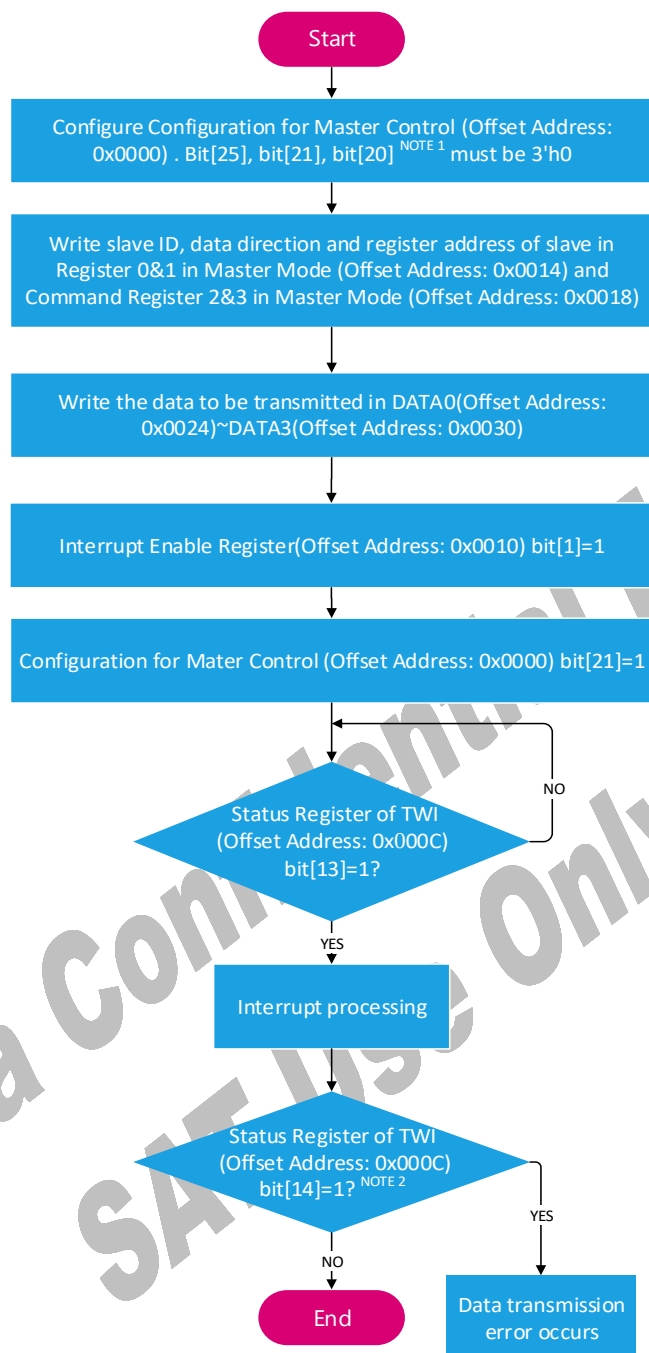


Figure 8-13 Configuration Flow of Master-mode as transmitter

NOTE

- 1) When TWI controller works as master-transmitter, the bit[20]- tx_rx in *Configuration for Master Control (Offset Address: 0x0000)* must be 0.
- 2) During interrupt processing, bit[14] in *Status Register of TWI(Offset Address: 0x000C)* read as 1 indicates that during data transmission, NACK has been responded and data sent by TWI master was not received successfully by TWI slave.

8.2.4.1.2 Read Operation

When TWI controller works in master mode as receiver, programmers should operate according to the following steps:

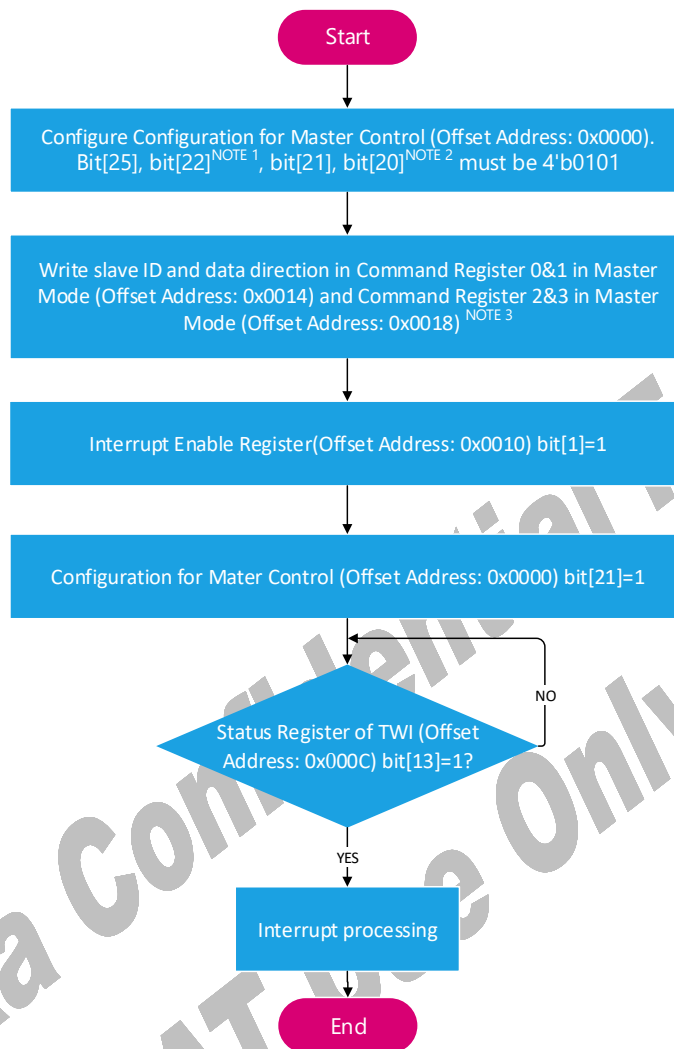


Figure 8-14 Configuration Flow of Master-mode as receiver

NOTE

- 1) When TWI controller works as master-receiver, the bit[22] and bit[20] in *Configuration for Master Control (Offset Address: 0x0000)* must be 1.
- 2) When master-mode TWI works as receiver, only one “cmd_data” can be configured to generate “start condition” and send slave ID as well as data direction bit to external TWI slave. Specifically, once a certain “cmd_data” among cmd1_data~cmd4_data in *Command Register 0&1 in Master Mode (Offset Address: 0x0014)* and *Command Register 2&3 in Master Mode (Offset Address: 0x0018)* is enabled, the other three “cmd_data” must be disabled.

8.2.4.1.3 Combined Operation

When TWI firstly works in master-transmitter mode and then switches to master-receiver mode, programmers should operate according to the following steps.

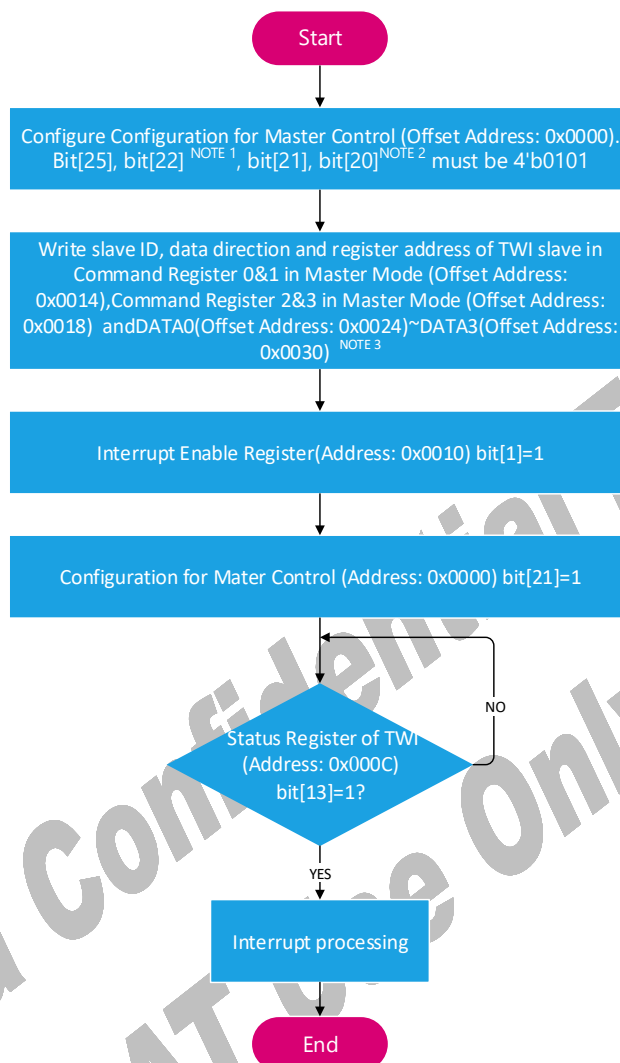


Figure 8-15 Configuration Flow of Master-mode with a combined format

NOTE

- 1) When TWI tranfers data in combined format, bit[22] and bit[20] in *Configuration for Master Control (Offset Address: 0x0000)* must be 1.
- 2) When TWI switches from master-transmitter mode to master-receiver mode, repeated START condition will be generated and the following slave ID will be sent to TWI slave again. And the data direction bit(R/W) will be changed from 0 to 1.

8.2.4.2 Slave mode

When TWI works in slave mode, data transfer formats depend on the control of external TWI master. Therefore, programmers only need to enable the slave mode, configure the related parameters and then wait until data transmission is finished. Programmers should operate according to the following steps.

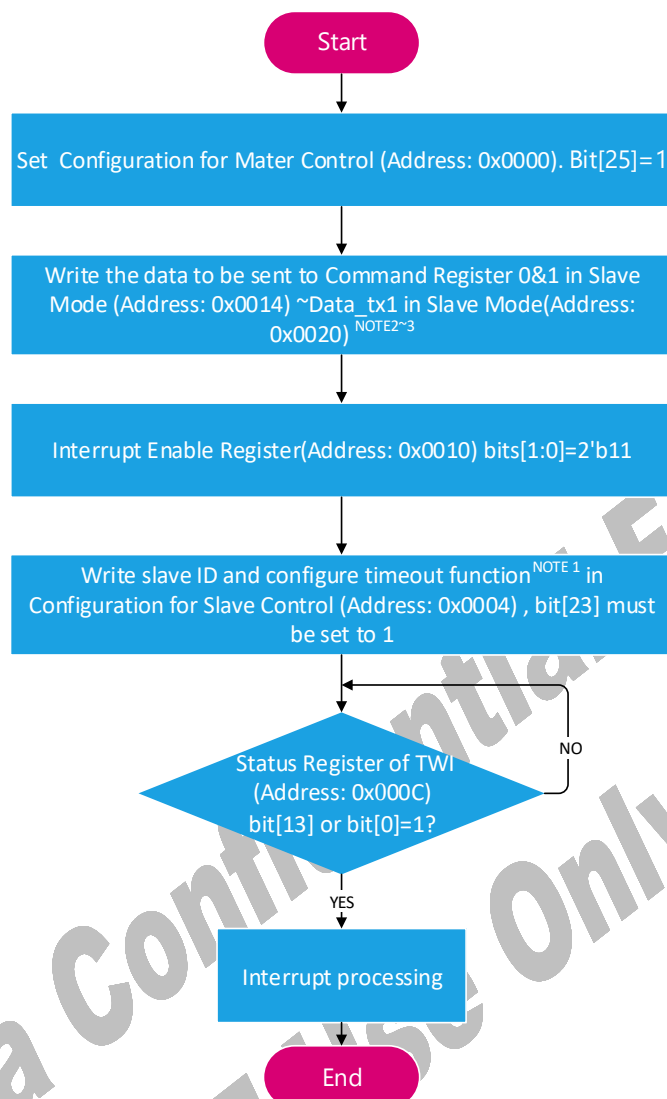


Figure 8-16 Configuration Flow of Slave-mode TWI

NOTE

- 1) Programmers should configure bits[15:1] of *Configuration for Slave Control (Offset Address: 0x0004)* to 15'h0 to disable timeout function before TWI controller needs to switch from slave mode to master mode.
- 2) *Data_tx0(Offset Address: 0x001C)* and *Data_tx0(Offset Address: 0x0020)* can only be used in slave mode.
- 3) *Command Register 0&1 in Slave Mode (Address: 0x0014)~ Data_tx1 in Slave Mode (Offset Address: 0x0020)* are intended for sending data to TWI Bus in slave mode.
- 4) *DATA0(Offset Address: 0x0024)~DATA3(Offset Address: 0x0030)* are intended for receiving data from TWI bus in slave mode.

8.2.5 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the

detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

NOTE

The three TWI ports (TWI0~2) have same register definitions.

- 1) Base address of TWI0 is 0x2015_0000.
- 2) Base address of TWI1 is 0x2019_0000.
- 3) Base address of TWI2 is 0x201C_0000.

Register No.	Offset Address	Register Name and Description
1	0x0000	<i>Configuration for Mater Control</i>
2	0x0004	<i>Configuration for Slave Control</i>
3	0x0008	<i>Delay Configuration Register of TWI</i>
4	0x000C	<i>Status Register of TWI</i>
5	0x0010	<i>Interrupt Enable Register</i>
6	0x0014	<i>Command Register 0&1 in Master Mode</i>
7		<i>Command Register 0&1 in Slave Mode</i>
8	0x0018	<i>Command Register 2&3 in Master Mode</i>
9		<i>Command Register 2&3 in Slave Mode</i>
10	0x001C	<i>Data_tx0 in Slave Mode</i>
11	0x0020	<i>Data_tx1 in Slave Mode</i>
12	0x0024	<i>DATA0</i>
13	0x0028	<i>DATA1</i>
14	0x002C	<i>DATA2</i>
15	0x0030	<i>DATA3</i>

8.2.6 Register Definition

- Configuration for Master Control (Offset Address: 0x0000)

Bits	Attributes	Reset	Description
[31:26]	RO	6'h0	Reserved
[25]	RW	1'h0	mode_sel 1: Slave mode is selected 0: Master mode is selected NOTE: This field is used to select either master mode or slave mode.
[24]	RW	1'h0	Lastack 1: "NACK" is responded by TWI controller when the last byte of data has been received from external slave-transmitter. 0: "ACK" is responded by TWI controller when the last byte of data has been received from external slave-transmitter. NOTE: Programmers need to set 1 to this bit if master-mode TWI controller needs to read data from external slave devices.
[23]	RW	1'h0	nostop: 1: "stop condition" will not be generated when TWI controller operates in master mode.

Bits	Attributes	Reset	Description
			0: "stop condition" will be generated when TWI controller operates in master mode.
[22]	RW	1'h0	force0: 1: "ACK" is responded by TWI controller when it works in master-receiver mode (i.e. TWI controller reads data from external slave). 0: "NACK" is responded by TWI controller when it operates in master-receiver mode. NOTE: This bit is valid only before the last data byte is read from the external slave-transmitter.
[21]	RW/SC	1'h0	start Write 1 to this bit to start the operation of master mode. This field will be cleared automatically at the moment of "(div_cnt_cfg+1) * GCLK cycles" after the start of master-mode operation.
[20]	RW	1'h0	tx_rx Transmit / Receive operation selection control 0: Transmit 1: Receive
[19:15]	RW	5'h0	length_of_bytes The length of data bytes to be transmitted or received ranges from 0~16
[14:13]	RW	2'h0	sda_delay Output delay of SDA = (5 * sda_delay[1:0]) * T _{asic_clk} NOTE: 1) This field must be set to 2'h3. 2) When a TWI slave uses the falling edge of SCL to sample SDA, which is sent by a TWI master, the register of " <i>sda_delay</i> " must be configured accordingly by programmers. However, for TWI slaves that utilize the rising edge of SCL to sample SDA, there is no configuration requirement for the register " <i>sda_delay</i> ".
[12:0]	RW	13'h0	div_cnt_cfg: Frequency division factor in master mode NOTE: $f_{scl} = f_{asic_clk} / (2 * (div_cnt_cfg + 1))$

● Configuration for Slave Control (Offset Address: 0x0004)

Bits	Attributes	Reset	Description
[31:24]	RO	8'h0	Reserved
[23]	RW	1'h0	s_mod_en TWI controller slave mode control 1: Enabled 0: Disabled
[22:16]	RW	7'h0	slave_addr Slave ID of TWI controller
[15:1]	RW	15'h0	time_out_cfg The number of asic_clk cycles configured to trigger the

Bits	Attributes	Reset	Description
			timeout function, which should be greater than $(T_{scl}/T_{asic_clk})/2 - 1$ NOTE: When this field is written as 0, timeout function is disabled. If TWI controller switches from slave mode to master mode, this field should be set to 0.
[0]	WO/SC	1'h0	slave_clr Write 1 to this bit to clear the internal status of TWI controller in slave mode. NOTE: 1) In order to restart the slave-mode operation, bit[23] of this register must be set to 1 after value of this bit is valid. 2) This field is valid only when the control signal "reg_wr" of the register bus remains high.

● Delay Configuration Register of TWI (Offset Address: 0x0008)

Bits	Attributes	Reset	Description
[31:12]	RO	20'h0	Reserved
[11:0]	RW	12'h0	Interval: Set the interval between the data bytes (8-bit data) transmitted or received by master-mode TWI controller. $T_{interval} = (interval+1) * (T_{scl}/2)$ NOTE: This field is only used in master mode.

● Status Register of TWI (Offset Address: 0x000C)

Bits	Attributes	Reset	Description
[31:15]	RO	17'h0	Reserved
[14]	R/W1C	1'h0	ack 1: NACK is responded by the slave device in the process of data transfer when master-mode TWI controller sends data to the bus. 0: No NACK is responded by the slave devices when TWI controller operates in master-transmitter mode. NOTE: 1) This bit is used in the master mode for indicating the status of TWI controller. 2) This bit must be checked by software when master-mode communication is over. 3) The high-level status of this bit represents that abnormal conditions happen during master-mode data transfer. A proper action must be taken to handle this situation.
[13]	R/W1C	1'h0	trans_finish_int Interrupt request due to the completion of data transfer in either master mode or slave mode: a read access returns the interrupt status 1: Interrupt triggered

Bits	Attributes	Reset	Description
			0: No interrupt NOTE: The interrupt request is triggered only when stop condition occurs in either master mode or slave mode.
[12]	WO/SC	1'h0	tx_byte_cnt_clr This field is used to clear bits[10:6]-“tx_byte_cnt”. NOTE: 1) This field is only used in slave mode. 2) This field is only effective when the control signal “reg_wr” of the register bus remains high.
[11]	WO/SC	1'h0	rx_byte_cnt_clr This field is used to clear bits[5:1]- “rx_byte_cnt” NOTE: 1) This field is only used in slave mode. 2) This field is only effective when the control signal “reg_wr” of the register bus remains high.
[10:6]	RO	5'h0	tx_byte_cnt The total bytes of data that have been sent to TWI bus when stop condition occurs. This field can be cleared by bit[12] -“tx_byte_cnt_clr” NOTE: This field is only used in slave mode.
[5:1]	RO	5'h0	rx_byte_cnt The total bytes of data which have been received from TWI bus when stop condition is detected. This field can be cleared by bit[11] -“rx_byte_cnt_clr”. NOTE: This field is only used in slave mode.
[0]	R/W1C	1'h0	time_out_int The missing falling edge or rising edge of SCL in the process of communication leads to the time-out interrupt request in slave mode of operation. 1: Interrupt triggered 0: No interrupt

● Interrupt Enable Register (Offset Address: 0x0010)

Bits	Attributes	Reset	Description
[31:2]	RO	30'h0	Reserved
[1]	RW	1'h0	trans_finish_int_en Enable the interrupt request caused by the completion of data transfer in either master mode or slave mode. 0: Disabled 1: Enabled
[0]	RW	1'h0	time_out_int_en Enable the interrupt request caused by timeout when TWI operates in slave-mode. 0: Disabled 1: Enabled

● Command Register 0&1 in Master Mode (Offset Address: 0x0014)

Bits	Attributes	Reset	Description
[31:26]	RO	6'h0	Reserved
[25]	RW	1'h0	cmd2_enable 1: Enable bits[23:16] - "cmd2_data" to be sent to TWI bus 0: Disable the transmission of bits[23:16] - "cmd2_data"
[24]	RW	1'h0	cmd2_start_prefix 1: "start condition" will be generated before bits[23:16] - "cmd2_data" is sent. 0: "start condition" will not be generated before bits[23:16] - "cmd2_data" is sent.
[23:16]	RW	8'h0	cmd2_data Data stored in bits[23:16]
[15:10]	RO	6'h0	Reserved
[9]	RW	1'h0	cmd1_enable 1: Enable bits[7:0] - "cmd1_data" to be sent to TWI bus. 0: Disable the transmission of bits[7:0] - "cmd1_data".
[8]	RW	1'h0	cmd1_start_prefix 1: "start condition" will be generated before bits[7:0] - "cmd1_data" is sent. 0: "start condition" will not be generated before bits[7:0] - "cmd1_data" is sent.
[7:0]	RW	8'h0	cmd1_data Data stored in bits[7:0]

NOTE

Command Register 0&1 in Master Mode (Address: 0x0014) is for the configuration of CMD1&2 when TWI controller operates in master mode.

● Command Register 0&1 in Slave Mode (Offset Address: 0x0014)

Bits	Attributes	Reset	Description
[31:24]	RW	8'h0	tx_byte3 The fourth byte of data to be transmitted in slave mode.
[23:16]	RW	8'h0	tx_byte2 The third byte of data to be transmitted in slave mode.
[15:8]	RW	8'h0	tx_byte1 The second byte of data to be transmitted in slave mode.
[7:0]	RW	8'h0	tx_byte0 The first byte of data to be transmitted in slave mode.

NOTE

Command Register 0&1 in Slave Mode (Address: 0x0014) is only intended for sending data to TWI Bus when TWI controller operates in slave mode.

● Command Register 2&3 in Master Mode (Offset Address: 0x0018)

Bits	Attributes	Reset	Description
[31:26]	RO	6'h0	Reserved

Bits	Attributes	Reset	Description
[25]	RW	1'h0	cmd4_enable 1: Enable bits[23:16] - "cmd4_data" to be sent to TWI bus 0: Disable the transmission of bits[23:16] - "cmd4_data"
[24]	RW	1'h0	cmd4_start_prefix 1: "start condition" will be generated before bits[23:16] - "cmd4_data" is sent. 0: "start condition" will not be generated before bits[23:16] - "cmd4_data" is sent.
[23:16]	RW	8'h0	Cmd4_data Data stored in bits[23:16]
[15:10]	RO	6'h0	Reserved
[9]	RW	1'h0	cmd3_enable 1: Enable bits[7:0] - "cmd3_data" to be sent to TWI bus. 0: Disable the transmission of bits[7:0] - "cmd3_data".
[8]	RW	1'h0	cmd3_start_prefix 1: "start condition" will be generated before bits[7:0] - "cmd3_data" is sent. 0: "start condition" will not be generated before bits[7:0] - "cmd3_data" is sent.
[7:0]	RW	8'h0	cmd3_data Data stored in bits[7:0]

NOTE

Command Register 2&3 in Master Mode (Address: 0x0018) is for the configuration of CMD3&4 when TWI controller operates in master mode.

- Command Register 2&3 in Slave Mode (Offset Address: 0x0018)

Bits	Attributes	Reset	Description
[31:24]	RW	8'h0	tx_byte7 The eighth byte of data to be transmitted in slave mode.
[23:16]	RW	8'h0	tx_byte6 The seventh byte of data to be transmitted in slave mode.
[15:8]	RW	8'h0	tx_byte5 The sixth byte of data to be transmitted in slave mode.
[7:0]	RW	8'h0	tx_byte4 The fifth byte of data to be transmitted in slave mode.

NOTE

Command Register 2&3 in Slave Mode (Address: 0x0018) is only intended for sending data to TWI Bus when TWI controller operates in slave mode.

- Data_tx0 in Slave Mode (Offset Address: 0x001C)

Bits	Attributes	Reset	Description
[31:24]	RW	8'h0	tx_byteb The twelfth byte of data to be transmitted in slave mode.

Bits	Attributes	Reset	Description
[23:16]	RW	8'h0	tx_bytea The eleventh byte of data to be transmitted in slave mode.
[15:8]	RW	8'h0	tx_byte9 The tenth byte of data to be transmitted in slave mode.
[7:0]	RW	8'h0	tx_byte8 The ninth byte of data to be transmitted in slave mode.

NOTE

Data_tx0 in Slave Mode (Address: 0x001C) is only intended for sending data to TWI Bus in slave mode.

- Data_tx1 in Slave Mode (Offset Address: 0x0020)

Bits	Attributes	Reset	Description
[31:24]	RW	8'h0	tx_bytEf The sixteenth byte of data to be transmitted in slave mode.
[23:16]	RW	8'h0	tx_bytEe The fifteenth byte of data to be transmitted in slave mode.
[15:8]	RW	8'h0	tx_bytEd The fourteenth byte of data to be transmitted in slave mode.
[7:0]	RW	8'h0	tx_bytEc The thirteenth byte of data to be transmitted in slave mode.

NOTE

Data_tx1 in Slave Mode (Address: 0x0020) is only intended for sending data to TWI Bus in slave mode.

- DATA0 (Offset Address: 0x0024)

Bits	Attributes	Reset	Description
[31:24]	RW	8'h0	byte3 in master mode data0[31:24] in slave mode
[23:16]	RW	8'h0	byte2 in master mode data0[23:16] in slave mode
[15:8]	RW	8'h0	byte1 in master mode data0[15:8] in slave mode
[7:0]	RW	8'h0	byte0 in master mode data0[7:0] in slave mode

NOTE

DATA0 (Address: 0x0024) is a data storing register which can be used in both master mode and slave mode. It can be configured to send data to TWI bus or receive data from TWI bus in master mode. While in slave mode, it is only intended for receiving data from TWI bus.

● DATA1 (Offset Address: 0x0028)

Bits	Attributes	Reset	Description
[31:24]	RW	8'h0	byte7 in master mode data1[31:24] in slave mode
[23:16]	RW	8'h0	byte6 in master mode data1[23:16] in slave mode
[15:8]	RW	8'h0	byte5 in master mode data1[15:8] in slave mode
[7:0]	RW	8'h0	byte4 in master mode data1[7:0] in slave mode

NOTE

DATA1 (Address: 0x0028) is a data storing register, which can be used in both master mode and slave mode. It can be configured to send data to TWI bus or receive data from TWI bus in master mode. While in slave mode, it is only intended for receiving data from TWI bus.

● DATA2 (Offset Address: 0x002C)

Bits	Attributes	Reset	Description
[31:24]	RW	8'h0	byteb in master mode data2[31:24] in slave mode
[23:16]	RW	8'h0	bytea in master mode data2[23:16] in slave mode
[15:8]	RW	8'h0	byte9 in master mode data2[15:8] in slave mode
[7:0]	RW	8'h0	byte8 in master mode data2[7:0] in slave mode

NOTE

DATA2 (Address: 0x002C) is a data storing register, which can be used in both master mode and slave mode. It can be configured to send data to TWI bus or receive data from TWI bus in master mode. While in slave mode, it is only intended for receiving data from TWI bus.

● DATA3 (Offset Address: 0x0030)

Bits	Attributes	Reset	Description
[31:24]	RW	8'h0	bytef in master mode data3[31:24] in slave mode
[23:16]	RW	8'h0	bytee in master mode data3[23:16] in slave mode
[15:8]	RW	8'h0	byted in master mode data3[15:8] in slave mode
[7:0]	RW	8'h0	bytec in master mode data3[7:0] in slave mode

NOTE

DATA3 (Address: 0x0030) is a data storing register, which can be used in both master mode and slave mode. It can be configured to send data to TWI bus or receive data from TWI bus in master mode. While in slave mode, it is only intended for receiving data from TWI bus.

8.3 UART

8.3.1 Overview

UART is an asynchronous serial I/O port and supports full duplex data transmitting and receiving. UART is featured with converting the parallel data from peripheral devices into serial data, and transmit the data into internal bus. The UART module includes three independent asynchronous ports, each of which provides serial communication capability with external devices through an RS232 cable.

8.3.2 Features

Key features of UART module are as follows:

- Three asynchronous ports: UART0, UART1 and UART2
- UART2 supports hardware flow control
- Operating in polling-based mode or interrupt-based mode
- Programmable baud rate, ranging from 95bps to 6.0Mbps
- Data packet of 11 bits: 1 start bit (always “0”), 8 data bits, 1 parity bit (optional), and 1 stop bit (should be “1”)
- Select active voltages of TXD/RXD/CTS/RTS by configuration

8.3.3 Functional Description

8.3.3.1 Communication Protocol

UART is an asynchronous serial communication port, with only two lines transmitting and receiving data. The time interval between two words is not fixed in communication but fixed between two adjacent bits.

In UART protocol, there are two states which can be symbolled as high level (logic 1) and low level (logic 0). In idle state, the data line is kept at the high level. The whole transmitting procedure of UART consists of four parts, including a logic low Start Bit, Data Bits, possibly a Parity Bit and one or more Stop Bits in sequence.

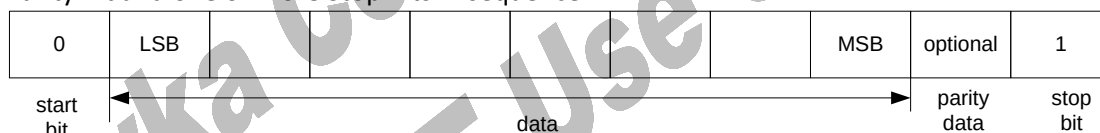


Figure 8-17 Data Framing

In idle state, the data line is always at high level, which means there is no data being transmitted.

● Start Bit

Start Bit is a logic 0 signal which signals the Receiver the beginning of a data transmitting.

● Data Bits

Data bits contain 5~9(depending on the code set employed) bits of logic 0 or 1, so transmitter and receiver must define the same data width. The transmitting starts from the least significant bit.

● Parity Bit

This bit is used to check the correction of the whole transmission. As an optional configuration item, parity bit has five configuration ways: none, odd, even, mark, space.

● Stop Bits

These bits are the ending sign of data transmission which can be configured as 1bit, 1.5bits or 2bits. Stop bit ensures sufficient time when transmitter and receiver are doing hardware operation.

UART belongs to asynchronous serial communication, so there is no clock lane for receiver to sample correctly. Receiver must configure the same baud rate and word length with transmitter for proper operation, when it samples and decodes the pin signal.

As soon as the sending system deposits a word in the shift register, the transmitter of UART will generate a Start Bit, a Parity Bit (if used) and Stop Bits to create a data packet. Next, the data packet is transmitted in serial form bit by bit on Tx pin. The receiver of UART reads data packet bit by bit on Rx pin. Then, the receiver of UART converts the data back to parallel data form and deletes the Start Bit, Parity Bit and Stop Bit.

8.3.3.2 Data Transmitting

Data transmitting of UART relies on UART bus, which is used to send data to UART from CPU, storage, micro-controller and other devices.

As described in the previous chapter, L2 memory is a bridge connecting DMA controller and every kind of interfaces. Data is transmitted to UART from data bus in parallel form. UARTn gets the parallel data to be sent from L2 memory, converts into serial data format, and then sends to receiver via UTDx pin; it receives serial data via the URDx pin, converts into parallel data format, and then saves temporarily in L2 memory. The data width between L2 memory and UARTn is 32 bits.

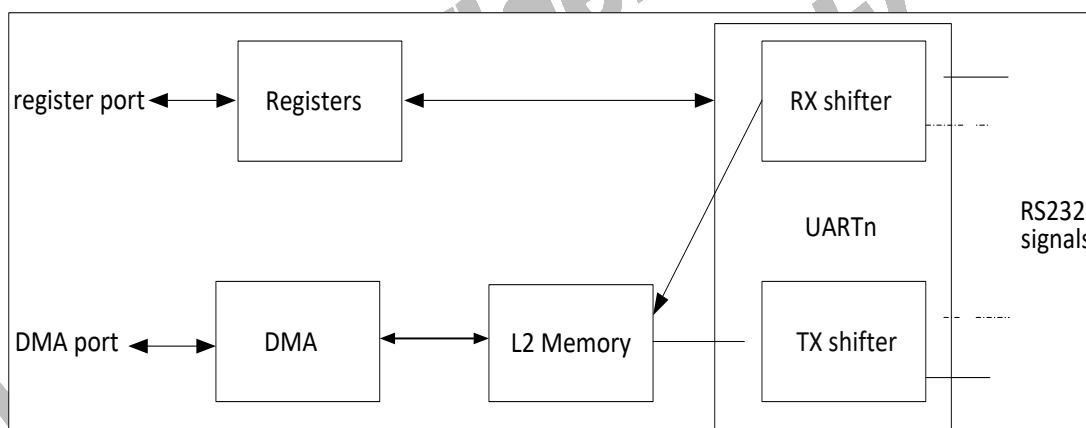


Figure 8-18 UART Module Block Diagram

NOTE

->: hardware flow control signals, available in UART2.

RX shifter: from serial transmitting to parallel transmitting

TX shifter: from parallel transmitting to serial transmitting

8.3.3.3 Hardware Flow Control

UART2 supports hardware flow control:

PIN NAME	TYPE	DESCRIPTION
UART2_RTS	O	Request to Send Signal from UART, active low.
UART2_CTS	I	Clear to Send Signal of UART, active low.

● UART2 receiving data

UART2 asserts the RTS signal to indicate that it is ready to receive data. If the data in L2 memory is full, or UART2 is not allowed to be used, RTS signal is de-asserted.

● UART2 transmitting data

When UART2 wants to send data, it detects the CTS signal. If CTS signal is asserted, it begins to transmit data. Otherwise, the data in L2 memory remains undisturbed.

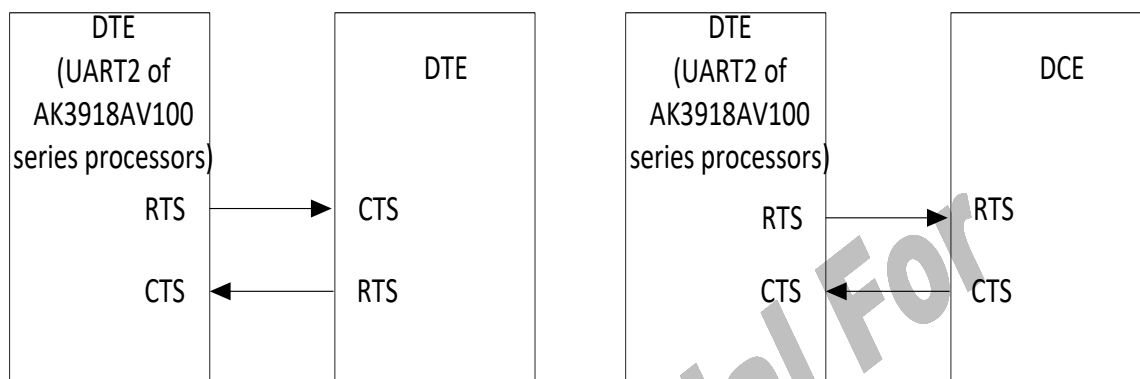


Figure 8-19 Flow Control of UART2

8.3.4 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Offset Address	Register Name
1	0x0000	<i>Configuration Register 0 of UART</i>
2	0x0004	<i>Configuration Register 1 of UART</i>
3	0x0008	<i>Data Configuration Register of UART</i>
4	0x000C	<i>TX/RX Buffer Threshold Register</i>
5	0x0010	<i>RX Buffer Register</i>
6	0x0014	<i>RX Backup Buffer Register</i>
7	0x0018	<i>Stop Bit and Time out Configuration Register</i>
8	0x001C	<i>Division Factor Configuration Register</i>

NOTE

- 1) The three asynchronous ports (UART0~2) have the same register definitions.
- 2) Base address of UART0 is 0x2013_0000.
- 3) Base address of UART1 is 0x2013_8000.
- 4) Base address of UART2 is 0x201A_0000.

8.3.5 Register Definition

- Configuration Register 0 of UART (Address: 0x0000)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	tx_add_clr

Bits	Attributes	Reset	Description
			0: No instruction 1: To clear the read address output to L2 FIFO. NOTE: This bit will be automatically cleared after being read.
[30]	RW	1'h0	rx_add_clr 0: No instruction 1: To clear the write address output to L2 FIFO. NOTE: This bit will be automatically cleared after being read.
[29]	RW	1'h0	rx_sta_clr 0: No instruction 1: To clear the internal state of receive circuitry. NOTE: This bit will be automatically cleared after being read.
[28]	RW	1'h0	tx_sta_clr 0: No instruction 1: To clear the internal state of transmit circuitry. NOTE: This bit will be automatically cleared after being read.
[27]	RW	1'h0	endian_sel 0: Little endian 1: Big endian
[26]	RW	1'h0	par_en 0: To disable parity function 1: To enable parity function
[25]	RW	1'h0	par_sel 0: To employ odd parity. 1: To employ even parity. NOTE: If par_en(bit[26] of this register) is disabled, configuration of this bit will be ignored.
[24]	RO	1'h0	Reserved
[23]	RW	1'h1	timeout_en 0: To disable the timeout function of receive circuitry. 1: To enable the timeout function of receive circuitry. NOTE: When UART module is in receive data operation but no more data is received for the time which is equal to the waiting time (configured by bits[31:16] of Stop Bit and Time out Configuration Register), an interrupt is generated (if the interrupt function is enabled). This function is called "timeout function" for short.
[22]	RW	1'h0	div_adj 0: To disable baud rate adjustment function. 1: To enable baud rate adjustment function. NOTE: 1) When baud rate adjustment function is enabled, the bit cycle of data frame will be extended one more GCLK cycle than that configured.

Bits	Attributes	Reset	Description
			2) This function is used to reduce the tolerance caused by the inaccurate ratio of frequency dividing.
[21]	RW	1'h0	uart_en 0: To disable UARTn interface. 1: To enable UARTn interface.
[20]	RW	1'h1	rts_cfg The actual RTSn signal is generated by performing an AND operation of the RTS signal configured by CPU (this bit) and the RTS signal generated by the circuitry.
[19]	RW	1'h0	rts_sel 0: To output the RTS signal directly. 1: To output the RTS signal inversely.
[18]	RW	1'h0	cts_sel 0: To input the CTS signal directly. 1: To input the CTS signal inversely.
[17]	RW	1'h0	rx_d_sel 0: To input the RXD signal directly. 1: To input the RXD signal inversely.
[16]	RW	1'h0	tx_d_sel 0: To output the TXD signal directly. 1: To output the TXD signal inversely.
[15:0]	RW	16'h14	div_cnt Baud rate of UARTn $\text{Baud rate} = \frac{GCLK}{DIV_CNT + 1}$ NOTE: If the baud rate is not precise enough, programmers may enable the baud rate adjustment function (bit[22] of this register) to reduce the tolerance.

● Configuration Register 1 of UART (Address: 0x0004)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	tx_th_int_sta 0: The tx_th interrupt is not generated. 1: A tx_th interrupt is generated. NOTE: 1) Programmers may set a threshold of the transmit buffer. When the number of transmitted data is equal to the threshold value, an interrupt is generated. This kind of interrupt is called tx_th interrupt. 2) The transmit buffer threshold value is set by bits[10:6] of <i>TX/RX Buffer Threshold Register (Address: 0x000C)</i> . 3) This bit is cleared by being written with 1 or when tx_th_clr, bit[11] of <i>TX/RX Buffer Threshold Register (Address: 0x000C)</i> , is 1.
[30]	RW	1'h0	rx_th_int_sta

Bits	Attributes	Reset	Description
			0: The rx_th interrupt is not generated. 1: A rx_th interrupt is generated. NOTE: 1) Programmers may set a threshold of the receive buffer. When the number of received data is equal to the threshold value, an interrupt is generated. This kind of interrupt is called rx_th interrupt. 2) The receive buffer threshold value is set by bits[4:0] of <i>TX/RX Buffer Threshold Register (Address: 0x000C)</i>. 3) This bit is cleared by being written with 1 or when rx_th_clr, bit[5] of <i>TX/RX Buffer Threshold Register (Address: 0x000C)</i>, is 1.
[29]	RW	1'h0	tx_th_int_en 0: To disable tx_th interrupt. 1: To enable tx_th interrupt. NOTE: 1) Programmers may set a threshold of the transmit buffer. When the number of transmitted data is equal to the threshold value, an interrupt is generated. This kind of interrupt is called tx_th interrupt. 2) The transmit buffer threshold value is set by bits[10:6] of <i>TX/RX Buffer Threshold Register (Address: 0x000C)</i>.
[28]	RW	1'h0	rx_th_int_en 0: To disable rx_th_int interrupt. 1: To enable rx_th_int interrupt. NOTE: 1) Programmers may set a threshold of the receive buffer. When the number of received data is equal to the threshold value, an interrupt is generated. This kind of interrupt is called rx_th interrupt. 2) The receive buffer threshold value is set by bits[4:0] of <i>TX/RX Buffer Threshold Register (Address: 0x000C)</i>.
[27]	RW	1'h0	tx_end_int_en 0: To disable tx_end interrupt. 1: To enable tx_end interrupt. NOTE: When all the data to be sent has been sent, an interrupt is generated. This kind of interrupt is called tx_end interrupt.
[26:25]	RO	2'h0	Reserved
[24]	RW	1'h0	tx_buf_emp_int_en 0: To disable tx_buf_emp interrupt. 1: To enable tx_buf_emp interrupt. NOTE: When the transmit buffer is empty, an interrupt is generated. This kind of interrupt is called tx_buf_emp

Bits	Attributes	Reset	Description
			interrupt.
[23]	RW	1'h0	r_err_int_en 0: To disable r_err interrupt. 1: To enable r_err interrupt. NOTE: When an error occurs in the received data, an interrupt is generated. This kind of interrupt is called r_err interrupt.
[22]	RW	1'h0	timeout_int_en 0: To disable timeout interrupt. 1: To enable timeout interrupt. NOTE: Please refer to bit[23] of Configuration Register 0 of UART for the definition of timeout function.
[21]	RW	1'h0	uart_wmem_nrdy_int_en 0: To disable uart_wmem_nrdy interrupt. 1: To enable uart_wmem_nrdy interrupt. NOTE: When the receive buffer is not ready to receive data, an interrupt is generated. This kind of interrupt is called uart_wmem_nrdy interrupt.
[20]	RO	1'h0	Reserved
[19]	RO	1'h1	tx_end 1: Data transmission in TX buffer has finished. 0: Data transmission in TX buffer has not finished.
[18]	RO	1'h0	This bit is Reserved and should be read as 0.
[17]	RO	1'h0	uart_wmem_Nrdy 0: The receive buffer is ready to receive new data. 1: There is no enough buffer to store new data.
[16]	WO	1'h0	tx_byt_cnt_vld 0: The value hold by tx_byt_cnt_vld (bits[15:4]) is invalid. 1: The value hold by tx_byt_cnt_vld (bits[15:4]) is valid. NOTE: This bit is cleared automatically when the configuration is finished (i.e.: The value hold by tx_byt_cnt (bits[15:4] of this register) has been added to tx_byt_sum (bits[12:0] of Data Configuration Register of UART (Address: 0x0008)).
[15:4]	WO	12'h0	tx_byt_cnt The number of data to be sent (unit: byte) NOTE: 1) Programmers should write the number of data to be sent into these bits. The number written into these bits will be added to tx_byt_sum (bits[12:0] of Data Configuration Register of UART (Address: 0x0008)). 2) If tx_byt_cnt_vld, bit[16] of this register, is invalid, these bits are invalid.
[3]	RW/W1C	1'h0	r_err 1: An error occurs in the received data. 0: No error occurs in the received data. NOTE: This bit is cleared by being written with 1.
[2]	RW/W1C	1'h0	timeout

Bits	Attributes	Reset	Description
			1: The receiving timeout occurs. 0: No receiving timeout occurs. NOTE: 1) Please refer to bit[23] of <i>Configuration Register 0 of UART</i> for the definition of timeout function. 2) This bit is cleared by being written with 1.
[1]	RW	1'h0	rx_fifo_full 1: The receive FIFO is full. 0: The receive FIFO is not full. NOTE: This bit is cleared by being written with 1.
[0]	RO	1'h1	tx_fifo_empty 1: The transmit buffer is empty. 0: The transmit buffer is not empty.

● Data Configuration Register of UART (Address: 0x0008)

Bits	Attributes	Reset	Description
[31:25]	RW	7'h0	rx_th_cfg_h The higher configuration bits of receive buffer threshold If these bits are not used: Receive buffer threshold value = (RX_th_cfg_l+1) x 1 (byte) Otherwise, Receive buffer threshold value = RX_th_cfg_h x 32(byte) + (RX_th_cfg_l+1) x 1 (byte) NOTE: RX_th_cfg_l is configured by bits[4:0] of <i>TX/RX Buffer Threshold Register (Address: 0x000C)</i> .
[24:20]	RO	5'h0	tx_adr The address of transmit buffer accessed now. (unit: 4bytes)
[19:13]	RO	7'h0	rx_adr The address of receive buffer will be accessed in next UART receive operation. (unit: byte) NOTE: If the receive buffer is not ready, the address configured here and the actual address of L2 buffer into which data is written will have a difference within 4 bytes.
[12:0]	RO	13'h0	tx_byt_sum The total number of data to be sent (unit: byte) NOTE: When all the data has been sent, these bits are set to 0s; bit[19] of <i>Configuration Register 1 of UART (Address: 0x0004)</i> is set to 1 at the same time; and an interrupt is generated if the TX_end interrupt function is enabled (bit[27] of <i>Configuration Register 1 of UART (Address: 0x0004)</i>).

● TX/RX Buffer Threshold Register (Address: 0x000C)

Bits	Attributes	Reset	Description
[31]	RW	1'h1	uart_rx_start

Bits	Attributes	Reset	Description
			1: UART starts to receive data. NOTE: 1) After UART receives the first byte, this bit will be cleared automatically. 2) If time out function is enabled, software needs to set this bit to 1 every time out event to prevent the UART from generating time out event continuously.
[30:29]	RO	2'h0	Bfifo_byte_num The data bytes left in receive backup FIFO.
[28:24]	RO	5'h0	tx_th_cnt Transmit threshold count These bits are added "1" every time the transmit buffer has transferred 4-byte data. When the number it counts is equal to the value of tx_th_cfg (bits[10:6] of this register), an interrupt is generated (if interrupt function is enabled) and these bits are set to 0s at the same time.
[23:12]	RO	12'h0	rx_th_cnt Receive data threshold count. These bits are added "1" every time the receive buffer has received 1 byte data. When the number it counts equals receive buffer threshold value (configured by rx_th_cfg_h and rx_th_cfg_l), an interrupt is generated (if interrupt function is enabled) and these bits are set to 0s at the same time.
[11]	RW	1'h1	tx_th_clr 1: To clear tx_th interrupt, and prevent the threshold from counting. NOTE: Please refer to bit[29] of Configuration Register 1 of UART for the definition of tx_th interrupt.
[10:6]	RW	5'h0	tx_th_cfg Transmit buffer threshold configuration Transmit buffer threshold value = (tx_th_cfg+1) x 4 (byte) NOTE: Please refer to bits[28:24] for more details on transmit buffer threshold.
[5]	RW	1'h1	rx_th_clr 1: To clear Rx_th interrupt, and prevent the threshold from counting. NOTE: Please refer to bit[28] of Configuration Register 1 of UART for the definition of rx_th interrupt.
[4:0]	RW	5'h0	rx_th_cfg_l The lower configuration bits of receive buffer threshold If rx_th_cfg_h (bits[31:25] of Data Configuration Register of UART (Address: 0x0008)) are not used: Receive buffer threshold value = (rx_th_cfg_l+1) x 1 (byte) Otherwise, Receive buffer threshold value = rx_th_cfg_h x 32(byte) + (rx_th_cfg_l+1) x 1 (byte)

- RX Buffer Register (Address: 0x0010)

This register holds the current data received by UART. The software can read it directly if needed. The data UART received will be stored in this register temporarily and then transferred to L2 memory. Every time UART receives a data byte and stores it in this register, a writing request to L2 memory will be sent. Once the request is responded, the data stored here will be transferred to L2 memory. If this register is full of four data bytes and all the writing requests sent when the first three data bytes are received are not responded, an interrupt is generated and the RX Backup Buffer Register will be used.

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	rx_data The current data received by UART

- RX Backup Buffer Register (Address: 0x0014)

When an interrupt is generated because RX Buffer Register is full, this register will be used. When more three data bytes are stored in this register and the L2 memory doesn't respond to the writing request, the UART stops receiving data.

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	bfifo_data The current data received by UART backup

- Stop Bit and Time out Configuration Register (Address: 0x0018)

This register is used to configure the stop bit cycle of every UART byte transmission and the time out cycle of UART receive operation.

Bits	Attributes	Reset	Description
[31:16]	RW	16'h1F	time_out_cfg Actual time out cycle = Time_out_cfg + 1. NOTE: 1) Please refer to bit[23] of Configuration Register 0 of UART for time out function definition. 2) For UART0, UART1, Time_out_cfg is configurable.
[15:9]	RO	7'h0	Reserved
[8:0]	RW	9'h0	stop_bit_cfg These bits configure the period of time the UART will stop transmitting after it finishes transmitting a data package. Actual stop bit cycle = Stop_bit_cfg + 1.

- Division Factor Configuration Register (Address: 0x001C)

Bits	Attributes	Reset	Description
[31:5]	RO	27'h0	Reserved
[4:0]	RW	5'h0	div_cnt_cfg_h The higher configuration bits of frequency division coefficients NOTE: 1) This field, together with the lower configuration bits(bits[15:0] in Configuration Register 0 of UART

Bits	Attributes	Reset	Description
			(Address: 0x0000) forms the real division factor. div_cnt_cfg_real[20:0]= {Div_cnt_cfg_h,Div_cnt_cfg} 2) By configuring both Div_cnt_cfg_h and Div_cnt_cfg the minimum Baud rate can be as low as 95bps.

8.3.6 Programmer's Guide

8.3.6.1 Data Transmitting

Programmers should operate according to the following steps:

1. Configure the clock of L2 buffer and UART, then release the reset signals of these two modules.
2. Configure the corresponding pin-mux of UART.
3. Assign buffer room in L2 Buffer for data transmitting from peripheral devices.
4. Configure polarity of UART port, baudrate of data transmission and other parameters about data transmitting.
5. Set the total length of data to send, and start UART.
6. Wait until data transmitting is over.

8.3.6.2 Data Receiving

Programmers should operate according to the following steps:

1. Configure the clock of L2 buffer and UART, then release reset signals of this two modules.
2. Configure the corresponding pin-mux of UART.
3. Assign buffer room in L2 Buffer for data receiving into peripheral devices.
4. Configure polarity of UART port, baudrate of data transmission and other parameters about data receiving.
5. Set receiving threshold interrupt, offline interrupt and rx_start signals, then start UART receiving.
6. Wait for receiving interrupt signal.

8.4 SPI1/SPI2 Controller

8.4.1 Overview

AK3918AV100 series processors support SPI1 and SPI2 for general purposes. SPI1/SPI2 interface allows fast synchronous serial communication of 8-bit data length. When SPI1/SPI2 interface is used as master, transmission frequency is configured by setting the clock divider ratio, and a byte transmit/receive operation starts with host writing data into the transmit FIFO. When SPI1/SPI2 interface is used as slave, external bus master supplies the clock.

8.4.2 Features

Features of SPI controller are as follows:

- Compatible with SPI Protocol, Version 2.11
- Supports both master mode and slave mode
- Supports full-duplex transmission
- Supports polling mode and interrupt mode
- Supports CPU mode and DMA mode (through L2 memory)
- Supports Transmit Garbage Data Mode (TGDM) and Auto Reject Received Data Mode (ARRM)
- Supports only Standard SPI mode

- Supports four timing modes: mode 0 (CPOL = 0, CPHA = 0) , mode 1 (CPOL = 0, CPHA = 1), mode 2 (CPOL = 1, CPHA = 0), and mode 3 (CPOL = 1, CPHA = 1)
- SPI2 controller supports one or two SPI slave device(s) via #CS signal

8.4.3 Functional Description

SPI1/2 controller can operate as both a master and a slave. Set bit[4] of *Control Register of SPI1/SPI2 (Address: 0x0000)* to 1 for spi1/2 to function as a master device and 0 for SPI1/2 to operate as a slave.

In master mode, SPI1/2 controller initiates a data transmission operation and selects the slave device to communicate by propagating the clock signal and driving the SPI1/2_CS low. For SPI2 controller, when communicating with multiple slave devices, the slave selection pins (SPI2_CS0 and SPI2_CS1) are used. Transmission frequency can be controlled by setting the bits[15:8] of *Control Register of SPI1/2 (Address: 0x0000)*.

In slave mode, SPI1/2 works under the control of the clock signals provided by the external master and the SPI1/2_CS should keep low until the transmission is completed. Its frequency can be modified to adjust the baud rate data register value.

8.4.3.1 Block Diagram

As shown in the following figure, the system can send/receive 8-bit data simultaneously to/from an external device. A serial clock line is synchronized with the two data lines for shifting and sampling of the information.

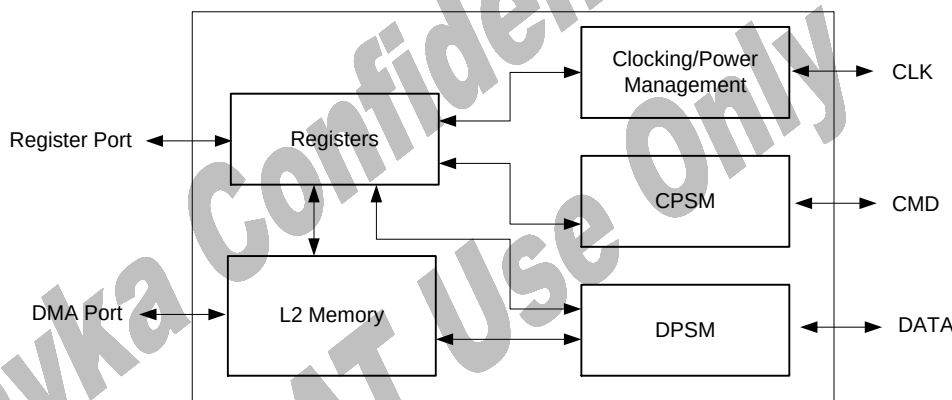


Figure 8-20 Block Diagram of SPI1/SPI2 Controller

8.4.3.2 SPI Operations

- Auto-reload in Slave Mode

When SPI1/2 works in slave mode, the chip supports auto-reload of data in an effort to avoid data corruption if CPU is busy with other operations. Without CPU interference, SPI1/2 in slave mode automatically reloads the configured value of last operation once the counter reaches 0. Thus, data reception can progress successfully. To enable auto-reload of SPI1/2 in slave mode, please set bit[7] of *Control Register of SPI1/2 (Address: 0x0000)* as 1 prior to data transmission .

- SPI_CS Function

In master mode, SPI2 supports two #CS signals (SPI2_CS0 and SPI2_CS1), which allows exclusive write/read operations of up to two slave devices. When bit[18] of *Control Register of SPI1/2 (Address: 0x0000)* is set to 0, the SPI2_CS0 externally connected slave is selected; while bit[18] is 1, the SPI2_CS1 externally connected slave is selected.

In slave mode, if SPI1/2_CS is brought from low level to high, data exchanges in SPI1/2 are finished. When SPI1/2_CS signal is detected as the rising edge, cs_end_int will be reported. Bit[11] of [Status Register of SPI1/SPI2 \(Address: 0x0004\)](#) will become 1 and value of this field will be cleared automatically after being read by software.

•Threshold Interrupt in Slave Mode

SPI1/2 supports threshold interrupt report in slave mode, which means an interrupt will be generated once the number of received data reaches the threshold value (1Byte ~ 5512Byte, configurable by software).

•Less_word Interrupt & timeout function in Slave Mode

SPI Interface processes 4-byte data in a data transfer operation. If the number of data in RX buffer is less than 4 bytes, the data cannot be accessed. Therefore, in slave mode a specific wait time([Less_word and timeout Register of SPI1/SPI2 \(Address: 0x0020\)](#)) should be set prior to data transmission. If no more data is received when the wait time is out, an interrupt is generated (if the interrupt function is enabled) and the data left is accessed. This function is called “less_word” function for short.

The timeout function is valid in slave mode only, as the SPI1/2 Interface can know well the total number of data to be received when in master mode.

8.4.3.3 Transmission Format

During a SPI1/2 transmission, data are transmitted and received simultaneously. SPI1/SPI2 interface of AK3918AV100 series processors provide the flexibility to program four different CPOL and CPHA combinations: mode0 ~ mode3. The following figure illustrates the SPI-standard defined timing diagram of SPI1/SPI2.

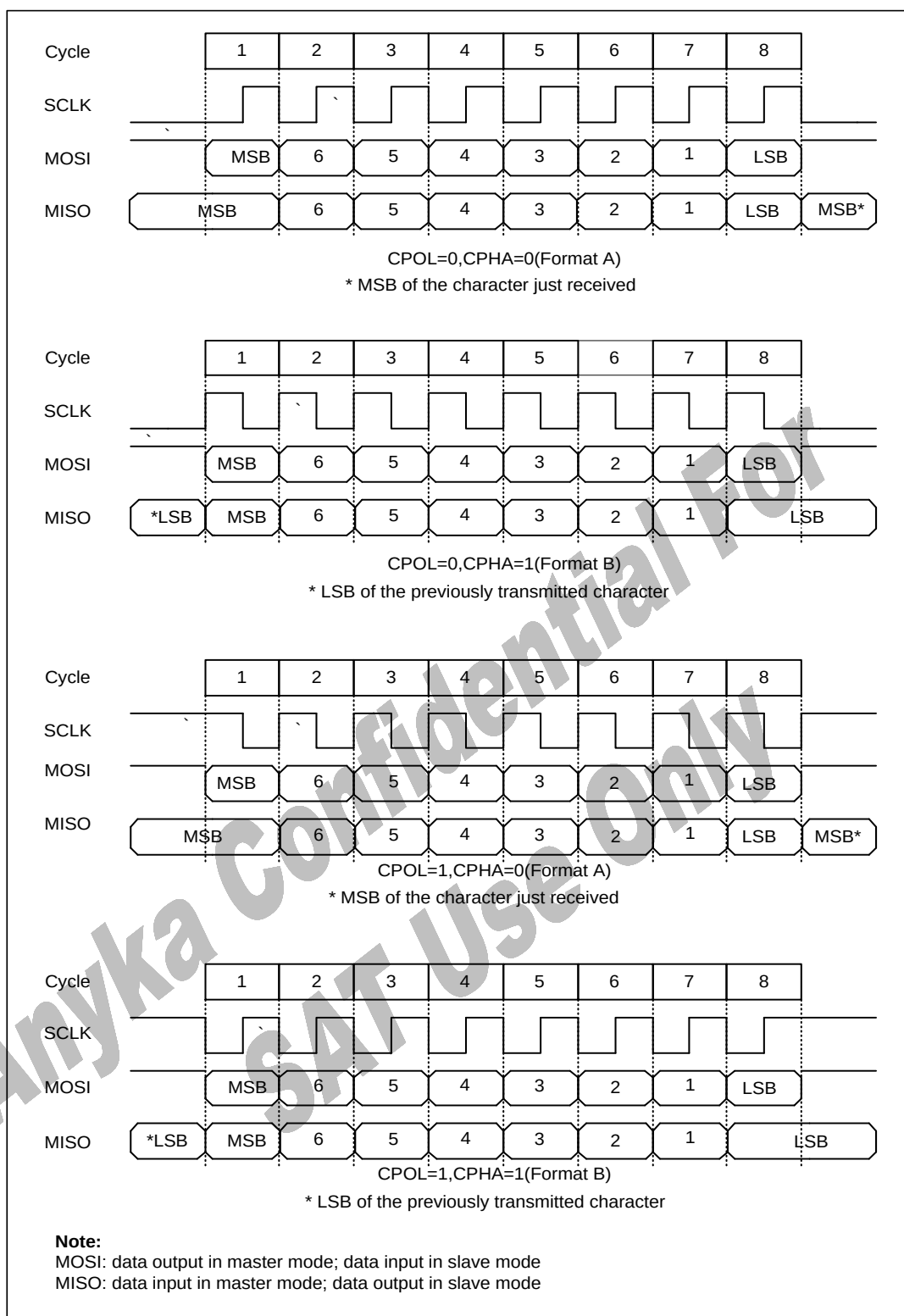


Figure 8-21 Timing Diagram of transmission formats of SPI1/SPI2

8.4.3.4 Interface Timing

Timings of SPI1 and SPI2 master interface are shown in the following figures.

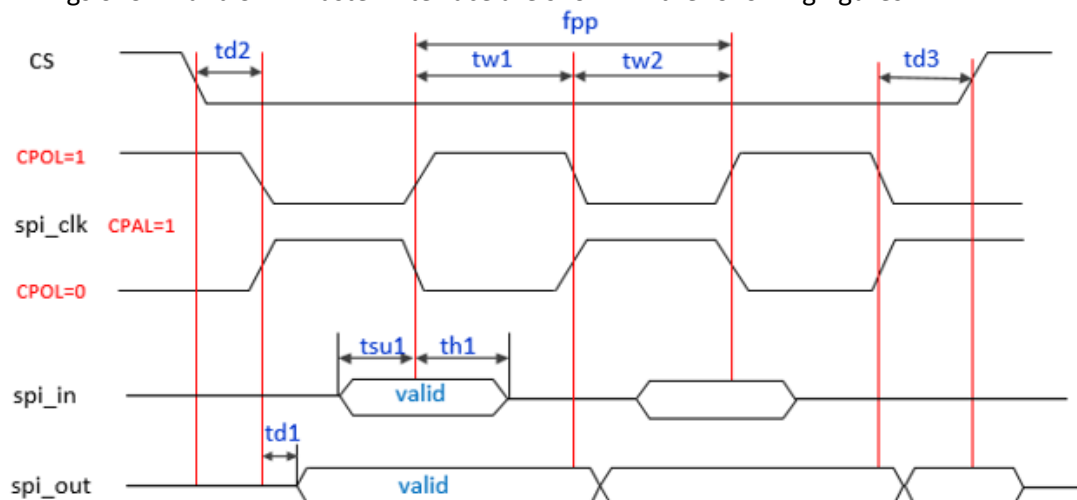


Figure 8-22 Timing of SPI1 and SPI2 master interface (Mode1 and Mode 3)

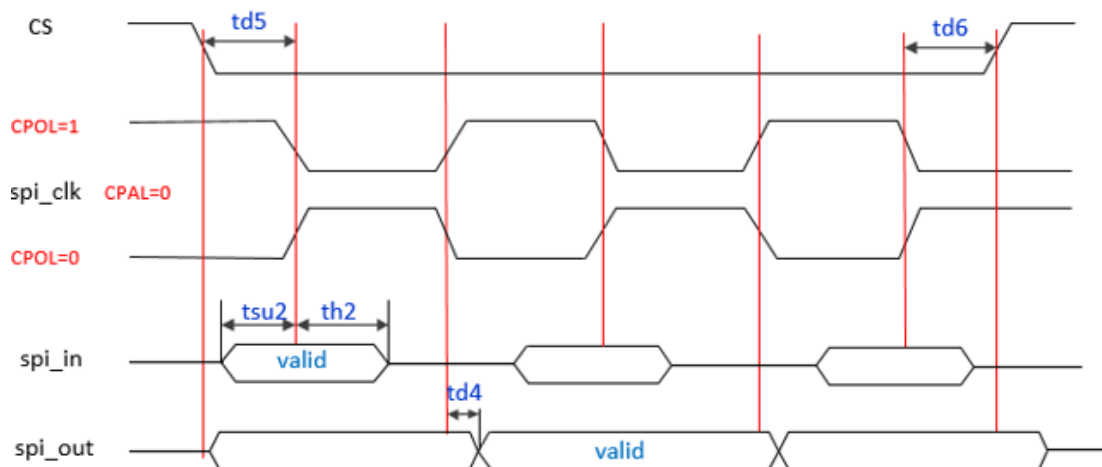


Figure 8-23 Timing of SPI1 and SPI2 master interface (Mode 0 and Mode 2)

Parameters of SPI1 and SPI2 mater interface timing are shown in the following table.

Table 8-4 Timing Parameters of SPI1 and SPI2 Master Interface

Parameter	Symbol	Min	Max	Unit
Clock frequency	fpp	0	50	Mhz
Clock high time	tw1	10	-	ns
Clock low time	tw2	10	-	ns
Output delay time	td1	0.4	2	ns
Delay time (cs falling edge to first spi1/spi2_clk rising or falling edge)	td2	1/fpp	-	ns
delay time (cs rising edge to first spi1/spi2_clk rising or falling edge)	td3	1/2fpp	-	ns

Parameter	Symbol	Min	Max	Unit
Input set up time	tsu1	9.2	-	ns
Input hold time	th1	0	-	ns
Input set up time	tsu2	9.2	-	ns
Input hold time	th2	0	-	ns
Output delay time	td4	0.4	2	ns
Delay time (cs falling edge to first spi1/spi2_clk rising or falling edge)	td5	1/2fpp	-	ns
Delay time (cs rising edge to first spi1/spi2_clk rising or falling edge)	td6	1/fpp	-	ns

8.4.3.5 Data Transmission inside the Chip

SPI1/SPI2 supports two data transmission modes, the process of which can be explained as follows.

1) DMA mode

TX operation: external RAM → through DMA bus → L2 memory → TX buffer of SPI1/SPI2 Interface → TX data line

RX operation: RX data line → RX buffer of SPI1/SPI2 Interface → L2 memory → through DMA bus → external RAM

2) CPU mode

TX operation: CPU writes data into TX buffer of SPI1/SPI2 Interface → TX data line

RX operation: RX data line → CPU reads data from RX buffer of SPI1/SPI2 Interface

The Output Data Register holds the data CPU writes into, and the Input Data Register holds the data from RX data lines.

8.4.4 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

NOTE

- 1) Registers with the base address 0x2012_8000 are the address space of SPI1 controller.
- 2) Registers with the base address 0x2016_0000 are the address space of SPI2 controller.

Register No.	Offset Address	Register Name
1	0x0000	<i>Control Register of SPI1/SPI2</i>
2	0x0004	<i>Status Register of SPI1/SPI2</i>
3	0x0008	<i>SPI1/SPI2 Interrupt Enable Register</i>
4	0x000C	<i>Data Count Register of SPI1/SPI2</i>
5	0x0010	<i>External TX Buffer Register of SPI1/SPI2</i>
6	0x0014	<i>External RX Buffer Register of SPI1/SPI2</i>
7	0x0018	<i>Output Data Register of SPI1/SPI2</i>
8	0x001C	<i>SPI1/SPI2 Input Data Register</i>

Register No.	Offset Address	Register Name
9	0x0020	Less_word and timeout Register of SPI1/SPI2
10	0x0024	Slave Output Data Delay Configuration Register of SPI1/SPI2
11	0x0028	Threshold count register of SPI1/SPI2

8.4.5 Register Definition

- Control Register of SPI1/SPI2 (Address: 0x0000)

This register is used to configure the SPI controller, such as timing format, working frequency, and etc.

Bits	Attributes	Reset	Description
[31:24]	RO	8'h0	Reserved
[23]	RW	1'h0	dma_modify_en NOTE: This bit must be configured as 0.
[22]	RW	1'h0	under_run_modify_en NOTE: This bit must be configured as 0.
[21]	RW	1'h0	modify_en NOTE: This bit must be configured as 0.
[20]	RW	1'h0	less_word_en This bit is used to enable the upload of data which is less than a Word. 1: Enable 0: Disable
[19]	RW	1'h0	threshold_en 0: Enable threshold 1: Disable threshold
[18]	RW	1'h0	cs_mux SPI2_CS selection in master mode 0: CS0 is selected, active low. 1: CS1 is selected, active low. NOTE: spi2_cs_mux is used to select two external SPI devices.
[17:16]	RW	2'h0	master_d_m 1/2/4-wire data selection for master mode 00: Use 1-wire data mode 01: Use 2-wire data mode 10: Use 4-wire data mode 11: Use 1-wire data mode NOTE: For slave mode, these bits must be 2'h0.
[15:8]	RW	8'h0	clk_div SPI clock divider $SPI_CLK = GCLK / (2 * (CLK_DIV + 1))$ NOTE: Master CLK_DIV >= 0, slaver's input clock must be less than or equal to GCLK/8.
[7]	RW	1'h0	auto_reload_en SPI automatic reload function enable control 1: Enable 0: Disable NOTE: SPI automatic reload function can only be

Bits	Attributes	Reset	Description
			used in slave mode.
[6]	RW	1'h0	spi_en SPI enable control 0: Disable 1: Enable
[5]	RO	1'h0	cs_force 0: No instruction 1: To forces #CS signal in master mode NOTE: Normally, the #CS signal is HIGH after a transmission of 8bits data. If this bit is set, the #CS signal is LOW until the all the data transmission has been finished.
[4]	RW	1'h0	master/slave 1: SPI works in master mode 0: SPI works in slave mode
[3]	RW	1'h0	cpha Initial phase of clock (CPHA) selection 0: Format A 1: Format B
[2]	RW	1'h0	cpol Active high or active low clock (CPOL) selection 0: Active high 1: Active low
[1]	RW	1'h0	arrm ARRM mode enable control 0: Disable 1: Enable NOTE: Once enabled, the SPI interface transmits and receives data simultaneously. If only data transmission is required, the ARRM (Auto Reject Received Data) mode can be enabled to discard the received data.
[0]	RW	1'h0	tgdm TGDM mode enable control 0: Disabled 1: Enabled NOTE: Once enabled, the SPI interface transmits and receives data simultaneously. If only data reception is required, the TGDM (Transmit Garbage Data Mode) mode can be enabled to transmit dummy data on transmit line.

● Status Register of SPI1/SPI2 (Address: 0x0004)

This register indicates current status of SPI1/SPI2 controller, such as TX/RX buffer status, data transmission operation status, and etc.

Bits	Attributes	Reset	Description
[31]	RW	1'h0	abnormal_flag

Bits	Attributes	Reset	Description
			0: It is normal when the received data is more than 1 byte. 1: It is abnormal when the received data is less than 1 byte. NOTE: Once an interrupt is generated, no matter When CS signal is HIGH or a timeout event occurs, abnormal flag will be valid and indicate whether the received data is abnormal or not.
[30:18]	RO	13'h0	Reserved
[17:14]	RO	4'h0	rx_buf_cnt Data size of SPI RX buffer(0byte~8byte)
[13]	RO	1'h0	timeout_int 0: Timeout occurs 1: No timeout occurs. NOTE: This bit is only valid in slave mode.
[12]	RO	1'h0	threshold_int 0: Received data has not reached the threshold 1: Received data has reached the threshold. NOTE: This bit is only valid for receiving data in slave mode. And the threshold is configurable. (unit: byte, range: 1byte~512 bytes)
[11]	RO	1'h0	cs_end_int 0: CS signal is low. 1: CS signal is high.
[10]	RO	1'h0	less_word_int 1: There is less than 1word data left in buffer, and the data will be uploaded to L2 memory after a while, then less_word interrupt will be generated. NOTE: 1) SPI Interface processes 4bytes data in a data transfer operation. If the number of data in RX buffer is less than 4 bytes, the data cannot be accessed. Therefore, a specific wait time(configured by Less_word and Timeout Register of SPI(Address:0x0020)) should be set prior to data transmission. If no data is received when the wait time is out, an interrupt will be generated (if the interrupt function is enabled) and the data left is accessed. This function is called "less_word" function for short. 2) The timeout function is valid in slave mode only, as the SPI Interface can know well the total number of data to be received in master mode.
[9]	RO	1'h0	m_in_progress_int 0: The data transmission operation is finished in master mode. 1: The data transmission operation is in progress in master mode.

Bits	Attributes	Reset	Description
[8]	RO	1'h0	trans_finish_int 0: The transmission operation has not been finished. 1: The transmission operation has been finished.
[7]	RO	1'h0	rx_buf_overnrun_int 1: RX buffer is written into even when it is full.
[6]	RO	1'h0	rx_buf_hfull_int 1: RX buffer of SPI interface is half full.
[5]	RO	1'h0	rx_buf_full_int 1: RX buffer of SPI interface is full.
[4]	RO	1'h1	rx_buf_emp_int 1: RX buffer of SPI interface is empty.
[3]	RO	1'h0	tx_buf_underrun_int 1: TX buffer is read even when it is empty.
[2]	RO	1'h1	tx_buf_hemp_int 1: TX buffer of SPI interface is half empty.
[1]	RO	1'h0	tx_buf_full_int 1: TX buffer of SPI interface is full.
[0]	RO	1'h1	tx_buf_emp_int 1: TX buffer of SPI interface is empty.

NOTE

This register is cleared after being read.

- SPI1/SPI2 Interrupt Enable Register (Address: 0x0008)

The SPI1/SPI2 controller supports various interrupts, each of which can be enabled or disabled independently by a specific bit of this register.

Bits	Attributes	Reset	Description
[31:14]	RO	20'h0	Reserved
[13]	RW	1'h0	timeout_int_en Timeout interrupt enable control 0: Disable 1: Enable NOTE: This bit is only valid in slave mode.
[12]	RW	1'h0	threshold_int_en Threshold interrupt enable control 0: Disable 1: Enable
[11]	RW	1'h0	cs_end_int_en 0: CS signal is high and interrupt function is disabled 1: CS signal is HIGH and interrupt function is enabled.
[10]	RW	1'h0	less_word_int_en less_word interrupt enable control 0: Disable 1: Enable NOTE: 1) SPI Interface processes 4bytes data in a data

Bits	Attributes	Reset	Description
			transfer operation. If the number of data in RX buffer is less than 4 bytes, the data cannot be accessed. Therefore, a specific wait time (configured by Less_word and Timeout Register of SPI(Address: 0x0020)) should be set prior to data transmission. If no more data is received when the wait time is out, an interrupt will be generated, which is called as “less_word interrupt” for short. And the data left is accessed. 2) The less_word function is valid in slave mode only, as the SPI Interface can know well the total number of data to be received in master mode.
[9]	RW	1'h0	m_in_progress_int_en m_in_progress interrupt enable control 0: Disable 1: Enable NOTE: The m_in_progress interrupt is generated when the data transmission operation is in progress in master mode.
[8]	RW	1'h0	trans_finish_int_en 0: To disable trans_finish interrupt 1: To enable trans_finish i interrupt NOTE: The trans_finish interrupt is generated when the transmission operation has been finished.
[7]	RW	1'h0	rx_buf_overflow_int_en 0: To disable rx_buf_overflow interrupt 1: To enable rx_buf_overflow interrupt NOTE: The rx_buf_overflow interrupt is generated if the RX buffer is written into when it is full.
[6]	RW	1'h0	rx_buf_hfull_int_en 0: To disable rx_buf_hfull interrupt 1: To enable rx_buf_hfull interrupt NOTE: The rx_buf_hfull interrupt is generated when the RX buffer of SPI Interface is half full.
[5]	RW	1'h0	rx_buf_full_int_en 0: To disable rx_buf_full interrupt 1: To enable rx_buf_full interrupt NOTE: The rx_buf_full interrupt is generated when the RX buffer of SPI Interface is full.
[4]	RW	1'h0	rx_buf_emp_int_en 0: To disable rx_buf_emp interrupt 1: To enable rx_buf_emp interrupt NOTE: The rx_buf_emp interrupt is generated when the RX buffer of SPI Interface is empty.
[3]	RW	1'h0	tx_buf_underrun_int_en 0: To disable tx_buf_underrun interrupt 1: To enable tx_buf_underrun interrupt

Bits	Attributes	Reset	Description
			NOTE: The tx_buf_underrun interrupt is generated if the TX buffer is read when it is empty.
[2]	RW	1'h0	tx_buf_hemp_int_en 0: To disable tx_buf_hemp interrupt 1: To enable tx_buf_hemp interrupt NOTE: The tx_buf_hemp interrupt is generated when the TX buffer of SPI Interface is half empty.
[1]	RW	1'h0	tx_buf_full_int_en 0: To disable tx_buf_full interrupt 1: To enable tx_buf_full interrupt NOTE: The tx_buf_full interrupt is generated when the TX buffer of SPI Interface is full.
[0]	RW	1'h0	tx_buf_emp_int_en 0: To disable tx_buf_emp interrupt 1: To enable tx_buf_emp interrupt NOTE: The tx_buf_emp interrupt is generated when the TX buffer of SPI Interface is empty.

● Data Count Register of SPI1/SPI2 (Address: 0x000C)

This register defines the total number of data to be sent or having been received in CPU/DMA mode.

Bits	Attributes	Reset	Description
[31:25]	RO	7'h0	Reserved
[24:16]	RO	9'h0	rxbyte_addr_l2_x The address in which the received data to be written in slave mode. (address: 0~511)
[15:0]	RW	16'h0	data_cnt Master mode: (unit: byte) To transmit data: total number of data to be sent. To receive data: total number of data to be received. Slave mode: To transmit data: total number of data to be sent. To receive data: total number of data having been received. NOTE: The value of these bits is decreased by step length of "1" if one byte data has been sent/received when the SPI Interface works in the first three modes; and the value of these bits is increased by step length of "1" if one byte of data has been received when the SPI Interface works in the fourth mode.

● External TX Buffer Register of SPI1/SPI2 (Address: 0x0010)

This register configures the relative parameters of external TX buffer (in L2 memory), such as buffer size, start address etc.

Bits	Attributes	Reset	Description
[31:17]	RW	15'h0	buf_len

Bits	Attributes	Reset	Description
			The size of the TX buffer (in L2 memory) assigned to SPI Interface.
[16]	RW	1'h0	dma_en 0: No instruction. 1: Enable DMA operation. (SPI interface is ready to access the TX buffer in L2 memory.)
[15:8]	RO	8'h0	Reserved
[7:1]	RW	7'h0	saddr Physical address of external buffer in L2 memory, from which the data will be sent.
[0]	RW	1'h0	buf_en 0: Disable the TX buffer in L2 memory (i.e. only the 8-byte TX buffer in SPI Interface is utilized.) 1: Enable the TX buffer in L2 memory.

● External RX Buffer Register of SPI1/SPI2 (Address: 0x0014)

Bits	Attributes	Reset	Description
[31:17]	RW	15'h0	buf_len The size of the RX buffer (in L2 memory) assigned to SPI Interface.
[16]	RW	1'h0	dma_en 0: No instruction. 1: Enable DMA operation. (SPI interface is ready to access the RX buffer in L2 memory.)
[15:8]	RO	8'h0	Reserved
[7:1]	RW	7'h0	saddr Physical address of external buffer in L2 memory, from which the data will be stored.
[0]	RW	1'h0	buf_en 0: Disable the RX buffer in L2 memory (i.e. only the 8-byte RX buffer in SPI Interface is utilized.) 1: Enable the RX buffer in L2 memory.

There are two data transmission ways:

3) DMA mode

TX operation: external RAM → through DMA bus → L2 memory → TX buffer of SPI1/SPI2 Interface → TX data line

RX operation: RX data line → RX buffer of SPI1/SPI2 Interface → L2 memory → through DMA bus → external RAM

4) CPU mode

TX operation: CPU writes data into TX buffer of SPI1/SPI2 Interface → TX data line

RX operation: RX data line → CPU reads data from RX buffer of SPI1/SPI2 Interface

The Output Data Register holds the data CPU writes into, and the Input Data Register holds the data from RX data lines.

● Output Data Register of SPI1/SPI2 (Address: 0x0018)

The Output Data Register holds the data CPU writes into, and the Input Data Register holds the data from RX data lines.

Bits	Attributes	Reset	Description
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Bits	Attributes	Reset	Description
[31:0]	WO	32'h0	output data CPU writes 4 bytes data to be transferred into this register in a writing operation. NOTE: 1) As the TX buffer of the SPI interface is 8-byte wide, the 4 bytes data CPU first writes into is stored in the lower 4 bytes of the buffer; next 4 bytes data is stored in the higher 4 bytes of the buffer; the third 4 bytes is stored in the lower 4 bytes of the buffer; and the fourth 4bytes is stored in the higher 5 bytes of the buffer; ... 2) High-order alignment is utilized. Bits[31:24] corresponds with Byte 3 and Byte 7; Bits[23:16], Byte 2 and Byte 6; Bits[15:8], Byte 1 and Byte 5; Bits[7:0], Byte 0 and Byte 4.

- SPI1/SPI2 Input Data Register (Address: 0x001C)

The Output Data Register holds the data CPU writes into, and the Input Data Register holds the data from RX data lines.

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	input data CPU reads 4-byte received data

- Less_word and timeout Register of SPI1/SPI2 (Address: 0x0020)

SPI1/SPI2 Interface processes 4bytes data in a data transfer operation. If the number of data in RX buffer is less than 4 bytes, the data cannot be accessed. Therefore, a specific wait time (configured by this register) should be set prior to data transmission. If no more data is received when timeout happens, an interrupt is generated (if the interrupt function is enabled) and the data left is accessed, which is called as timeout function.

The timeout function is valid in slave mode only, as the SPI1/SPI2 Interface can know well the total number of data to be received in master mode.

Bits	Attributes	Reset	Description
31:16	RW	16'd31	timeout The intervals of timeout
[15:0]	RW	16'hFFFF	less_word_cnt The waiting time for the data less than a word to be uploaded to L2 memory.

- Slave Output Data Delay Configuration Register of SPI1/SPI2 (Address: 0x0024)

This register is used to configure the delay of output data when the SPI1/SPI2 controller works in slave mode.

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15:0]	RW	16'h0	delay Output data delay when SPI in slave mode. (unit: GCLK)

● Threshold count register of SPI1/SPI2 (Address: 0x0028)

Bits	Attributes	Reset	Description
[31:10]	RO	22'h0	Reserved
[9:0]	RW	10'h0	threshold_cnt Threshold configuration of received data in slave mode. (Maximum value: 512 bytes) NOTE: If the threshold is set as n, a threshold interrupt will be generated as long as (n+1) bytes is transmitted.

8.4.6 Programmer's Guide

8.4.6.1 DMA mode

8.4.6.1.1 Master mode

Programmers should operate according to the following steps:

7. To set SPI1/SPI2 Interface to work in master mode and configure relative parameters (*Control Register of SPI1/SPI2 (Address: 0x0000)*);
8. To enable trans_fini interrupt (bit[8] of *SPI1/SPI2 Interrupt Enable Register (Address: 0x0008)*);
9. To write the total number of data to be sent/received into *Data Count Register of SPI1/SPI2 (Address: 0x000C)*;
10. To transmit data: to configure the TX buffer in L2 memory (*External TX Buffer Register of SPI1/SPI2 (Address: 0x0010)* and relative registers of L2 memory);
11. To receive data: to configure the RX buffer in L2 memory (*External RX Buffer Register of SPI1/SPI2 (Address: 0x0014)* and relative registers of L2 memory);
12. To check *Data Count Register of SPI1/SPI2 (Address: 0x000C)* to learn whether the data transmission has been finished or not;
13. To check bit[8] of *Status Register of SPI1/SPI2 (Address: 0x0004)* to learn whether the data transmission has been finished or not (this step is optional).

8.4.6.1.2 Slave Mode

Programmers should operate according to the following steps:

1. To set SPI/SPI2 Interface to work in slave mode and configure relative parameters (*Control Register of SPI1/SPI2 (Address: 0x0000)*);
2. To enable trans_fini interrupt and timeout interrupt (bit[10] and bit[8] of *SPI1/SPI2 Interrupt Enable Register (Address: 0x0008)*);
3. To transmit data: to write the total number of data to be sent into *Data Count Register of SPI1/SPI2 (Address: 0x000C)*;
4. To receive data: to set timeout value in *Less_word and Timeout Register of SPI1/SPI2 (Address: 0x0020)*;
5. To transmit data: to configure the TX buffer in L2 memory (*External TX Buffer Register of SPI1/SPI2 (Address: 0x0010)* and relative registers of L2 memory);
6. To receive data: to configure the RX buffer in L2 memory (*External RX Buffer Register of SPI1/SPI2 (Address: 0x0014)* and relative registers of L2 memory);
7. To check *Data Count Register of SPI1/SPI2 (Address: 0x000C)* to learn whether the data transmission has been finished or not;
8. To check bit[8] of *Status Register of SPI1/SPI2 (Address: 0x0004)* to learn whether the data transmission has been finished or not (this step is optional).

8.4.6.2 CPU Mode

8.4.6.2.1 Master mode

Programmers should operate according to the following steps:

1. To set SPI1/SPI2 Interface to work in master mode and configure relative parameters (*Control Register of SPI1/SPI2 (Address: 0x0000)*);
2. To enable trans_fini interrupt (bit[8] of *SPI1/SPI2 Interrupt Enable Register (Address: 0x0008)*); (This step is optional)
3. To transmit data: to enable TX_buf_hemp interrupt (bit[2] of *SPI1/SPI2 Interrupt Enable Register (Address: 0x0008)*);
4. To receive data: to enable RX_buf_hfull interrupt (bit[6] of *SPI1/SPI2 Interrupt Enable Register (Address: 0x0008)*);
5. To write the total number of data to be sent/received into *Data Count Register of SPI1/SPI2 (Address: 0x000C)*;
6. To transmit data: to write data into *Output Data Register of SPI1/SPI2 (Address: 0x0018)*;
7. To receive data: to read data from *SPI1/SPI2 Input Data Register (Address: 0x001C)* when a RX_buf_hfull interrupt is generated;
8. To check *Data Count Register of SPI1/SPI2 (Address: 0x000C)* to learn whether the data transmission has been finished or not;
9. To check bit[8] of *Status Register of SPI1/SPI2 (Address: 0x0004)* to learn whether the data transmission has been finished or not (this step is optional).

8.4.6.2.2 Slave Mode

Programmers should operate according to the following steps:

1. To set SPI1/SPI2 Interface to work in slave mode and configure relative parameters (*Control Register of SPI1/SPI2 (Address: 0x0000)*);
2. To enable trans_fini interrupt (bit[8] of *SPI1/SPI2 Interrupt Enable Register (Address: 0x0008)*); (This step is optional)
3. To transmit data: to enable TX_buf_hemp interrupt (bit[2] of *SPI1/SPI2 Interrupt Enable Register (Address: 0x0008)*);
4. To receive data: to enable RX_buf_hfull interrupt (bit[6] of *SPI1/SPI2 Interrupt Enable Register (Address: 0x0008)*);
5. To transmit data: to write the total number of data to be sent into *Data Count Register of SPI1/SPI2 (Address: 0x000C)*;
6. To receive data: to set timeout value in *Less_word and Timeout Register of SPI1/SPI2 (Address: 0x0020)*, and to enable timeout interrupt (bit[10] of *SPI1/SPI2 Interrupt Enable Register (Address: 0x0008)*);
7. To transmit data: to write data into *Output Data Register of SPI1/SPI2 (Address: 0x0018)*;
8. To receive data: to read data from *SPI1/SPI2 Input Data Register (Address: 0x001C)* when a RX_buf_hfull interrupt or a timeout interrupt has been generated;
9. To check *Data Count Register of SPI1/SPI2 (Address: 0x000C)* to learn whether the data transmission has been finished or not;
10. To check bit[8] of *Status Register of SPI1/SPI2 (Address: 0x0004)* to learn whether the data transmission has been finished or not (this step is optional).

8.5 USB

8.5.1 Overview

The USB 2.0 HS Host and Slave controller, fully compliant with *USB Specification Version 2.0 (HS)*, provides a 'dual-role' USB controller operating either as the host in point-to-point

communications with another USB function or as a function controller for a high-speed/full-speed USB peripheral.

The USB controller provides all the data transmission and receiving in the format of USB packets on the bus. That is, the USB controller interrupts the CPU only when endpoint data have been successfully transferred. When acting as the host for point-to-point communications, the controller additionally maintains a frame counter and automatically schedules SOF, Isochronous, Interrupt and Bulk transfers.

8.5.2 Features

Features of USB 2.0 HS Host and Slave controller are as follows:

- Compliant with USB Specification Version 2.0 (HS)
- Operates as the host in point-to-point communications with another USB function or as a function controller for a USB peripheral
- Supports UTMI+ Level 2 Transceiver Interface
- 6 Transmit/Receive endpoints in addition to Endpoint 0
- 6 DMA channels

The function and FIFO size of each endpoint are as follows:

ENDPOINT	FUNCTION(usually)	FIFO SIZE
Endpoint 0	Control endpoint	64 Bytes
Endpoint 1	Interrupt/Bulk/Isochronous endpoint	512 Bytes
Endpoint 2	Interrupt/Bulk/Isochronous endpoint	512 Bytes
Endpoint 3	Interrupt/Bulk/Isochronous endpoint	512 Bytes
Endpoint 4	Interrupt/Bulk/Isochronous endpoint	2048 Bytes
Endpoint 5	Interrupt/Bulk/Isochronous endpoint	512 Bytes
Endpoint 6	Interrupt/Bulk/Isochronous endpoint	512 Bytes

In addition to endpoint 0, other endpoints can be individually configured by software to handle either Bulk transfers (which also allows them to handle Interrupt transfers) or isochronous transfers.

8.5.3 Functional Description

8.5.3.1 Packet

Packet is the basic unit for USB data receiving and transmission. It is composed of SOP, SYNC, packet content and EOP. And a packet includes PID (packet identifier), address, frame number, data and CRC.

There are four kinds of USB packets: token packet, SOF Packet, data packet and handshake packet. Different packets consist of different parts, please refer to the following for more details.

➤Data packet

There are four kinds of PIDs for options: DATA0, DATA1, DATA2, DATAM. Depending on the transfer type, the maximum length of data packet is different. The address and frame number can be omitted.

➤Token Packet

The token packet is used to initiate a USB transferring.

OUT Token Packet: notify the device that a data packet is about to be sent out.

IN Token Packet: inform the device to return a data packet.

SETUP Token Packet is only used in Control Transfer. Like the OUT token packet, it also informs the device that a data packet is about to be sent out.

Only the DATA0 data packet is used after the SETUP token packet. It can only be sent to the control endpoint of the device and the device must receive it, while the OUT token packet does not have these restrictions.

➤SOF Packet

SOF packet is sent at the beginning of each frame in the form of broadcast. All USB full-speed devices and high-speed devices can receive SOF packets. For this kind of packet, PID equals to SOF.

➤Handshake packet

ACK: For IN transactions, handshake packet will be initiated by the host; For OUT, SETUP and PING transactions, it will be done by the device.

NAK: In the data phase, for IN transactions, handshake packet will be issued by the device. In the handshake phase, for OUT and PING transactions, it will also be issued by the device; the host never sends NAK packets.

8.5.3.2 Transferring type

There are four types of transfers: bulk transfer, isochronous transfer, interrupt transfer, and control transfers.

➤Bulk Transfer: aperiodic

Bulk transfer is suitable for communication of large-capacity data which can occupy any kinds of bandwidth and can tolerate delay, such as USB printers, scanners, mass storage devices, etc.

➤Interrupt Transfer: Periodic, low frequency

Interrupt transfer allows limited-latency communication, such as mouse, keyboard, trackball, etc.

An interrupt transmission guarantees the frequency of polling. The interrupt endpoint should report its polling interval in the endpoint descriptor, and the host should ensure that a transfer is scheduled within a range less than this interval.

➤Isochronous Transfer: Periodic

Isochronous transfer means continuous transmission and is used to transmit information related to timing, and save time stamp information in the data, such as audio and video equipment.

The isochronous transfer is used in occasions with a large amount of data and high requirements to real-time communication such as audio equipment, video equipment, which are sensitive to data delay. The 100% correctness of audio or video equipment data is not required and a small amount of data errors can be tolerated, mainly to ensure that the data cannot be paused. When the datum is wrong, the operation stops. Therefore, there is no response packet in the isochronous transfer and whether the data is correct or not is confirmed by the CRC check of the data.

➤Control Transfers: Aperiodic

This kind of transfer is used for command and status transmissions.

Control transfer can be divided into three stages: (1) setup process (2) data process (optional) (3) status process. Every USB device must have a control endpoint that supports control transfers for command and status transfer. The USB host driver will communicate with the control endpoint of the USB device through control transfer to complete the enumeration and configuration of the USB device.

The control transfer is bidirectional and there must be control endpoints with specific endpoint numbers of IN and OUT to complete the control transfer in both directions.

8.5.4 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

The USB Controller registers accessible to the CPU are defined in this section.

NOTE

Set means that the bit can only be written to set it.

Clear means that the bit can only be written to clear it.

The following are USB control registers.

Register No.	Register Address	Register Name
1	0x0800,0058	Refer to Section 3.2.5 USB Operate Mode and I2S Mode Control
2	0x2020,0000	Function Address Register (FADDR)
3	0x2020,0001	Power Control Register (POWER)
4	0x2020,0002	Interrupt Register 0 (IntrTX)
5	0x2020,0004	Interrupt Register 1 (IntrRX)
6	0x2020,0006	Interrupt Enable Register for IntrTX (IntrTXE)
7	0x2020,0008	Interrupt Enable Register for IntrRX (IntrRXE)
8	0x2020,000A	Interrupt Register for Common USB Interrupts (IntrUSB)
9	0x2020,000B	Interrupt Enable Register for IntrUSB(IntrUSBE)
10	0x2020,000C	Frame Number Register (FRAME)
11	0x2020,000E	Index Register (INDEX)
12	0x2020,000F	Testmode Register (TESTMODE)
13	0x2020,0060	USB Device Control Register (DevCtl)
14	0x2020,0010	Maximum Packet Size For Host/Client TX Endpoint Register (TXMAXP)
15	0x2020,0012	Control and Status Register 0 for Endpoints (TXCSR)
16	0x2020,0014	Maximum Packet Size For Host/Client RX Endpoint Register (RXMAXP)
17	0x2020,0016	Control and Status Register 1 for Endpoints (RXCSR)
18	0x2020,0304 ~ 0x2020,0318	Request Packet Count Register for RX Endpoints
19	0x2020,0018	EndpointX Count Register

Register No.	Register Address	Register Name
20	0x2020,001A	TX Endpoint Type Register (TXTYPE)
21	0x2020,001B	NAK Response Timeout on Endpoint 0(NAKLIMIT0)/TX interval Register (TXINTERVAL)
22	0x2020,001C	RX Endpoint Type Register (RXTYPE)
23	0x2020,001D	RX Interval Register (RXINTERVAL)
24	0x2020,001F	Configuration Information Register (ConfigData)/FIFO Size Register(FIFOSIZE)

8.5.5 Register Definition

- Function Address Register (FADDR) (Address: 0x2020,0000)

FADDR is an 8-bit register that should be written with the 7-bit address of the peripheral part of the transaction.

When the USB controller is being used in host mode (Bit[2] of [USB Device Control Register \(Address: 0x2020,0060\)](#) is set to 1), this register should be set to the value sent in a SET_ADDRESS command during device enumeration as the address for the peripheral device. When the USB controller is being used in client mode (Bit[2] of [USB Device Control Register \(Address: 0x2020,0060\)](#) is cleared), this register should be written with the address received through a SET_ADDRESS command, which will then be used for decoding the function address in subsequent token packets.

The new address will not take effect immediately as the host will still be using the old address for the Status stage of the device request. The USB controller will continue to use the old address for decoding packets until the device request has completed.

Bits	Attributes	Reset	Description
[7]	RO	1'h0	Reserved
[6:0]	RW	7'h0	faddr The function address

- Power Control Register (POWER) (Address: 0x2020,0001)

POWER is an 8-bit register that is used for controlling SUSPEND and RESUME signalings, also for some basic operational instructions of the USB controller.

Bits	Attributes	Reset	Description
[7]	RW	1'h0	iso update 0: No instruction 1: The controller will wait for an SOF token from the time TxPktRdy is set before sending the packet. If an IN token is received before an SOF token, a zero length data packet will be sent. NOTE: 1) Only valid in client mode. 2) This bit only affects endpoints performing Isochronous transfers. 3) This bit affects all IN Isochronous endpoints. It is normally used as a method of ensuring a "clean" start-up of an IN Isochronous pipe.
[6]	RO	1'h0	Reserved

Bits	Attributes	Reset	Description
[5]	RW	1'h1	hs enab 0: To operate in full-speed mode. 1: The controller will negotiate for high-speed mode when the device is reset by the hub.
[4]	RO	1'h0	hs mode 1: High-speed mode is successfully negotiated during USB reset. NOTE: In client mode, this bit becomes valid when USB reset is completed (as indicated by USB reset interrupt). In host mode, this bit becomes valid when Reset bit is cleared. This bit remains valid for the duration of the session.
[3]	RO	1'h0	reset 0: No instruction 1: Reset signaling is present on the bus. NOTE: 1) This bit can be used to determine when Reset signaling is present on the USB. It will go high when Reset signaling is detected and remain high until the bus reverts to an idle state. 2) This bit is read-write in host mode.
[2]	RW	1'h0	resume 0: No instruction. 1: To generate Resume signaling when the function is in Suspend mode. The CPU should clear this bit after 10 ms (a maximum of 15 ms) to end Resume signaling. NOTE: In host mode, this bit is also automatically set when Resume signaling from the target is detected while the USB controller is suspended.
[1]	RO	1'h0	sus mode In host mode, this bit is set to enter suspend mode. In client mode, this bit is set on entry into suspend mode. It is cleared when the CPU reads the interrupt register, or sets the Resume bit.
[0]	RW	1'h0	en_suspendm 0: No instruction 1: To enable the SUSPENDM signal. NOTE: This bit enables the SUSPENDM signal to indicate when the USB controller is in Suspend mode. With this set to 1, the SUSPENDM signal will be low whenever the USB controller is in Suspend mode and high at all other times.

● Interrupt Register 0 (IntrTX) (Address: 0x2020,0002)

Interrupt Register 0 indicates which interrupts are currently active for Endpoint 0, TX Endpoint 1 ~ TX Endpoint 6.

NOTE

- 1) Bits relating to endpoints that have not been configured will always remain 0s.
- 2) All active interrupts are cleared when this register is read.

Bits	Attributes	Reset	Description
[15:7]	RO	9'h0	Reserved
[6]	RO	1'h0	intrtx 0: Endpoint 6 interrupt is inactive. 1: Endpoint 6 interrupt is active.
[5]	RO	1'h0	intrtx 0: Endpoint 5 interrupt is inactive. 1: Endpoint 5 interrupt is active.
[4]	RO	1'h0	intrtx 0: Endpoint 4 interrupt is inactive. 1: Endpoint 4 interrupt is active.
[3]	RO	1'h0	intrtx 0: Endpoint 3 interrupt is inactive. 1: Endpoint 3 interrupt is active.
[2]	RO	1'h0	intrtx 0: Endpoint 2 interrupt is inactive. 1: Endpoint 2 interrupt is active.
[1]	RO	1'h0	intrtx 0: Endpoint 1 interrupt is inactive. 1: Endpoint 1 interrupt is active.
[0]	RO	1'h0	intrtx 0: Endpoint 0 interrupt is inactive. 1: Endpoint 0 interrupt is active.

● Interrupt Register 1 (IntrRX) (Address: 0x2020,0004)

Interrupt Register 1 indicates which interrupts are currently active for RX Endpoint 1 ~ RX Endpoint 6.

Programmers should note:

- 1) Bits relating to endpoints that have not been configured will always remain 0s.
- 2) All active interrupts are cleared when this register is read.

Bits	Attributes	Reset	Description
[15:7]	RO	9'h0	Reserved
[6]	RW	1'h0	intrrx 0: RX Endpoint 6 interrupt is inactive. 1: RX Endpoint 6 interrupt is active.
[5]	RW	1'h0	intrrx 0: RX Endpoint 5 interrupt is inactive. 1: RX Endpoint 5 interrupt is active.
[4]	RW	1'h0	intrrx 0: RX Endpoint 4 interrupt is inactive. 1: RX Endpoint 4 interrupt is active.
[3]	RW	1'h0	intrrx 0: RX Endpoint 3 interrupt is inactive. 1: RX Endpoint 3 interrupt is active.
[2]	RW	1'h0	intrrx 0: RX Endpoint 2 interrupt is inactive.

Bits	Attributes	Reset	Description
			1: RX Endpoint 2 interrupt is active.
[1]	RW	1'h0	intrrx 0: RX Endpoint 1 interrupt is inactive. 1: RX Endpoint 1 interrupt is active.
[0]	RO	1'h0	Reserved

● Interrupt Enable Register for IntrTX (IntrTXE) (Address: 0x2020,0006)

IntrTXE register provides interrupt enable bits for the interrupts in IntrTx. On reset, the bits corresponding to Endpoint 0 and the TX endpoints are set to 1s, while the remaining bits are set to 0s.

Programmers should note:

Bits relating to endpoints that have not been configured will always remain 0s.

Bits	Attributes	Reset	Description
[15:7]	RO	9'h0	Reserved
[6]	RO	1'h1	intrtxe 0: To disable TX Endpoint 6 interrupt. 1: To enable TX Endpoint 6 interrupt.
[5]	RO	1'h1	intrtxe 0: To disable TX Endpoint 5 interrupt. 1: To enable TX Endpoint 5 interrupt.
[4]	RO	1'h1	intrtxe 0: To disable TX Endpoint 4 interrupt. 1: To enable TX Endpoint 4 interrupt.
[3]	RO	1'h1	intrtxe 0: To disable TX Endpoint 3 interrupt. 1: To enable TX Endpoint 3 interrupt.
[2]	RO	1'h1	intrtxe 0: To disable TX Endpoint 2 interrupt. 1: To enable TX Endpoint 2 interrupt.
[1]	RO	1'h1	intrtxe 0: To disable TX Endpoint 1 interrupt. 1: To enable TX Endpoint 1 interrupt.
[0]	RO	1'h1	intrtxe 0: To disable Endpoint 0 interrupt. 1: To enable Endpoint 0 interrupt.

● Interrupt Enable Register for IntrRX (IntrRXE) (Address: 0x2020,0008)

IntrRXE register provides interrupt enable bits for the interrupts in IntrRx. On reset, the bits corresponding to the Rx endpoints are set to 1s, while the remaining bits are set to 0s.

Programmers should note:

Bits relating to endpoints that have not been configured will always remain 0s.

Bits	Attributes	Reset	Description
[15:7]	RO	9'h0	Reserved
[6]	RO	1'h1	intrrx RX Endpoint 6 interrupt control 0: Disable 1: Enable

Bits	Attributes	Reset	Description
[5]	RO	1'h1	intrrx RX Endpoint 5 interrupt control 0: Disable 1: Enable
[4]	RO	1'h1	intrrx RX Endpoint 4 interrupt control 0: Disable 1: Enable
[3]	RO	1'h1	intrrx RX Endpoint 3 interrupt control 0: Disable 1: Enable
[2]	RO	1'h1	intrrx RX Endpoint 2 interrupt control 0: Disable 1: Enable
[1]	RO	1'h1	intrrx RX Endpoint 1 interrupt control 0: Disable 1: Enable
[0]	RO	1'h0	Reserved

- Interrupt Register for Common USB Interrupts (IntrUSB) (Address: 0x2020,000A)
This register indicates which USB interrupts are currently active. All active interrupts will be cleared when this register is read.

Bits	Attributes	Reset	Description
[7]	RO	1'h1	vbus error 0: VBus does not drop below the VBus Valid threshold during a session. 1: VBus drops below the VBus Valid threshold during a session. NOTE: Valid only when the USB controller is "A" device.
[6]	RO	1'h1	sess req 0: Session Request signaling has not been detected. 1: Session Request signaling has been detected. NOTE: Valid only when the USB controller is "A" device.
[5]	RO	1'h1	discon In host mode: 0: A device disconnect has not been detected. 1: A device disconnect has been detected. In client mode: 0: A session does not end. 1: A session ends.
[4]	RO	1'h1	conn 0: A device connection has not been detected.

Bits	Attributes	Reset	Description
			1: A device connection has been detected. NOTE: Only valid in host mode.
[3]	RO	1'h1	sof 0: No new frame 1: A new frame starts.
[2]	RO	1'h1	reset/babble In host mode: 0: Babble has not been detected. 1: Babble has been detected. NOTE: Only active after first SOF has been sent. In client mode: 0: Reset signaling has not been detected on the bus. 1: Reset signaling has been detected on the bus.
[1]	RO	1'h1	resume 0: Resume signaling has not been detected on the bus when the USB controller is in suspend mode. 1: Resume signaling has been detected on the bus when the USB controller is in suspend mode.
[0]	RO	1'h1	suspend 0: Suspend signaling has not been detected on the bus. 1: Suspend signaling has been detected on the bus. NOTE: Only valid in client mode.

- Interrupt Enable Register for IntrUSB(IntrUSBE) (Address: 0x2020,000B)
This register provides interrupt enable bits for each of the interrupts in IntrUSB.

Bits	Attributes	Reset	Description
[7]	RW	1'h0	vbus error 0: To disable Vbus Error interrupt. 1: To enable Vbus Error interrupt.
[6]	RW	1'h0	sess req 0: To disable Sees Req interrupt. 1: To enable Sees Req interrupt.
[5]	RW	1'h0	discon 0: To disable Discon interrupt. 1: To enable Discon interrupt.
[4]	RW	1'h0	conn 0: To disable Conn interrupt. 1: To enable Conn interrupt.
[3]	RW	1'h0	sof 0: To disable SOF interrupt. 1: To enable SOF interrupt.
[2]	RW	1'h1	reset/babble 0: To disable Reset/Babble interrupt. 1: To enable Reset/Babble interrupt.
[1]	RW	1'h1	resume 0: To disable Resume interrupt. 1: To enable Resume interrupt.

Bits	Attributes	Reset	Description
[0]	RW	1'h0	suspend 0: To disable Suspend interrupt. 1: To enable Suspend interrupt.

● Frame Number Register (FRAME) (Address: 0x2020,000C)

This register holds the last received frame number.

Bits	Attributes	Reset	Description
[15:11]	RO	5'h0	Reserved
[10:0]	RO	11'h0	fra_num Frame number Bit[10] is the MSB, and bit[0] is the LSB.

● Index Register (INDEX) (Address: 0x2020,000E)

INDEX is a 4-bit register that determines which endpoint control/status registers are accessed. Before accessing an endpoint's control/status registers, the endpoint number should be written to this register to ensure that the correct control/status registers appear in the memory map.

Bits	Attributes	Reset	Description
[7:4]	RO	4'h0	Reserved
[3:0]	RW	4'h0	index Index for selecting the endpoint's control/status registers. Bit[3] is the MSB, and bit[0] is the LSB.

● Testmode Register (TESTMODE) (Address: 0x2020,000F)

This register is primarily used to put the USB controller into one of the four test modes for High-speed operation described in the USB 2.0 specification – in response to a SET FEATURE: TESTMODE command. It is not used in normal operation.

Bits	Attributes	Reset	Description
[7]	RW	1'h0	force_host The CPU sets this bit to instruct the USB controller to enter host mode when the Session bit is set, regardless of whether it is connected to any peripheral. The state of the CID input, Host Disconnect and LineState signals are ignored. The USB controller will then remain in host mode until the Session bit is cleared, even if a device is disconnected. And if this bit remains set, the USB controller will re-enter host mode the next time the Session bit is set. While in this mode, the status of the HOSTDISCON signal from the PHY may be read from bit 7 of the DevCtl register.
[6]	RW	1'h0	fifo_access 0: No instruction 1: To transfer the packet in the Endpoint 0 TX FIFO to the Endpoint 0 RX FIFO. It is cleared automatically.

Bits	Attributes	Reset	Description															
[5]	RW	1'h0	The CPU sets this bit either in conjunction with Force_Host (bit[7]) or to force the USB controller into full-speed mode when it receives a USB reset. The operating speed is determined from the Force_HS and Force_FS bits as follows: <table><tr><th>Force_HS</th><th>Force_FS</th><th>Operating speed</th></tr><tr><td>0</td><td>0</td><td>Low speed</td></tr><tr><td>0</td><td>1</td><td>Full speed</td></tr><tr><td>1</td><td>0</td><td>High speed</td></tr><tr><td>1</td><td>1</td><td>undefined</td></tr></table>	Force_HS	Force_FS	Operating speed	0	0	Low speed	0	1	Full speed	1	0	High speed	1	1	undefined
Force_HS	Force_FS	Operating speed																
0	0	Low speed																
0	1	Full speed																
1	0	High speed																
1	1	undefined																
[4]	RW	1'h0	Reserved															
[3]	RW	1'h0	test_packet (High-speed mode) 0: No instruction 1: To enter the Test_Packet test mode. In this mode, the USB controller repetitively transmits on the bus a 53-byte test packet, the form of which is defined in the Universal Serial Bus Specification Revision 2.0, Section 7.1.20. NOTE: The test packet has a fixed format and must be loaded into the Endpoint 0 FIFO before the test mode is entered.															
[2]	RW	1'h0	test_k (High-speed mode) 0: No instruction 1: To enter the Test_K test mode. In this mode, the USB controller transmits continuous Ks on the bus.															
[1]	RW	1'h0	test_j (High-speed mode) 0: No instruction 1: To enter the Test_J test mode. In this mode, the USB controller transmits continuous Js on the bus.															
[0]	RW	1'h0	test_se0_nak (High-speed mode) 0: No instruction 1: To enter the Test_SE0_NAK test mode. In this mode, the USB controller remains in high-speed mode but responds to any valid IN token with a NAK.															

NOTE

Only one of Bits[6:0] can be set at any time.

- USB Device Control Register (DevCtl) (Address: 0x2020,0060)

This register is used to select whether the USB controller is operating in Peripheral mode or in Host mode, and for controlling and monitoring the USB VBus line.

Bits	Attributes	Reset	Description
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Bits	Attributes	Reset	Description
[7]	RO	1'h0	b-device 0: "A" device. (The USB controller is used as the host.) 1: "B" device. (The USB controller is used as the client.) NOTE: 1) This is only valid when a session is in progress. 2) If the USB controller is in Force_Host mode (i.e. a session has been started with Testmode.), this bit will indicate the state of the HOSTDISCON input signal from the PHY.
[6]	RO	1'h0	fsdev 0: A full-speed or high-speed device has not been detected in being connected to the port. 1: A full-speed or high-speed device has been detected in being connected to the port. (High-speed devices are distinguished from full-speed by checking for high-speed chirps when the device is reset.) NOTE: Only valid in host mode.
[5]	RO	1'h0	Reserved
[4:3]	RO	2'h0	vbus Encoding the current VBus level. 2'h0: Below SessionEnd. 2'h1: Above SessionEnd, below AValid. 2'h2: Above AValid, below VBusValid. 2'h3: Above VBusValid
[2]	RO	1'h0	host mode 0: In client mode 1: In host mode
[1]	RO	1'h0	host req 0: No instruction 1: The USB controller will initiate the Host Negotiation when suspend mode is entered. It is cleared when Host Negotiation is completed. NOTE: "B" device only.
[0]	RW	1'h0	session In host mode: 0: To end a session. 1: To start a session. In client mode: 0: A session is ended. 1: A session is started. NOTE: 1) When operating as an "A" device, this bit is set or cleared by the CPU. When operating as a "B" device, this bit is set or cleared by the USB controller. It is also set by the CPU to initiate the Session Request Protocol, or cleared by the CPU when in suspend mode to perform a software disconnect.

Bits	Attributes	Reset	Description
			2) Clearing this bit when the USB controller is not suspended will result in undefined behavior.

- Maximum Packet Size For Host/Client TX Endpoint Register (TXMAXP) (Address: 0x2020,0010)

This register defines the maximum amount of data that can be transferred through the selected TX Endpoint in a single operation. There is a TXMAXP register for each TX endpoint (except Endpoint 0).

If this register is changed after packets have been sent from the endpoint, the TX endpoint FIFO should be completely flushed (using the FlushFIFO bit in TXCSR) after writing the new value to this register.

Bits	Attributes	Reset	Description
[15:11]	RO	5'h0	Reserved
[10:0]	RW	11'h0	The maximum payload transmitted in a single transaction. (unit: byte) The value can be up to 1024 bytes but is subject to the constraints placed by the USB Specification on packet sizes for Bulk, Interrupt and Isochronous transfers in full-speed and high-speed operations. NOTE: 1) The value written into these bits must match the value given in the wMaxPacketSize field of the Standard Endpoint Descriptor for the associated endpoint. A mismatch could cause unexpected results. 2) The total amount of data represented by the value written to this register must not exceed the FIFO size for the TX endpoint, and should not exceed half the FIFO size if double-buffering is required.

- Control and Status Register 0 for Endpoints (TXCSR) (Address: 0x2020,0012)

This register has different bit definitions with [Index Register \(Address: 0x2020,000E\)](#) value set to 0 and set to 1~6, in host mode and in client mode.

With Index Register Value set to 0 in HOST mode:

Bits	Attributes	Reset	Description
[15:12]	RO	4'h0	Reserved
[11]	RW	1'h0	dis ping 0: No instruction 1: To instruct the USB controller not to issue PING tokens in data and status phases of a high-speed control transfer (for use with devices that do not respond to PING).
[10:9]	RO	2'h0	Reserved
[8]	RW	1'h0	flushfifo 0: No instruction 1: To flush the next packet to be transmitted/read from the Endpoint 0 FIFO. The FIFO pointer is reset

Bits	Attributes	Reset	Description
			and the TxPktRdy/RxPktRdy bit (Bit[1]/Bit[0]) is cleared. NOTE: This bit should only be used when TxPktRdy/RxPktRdy (Bit[1]/Bit[0]) is set.
[7]	RO	1'h0	nak timeout 1: Endpoint 0 is halted following the receipt of NAK responses for longer than the time set by the NAKLimit0 register. The CPU should clear this bit to allow the endpoint to continue.
[6]	RW	1'h0	statuspkt 0: No instruction 1: To perform a status stage transaction. NOTE: This bit is set at the same time as the TxPktRdy Bit (Bit[1]) or ReqPkt bit (Bit[5]) is set. Setting this bit ensures that the data toggle is set to 1 so that a DATA1 packet is used for the Status Stage transaction.
[5]	RW	1'h0	reqpkt 0: No instruction 1: To request an IN transaction. It is cleared when RxPktRdy Bit (Bit[0]) is set.
[4]	RO	1'h0	error 0: No error. 1: Three attempts have been made to perform a transaction with no response from the peripheral. NOTE: The CPU should clear this bit. An interrupt is generated when this bit is set.
[3]	RW	1'h0	setuppkt 0: No instruction 1: To send a SETUP token instead of an OUT token for the transaction. NOTE: This bit is set at the same time as the TxPktRdy Bit (Bit[1]) is set.
[2]	RO	1'h0	rxstall 0: A STALL handshake has not been received. 1: A STALL handshake has been received. NOTE: The CPU should clear this bit.
[1]	RO	1'h0	txpktrdy 0: The CPU has not loaded a data packet into the FIFO. 1: The CPU has loaded a data packet into the FIFO. NOTE: This bit is cleared automatically when the data packet has been transmitted. An interrupt is generated (if enabled) when the bit is cleared.
[0]	RO	1'h0	rxpktrdy 0: A data packet has not been received. 1: A data packet has been received. NOTE: An interrupt is generated when this bit is set. The CPU clears this bit when the packet has been read from the FIFO.

With Index Register Value set to 0 in CLIENT mode:
(Address: 0x2020,0012)

Bits	Attributes	Reset	Description
[15:9]	RO	7'h0	Reserved
[8]	RW	1'h0	flushfifo 0: No instruction 1: To flush the next packet to be transmitted/read from the Endpoint 0 FIFO. The FIFO pointer is reset and the TxPktRdy/RxPktRdy Bit (Bit[1] /Bit[0]) is cleared. NOTE: This bit should only be used when TxPktRdy/RxPktRdy (Bit[1] /Bit[0]) is set.
[7]	RW	1'h0	servicedsetupend 0: No instruction 1: To clear the SetupEnd Bit(Bit[4]). NOTE: This bit is cleared automatically.
[6]	RW	1'h0	servicedrxpktrdy 0: No instruction 1: To clear the RxPktRdy Bit (Bit[0]). NOTE: This bit is cleared automatically.
[5]	RW	1'h0	sendstall 0: No instruction 1: To terminate current transaction. The STALL handshake will be transmitted and then this bit will be cleared automatically. NOTE: The FIFO should be flushed before this bit is set.
[4]	RO	1'h0	setupend 1: A control transaction ends before the DataEnd bit has been set. An interrupt will be generated and the FIFO flushed at this time. NOTE: This bit is cleared by the CPU writing a 1 to the ServicedSetupEnd Bit (Bit[7]).
[3]	RW	1'h0	dataend The CPU set this bit: 1. when setting TxPktRdy (Bit[1]) for the last data packet; 2. when clearing RxPktRdy (Bit[0]) after unloading the last data packet; 3. when setting TxPktRdy (Bit[1]) for a zero length data packet. NOTE: It is cleared automatically.
[2]	RO	1'h0	sentstall 0: A STALL handshake has not been transmitted. 1: A STALL handshake has been transmitted. NOTE: The CPU should clear this bit.
[1]	RO	1'h0	txpktrdy 0: The CPU has not loaded a data packet into the FIFO. 1: The CPU has loaded a data packet into the FIFO.

Bits	Attributes	Reset	Description
			NOTE: It is cleared automatically when the data packet has been transmitted. An interrupt is generated (if enabled) when this bit is cleared.
[0]	RO	1'h0	rxpktrdy 0: A data packet has not been received. 1: A data packet has been received. NOTE: An interrupt is generated when this bit is set. The CPU clears this bit by setting the ServicedRxPktRdy Bit (Bit[6]).

With Index Register Value set to 1~6 in HOST mode:
(Address: 0x2020,0012)

Bits	Attributes	Reset	Description
[15]	RW	1'h0	autoset 0: No instruction 1: TxPktRdy will be automatically set when data of the maximum packet size (value in TXMAXP) is loaded into the TX FIFO. NOTE: 1) If a packet of less than the maximum packet size is loaded, then TxPktRdy will have to be set manually. 2) This bit should not be set for either high-bandwidth isochronous endpoints or high-bandwidth Interrupt endpoints.
[14]	RO	1'h0	Reserved
[13]	RW	1'h0	mode Endpoint direction. 0: RX 1: TX NOTE: This bit will not have any effect where the same endpoint FIFO is used for both TX and RX transactions.
[12]	RW	1'h0	damreqenab 0: To disable the DMA request. 1: To enable the DMA request for the TX Endpoint.
[11]	RW	1'h0	frcdatatog 0: No instruction 1: To force the endpoint data toggle to switch and the data packet to be cleared from the FIFO, regardless of whether an ACK was received. This can be used by Interrupt Tx endpoints that are used to communicate rate feedback for Isochronous endpoints.
[10]	RO	1'h0	Reserved
[9]	RO	1'h0	datatog This bit indicates the current data toggle status of the endpoint.
[8]	WO	1'h0	setdatatog

Bits	Attributes	Reset	Description
			1: To set the endpoint data toggle to 1.
[7]	RO	1'h0	nak timeout/incomptx Bulk Endpoints: 1: TX endpoint is halted following the receipt of NAK responses for longer than the time set as the NAKLimit by the TxInterval register. NOTE: The CPU should clear this bit to allow the endpoint to continue. High-bandwidth Interrupt endpoints: 1: No response is received from the device to which the packet is being sent.
[6]	RW	1'h0	clrdatatog 0: No instruction 1: To reset the endpoint data toggle to 0.
[5]	RO	1'h0	rxstall 0: A STALL handshake has not been received. 1: A STALL handshake has been received. NOTE: When this bit is set, any DMA request that is in progress is stopped, the FIFO is completely flushed and the TxPktRdy bit (bit[0]) is cleared. The CPU should clear this bit.
[4]	RO	1'h0	Reserved
[3]	RW	1'h0	flushfifo 0: No instruction 1: To flush the latest packet from the endpoint TX FIFO. The FIFO pointer is reset, the TxPktRdy Bit (Bit[0]) is cleared, and an interrupt is generated. NOTE: 1) FlushFIFO should only be used when TxPktRdy (Bit[0]) is set. 2) If the FIFO is double-buffered, FlushFIFO may need to be set twice to completely clear the FIFO.
[2]	RO	1'h0	error 0: No error 1: Three attempts have been made to send a packet and no handshake packet has been received. NOTE: 1) When this bit is set, an interrupt is generated, TxPktRdy bit (bit[0]) is cleared and the FIFO completely flushed. The CPU should clear this bit. 2) Valid only when the endpoint is operating in Bulk or Interrupt mode.
[1]	RO	1'h0	fifonotempty 1: There is at least 1 packet in the TX FIFO.
[0]	RO	1'h0	txpktrdy The CPU sets this bit after loading a data packet into the FIFO. It is cleared automatically when a data packet has been transmitted. An interrupt is also

Bits	Attributes	Reset	Description
			generated at this point (if enabled). This bit is also automatically cleared prior to loading a second packet into a double-buffered FIFO.

With Index Register Value set to 1~6 in CLIENT mode:
(Address: 0x2020,0012)

Bits	Attributes	Reset	Description
[15]	RW	1'h0	autoset 0: No instruction 1: TxPktRdy will be automatically set when data of the maximum packet size (value in TXMAXP register (Address: 0x2020,0010)) is loaded into the TX FIFO. NOTE: 1) If a packet of less than the maximum packet size is loaded, then TxPktRdy (Bit[0]) will have to be set manually. 2) This bit should not be set for either high-bandwidth Isochronous endpoints or high-bandwidth Interrupt endpoints.
[14]	RW	1'h0	iso 0: To enable the TX endpoint for Bulk or Interrupt transfers. 1: To enable the TX endpoint for Isochronous transfers.
[13]	RW	1'h0	mode Endpoint direction 0: RX 1: TX NOTE: This bit will not have any effect where the same endpoint FIFO is used for both TX and RX transactions.
[12]	RW	1'h0	damreqenab 0: To disable the DMA request. 1: To enable the DMA request for the Tx Endpoint.
[11]	RW	1'h0	frcdatatog 0: No instruction 1: To force the endpoint data toggle to switch and the data packet to be cleared from the FIFO, regardless of whether an ACK was received. This can be used by Interrupt TX endpoints that are used to communicate rate feedback for Isochronous endpoints.
[10]	RO	1'h0	Reserved
[9]	RO	1'h0	datatog This bit indicates the current data toggle status of the endpoint.
[8]	WO	1'h0	setdatatog 1: To set the endpoint data toggle to 1.
[7]	RO	1'h0	incomptx

Bits	Attributes	Reset	Description
			When the endpoint is being used for high-bandwidth Isochronous /Interrupt transfers, this bit is set to indicate where a large packet has been split into 2 or 3 packets for transmission but insufficient IN tokens have been received to send all the parts. NOTE: In anything other than a high-bandwidth transfer, this bit will always return 0.
[6]	SET	1'h0	clrdatatog 0: No instruction 1: To reset the endpoint data toggle to 0.
[5]	RO	1'h0	sentstall 0: A STALL handshake has not been transmitted. 1: A STALL handshake has been transmitted. The FIFO is flushed and the TxPktRdy Bit (Bit[0]) is cleared. NOTE: The CPU should clear this bit.
[4]	RW	1'h0	sendstall 0: No instruction 1: To issue a STALL handshake to an IN token. NOTE: 1) The CPU clears this bit to terminate the stall condition. 2) The FIFO should be flushed before SendStall is set. 3) This bit has no effect where the endpoint is being used for Isochronous transfers.
[3]	SET	1'h0	flushfifo 0: No instruction 1: To flush the latest packet from the endpoint TX FIFO. The FIFO pointer is reset, the TxPktRdy Bit (Bit[0]) is cleared, and an interrupt is generated. NOTE: 1) FlushFIFO should only be used when TxPktRdy (Bit[0]) is set. 2) If the FIFO is double-buffered, FlushFIFO may need to be set twice to completely clear the FIFO. 3) This bit may be set simultaneously with TxPkeRdy bit (bit[0]) to abort the packet that is currently being loaded into the FIFO.
[2]	RO	1'h0	underrun 1: An IN token is received when the TxPktRdy Bit has not been set. NOTE: The CPU should clear this bit.
[1]	RO	1'h0	fifonotempty 1: There is at least 1 packet in the TX FIFO.
[0]	RO	1'h0	txpktRdy The CPU sets this bit after loading a data packet into the FIFO. It is cleared automatically when a data packet has been transmitted. An interrupt is also generated at this point (if enabled). This bit is also

Bits	Attributes	Reset	Description
			automatically cleared prior to loading a second packet into a double-buffered FIFO.

- Maximum Packet Size for Host/Client RX Endpoint Register (RXMAXP) (Address: 0x2020,0014)

This register defines the maximum amount of data that can be transferred through the selected RX Endpoint in a single operation. There is a RXMAXP register for each RX endpoint (except Endpoint 0).

Bits	Attributes	Reset	Description
[15:11]	RO	5'h0	Reserved
[10:0]	RW	11'h0	maximum payload /transaction The maximum payload transmitted in a single transaction. (unit: byte) The value can be up to 1024 bytes but is subject to the constraints placed by the USB Specification on packet sizes for Bulk, Interrupt and Isochronous transfers in full-speed and high-speed operations. NOTE: 1) The value written into these bits must match the value given in the wMaxPacketSize field of the Standard Endpoint Descriptor for the associated endpoint. A mismatch could cause unexpected results. 2) The total amount of data represented by the value written to this register must not exceed the FIFO size for the OUT endpoint, and should not exceed half the FIFO size if double-buffering is required.

- Control and Status Register 1 for Endpoints (RXCSR) (Address: 0x2020,0016)

This register has different bit definitions in host mode and in client mode with [Index Register \(Address: 0x2020,000E\)](#) value set to 1~6.

In HOST mode:

(Address: 0x2020,0016)

Bits	Attributes	Reset	Description
[15]	RW	1'h0	autoclr 0: No instruction 1: RxPktRdy Bit (Bit[0]) will be automatically cleared when a packet of RXMAXP bytes has been unloaded from the RX FIFO. NOTE: 1) When packets of less than the maximum packet size are unloaded, RxPktRdy will have to be cleared manually. 2) This bit should not be set for high-bandwidth isochronous endpoints.
[14]	RW	1'h0	autoreq 0: No instruction

Bits	Attributes	Reset	Description
			1: The ReqPkt Bit (Bit[5]) will be automatically set when the RxPktRdy Bit (Bit[0]) is cleared. NOTE: This bit is automatically cleared when a short packet is received.
[13]	RW	1'h0	damregenab 0: To disable the DMA request. 1: To enable the DMA request for the RX Endpoint.
[12]	RW	1'h0	pid error 1: A PID error occurs in the received packet. NOTE: Valid in ISO Transactions only.
[11]	RW	1'h0	DMAReqMod The CPU sets this bit to select DMA Request Mode 1 and clears it to select DMA Request Mode 0. NOTE: This bit should not be cleared in the same cycle as RxPktRdy is cleared.
[10]	WO	1'h0	setdatatog 1: To set the endpoint data toggle to 1.
[9]	RO	1'h0	datatog This bit indicates the current data toggle status of the endpoint.
[8]	RO	1'h0	incomprx This bit will be set in a high-bandwidth Isochronous transfer if the packet received is incomplete. It will be cleared when RxPktRdy Bit (Bit[0]) is cleared. NOTE: If USB protocols are followed correctly, this bit should never be set. The bit becoming set indicates a failure of the associated peripheral device to behave correctly. (In anything other than a high-bandwidth Isochronous transfer, this bit will always return 0.)
[7]	SET	1'h0	clrdatatog 0: No instruction. 1: To reset the endpoint data toggle to 0.
[6]	RO	1'h0	rxstall 0: A STALL handshake has not been received. 1: A STALL handshake has been received and an interrupt is generated. NOTE: The CPU should clear this bit.
[5]	RW	1'h0	reqpkt 0: No instruction 1: To request an IN transaction. NOTE: It is cleared when RxPktRdy bit (bit[0]) is set.
[4]	SET	1'h0	flushfifo 0: No instruction. 1: To flush the next packet to be read from the endpoint RX FIFO. The FIFO pointer is reset and the RxPktRdy Bit is cleared. NOTE: 1) FlushFIFO should only be used when RxPktRdy bit

Bits	Attributes	Reset	Description
			(bit[0]) is set. 2) If the FIFO is double-buffered, FlushFIFO may need to be set twice to completely clear the FIFO.
[3]	RO	1'h0	dataerror/nak timeout In ISO mode: This bit is set when RxPktRdy bit (bit[0]) is set if the data packet has a CTC or bit-stuff error and cleared when RxPktRdy bit (bit[0]) is cleared. In Bulk mode: This bit will be set when the Rx endpoint is halted following the receipt of NAK responses for longer than the time set as the NAKLimit by the RxInterval register. The CPU should clear this bit to allow the endpoint to continue.
[2]	RO	1'h0	error 0: No error 1: Three attempts have been made to receive a packet and no data packet has been received. NOTE: 1) The CPU should clear this bit. An interrupt is generated when the bit is set. 2) Valid only when the endpoint is operating in Bulk or Interrupt mode.
[1]	RO	1'h0	fifofull 1: No more packets can be loaded into the RX FIFO.
[0]	RO	1'h0	rxpktrdy 0: The packet has been unloaded from the RX FIFO. 1: A data packet has been received. NOTE: An interrupt is generated when this bit is set.

In CLIENT mode:
(Address: 0x2020,0016)

Bits	Attributes	Reset	Description
[15]	RW	1'h1	autoclr 0: No instruction. 1: RxPktRdy Bit (Bit[0]) will be automatically cleared when a packet of RXMAXP bytes has been unloaded from the RX FIFO. NOTE: 1) When packets of less than the maximum packet size are unloaded, RxPktRdy will have to be cleared manually. 2) This bit should not be set for high-bandwidth isochronous endpoints.
[14]	RW	1'h1	iso 0: To enable the RX endpoint for Bulk or Interrupt transfers.

Bits	Attributes	Reset	Description
			1: To disable the RX endpoint for Bulk or Interrupt transfers.
[13]	RW	1'h1	damregenab 0: To disable the DMA request. 1: To enable the DMA request for the RX Endpoint.
[12]	RW	1'h1	disnyet/pid error Bulk/Interrupt Transactions: 0: No instruction 1: To disable the sending of NYET handshakes. When set, all successfully received RX packets are ACK'd including at the point at which the FIFO becomes full. NOTE: This bit will not have any effect in high-speed mode, in which mode it should be set for all Interrupt endpoints. ISO transactions: 1: A PID error occurs in the received packet.
[11]	RW	1'h0	DMAReqMod The CPU sets this bit to select DMA Request Mode 1 and clears it to select DMA Request Mode 0. NOTE: This bit should not be cleared in the same cycle as RxPktRdy is cleared.
[10]	WO	1'h1	setdatatog 1: To set the endpoint data toggle to 1.
[9]	RO	1'h1	datatog This bit indicates the current data toggle status of the endpoint.
[8]	RO	1'h1	incomprx This bit will be set in a high-bandwidth Isochronous transfer if the packet in the RX FIFO is incomplete because parts of the data were not received. It is cleared when RxPktRdy is cleared. NOTE: In anything other than a high-bandwidth Isochronous transfer, this bit will always return 0.
[7]	SET	1'h1	clrdatatog 0: No instruction 1: To reset the endpoint data toggle to 0.
[6]	RO	1'h1	sentstall 0: A STALL handshake has not been transmitted. 1: A STALL handshake has been transmitted. NOTE: The CPU should clear this bit.
[5]	RW	1'h1	sendstall 0: No instruction. 1: To issue a STALL handshake. NOTE: 1) The CPU clears this bit to terminate the stall condition. 2) The FIFO should be flushed before this bit is set. 3) This bit has no effect where the endpoint is being

Bits	Attributes	Reset	Description
			used for isochronous transfers.
[4]	SET	1'h1	flushfifo 0: No instruction 1: To flush the next packet to be read from the endpoint RX FIFO. The FIFO pointer is reset and the RxPktRdy Bit is cleared. NOTE: 1) FlushFIFO should only be used when RxPktRdy is set. 2) If the FIFO is double-buffered, FlushFIFO may need to be set twice to completely clear the FIFO.
[3]	RO	1'h1	dataerror 0: RxPktRdy is cleared. 1: RxPktRdy is set if the data packet has a CRC or bit-stuff error. NOTE: This bit is only valid when the endpoint is operating in ISO mode. In Bulk mode, it always returns zero.
[2]	RO	1'h1	overrun 1: An OUT packet cannot be loaded into the Rx FIFO. NOTE: 1) The CPU should clear this bit. 2) This bit is only valid when the endpoint is operating in ISO mode. In Bulk mode, it always returns zero.
[1]	RO	1'h1	fifofull 1: No more packets can be loaded into the RX FIFO.
[0]	RO	1'h1	rxpktrdy 0: The packet has been unloaded from the RX FIFO. 1: A data packet has been received NOTE: An interrupt is generated when this bit is set.

- Request Packet Count Register for RX Endpoints (Address: 0x2020,0304 ~ 0x2020,0318)
 Registers 0x2020,0304 ~ 0x2020,0314 are Request Packet Count Register for endpoints 1 ~ 6. These registers are used in Host mode to specify the number of packets that are to be transferred in a block transfer of one or more Bulk packets of length MaxP to Rx Endpoints. The core uses the value recorded in this register to determine the number of requests to issue where the AutoReq option (included in the RxCSR register) has been set. These six register have the same definition, following only presents the definition of one register.

Bits	Attributes	Reset	Description
[15:0]	RW	16'h0	The number of packets of size RXMAXP that are to be transferred in a block transfer. NOTE: These bits are only used in Host mode when AutoReq is set and have no effect in Peripheral mode or when AutoReq is not set.

- EndpointX Count Register (Address: 0x2020,0018)

This register indicates the number of received data bytes in the Endpoint 0/RX Endpoint FIFO. The value returned changes as the contents of the FIFO change and is only valid while RxPktRdy is set.

This register has different bit definitions with [Index Register \(Address: 0x2020,000E\)](#) value set to 0 and set to 1~6.

With Index Register value set to 0:
(Address: 0x2020,0018)

Bits	Attributes	Reset	Description
[15:7]	RO	9'h0	Reserved
[6:0]	RO	7'h0	count 0 The number of received data bytes in the Endpoint 0 FIFO.

With Index Register value set to 1~6:
(Address: 0x2020,0018)

Bits	Attributes	Reset	Description
[15:7]	RO	3'h0	Reserved
[6:0]	RO	13'h0	rx count The number of received data bytes in the RX FIFO.

- TX Endpoint Type Register (TXTYPE) (Address: 0x2020,001A)

(host mode only) (with [Index Register \(Address: 0x2020,000E\)](#) value set to 1~6)

This register should be written with the endpoint number to be targeted by the endpoint in the lower 4 bits, and the transaction protocol to use for the currently-selected TX endpoint in the upper 2 bits. There is a TXTYPE register for each configured TX endpoint (except Endpoint 0).

Bits	Attributes	Reset	Description
[7:6]	RO	2'h0	Reserved
[5:4]	RW	2'h0	protocol Tx endpoint protocol selection. 2'h0: Illegal 2'h1: Isochronous 2'h2: Bulk 2'h3: Interrupt
[3:0]	RW	4'h0	ep_num The CPU should set this value to the endpoint number contained in the TX endpoint descriptor returned to the USB controller during device enumeration.

- NAK Response Timeout on Endpoint 0(NAKLIMIT0)/TX interval Register (TXINTERVAL) (Address: 0x2020,001B)

When the [Index Register \(Address: 0x2020,000E\)](#) value is set to 0, this register sets the number of frames/microframes (High-Speed transfers) after which Endpoint 0 should timeout on receiving a stream of NAK responses; When the [Index Register \(Address: 0x2020,000E\)](#) value is set to 1~6, this register defines the polling interval for the currently-selected TX endpoint for Interrupt and Isochronous transfers, and sets the number of

frames/microframes after which the endpoint should timeout on receiving a stream of NAK responses for Bulk endpoints. There is a TXINTERVAL register for each configured TX endpoint (except Endpoint 0).

NOTE: This register is used in host mode only.

With Index Register value set to 0:

(Address: 0x2020,001B)

Bits	Attributes	Reset	Description
[7:5]	RO	3'h0	Reserved
[4:0]	RW	5'h0	Valid values ranges from 2 to 16. The number of frames/microframes selected is $2^{(m-1)}$. If the host receives NAK responses from the target for more frames than the number represented by the limit set in this register, the endpoint will be halted. NOTE: A value of 0 or 1 disables the NAK timeout function.

With Index Register value set to 1~6:

(Address: 0x2020,001B)

Bits	Attributes	Reset	Description																			
[7:0]	RW	8'h0	In each case the value that is set defines a number of frames/microframes (High Speed transfers), as follows:																			
			<table><tr><th>TRANSFER TYPE</th><th>SPEED</th><th>VALID VALUES(m)</th><th>INTERPRETATION</th></tr><tr><td rowspan="2">Interrupt</td><td>full speed</td><td>1~255</td><td>Polling interval is m frames.</td></tr><tr><td>high speed</td><td>1~16</td><td>Polling interval is 2^(m-1) microframes.</td></tr><tr><td>Isochronous</td><td>full speed or high speed</td><td>1~16</td><td>Polling interval is 2^(m-1) frames/microframes.</td></tr><tr><td>Bulk</td><td>full speed or high speed</td><td>2~16</td><td>NAK limit is 2^(m-1) frames/microframes. NOTE: A value of 0 or 1 disables the NAK timeout function.</td></tr></table>	TRANSFER TYPE	SPEED	VALID VALUES(m)	INTERPRETATION	Interrupt	full speed	1~255	Polling interval is m frames.	high speed	1~16	Polling interval is 2 ^(m-1) microframes.	Isochronous	full speed or high speed	1~16	Polling interval is 2 ^(m-1) frames/microframes.	Bulk	full speed or high speed	2~16	NAK limit is 2 ^(m-1) frames/microframes. NOTE: A value of 0 or 1 disables the NAK timeout function.
			TRANSFER TYPE	SPEED	VALID VALUES(m)	INTERPRETATION																
			Interrupt	full speed	1~255	Polling interval is m frames.																
				high speed	1~16	Polling interval is 2 ^(m-1) microframes.																
Isochronous	full speed or high speed	1~16	Polling interval is 2 ^(m-1) frames/microframes.																			
Bulk	full speed or high speed	2~16	NAK limit is 2 ^(m-1) frames/microframes. NOTE: A value of 0 or 1 disables the NAK timeout function.																			

● RX Endpoint Type Register (RXTYPE) (Address: 0x2020,001C)

(host mode only) (with [Index Register \(Address: 0x2020,000E\)](#) value set to 1~6)

This register should be written with the endpoint number to be targeted by the endpoint in the lower 4 bits, and the transaction protocol to use for the currently-selected TX endpoint in the upper 2 bits. There is an RXTYPE register for each configured RX endpoint (except Endpoint 0).

Bits	Attributes	Reset	Description
[7:6]	RO	2'h0	Reserved
[5:4]	RW	2'h0	protocol RX endpoint protocol selection. 2'h0: Illegal

Bits	Attributes	Reset	Description
			2'h1: Isochronous 2'h2: Bulk 2'h3: Interrupt
[3:0]	RW	4'h0	ep_num The CPU should set this value to the endpoint number contained in the RX endpoint descriptor returned to the USB controller during device enumeration.

● RX Interval Register (RXINTERVAL) (Address: 0x2020,001D)

This register defines the polling interval for the currently-selected RX endpoint for Interrupt and Isochronous transfers, and sets the number of frames/microframes after which the endpoint should timeout on receiving a stream of NAK responses for Bulk endpoints. There is a RXINTERVAL register for each configured RX endpoint (except Endpoint 0).

NOTE: This register is used in host mode only.

Bits	Attributes	Reset	Description																			
[7:0]	RW	8'h0	In each case the value that is set defines a number of frames/microframes (High Speed transfers), as follows:																			
			<table><tr><th>TRANSFER TYPE</th><th>SPEED</th><th>VALID VALUES(m)</th><th>INTERPRETATION</th></tr><tr><td rowspan="2">Interrupt</td><td>full speed</td><td>1~255</td><td>Polling interval is m frames.</td></tr><tr><td>high speed</td><td>1~16</td><td>Polling interval is 2^(m-1) microframes.</td></tr><tr><td>Isochronous</td><td>full speed or high speed</td><td>1~16</td><td>Polling interval is 2^(m-1) frames/microframes.</td></tr><tr><td>Bulk</td><td>full speed or high speed</td><td>2~16</td><td>NAK limit is 2^(m-1) frames/microframes. NOTE: A value of 0 or 1 disables the NAK timeout function.</td></tr></table>	TRANSFER TYPE	SPEED	VALID VALUES(m)	INTERPRETATION	Interrupt	full speed	1~255	Polling interval is m frames.	high speed	1~16	Polling interval is 2 ^(m-1) microframes.	Isochronous	full speed or high speed	1~16	Polling interval is 2 ^(m-1) frames/microframes.	Bulk	full speed or high speed	2~16	NAK limit is 2 ^(m-1) frames/microframes. NOTE: A value of 0 or 1 disables the NAK timeout function.
			TRANSFER TYPE	SPEED	VALID VALUES(m)	INTERPRETATION																
			Interrupt	full speed	1~255	Polling interval is m frames.																
				high speed	1~16	Polling interval is 2 ^(m-1) microframes.																
Isochronous	full speed or high speed	1~16	Polling interval is 2 ^(m-1) frames/microframes.																			
Bulk	full speed or high speed	2~16	NAK limit is 2 ^(m-1) frames/microframes. NOTE: A value of 0 or 1 disables the NAK timeout function.																			

● Configuration Information Register (ConfigData)/FIFO Size Register(FIFOSIZE) (Address: 0x2020,001F)

When [Index Register \(Address: 0x2020,000E\)](#) value is set to 0, this register returns information about the selected core configuration.

When [Index Register \(Address: 0x2020,000E\)](#) value is set to 1~6, this register returns the sizes of the FIFOs associated with the selected additional TX/RX endpoints. The lower nibble encodes the size of the selected TX endpoint FIFO; the upper nibble encodes the size of the selected RX endpoint FIFO.

With [Index Register \(Address: 0x2020,000E\)](#) Value set to 0:
(Address: 0x2020,001F)

Bits	Attributes	Reset	Description
[7]	RW	configurable	mprxe 0: Automatic amalgamation of bulk packets is not

Bits	Attributes	Reset	Description
			selected. 1: Automatic amalgamation of bulk packets is selected.
[6]	RW	configurable	mptxe 0: Automatic splitting of bulk packets is not selected. 1: Automatic splitting of bulk packets is selected.
[5]	RW	configurable	bigendian 0: Big Endian ordering is not selected. 1: Big Endian ordering is selected.
[4]	RW	configurable	hbrxe 0: High-bandwidth RX ISO endpoint support is not selected. 1: High-bandwidth RX ISO endpoint support is selected.
[3]	RW	configurable	hbtxe 0: High-bandwidth TX ISO endpoint support is not selected. 1: High-bandwidth TXISO endpoint support is selected.
[2:1]	RW	configurable	Reserved
[0]	RW	configurable	utmi datawidth Selected UTMI data width. 0: 8 bits. 1: 16 bits.

With *Index Register (Address: 0x2020,000E)* Value set to 1~6:
(Address: 0x2020,001F)

Bits	Attributes	Reset	Description
[7:4]	RO	configurable	rxfifo The size of the selected RX endpoint FIFO. Values of 3-13 correspond to a FIFO size of 2 ⁿ bytes (8-8192 bytes). If an endpoint has not been configured, a value of 0 will be displayed.
[3:0]	RO	configurable	txfifo The size of the selected TX endpoint FIFO. Values of 3-13 correspond to a FIFO size of 2 ⁿ bytes (8-8192 bytes). If an endpoint has not been configured, a value of 0 will be displayed.

8.5.6 Programmer's Guide

8.5.6.1 FIFOs for Endpoint 0~6

This address range provides 7 addresses for CPU access to the FIFOs for each endpoint. Writing to these addresses loads data into the TXFIFO for the corresponding endpoint. Reading from these addresses unloads data from the RXFIFO for the corresponding endpoint. The FIFOs are located on 32-bit double-word boundaries (Endpoint 0 at 0x2020,0020, Endpoint 1 at 0x2020,0024, Endpoint 2 at 0x2020,0028, Endpoint 3 at 0x2020,002C, Endpoint 4 at 0x2020,0030, Endpoint 5 at 0x2020,0034, Endpoint 6 at 0x2020,0038).

NOTE

- 1) Transfers to and from FIFOs may be 8-bit, 16-bit or 32-bit as required, and any combination of access is allowed provided the data accessed is contiguous. However, all the transfers associated with one packet must be of the same width so that the data is consistently byte-, word- or double-word-aligned. The last transfer may however contain fewer bytes than the previous transfers in order to complete an odd-byte or odd-word transfer.
- 2) Depending on the size of the FIFO and the expected maximum packet size, the FIFOs support either single-packet or double-packet buffering. However, burst writing of multiple packets is not supported as flags need to be set after each packet is written.

8.5.6.2 Data Transmitting and Receiving

Prior to utilizing the USB controller, programmers should pay special attention to the following items:

The data is stored in USB FIFO. The FIFO of endpoint 0 only can be accessed by CPU. The FIFO of other 6 endpoints can be accessed by DMA mode 0, DMA mode 1 or CPU. DMA mode 0 and DMA mode 1 can only transmit data between USB FIFO and L2 buffer. The data length must be multiple of 512bytes when transmitted by DMA mode 1. If the data length is less than 512bytes the data must be access by CPU.

Register bits of L2 memory is dynamically assigned to Endpoint n, which refers to endpoint 1~6. The different value of DMA Address Register can access different L2 buffer. The following list shows the relationship of endpoint number, register bits of L2 and the value of DMA Address Register.

Endpoint Number	Register bits of L2 memory	Value of DMA Address Register
Endpoint n (n=1~6)	USB_EPn_ADD (bit[2:0] of <i>Buffer Assignment Register 0</i> (Address:0x2014,0090))	32'h70000000
	USB_EPn_ADD (bit[5:3] of <i>Buffer Assignment Register 0</i> (Address:0x2014,0090))	32'h71000000
	USB_EPn_ADD (bit[8:6] of <i>Buffer Assignment Register 0</i> (Address:0x2014,0090))	32'h72000000
	USB_EPn_ADD (bit[17:15] of <i>Buffer Assignment Register 1</i> (Address:0x2014,0094))	32'h73000000
	USB_EPn_ADD (bit[20:18] of <i>Buffer Assignment Register 1</i> (Address:0x2014,0094))	32'h74000000
	USB_EPn_ADD (bit[23:21] of <i>Buffer Assignment Register 1</i> (Address:0x2014,0094))	32'h75000000

NOTE

- 1) The value of DMA address register is assigned to the fixed register bits of L2 memory, but USB_EPn_ADD can be dynamically assigned to the different register bits of L2 memory in the table above.
- 2) USB_EPn_ADD, which refers to USB_EP1_ADD, USB_EP2_ADD, USB_EP3_ADD, USB_EP4_ADD, USB_EP5_ADD and USB_EP6_ADD, can be configured to different buffers from Buffer 0 to Buffer 7.

8.5.6.2.1 Data Transmitting

1. Endpoint 0 transmitting data
 - 1) To write the data to be transferred into FIFO of Endpoint0 by CPU;
 - 2) To set TxPktRdy bit (bit[1]) of *TXCSR Register (Address: 0x2020,0012)* to indicate that the USB controller is ready to transfer data;
 - 3) To send data when an IN token is received and generate a MCU interrupt;
 - 4) To jump to Step 1 if more data is needed to be transferred.
2. Other endpoints transmitting data
 - 1) To set bit mode (D13) of *TXCSR Register (Address: 0x2020,0012)* to select the data transmission direction;
 - 2) To set AutoSet (D15), DMAReqEnab (D12) and the DMAReqMode (D10) bits of the appropriate *TXCSR Register (Address: 0x2020,0012)* register as 1;
 - 3) To write 1000000000B (512 bytes) into *TXMAXP Register (Address: 0x2020,0010)*;
 - 4) To configure the buffer of L2 memory assigned to endpoint (the data number must be multiple 512 bytes and less than total data that will to be sent);
 - 5) To configure built-in DMA controller as DMA mode1 (To write the number of data to be transferred into *DMA Count Register (Address: 0x2020,02XC)*, the data number must be multiple 512 byte and less than total data that will to be sent. To write the right address into *DMA Address Register (Address: 0x2020,02X8)*. To configure DMA mode1 to *DMA Control Register (Address: 0x2020,02X4)* and start DMA operation;
 - 6) When the data that has been configured to *DMA Count Register* have been transferred there will be a USB DMA interrupt;
 - 7) To set AutoSet (D15), DMAReqEnab (D12) and the DMAReqMode (D10) bits of the appropriate TxCSR register as 0;
 - 8) To write the residual data that less than 512 bytes into FIFO of according endpoint by CPU;
 - 9) To set TxPktRdy bit (bit [1]) of *TXCSR Register (Address: 0x2020,0012)* as 1 to indicate that the USB controller is ready to transfer data;
 - 10) To send data when an IN token is received and generate a MCU interrupt.

8.5.6.2.2 Data Receiving

1. Endpoint 0 receiving data
 - 1) To generate a MCU interrupt when the data packet is received;
 - 2) To read the received data from FIFO of Endpoint0 by CPU;
 - 3) To clear RxPktRdy bit ((bit [0]) of *RXCSR Register (Address: 0x2020,0016)*).
 - 4) To jump to Step 1 to receive data.
2. Other endpoints receiving data
 - 1) To set AutoClear (D15), DMAReqEnab (D13) and DMAReqMode (D11) bits of the appropriate *RXCSR Register (Address: 0x2020,0016)* as 1.
 - 2) To write 1000000000B (512 bytes) into *RXMAXP Register (Address: 0x2020,0014)*;
 - 3) To configure the buffer of L2 memory assigned to Endpoint (the data number must be multiple 512 bytes and less than total data that will to be received);
 - 4) To configure built-in DMA controller as DMA mode1 (To write the number of data to be received into *DMA Count Register (Address: 0x2020,02XC)*, the data number must be multiple 512 bytes and less than total data that will to be received. To write the right address into *DMA Address Register (Address: 0x2020,02X8)*. To configure DMA mode1 to *DMA Control Register (Address: 0x2020,02X4)* and start DMA operation;

- 5) When the data that has been configured to *DMA Count Register* have been received there will be a USB DMA interrupt;
- 6) To generate a MCU interrupt when the residual data that less than 512-byte packet is received;
- 7) To read the received data from FIFO of according endpoint by CPU;
- 8) To clear RxPktRdy bit ((bit[0]) of *RXCSR Register (Address: 0x2020,0016)*).

8.5.6.3 USB Reset

8.5.6.3.1 Host Mode

If the Reset bit in the Power Register is set while the USB controller is in host mode, the USB controller will generate Reset signaling on the bus. If the HS Enab bit in the *Power Control Register* (bit[5]) was set, it will also try to negotiate for high-speed operation.

The CPU should keep the Reset bit set for at least 20 ms to ensure correct resetting of the target device.

After the CPU has cleared the bit, the USB controller will start its frame counter and transaction scheduler. Whether high-speed operation is selected will be indicated by HS Mode bit (Bit[4] of *Power Control Register (Address: 0x2020,0001)*).

8.5.6.3.2 Client Mode

When the USB controller is acting as a client and a reset condition is detected on the USB, it will perform the following actions:

1. Set every corresponding bit of *Function Address Register (FADDR) (Address: 2020,0000)* to 0.
2. Set every corresponding bit of *Index Register (Address: 0x2020,000E)* to 0.
3. Flush all endpoint FIFOs.
4. Clear all control/status registers.
5. Enable all endpoint interrupts.
6. Generate a Reset interrupt.

If the HS Enab bit in the *Power Control Register* (bit[5]) was set, the USB controller also tries to negotiate for high-speed operation. Whether high-speed operation is selected is indicated by HS Mode bit (bit[4] of *Power Control Register (Address: 0x2020,0001)*).

When the application software driving the USB controller receives a Reset interrupt, it should close any open pipes and wait for bus enumeration to begin.

8.5.7 DMA Controller

8.5.7.1 Overview

The USB controller includes a six-channel DMA controller, which has higher priority than the DMA controller handling the DMA requests of other DMA-capable functional blocks. That is, the USB controller has the highest priority when multiple DMA-capable functional blocks are sending requests.

This DMA controller can handle packet sizes up to 8K. And the controller can further be programmed to conduct transfers using INCR4, INCR8 and INCR16 4/8/16-beat incrementing bursts rather than bursts of unspecified length.

The DMA controller can be only programmed to load/unload one packet, so processor intervention is required for each packet transferred over the USB. This mode can be used with any endpoint, whether it uses Control, Bulk, Isochronous, or Interrupt transfers.

8.5.7.2 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

The DMA controller has one interrupt register which indicates which channels have a pending interrupt, and a set of two control registers for each configured channel.

Register No.	Register Address	Register Name
1	0x2020,0200	<i>DMA Interrupt Status Register</i>
2	0x2020,0204, ~ 0x2020,0254	<i>DMA Control Register 0~5</i>
3	0x2020,0208, ~ 0x2020,0258	<i>DMA Address Register 0~5</i>
4	0x2020,020C, ~ 0x2020,025C	<i>DMA Address Register 0~5</i>

8.5.7.3 Register Definition

- DMA Interrupt Status Register (Address: 0x2020,0200)

This register indicates which channel(s) is (are) having a pending interrupt.

Bits	Attributes	Reset	Description
[31:6]	RO	26'h0	Reserved
[5]	RO	1'h0	ch5 1: DMA channel 5 is having a pending interrupt. NOTE: This bit is cleared after having been read.
[4]	RO	1'h0	ch4 1: DMA channel 4 is having a pending interrupt. NOTE: This bit is cleared after having been read.
[3]	RO	1'h0	ch3 1: DMA channel 3 is having a pending interrupt. NOTE: This bit is cleared after having been read.
[2]	RO	1'h0	ch2 1: DMA channel 2 is having a pending interrupt. NOTE: This bit is cleared after having been read.
[1]	RO	1'h0	ch1 1: DMA channel 1 is having a pending interrupt. NOTE: This bit is cleared after having been read.
[0]	RO	1'h0	ch0 1: DMA channel 0 is having a pending interrupt. NOTE: This bit is cleared after having been read.

DMA Control Register 0, 1, 2, 3, 4 & 5 are used to control the DMA channels, such as direction, DMA mode, endpoint number, burst mode, and etc. DMA Control Register 0 corresponds with DMA channel 0; DMA Control Register 1, DMA channel 1; DMA Control Register 2, DMA channel 2; DMA Control Register 3, DMA channel 3; DMA Control Register 4, DMA channel 4; DMA Control Register 5, DMA channel 5. All the six registers have the same bit definitions.

- DMA Control Register 0~5 (Address: 0x2020,0204~0x2020,0254)

Bits	Attributes	Reset	Description
[31:11]	RO	21'h0	Reserved
[10:9]	RW	2'h0	burst_mode 2'h0: Burst Mode 0, Bursts of unspecified length 2'h1: Burst Mode 1, INCR4 or unspecified length 2'h2: Burst Mode 2, INCR8, INCR4 or unspecified length 2'h3: Burst Mode 3, INCR16, INCR8, INCR4 or unspecified length
[8]	RW	1'h0	bus_err 1: A bus error occurs.
[7:4]	RW	4'h0	ep_no Endpoint number 4'h0: Endpoint 0 4'h1: Endpoint 1 4'h2: Endpoint 2 4'h3: Endpoint 3 4'h4: Endpoint 4 4'h5: Endpoint 5 4'h6: Endpoint 6 others: Reserved
[3]	RW	1'h0	int_en 0: To disable interrupt. 1: To enable interrupt.
[2]	RW	1'h0	DMA mode 0: DMA mode 0 1: DMA mode 1
[1]	RW	1'h0	dir 0: DMA write (Rx Endpoint) 1: DMA read (Tx Endpoint)
[0]	RW	1'h0	dma_en 0: To disable DMA. 1: To enable DMA.

DMA Address Register 0, 1, 2, 3, 4 & 5 are used to hold the DMA channel X AHB memory address. DMA Address Register 0 corresponds with DMA channel 0; DMA Address Register 1, DMA channel 1; DMA Address Register 2, DMA channel 2; DMA Address Register 3, DMA channel 3; DMA Address Register 4, DMA channel 4; DMA Address Register 5, DMA channel 5. All the six registers have the same bit definitions.

● DMA Address Register 0~5 (Address: 0x2020,0208~0x2020,0258)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	addr1 DMA channel 1 AHB memory address

DMA Count Register 0, 1, 2, 3, 4 & 5 are used to hold the byte number that will be transferred in a DMA operation. DMA Count Register 0 corresponds with DMA channel 0; DMA Count Register 1, DMA channel 1; DMA Count Register 2, DMA channel 2; DMA Count Register 3, DMA channel 3; DMA Count Register 4, DMA channel 4; DMA Count Register 5, DMA channel 5. All the six registers have the same bit definitions.

If a DMA is enabled with a count of 0, the bus will not be requested and a DMA interrupt will be generated.

● DMA Count Register 0~5 (Address: 0x2020,020C~0x2020,025C)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	cnt1 DMA channel 1 byte count

8.5.7.4 Programmer's Guide

8.5.7.4.1 DMA Bus Cycles

The DMA controller uses incrementing bursts on the AHB. A new burst is started when the USB controller is first granted bus mastership (whether at the start of a USB packet or when regaining the bus after being thrown off part way through a packet), and when the AHB address starts a new 1Kbyte block.

NOTE

The requirement to start a new burst at 1K boundaries is handled automatically by the DMA controller. The user does not need to take these boundaries into account in programming the DMA controller.

These bursts may be either 4-beat, 8-beat, 16-beat or of unspecified length, according to how the DMA channel is programmed, the size of packet being transferred and the location relative to the next 1K boundary (as burst operations are not permitted to cross 1K boundaries). Bits[10:9] of the *DMA Control Register X (Address: 0x2020,02X4)* select Burst Mode 0 – 3 which in turn define which burst types may be used. For example, selection of Burst Mode 2 allows use of 8-beat (INCR8), 4-beat (INCR4) bursts and bursts of unspecified length but not 16-beat (INCR16) bursts. There is no restriction on the Burst Mode selected for transfers in either DMA Mode 0 or DMA Mode 1.

Each transfer of a packet is generally carried out using word transfers (32 bits) but there may be additional byte or half-word transfers at the beginning or at the end of the packet transfer. If the start address (written to the *DMA Address Register X (Address: 0x2020,02X8)*) is word aligned, the packet transfer will start with a word transfer, but half-word and/or byte transfers may be added at the end to handle any residue. If the start address is byte or half-word aligned, the transfer will start with a byte and/or a half-word access but will switch to word transfers when the address becomes word-aligned, only reverting to byte/half-word

transfers to handle any residue at the end of the packet. AHB Split transactions and retries are supported.

8.5.7.4.2 Bus Errors

If a bus error occurs while the DMA controller is accessing memory on the AHB, the DMA controller will immediately terminate the DMA transfer and interrupt the processor with the Bus Error bit (Bit[8] of *DMA Control Register X (Address: 0x2020,02X4)*) set.

NOTE

The generation of this interrupt is NOT affected by the setting of the Interrupt Enable bit (Bit[3]) in the *DMA Control Register X (Address: 0x2020,02X4)*. The interrupt will still be generated when this bit is set to 0.

8.5.7.4.3 Transferring Packets

Use of the built-in DMA controller to access the FIFOs requires both the DMA controller and the endpoint to be appropriately programmed. Many variations are possible. The following sections detail the standard set-ups used for the basic actions of transferring individual packets.

Individual Packet: RX Endpoint

For this, the RX endpoint should be programmed as follows:

1. The relevant interrupt enable bit in the *IntrRxE register* set to 1.
2. The DMAReqEnab bit (Bit[13]) of the appropriate *RxCSR register* set to 0.

NOTE

- 1) There is no need to set the USB controller to generate DMA requests for this operation.
- 2) When a packet has been received by the USB controller, it will generate the appropriate Endpoint interrupt. The processor should then program selected channel of the DMA controller as follows:
 - a. ADDR: Memory address to store packet
 - b. COUNT: Size of packet (determined by reading the RxCount register)
 - c. CNT: DMA Enable (bit[0]) =1; Direction (bit[1]) =0; DMA Mode (bit[2]) =0; Interrupt Enable (bit[3]) =1; Required Burst Mode (bits[10:9]).
- 3) The DMA controller will then request bus mastership and transfer the packet to memory. When it has completed the transfer, it will generate a DMA interrupt (DMA_NINT taken low). The processor should then clear the RxPktRdy bit in the *RxCSR register*

Individual Packet: TX Endpoint

A TX endpoint should be programmed as follows:

The relevant interrupt enable bit in the *IntrTxE register* set to 1.

The DMAReqEnab bit (bit[12]) of the appropriate *TxCSR register* set to 0.

NOTE

- 1) There is no need to set the USB controller to generate DMA requests for this operation.
- 2) When the FIFO in the USB controller becomes available, the USB controller will interrupt the processor with the appropriate TX Endpoint interrupt. The processor should then program the DMA controller as follows:
 - a. ADDR: Memory address of packet to send
 - b. COUNT: Size of packet to be sent

- c. CNT: DMA Enable (bit[0]) =1; Direction (bit[1]) =1; DMA Mode (bit[2]) =0; Interrupt Enable (bit[3]) =1; Required Burst Mode (bits[10:9]).
- 3) The DMA controller will then request bus mastership and transfer the packet to the USB controller FIFO. When it has completed the transfer, it will generate a DMA interrupt. The processor should then set the TxPktRdy bit in the [TxCSR register](#).

Multiple Packets: RX Endpoint

The transfer of multiple packets will normally be carried out using DMA Mode 1. Where multiple packets are to be received using DMA Mode 1, the DMA Controller should be programmed as follows:

ADDR: Memory address of the buffer in which to store transfer

COUNT: Maximum size of data buffer

CNTL: DMA Enable (D0) =1; Direction (D1) =0; DMA Mode (D2) =1; Interrupt Enable (D3) =1; Required Burst Mode (D10–9).

And the MUSBHDRC RX endpoint should be programmed as follows:

The relevant interrupt enable bit in the [IntrRx register](#) should be set to 1.

The AutoClear (D15), DMAReqEnab (D13) and DMAReqMode (D11) bits of the appropriate [RxCSR register](#) should be set to 1. In Host mode, the AutoReq (D14) bit should also be set to 1 and the [RqPktCount register](#) should be programmed with the number of packets in the transfer.

As each packet is received by the MUSBHDRC, the DMA controller will request bus mastership and transfer the packet to memory. With AutoClear set, the MUSBHDRC will automatically clear the RxPktRdy bit.

In Peripheral mode or where RqPktCount is zero, this process will continue automatically until the MUSBHDRC receives a 'short packet' (one of less than the maximum packet size for the endpoint) signifying the end of the transfer. This 'short packet' will not be transferred by the DMA controller: instead the MUSBHDRC will interrupt the processor by generating the appropriate Endpoint interrupt. The processor can then read the MUSBHDRC RxCount register to see the size of the 'short packet' and either unload it manually or reprogram the DMA controller in Mode 0 to unload the packet.

In Host mode with AutoReq set and RqPktCount non-zero, the core will decrement the value in the [RqPktCount register](#) following each request. When the value decreases from 1 to 0, the AutoReq bit is cleared to prevent any further transactions being attempted.

The DMA controller ADDR register which have been incremented as the packets were unloaded so the processor can determine the size of the transfer by comparing the current value of ADDR against the start address of the memory buffer.

NOTE

If the size of the transfer exceeds the data buffer size, the DMA controller will stop unloading the FIFO and interrupt the processor via the DMA_NINT line.

Multiple Packets: TX Endpoint

To carry out this operation using DMA Mode 1, the DMA controller should be programmed as follows:

ADDR: Memory address of data block to send

COUNT: Size of data block

CNTL: DMA Enable (D0) =1; Direction (D1) =1; DMA Mode (D2) =1; Interrupt Enable (D3) =1; Required Burst Mode (D10–9).

And the MUSBHDRC Tx endpoint should be programmed as follows:

The relevant interrupt enable bit in the IntrTxE register should be set to 1 (simply so that errors can be detected).

The AutoSet (D15), DMAReqEnab (D12) and the DMAReqMode (D10) bits of the appropriate [TxCSR register](#) should be set to 1.

When the FIFO in the MUSBHDRC becomes available, the DMA controller will request bus mastership and transfer a packet to the FIFO. With AutoSet set, the MUSBHDRC will automatically set the TxPktRdy bit. This process will continue until the entire data block has been transferred to the MUSBHDRC. The DMA controller will then interrupt the processor by taking DMA_NINT low. If the last packet to be loaded was less than the maximum packet size for the endpoint, the TxPktRdy bit will not have been set for this packet: the processor should therefore respond to the DMA interrupt by setting the TxPktRdy bit to allow the last 'short packet' to be sent. If the last packet to be loaded was of the maximum packet size, then the action to take depends on whether the transfer is under the control of an application such as the mass storage software on a Windows system that keeps count of the individual packets sent. If the transfer isn't under such control, the processor should still respond to the DMA interrupt by setting the TxPktRdy bit. This has the effect of sending a null packet for the receiving software to interpret as indicating the end of the transfer.

8.6 MMC/SD/SDIO

8.6.1 Overview

AK3918AV100 series processors supports three MMC/SD/SDIO interfaces, which are used for external MMC, SD card, and SDIO WiFi.

8.6.2 Features

Features of MMC/SD/SDIO interfaces are as follows:

- Conforms to The Multimedia Card Specification, Version 4.2, SD Memory Card Specification, Version 2.0, SDIO Specification, Version 2.0
- Maximum reading/writing speed is half of GCLK
- Supports little endian only
- 32-bit local buffer
- Supports single block operation, multiple block operation and stream mode (MMC card only)
- Supports SD card plug-in/off monitoring via pin SD_DATA3/SD_MCLK or certain GPIO port selected by software

NOTE

As defined in the *The Multimedia Card Specification*, the data will not be followed by the CRC bits when it is transferred in stream mode.

8.6.3 Functional Description

8.6.3.1 Block Diagram

The following figure shows the block diagram of MMC/SD/SDIO interface.

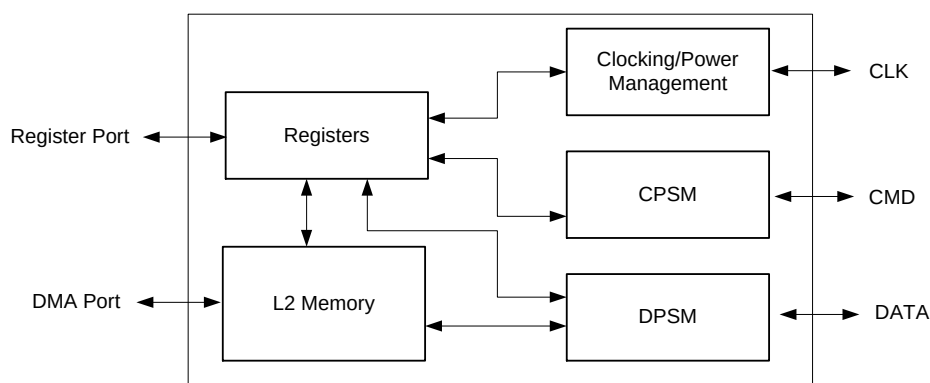


Figure 8-24 MMC/SD/SDIO Interface Block Diagram

8.6.3.2 Clock Generator/Power Management

The clock generation logic generates and controls the MCK signal. The MCK output can use a clock divider. The clock output is inactive after reset, during power off and power up phases, and during IDLE state in power saving mode.

The power management logic controls an external power supply unit, and disables the card bus output signals during power off or power up phases. The power up phase is a transition phase between the power off and power on phases, and allows an external power supply to reach the card bus operating voltage. A device driver is used to ensure that the MMC/SD controller remains in the power up phase until the external power supply reaches the operating voltage.

8.6.3.3 Command Path

The command path sub-unit sends commands to and receives responses from the cards. When the command register is written into and the enable bit is set, command transfer starts. When the command has been sent, the command path state machine sets the status flags and enters the IDLE state if a response is not required. If a response is required, it waits for the response. When the response is received, the received CRC code and the internally generated code will be compared, and the appropriate status flags will be set.

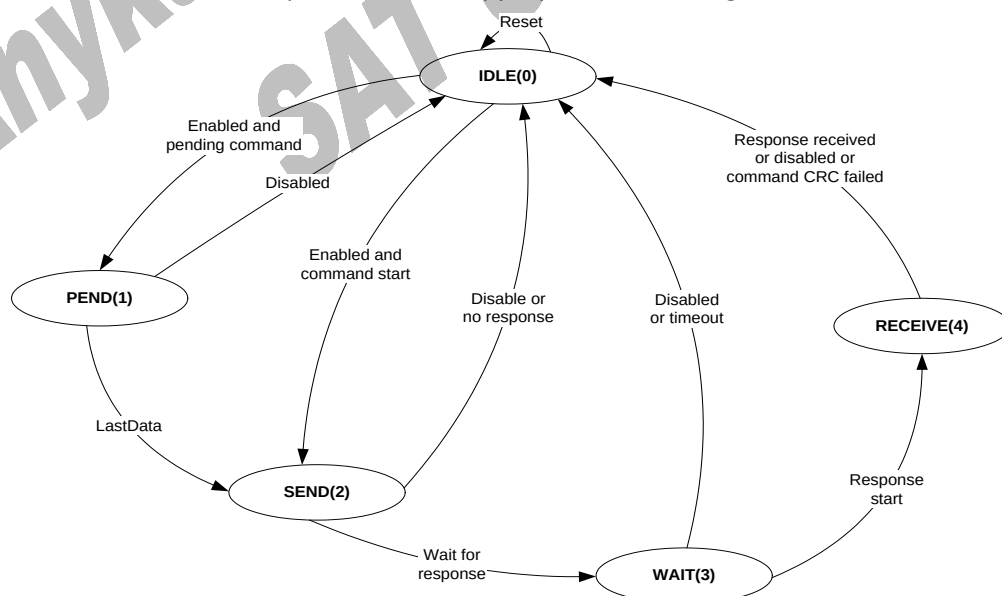


Figure 8-25 Command Path State Machine (CPSM)

When the WAIT state is entered, the command timer starts running. If the timeout is reached (64 MCK cycles) before the CPSM moves to the RECEIVE state, the timeout flag is set and the IDLE state is entered.

If a pending bit is set in the [command register](#), the CPSM enters the PEND state, and waits for a CmdPend signal from the data path sub-unit. When CmdPend is detected, the CPSM moves to the SEND state. This enables the data counter to trigger the stop command transmission.

The CPSM remains in the IDLE state for at least 8 MCK periods to meet Ncc and Nrc timing constraints.

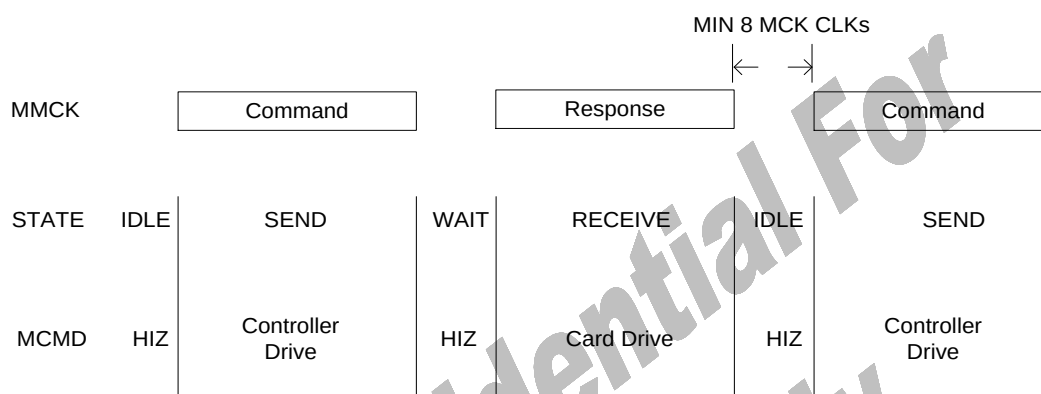


Figure 8-26 MMC/SD Controller Transfer

The command path operates in a half-duplex mode, so that commands and responses can either be sent or received. If the CPSM is not in the SEND state, the MCMD output is in HI-Z state, as shown in [Figure 8-26](#). Data on MCMD is synchronous to the rising edge of MCK. All commands have a fixed length of 48 bits, as shown in [Figure 8-32](#); while the response can be either 48 bits (short response) or 136 bits (long response), as shown in [Figure 8-33](#).

The command implements 5 status flags: CmdRespEnd, CmdCrcFail, CmdSent, CmdTimeOut, CmdActive.

The CRC generator calculates the CRC checksum for all bits before the CRC code. This includes the start bit, transmitter bit, command index, and command argument (or card status).

The CRC checksum is calculated for the first 120 bits of CID or CSD for the long response format. Note that the start bit, transmitter bit and the six reserved bits are not used in the CRC calculation. The CRC checksum is a 7-bit value:

$$\text{CRC}[6:0] = \text{Remainder}[(M(x) * x^7) / G(x)]$$

$$G(x) = x^7 + x^3 + 1$$

$$M(x) = (\text{start bit}) * x^{39} + \dots + (\text{last bit before CRC}) * x^0, \text{ or}$$

$$M(x) = (\text{start bit}) * x^{119} + \dots + (\text{last bit before CRC}) * x^0$$

8.6.3.4 Data Path

The data path sub-unit transfers data to and from cards.


```

graph TD
    IDLE0([IDLE(0)])
    BUSY6([BUSY(6)])
    WAIT_S4([WAIT_S(4)])
    SEND5([SEND(5)])
    WAIT_FOR_DATA7([WAIT_FOR_DATA(7)])
    WAIT_R1([WAIT_R(1)])
    RECEIVE2([RECEIVE(2)])
    WAIT_FOR_ROOM3([WAIT_FOR_ROOM(3)])

    IDLE0 -- Reset --> IDLE0
    IDLE0 -- "Enabled and not send" --> WAIT_R1
    IDLE0 -- "Disabled or CRC fail" --> WAIT_R1
    IDLE0 -- "Disable" --> WAIT_FOR_DATA7
    IDLE0 -- "Disabled or end of stream data" --> SEND5
    IDLE0 -- "Disabled or CRC fail or timeout" --> BUSY6
    IDLE0 -- "Disable or all data have been sent" --> WAIT_S4
    IDLE0 -- "Enable and send" --> WAIT_S4
    IDLE0 -- "Disabled or timeout or all data have been read out from the buffer" --> WAIT_R1

    BUSY6 -- "Not busy" --> WAIT_S4
    BUSY6 -- "End of packet" --> SEND5

    WAIT_S4 -- "Data ready" --> SEND5
    WAIT_S4 -- "Disable or all data have been sent" --> IDLE0
    WAIT_S4 -- "Enable and send" --> IDLE0

    SEND5 -- "Data ready" --> WAIT_FOR_DATA7
    SEND5 -- "End of packet" --> BUSY6
    SEND5 -- "No data ready" --> WAIT_FOR_DATA7

    WAIT_FOR_DATA7 -- "Data ready" --> SEND5
    WAIT_FOR_DATA7 -- "Disable" --> IDLE0

    WAIT_R1 -- "Start bit" --> RECEIVE2
    WAIT_R1 -- "End of packet or end of data, and no data need wait for write" --> RECEIVE2
    WAIT_R1 -- "Disabled or timeout or all data have been read out from the buffer" --> IDLE0
    WAIT_R1 -- "Enabled and not send" --> IDLE0

    RECEIVE2 -- "The room ready and end of packet or end of data" --> WAIT_FOR_ROOM3
    RECEIVE2 -- "The data need wait for write" --> WAIT_FOR_ROOM3
    RECEIVE2 -- "The room ready" --> WAIT_FOR_ROOM3
    RECEIVE2 -- "Disabled or CRC fail" --> WAIT_R1

    WAIT_FOR_ROOM3 -- "The room ready" --> RECEIVE2
    WAIT_FOR_ROOM3 -- "Disable" --> IDLE0
  
```

IDLE: The data path is inactive and MMC_data outputs HiZ. When the data control register is written and the enable bit is set, the DPSM loads the data counter with a new value and, depending on the data direction bit, moves to either the WAIT-S state or WAIT-R state.

RECEIVE: The DPSM moves to the IDLE state if the DPSM is disabled. If not, serial data received from a card is packed in bytes and written to the data buffer. Depending on the transfer mode bit in the data control register, the data transfer mode can be either block or stream:

In *stream mode*, the DPSM receives data while the data counter is not zero. When the counter is zero, the remaining data in the shift register is written to the data buffer, and the DPSM moves to the WAIT-R state.

During the receiving, if the buffer is full, the DPSM moves to the WAIT_FOR_ROOM state, and waits that the host or DMA to read the data buffer.

WAIT_FOR_ROOM: The DPSM moves to the IDLE state if the DPSM is disabled. If not, the DPSM disables SDCLK, and waits that there is a room in the data buffer. If the data buffer is read by the host or DMA, and the DPSM moves to RECEIVE state.

WAIT_S: The DPSM moves to the IDLE state if the DPSM is disabled, or if the data counter is zero. If not, it waits until the data is ready, and moves to the SEND state. It loads the block length with a new value if a block will be received.

SEND: The DPSM moves to the IDLE state if the DPSM is disabled. If not, the DPSM starts sending data to a card. Depending on the transfer mode bit in the data control register, the data transfer mode can be either block or stream:

In *block mode*, when the data block counter reaches zero, the DPSM sends an internally generated CRC code and end bit, and moves to the BUSY state.

In *stream mode*, the DPSM sends data to a card while the enable bit is HIGH and the data counter is not zero. It then moves to the IDLE state.

If the data is not ready for sending, the DPSM will move to the WAIT_FOR_DATA state, and wait for data.

WAIT_FOR_DATA: The DPSM moves to the IDLE state if the DPSM is disabled. If not, the DPSM disables SDCLK, and waits for data. If data is ready, the DPSM moves to SEND state.

BUSY: The DPSM moves to the IDLE state if the DPSM is disabled. If not, the DPSM waits for the CRC status flag:

- If it does not receive a positive CRC status, it moves to the IDLE state and sets the CRC fail status flag.
- If it receives a positive CRC status, it moves to the WAIT_S state if SDDAT is not LOW (the card is not busy).

If a timeout occurs while the DPSM is in the BUSY state, it sets the data timeout flag and moves to the IDLE state.

The data timer is enabled when the DPSM is in the WAIT_R or BUSY state, and generates the data timeout error:

- When transmitting data, the timeout occurs if the DPSM stays in the BUSY state for longer than the programmed timeout period.
- When receiving data, the timeout occurs if the end of the data is not true, and if the DPSM stays in the WAIT_R state for longer than the programmed timeout period.

DATA COUNTER: The data counter has two functions:

- To stop a data transfer when it reaches zero. This is the end of the data condition.
- To start transferring a pending command. This is used to send the stop command for a stream data transfer.

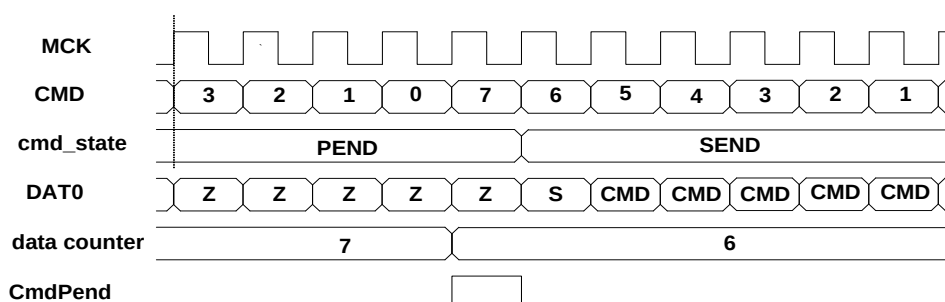


Figure 8-28 Pending Command Start

The data block counter determines the end of a data block. If the counter is zero, the end-of-data condition is TRUE.

Bus mode: In wide bus mode, depending on the data bus mode, four or eight data signals (MMC_data[3:0]) or MMC_data[7:0]) are used to transfer data, and the CRC code is calculated separately for each data signal. While transmitting data blocks to a card, only MMC_data[0] is used for the CRC token and busy signaling. The start bit must be transmitted on all four/eight data signals at the same time (during the same clock period). If the start bit is not detected on all data signals on the same clock edge while receiving data, the DPSM sets the start bit error flag and moves to the IDLE state.

The data path also operates in half-duplex mode, where data is either sent to a card or received from a card. While not being transferred, MMC_data[7:0] are in the HI-Z state. Data on these signals is synchronous to the falling edge of the clock period.

During sending, depending on the data bus mode, the corresponding SDDAT_OE outputs are driven high at the same time. If not, the corresponding MMC data line outputs are always in HI-Z state.

CRC token status: The CRC token status follows each write data block, and determines whether a card has received the data block correctly. When the token has been received, the card asserts a busy signal by driving MMC_data[0] LOW. Table 24-5 shows the CRC token status values.

Table 8-5 CRC Token Status

TOKEN	DESCRIPTION
010	Card has received error-free block.
101	Card has detected a CRC error.

CRC generator: The CRC generator calculates the CRC checksum only for the data bits in a single block, and is bypassed in data stream mode. The checksum is a 16-bit value:

$$\text{CRC}[15:0] = \text{Remainder}[(M(x) * x^{15}) / G(x)]$$

$$G(x) = x^{16} + x^{12} + x^5 + 1$$

$$M(x) = (\text{first data bit}) * x^n + \dots + (\text{last data bit}) * x^0$$

8.6.3.5 Data Packet Format

Communication over the MMC/SD bus is based on command and data bit streams which are initiated by a start bit and terminated by a stop bit. The MMC/SD controller first sends the command to the card and waits response before data transmitting.

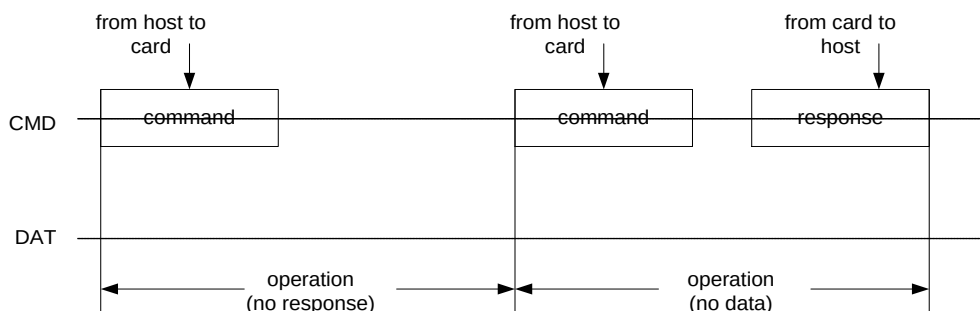


Figure 8-29 “No Response” and “No Data” Operations

Data transmission to/from the MMC/SD card is done in blocks, which are always followed by CRC bits. There are two kinds of block operations: Single Block Operation and Multiple Block Operations. (The Multiple-block operation mode is better for fast write operation.) A multiple block transmission is terminated when a stop command follows on the CMD line.

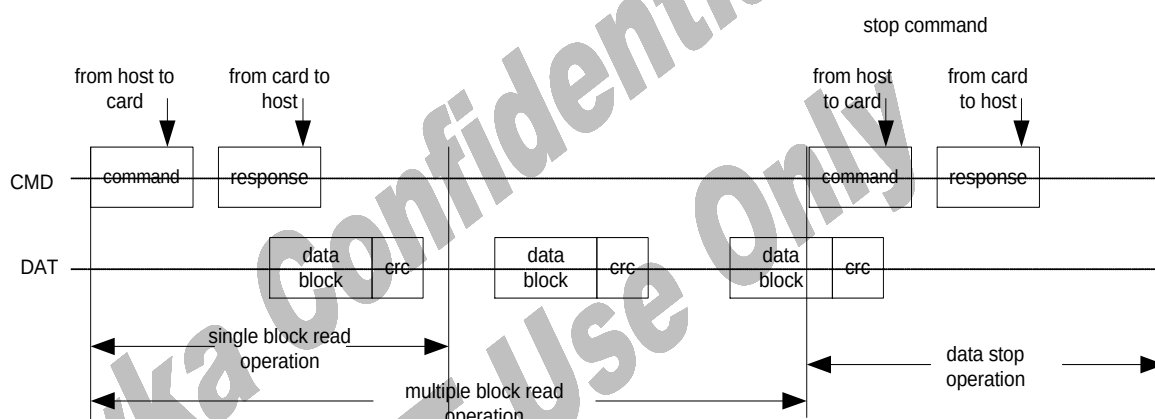


Figure 8-30 Single and Multiple Block Read Operation

The block write operation uses a simple busy signal on the DAT0 data line regardless of the number of data lines used for transferring the data.

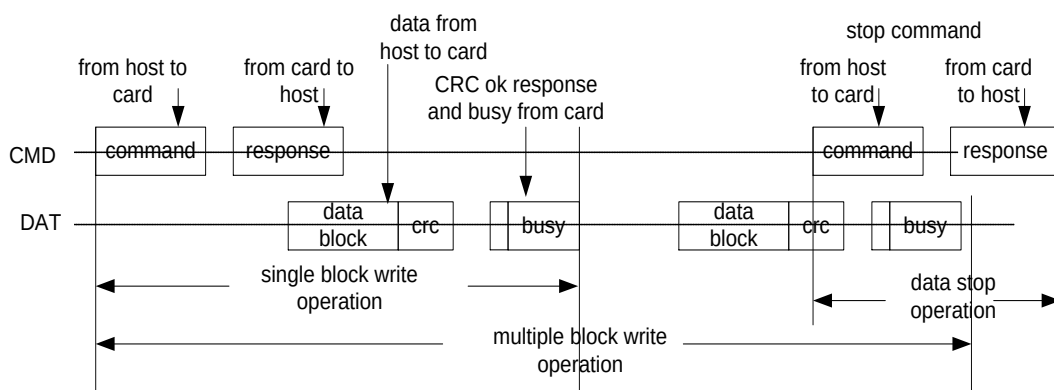


Figure 8-31 Single and Multiple Block Write Operation

Command tokens have the following coding scheme:

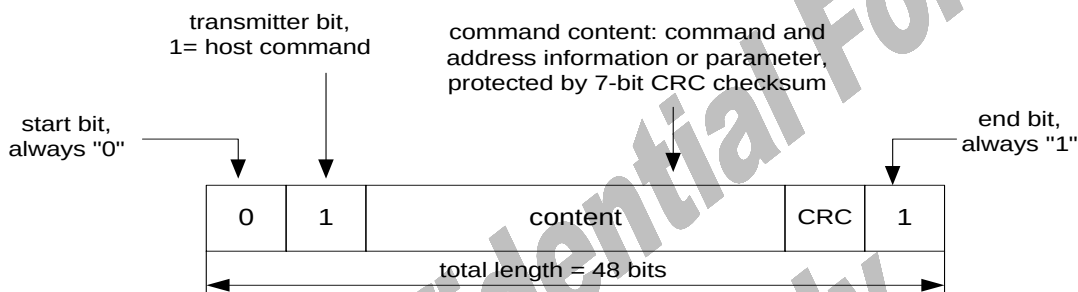


Figure 8-32 Command Token Format

Each command token is preceded by a start bit ("0") and succeeded by an end bit ("1"). The total length is 48 bits. Each token is protected by CRC bits so that transmission errors can be detected and the operation may be repeated.

Response tokens have four coding schemes depending on their content. The token length is either 48 bits or 136 bits. The CRC protection algorithm for block data is a 16-bit CCITT polynomial.

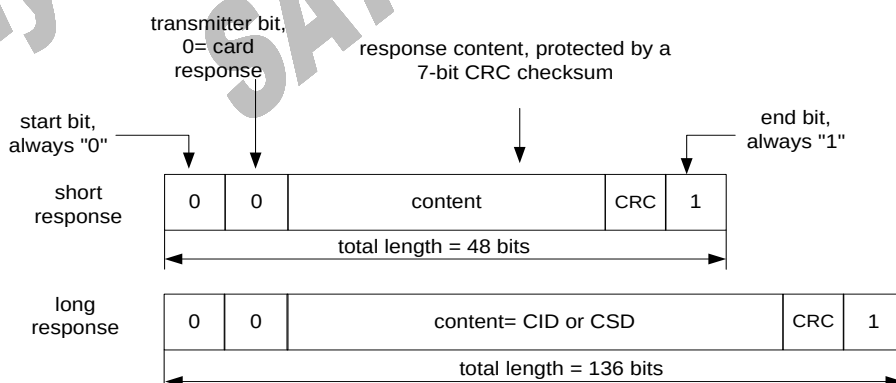


Figure 8-33 Response Token Format

In the CMD line the MSB is transmitted first and the LSB is the last.

When the 4-bit/8-bit option is used, the data is transferred 4 bits/8 bits at a time. Start and end bits, as well as the CRC bits, are transmitted for every data line. And CRC bits are

calculated and checked for every data line respectively. The CRC status response and Busy indication will be sent by the card to the host on data[0] only. (Data[1], data[2], ..., data[7] don't care during that period.)

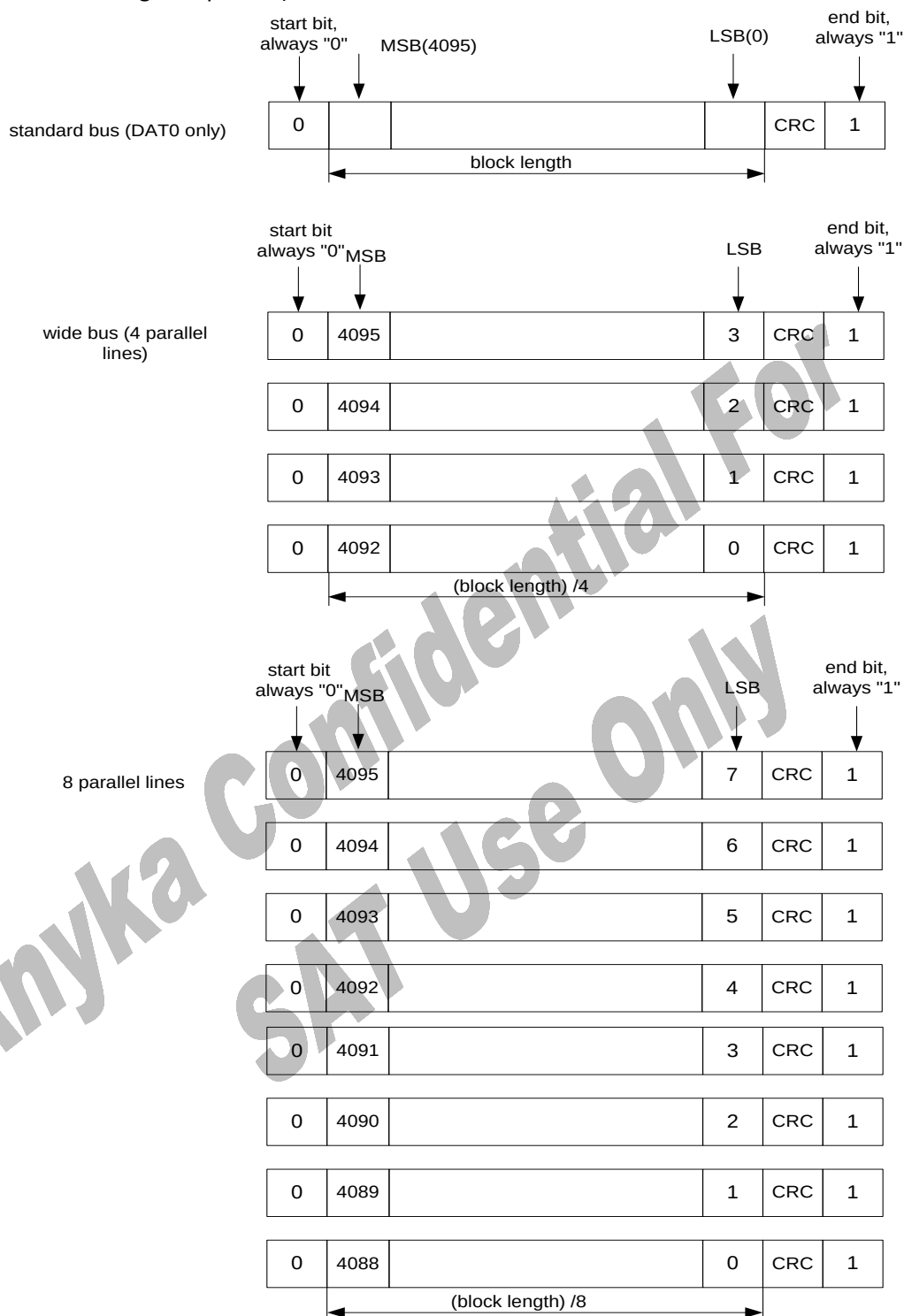


Figure 8-34 Data Packet Format

8.6.3.6 Interface Timing

Timing of MMC/SD/SDIO interface is shown in the following figure.

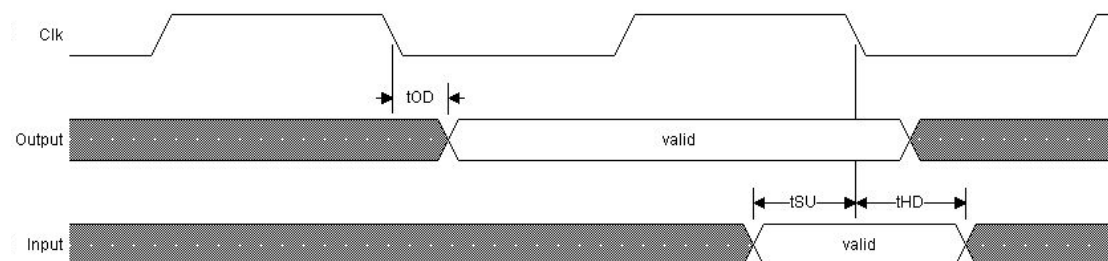


Figure 8-35 Timing of MMC/SD/SDIO

Parameters of MMC/SD/SDIO interface timing are shown in the following table.

Table 8-6 Timing Parameters of MMC/SD/SDIO Interface

Parameter	Symbol	Min	Max	Unit
Clock frequency Data Transfer Mode	Clk	0	62.5	MHz
Outputs CMD,DAT(referenced to CLK)				
Output delay time	tOD	-1	3	ns
Inputs CMD,DAT(referenced to CLK)				
Input set-up time	tSU	10	-	ns
Input hold time	tHD	0	-	ns

8.6.4 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

NOTE

- 1) The MMC/SD/SDIO interfaces have same register definitions.
- 2) Base address of eMMC is 0x2010_0000.
- 3) Base address of SDIO0 is 0x2010_8000.
- 4) Base address of SDIO1 is 0x201B_0000.

Register No.	Register Address	Register Name
1	0x0000	<i>SDIO Interrupt Control Register</i>
2	0x0004	<i>Clock Control Register</i>
3	0x0008	<i>Command Argument Register</i>
4	0x000C	<i>Command Register</i>
5	0x0010	<i>Command Response Register</i>
6	0x0014	<i>Response Register 0</i>
7	0x0018	<i>Response Register 1</i>
8	0x001C	<i>Response Register 2</i>
9	0x0020	<i>Response Register 3</i>
10	0x0024	<i>Data Timer Register</i>
11	0x0028	<i>Data Length Register</i>
12	0x002C	<i>Data Control Register</i>
13	0x0030	<i>Data Counter Register</i>
14	0x0034	<i>Status Register</i>
15	0x0038	<i>Interrupt Enable Register</i>
16	0x003C	<i>DMA Mode Register</i>
17	0x0040	<i>CPU Mode Register</i>
18	0x0800,00EC	<i>SD Plugged-in/Plugged-off Detection Configuration Register</i>
19	0x0800,00F0	<i>SD Plugged-in/Plugged-off Detection Status Register</i>

8.6.5 Register Definition

- SDIO Interrupt Control Register (Address: 0x0000)

Bit [8] of this register enables the MMC/SD controller to check if there is a SDIO interrupt generated or not.

Bits	Attributes	Reset	Description
[31:9]	RO	23'h0	Reserved
[8]	RW	1'h0	sdio_int_check 1: To enable SDIO controller to check SDIO interrupt.
[7:0]	RO	8'h0	Reserved

- Clock Control Register (Address: 0x0004)

This register controls the MCK output.

Programmers should note:

- 1) While the MMC/SD/SDIO controller is in identification mode, the CLK frequency must be less than 400 kHz. Programmers can change the clock frequency to the maximum card bus frequency after the identification mode.
- 2) Only when CPSM and DPSM are idle, this register can be set.

Bits	Attributes	Reset	Description
[31:21]	RO	11'h0	Reserved
[20]	RW	1'h0	mmc_en 0: To disable MMC/SD/SDIO interface. 1: To enable MMC/SD/SDIO interface.

Bits	Attributes	Reset	Description
[19]	RW	1'h0	falling_trigger 0: No instruction 1: To select the falling edge for triggering data out.
[18]	RO	1'h0	Reserved
[17]	RW	1'h0	pwrsave 0: Clock output is always enabled. 1: Clock output is enabled only when the bus is active.
[16]	RW	1'h0	en 0: To disable MMC/SD/SDIO bus clock. 1: To enable MMC/SD/SDIO bus clock.
[15:8]	RW	8'h0	clkdiv_h Higher 8 bits of MMC/SD/SDIO bus clock divider $MCK = \frac{GCLK}{(ClkDiv_H + 1) + (ClkDiv_L + 1)}$
[7:0]	RW	8'h0	clkdiv_l Lower 8 bits of MMC/SD/SDIO bus clock divider

● Command Argument Register (Address: 0x0008)

This register contains a 32-bit command argument that is sent to a card as part of a command message.

Programmers should note:

- 1) If a command contains an argument, it must be loaded into this register before writing a command into the command register.
- 2) Because the command argument is shifted out from the command argument register when the command is sending, it is invalid to read the value of the command argument.
- 3) Only when CPSM is idle, this register can be set. Namely, programmers must set this register before sending a command.

Bits	Attributes	Reset	Description
[31:0]	WO	32'h0	cmdarg Command argument.

● Command Register (Address: 0x000C)

This register contains the command index that is sent to a card as part of a command message, and controls the command types.

Programmers should note:

- 1) Because the command index is shifted out from the command index register when the command is sending, it is invalid to read the value of the command index.
- 2) Only when CPSM is idle, this register can be set.

Bits	Attributes	Reset	Description
[31:15]	RO	17'h0	Reserved
[14]	RW	1'h0	1: To restore SDIO interrupt timing caused by abort.
[13]	RW	1'h0	1: CPSM is cmd12 or io abort (for SDIO interrupt)
[12]	RW	1'h0	rspwaittime 1: CPSM waits 64 MCK cycles for a response. 1: CPSM waits 256 MCK cycles for a response.
[11]	RW	1'h0	cmdwithdata

Bits	Attributes	Reset	Description
			0: The command is not associated with data. 1: The command is associated with data.
[10]	RW	1'h0	rspcrnochk 1: CPSM doesn't check the CRC of the response.
[9]	RW	1'h0	pending 1: CPSM waits for CmdPend before it starts sending a command.
[8]	RW	1'h0	longrsp 0: CPSM does not wait for a long response. 1: CPSM waits for a long response.
[7]	RW	1'h0	rsp 0: CPSM does not wait for a response. 1: CPSM waits for a response.
[6:1]	RW	6'h0	cmdindex Command index.
[0]	RW	1'h0	en 0: To disable CPSM. 1: To enable CPSM.

Table 8-7 shows the response types.

Table 8-7 Command Response Types

Rsp(bit[7])	LongRsp (bit[8])	DESCRIPTION
0	0	No response, expect CmdSent flag
0	1	No response, expect CmdSent flag
1	0	Short response, expect CmdRespEnd and/or CmdCrcFail flag
1	1	Long response, expect CmdRespEnd or CmdCrcFail flag

● Command Response Register (Address: 0x0010)

This register contains the command index field of the last command response received.

Programmers should note:

- 1) If the command response transmission does not contain the command index field (long response), the RespCmdIndex field is unknown, although it must contain 111111 (the value of the Reserved field from the response).
- 2) The value of this register is invalid before the response has been received, and it must be read out before the next writing Command Register (Address: 0x000C).

Bits	Attributes	Reset	Description
[31:6]	RO	26'h0	Reserved
[5:0]	RO	6'h0	respcmd Response command index

Response Register 0, 1, 2 & 3

Response Register 0, 1, 2 & 3 contain the status of a card, which is part of the received response. When it is a short response, only **Response Register 0** is needed. When it is a long response, **Response Register 0** indicates bits[127:96] of card status; **Response Register 1**, bits[95:64] of card status; **Response Register 2**, bits[63:32] of card status; **Response Register 3**, bits[31:0] of card status.

Programmers should note:

- 1) The most significant bit of the card status is received first.
- 2) The LSB of *Response Register 3 (Address: 0x0020)* is always 0.
- 3) *Response Register 0, 1 & 2* must be read out before starting to receive the next response; but *Response Register 3 (Address: 0x0020)* must be read out before the next writing *Command Argument Register (Address: 0x0008)*.

● Response Register 0 (Address: 0x0014)

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	resp1 In short response mode, these bits describe card status. In long response mode, these bits indicate bits[127:96] of card status.

● Response Register 1 (Address: 0x0018)

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	resp2 In long response mode, these bits indicate bits[95:64] of card status. In short response mode, this register is reserved.

● Response Register 2 (Address: 0x001C)

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	resp3 In long response mode, these bits indicate bits[63:32] of card status. In short response mode, this register is reserved.

● Response Register 3 (Address: 0x0020)

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	resp4 In long response mode, these bits indicate bits[31:0] of card status. In short response mode, this register is reserved.

● Data Timer Register (Address: 0x0024)

This register contains the data timeout period, in card bus clock periods. A counter loads the value from this register, and starts decrementing when the DPSM enters the WAIT_R or BUSY state. If the timer reaches 0 while the DPSM is in either of these states, the timeout status flag is set.

Appropriate parameters must be written to the *Data Timer Register (Address: 0x0024)* and *Data Length Register (Address: 0x0028)* before setting the *Data Control Register (Address: 0x002C)*.

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	data time Data timeout period (unit: MCK clock)

● Data Length Register (Address: 0x0028)

This register contains the number of data bytes to be transferred. The value is loaded into the data counter when data transfer begins.

Programmers should note:

- 1) Because this register and [Data Counter Register \(Address: 0x0030\)](#) are the same register in hardware, and it is invalid to read the value of the data length.
- 2) For a block data transfer, the value in this register must be a multiple of the data block size.
- 3) Appropriate parameters must be written to the [Data Timer Register \(Address: 0x0024\)](#) and [Data Length Register \(Address: 0x0028\)](#) before setting the [Data Control Register \(Address: 0x002C\)](#).

Bits	Attributes	Reset	Description
[31:0]	WO	32'h0	Data length value

● Data Control Register (Address: 0x002C)

This register controls the DPSM. Data transfer starts if one is written into the enable bit. Depending on the direction bit, the DPSM moves to the WAIT_S or WAIT_R state. This bit does not need to be cleared after data transfer, and it is disabled automatically by the hardware after the DPSM is invoked.

Bits	Attributes	Reset	Description
[31:28]	RO	4'h0	Reserved
[27:16]	RW	12'h0	block length Data block length (unit: byte)
[15:7]	RO	9'h0	Reserved
[6]	RW	1'h0	abort_en_multiblock 0: Multiblock don't need abort cmd. 1: Multiblock needs abort cmd.
[5]	WO	1'h0	sdio_int_fix 0: To disable 4-wire SDIO interrupt before data transfer 1: To enable 4-wire SDIO interrupt before data transfer NOTE: This bit must be set to 0.
[4:3]	RW	2'h0	busmode 2'h0: One bit 2'h1: Four bits 2'h2: Eight bits 2'h3: Reserved
[2]	RW	1'h0	mode 0: Block data transfer. 1: Stream data transfer.
[1]	RW	1'h0	direction 0: From MMC/SD/SDIO interface to card. 1: From card to MMC/SD/SDIO interface.
[0]	RW	1'h0	en 0: To disable data transfer. 1: To enable data transfer.

- Data Counter Register (Address: 0x0030)

This register loads the value from the [Data Length Register \(Address: 0x0028\)](#) when the DPSM moves from the IDLE state to the WAIT_R state or WAIT_S state. As data is transferred, the counter decrements the value until it reaches zero. The DPSM then moves to the IDLE state and the data status end flag is set.

Bits	Attributes	Reset	Description
[31:0]	RO	32'h0	data counter Remaining number of data.

- Status Register (Address: 0x0034)

This register is a read-only register. It contains two types of flag:

STATIC[8:0]: These bits remain asserted until they are read by the CPU.

DYNAMIC[15:9]: These bits change state depending on the state of the underlying logic (e.g. FIFO full and empty flags are asserted and de-asserted when data is being written to the FIFO).

Bits	Attributes	Reset	Description
[31:18]	RO	14'h0	Reserved
[17]	RO	1'h0	sdioint 1: A SDIO interrupt has been generated. 0: A SDIO interrupt has not been generated.
[16]	RO	1'h0	datatransfinish 1: Data transmission is finished. 0: Data transmission is not finished.
[15]	RO	1'h0	databufhalfempty 1: Data buffer is half empty.
[14]	RO	1'h0	databufhalffull 1: Data buffer is half full.
[13]	RO	1'h0	databufempty 1: Data buffer is empty.
[12]	RO	1'h0	databuffull 1: Data buffer is full.
[11]	RO	1'h0	rxactive 1: It is receiving data.
[10]	RO	1'h0	txactive 1: It is transmitting data.
[9]	RO	1'h0	cmdactive 1: It is transferring command.
[8]	RO	1'h0	datastartbiterr 1: Start bit has not been detected on all data signals in wide bus mode.
[7]	RO	1'h0	datablockend 0: Data block has not been sent or received. 1: Data block has been sent/received (CRC check passed).
[6]	RO	1'h0	dataend 0: Data transfer does not end. 1: Data transfer ends. (Data counter is zero.)

Bits	Attributes	Reset	Description
[5]	RO	1'h0	cmdsent 0: Command has not been sent. 1: Command has been sent. (No response required)
[4]	RO	1'h0	cmdrespond 0: Command response has not been received. 1: Command response has been received. (CRC check has passed.)
[3]	RO	1'h0	datatimeout 0: Data is not timeout. 1: Data is timeout
[2]	RO	1'h0	cmdtimeout 0: Command response is not timeout. 1: Command response is timeout.
[1]	RO	1'h0	datacrcfail 0: Data block has not been sent or received. 1: Data block has been sent/received (CRC check failed).
[0]	RO	1'h0	cmdcrcfail 0: Command response has not been received. 1: Command response has been received (CRC check failed).

● Interrupt Enable Register (Address: 0x0038)

MMC/SD/SDIO interface supports various interrupts, each of which may be enabled/disabled independently by Interrupt Enable Register.

Bits	Attributes	Reset	Description
[31:20]	RO	12'h0	Reserved
[19]	RW	1'h0	ena19 0: To disable SDIO interrupt to ARM. 1: To enable SDIO interrupt to ARM.
[18]	RW	1'h0	ena18 0: To disable SDIO auto-detection interrupt. 1: To enable SDIO auto-detection interrupt. NOTE: The interrupt generated by the SDIO card is called SDIO interrupt.
[17]	RW	1'h0	ena17 0: To disable SDIO interrupt. 1: To enable SDIO interrupt. NOTE: The interrupt generated by the SDIO card is called SDIO interrupt.
[16]	RW	1'h0	ena16 0: To disable DataTransFinish interrupt. 1: To enable DataTransFinish interrupt. NOTE: The interrupt generated when the data transmission is finished is called DataTransFinish interrupt.
[15]	RW	1'h0	ena15

Bits	Attributes	Reset	Description
			0: To disable DataBufHalfEmpty interrupt. 1: To enable DataBufHalfEmpty interrupt. NOTE: The interrupt generated when the data buffer is half empty is called DataBufFull interrupt.
[14]	RW	1'h0	ena14 0: To disable DataBufHalfFull interrupt. 1: To enable DataBufHalfFull interrupt. NOTE: The interrupt generated when the data buffer is half full is called DataBufFull interrupt.
[13]	RW	1'h0	ena13 0: To disable DataBufEmpty interrupt. 1: To enable DataBufEmpty interrupt. NOTE: The interrupt generated when the data buffer is empty is called DataBufFull interrupt.
[12]	RW	1'h0	ena12 0: To disable DataBufFull interrupt. 1: To enable DataBufFull interrupt. NOTE: The interrupt generated when the data buffer is full is called DataBufFull interrupt.
[11]	RW	1'h0	ena11 0: To disable RxActive interrupt. 1: To enable RxActive interrupt. NOTE: The interrupt generated when it is receiving data is called RxActive interrupt.
[10]	RW	1'h0	ena0 10: To disable TxActive interrupt. 1: To enable TxActive interrupt. NOTE: The interrupt generated when it is transmitting data is called TxActive interrupt.
[9]	RW	1'h0	ena9 0: To disable CmdActive interrupt. 1: To enable CmdActive interrupt. NOTE: The interrupt generated when it is transmitting a command is called CmdActive interrupt.
[8]	RW	1'h0	ena8 0: To disable DataStartBitErr interrupt. 1: To enable DataStartBitErr interrupt. NOTE: The interrupt generated when start bit has not been detected on all data signals in wide bus mode is called DataStartBitErr interrupt.
[7]	RW	1'h0	ena7 0: To disable DataBlockEnd interrupt. 1: To enable DataBlockEnd interrupt. NOTE: The interrupt generated when Data block has been sent/received. is called DataBlockEnd interrupt.
[6]	RW	1'h0	ena6 0: To disable DataEnd interrupt. 1: To enable DataEnd interrupt.

Bits	Attributes	Reset	Description
			NOTE: The interrupt generated when Data ends (Data counter is zero) is called DataEnd interrupt.
[5]	RW	1'h0	ena5 0: To disable CmdSent interrupt. 1: To enable CmdSent interrupt. NOTE: The interrupt generated when Command has been sent (no response required) is called CmdSent interrupt.
[4]	RW	1'h0	ena4 0: To disable CmdRespEnd interrupt. 1: To enable CmdRespEnd interrupt. NOTE: The interrupt generated when Command response has been received (CRC check has passed) is called CmdRespEnd interrupt.
[3]	RW	1'h0	ena3 0: To disable DataTimeOut interrupt. 1: To enable DataTimeOut interrupt. NOTE: The interrupt generated when data transmission is timeout is called DataTimeOut interrupt.
[2]	RW	1'h0	ena2 0: To disable CmdTimeOut interrupt. 1: To enable CmdTimeOut interrupt. NOTE: The interrupt generated when command transmission is timeout is called DataTimeOut interrupt.
[1]	RW	1'h0	ena1 0: To disable DataCrcFail interrupt. 1: To enable DataCrcFail interrupt. NOTE: The interrupt generated when Data block has been sent/received (CRC check failed) is called DataCrcFail interrupt.
[0]	RW	1'h0	ena0 0: To disable CmdCrcFail interrupt. 1: To enable CmdCrcFail interrupt. NOTE: The interrupt generated when Command response has been received (CRC check failed) is called CmdCrcFail interrupt.

MMC/SD/SDIO interface can be programmed to be worked in DMA mode or CPU Mode. In DMA mode, the data is first sent to L2 memory, and then L2 memory sends a DMA request to DMA controller; or the data from DMA channel is stored in L2 memory first, and then sent to MMC/SD/SDIO interface; In CPU mode, the data is read by/written from CPU. DMA Mode Register configures relative parameters DMA mode needs; CPU Mode Register holds the data read by/written from CPU.

- DMA Mode Register (Address: 0x003C)

Bits	Attributes	Reset	Description
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Bits	Attributes	Reset	Description
[31:17]	RW	15'h0	buf_size The size of the buffer in L2 memory assigned to MMC/SD/SDIO interface.
[16]	RW	1'h0	dma_en 0: To disable DMA operation. 1: To enable DMA operation.
[15:1]	RW	15'h0	saddr Starting address of the buffer in L2 memory assigned to MMC/SD/SDIO interface. (physical address)
[0]	RW	1'h0	buf_en 0: To disable the buffer in L2 memory assigned to MMC/SD/SDIO interface. 1: To enable the buffer in L2 memory assigned to MMC/SD/SDIO interface.

● CPU Mode Register (Address: 0x0040)

Bits	Attributes	Reset	Description
[31:0]	RW	32'h0	In a reading operation, these bits hold the data from MMC/SD card or SDIO device; In a writing operation, these bits hold the data written by CPU. The MSB is sent first and stored in the lowest 8 bits; the second MSB is sent second and stored in the second lowest 8 bits; ...; the LSB is sent last and stored in the highest 8 bits.

● SD Plugged-in/Plugged-off Detection Configuration Register (Address: 0x0800,00EC)

Bits	Attributes	Reset	Description
[31:16]	RW	16'h00FF	The threshold value (unit: 12MHz clock) This field is used to define the threshold at which a SD plug-in/off triggered interrupt will be generated.
[15:12]	RW	4'h0	Reserved
[11]	RW	1'h0	SD0_DATA3 SD plug-in/off monitoring enable bit 0: Disabled 1: Enabled
[10]	RW	1'h0	SD1_DATA3 SD plug-in/off monitoring enable bit 0: Disabled 1: Enabled
[9]	RW	1'h0	SD2_DATA3 SD plug-in/off monitoring enable bit 0: Disabled 1: Enabled
[8]	RW	1'h0	SD0_MCLK SD plug-in/off monitoring enable bit 0: Disabled 1: Enabled
[7]	RW	1'h0	SD1_MCLK SD plug-in/off monitoring enable bit 0: Disabled 1: Enabled
[6]	RW	1'h0	SD2_MCLK SD plug-in/off monitoring enable bit

Bits	Attributes	Reset	Description
			0: Disabled 1: Enabled
[5]	RW	1'h1	SD0_DATA3 SD plug-in polarity select 0: An SD card is plugged in when the level is low. 1: An SD card is plugged in when the level is high.
[4]	RW	1'h1	SD1_DATA3 SD plug-in polarity select 0: An SD card is plugged in when the level is low. 1: An SD card is plugged in when the level is high.
[3]	RW	1'h1	SD2_DATA3 SD plug-in polarity select 0: An SD card is plugged in when the level is low. 1: An SD card is plugged in when the level is high.
[2]	RW	1'h0	SD0_MCLK SD plug-in polarity select 0: An SD card is plugged in when the level is low. 1: An SD card is plugged in when the level is high.
[1]	RW	1'h0	SD1_MCLK SD plug-in polarity select 0: An SD card is plugged in when the level is low. 1: An SD card is plugged in when the level is high.
[0]	RW	1'h0	SD2_MCLK SD plug-in polarity select 0: An SD card is plugged in when the level is low. 1: An SD card is plugged in when the level is high.

● SD Plugged-in/Plugged-off Detection Status Register (Address: 0x0800,00F0)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15]	RW	1'h0	SD0 plug-in triggered interrupt enable bit 0: Disabled 1: Enabled
[14]	RW	1'h0	SD0 plug-off triggered interrupt enable bit 0: Disabled 1: Enabled
[13]	RW	1'h0	SD1 plug-in triggered interrupt enable bit 0: Disabled 1: Enabled
[12]	RW	1'h0	SD1 plug-off triggered interrupt enable bit 0: Disabled 1: Enabled
[11]	RW	1'h0	SD2 plug-in triggered interrupt enable bit 0: Disabled 1: Enabled
[10]	RW	1'h0	SD2 plug-off triggered interrupt enable bit 0: Disabled 1: Enabled
[9]	RW	1'h0	SD0 plug-in triggered interrupt status 0: No interrupt has been generated. 1: An interrupt has been generated.
[8]	RW	1'h0	SD0 plug-off triggered interrupt status 0: No interrupt has been generated.

Bits	Attributes	Reset	Description
			1: An interrupt has been generated.
[7]	RW	1'h0	SD1 plug-in triggered interrupt status 0: No interrupt has been generated. 1: An interrupt has been generated.
[6]	RW	1'h0	SD1 plug-off triggered interrupt status 0: No interrupt has been generated. 1: An interrupt has been generated.
[5]	RW	1'h0	SD2 plug-in triggered interrupt status 0: No interrupt has been generated. 1: An interrupt has been generated.
[4]	RW	1'h0	SD2 plug-off triggered interrupt status 0: No interrupt has been generated. 1: An interrupt has been generated.
[3]	RO	1'h0	Reserved
[2]	RW	1'h0	SD0_plug-in/off status 0: An SD card is plugged off. 1: An SD card is plugged in.
[1]	RW	1'h0	SD1_plug-in/off status 0: An SD card is plugged off. 1: An SD card is plugged in.
[0]	RW	1'h0	SD2_plug-in/off status 0: An SD card is plugged off. 1: An SD card is plugged in.

8.6.6 Programmer's Guide

8.6.6.1 Configuring MMC/SD/SDIO interface

Programmers should operate according to the following steps to program this module.

- 1) Configure the prescaler value and control signals of MMC/SD/SDIO interface by programming [Clock Control Register \(Address: 0x0004\)](#) accordingly.
- 2) Configure [Interrupt Enable Register \(Address: 0x0038\)](#) accordingly.
- 3) Wait certain MCICLK clock cycle to initialize the card.

8.6.6.2 Configuring Command Path

Programmers should operate according to the following steps to configure the command path.

- 1) Write a 32-bit command argument to [Command Argument Register \(Address: 0x0008\)](#).
- 2) Determine command types and start command transmission by setting [Command Register \(Address: 0x000C\)](#).
- 3) Confirm the end of MCI command path operation when the specific flag of [Status Register \(Address: 0x0034\)](#) is set. There are four possible results:
 - a) The flag is CmdSent if the command type has no response;
 - b) The flag is CmdTimeOut if overtime occurs when receiving response;
 - c) The flag is CmdCrcFail if crc check for response fails;
 - d) The flag is CmdRespEnd if response is received successfully.
- 4) Clear the corresponding flag of MMC/SD/SDIO interface Status by reading [Status Register \(Address: 0x0034\)](#).

8.6.6.3 Configuring Data Path

Programmers should operate according to the following steps to configure the data path.

- 1) If DMA mode of this module is selected, configure bit[0], bits[31:17] and bits[15:1] of *DMA Mode Register (Address: 0x003C)* to set data buffer access mode, buffer size and start address of the data buffer. Otherwise, please jump to step2.
- 2) Program the value of data timeout period by setting *Data Timer Register (Address: 0x0024)*.
- 3) Program the total length of data by setting *Data Length Register (Address: 0x0028)*.
- 4) Configure bit[0], bit[1], bit[2], bits[4:3], bits[27:16] and other bits in *Data Control Register (Address: 0x002C)* to set data transfer direction, transfer mode, bus mode, block length and other parameters.
- 5) There are three ways of receiving/transmitting data between MMC/SD/SDIO interface and the external card.
 - a) Transmitting data
 - When using L2 buffer and CPU mode, CPU writes data to *CPU Mode Register (Address: 0x0040)* while data buffer is empty or half empty.
 - When using L2 buffer and DMA mode, data will be written into *CPU Mode Register (Address: 0x0040)* by DMA automatically while data buffer is full or half full.
 - When using CPU mode without L2 buffer, CPU writes data to *CPU Mode Register (Address: 0x0040)* while data buffer is empty.
 - b) Receiving data
 - When using L2 buffer and CPU mode, CPU reads data from *CPU Mode Register (Address: 0x0040)* while data buffer is full or half full.
 - When using L2 buffer and DMA mode, data will be read from *CPU Mode Register (Address: 0x0040)* by DMA automatically while data buffer is full or half full.
 - When using CPU mode without L2 buffer, CPU reads data from *CPU Mode Register (Address: 0x0040)* while data buffer is full.
- 6) Confirm the end of MCI DATA path operation when the specific flag of *Status Register (Address: 0x0034)* is set. There are five possible results:
 - a) The flag is DataTimeOut if timeout occurs when receiving data;
 - b) The flag is DataStartBitErr if start bit has not been detected on all data signals in 4/8-bit bus mode.;
 - c) The flag is DataCrcFail if data block has been sent/received (CRC check failed);
 - d) The flag is DataBlockEnd if data block has been sent/received (CRC check passed).
 - e) The flag is DataEnd if all data transfer ends.
- 7) Clear the corresponding flag of MMC/SD/SDIO interface Status by reading *Status Register (Address: 0x0034)*.

8.6.6.4 Enabling Card Plugged-in/off Detection Function of DATA3/MCK

Programmers should operate according to the following steps to enable card insertion/extraction detection function of SDx_DATA3/SDx_MCK pin.

- 1) Configure bits[31:16] of *SD Plugged-in/Plugged-off Detection Configuration Register (Address: 0x0800,00EC)* to set the minimum interval between two SD plug-in/off triggered interrupts.
- 2) Configure bits[5:0] in *SD Plugged-in/Plugged-off Detection Configuration Register (Address: 0x0800,00EC)* to set the polarity of SDx_DATA3/SDx_MCK pin.

- 3) Configure bits[11:6] in *SD Plugged-in/Plugged-off Detection Configuration Register (Address: 0x0800,00EC)* to enable card insertion/extraction detection function of SDx_DATA3/SDx_MCK pin. It should be noted that at one time, **only one** of SDx_DATA3 and SDx_MCK pin supports card insertion/extraction detection function.
- 4) Configure the corresponding bits of *SD Plugged-in/Plugged-off Detection Status Register (Address: 0x0800,00F0)* accordingly.
- 5) Wait until the corresponding interrupts have been generated.
- 6) Read the status of the card via reading bits[2:0] of *SD Plugged-in/Plugged-off Detection Status Register (Address: 0x0800,00F0)*.

8.7 Ethernet

8.7.1 Overview

The Ethernet module, including the Ethernet MAC and a stacked Fast Ethernet PHY transceiver, provides an efficient interface between device and network, and it enables the device to transmit and receive data over the Ethernet.

8.7.2 Features

The module has the following features:

- 10/100Mbps full-duplex/half-duplex mode
- Compliant with IEEE 802.3az-2010 (EEE)/10Base-T IEEE 802.3/ 100Base-TX IEEE 802.3u
- Supports Auto-Negotiation/ Auto-MDIX

8.7.3 Interface Timing

8.7.3.1 RMII Timing

Timing of RMII interface is shown in the following figure.

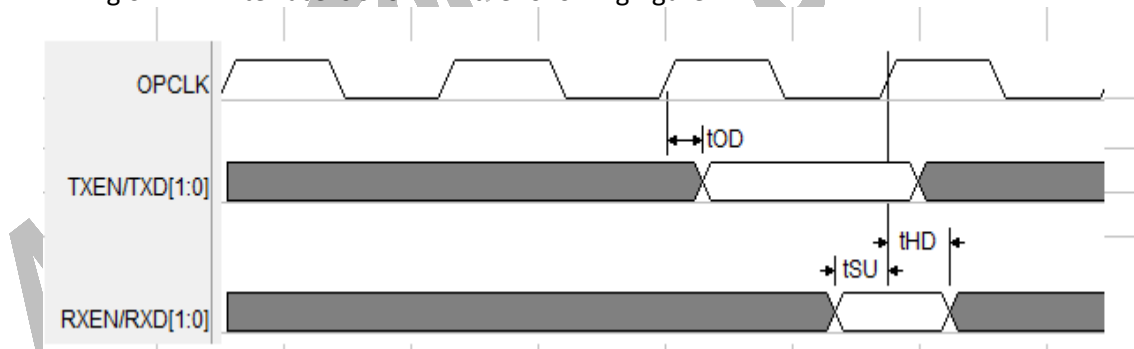


Figure 8-36 Timing of RMII interface

Parameters of RMII interface timing are shown in the following table.

Table 8-8 Timing Parameters of RMII Interface

Parameter	Symbol	Min.	Typ.	Max.	Unit
Output clock	OPCLK	-	50	-	MHz
Output delay time	tOD	4	-	10	ns
Input setup time	tSU	4	-	-	ns
Input hold time	tHD	1	-	-	ns

8.7.3.2 MDIO Timing

Timing of MDIO interface is shown in the following figure.

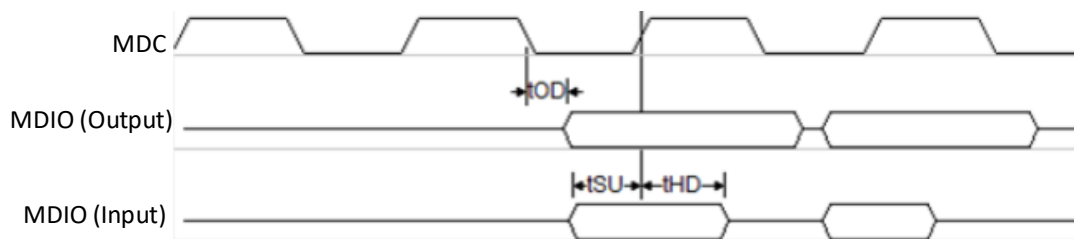


Figure 8-37 Timing of MDIO interface

Parameters of MDIO interface timing are shown in the following table.

Table 8-9 Timing Parameters of MDIO Interface

Parameter	Symbol	Min	Max	Unit
Output clock	MDC	-	-	MHz
Output delay time	tOD	10	12	ns
Input setup time	tSU	12	-	ns
Input hold time	tHD	1	-	ns

8.8 GPIO

8.8.1 Overview

AK3918AV100 series processors contain up to 61 GPIOs (GPIO[0:11], GPIO[16:27], GPIO[31:37], , GPIO[42:52], GPIO[55:56], GPIO[58], GPIO[65:71], GPIO[78:79] and GPIO[98:102]). GPIO37& GPIO50& GPIO98 are dedicated GPIOs and the rest are shared with other pins. For more details, please refer to [Table 2-1](#).

8.8.2 Features

GPIO provides the following features:

- Configurable input or output status
- Supports edge-triggered or level-triggered interrupt
- Programmable edge-triggered interrupt (HIGH or LOW active)
- Programmable level-triggered interrupt on a falling edge or a rising edge or both

8.8.3 Functional Description

8.8.3.1 Interrupt Function

The GPIO module supports interrupt function. Programmer can enable/disable the interrupt function, configure interrupt-triggered types and interrupts polarity, including level-triggered (HIGH or LOW active) and edge-triggered (rising edge or falling edge or double edge) interrupts. For more detailed information, please refer to Section [8.8.6 Programmer's Guide](#).

8.8.3.2 Wakeup Function

The AK3918AV100 series processors contain 45 GPIOs supporting wakeup function. Programmer can enable/disable the wakeup function, configure polarity of wakeup IO ports, check and clear the wakeup status. For more detailed information, please refer to Section [4.7 Working Modes](#).

8.8.4 Register Map

Register map in this section provides high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions. **Programmers can click the italic register name for the detailed description of a register.**

Register No.	Register Address	Register Name
1	0x2017,0000	<i>GPIO Direction Select Register 0</i>
2	0x2017,0004	<i>GPIO Direction Select Register 1</i>
3	0x2017,0008	<i>GPIO Direction Select Register 2</i>
4	0x2017,000C	<i>GPIO Direction Select Register 3</i>
5	0x2017,0014	<i>GPIO Output Data Value Register 0</i>
6	0x2017,0018	<i>GPIO Output Data Value Register 1</i>
7	0x2017,001C	<i>GPIO Output Data Value Register 2</i>
8	0x2017,0020	<i>GPIO Output Data Value Register 3</i>
9	0x2017,0028	<i>GPIO Input Data Value Register 0</i>
10	0x2017,002C	<i>GPIO Input Data Value Register 1</i>
11	0x2017,0030	<i>GPIO Input Data Value Register 2</i>
12	0x2017,0034	<i>GPIO Input Data Value Register 3</i>
13	0x2017,003C	<i>GPIO Interrupt Enable Register 0</i>
14	0x2017,0040	<i>GPIO Interrupt Enable Register 1</i>
15	0x2017,0044	<i>GPIO Interrupt Enable Register 2</i>
16	0x2017,0048	<i>GPIO Interrupt Enable Register 3</i>
17	0x2017,0050	<i>GPIO Interrupt Mode Select Register 0</i>
18	0x2017,0054	<i>GPIO Interrupt Mode Select Register 1</i>
19	0x2017,0058	<i>GPIO Interrupt Mode Select Register 2</i>
20	0x2017,005C	<i>GPIO Interrupt Mode Select Register 3</i>
21	0x2017,0064	<i>GPIO Interrupt Polarity Select Register 0</i>
22	0x2017,0068	<i>GPIO Interrupt Polarity Select Register 1</i>
23	0x2017,006C	<i>GPIO Interrupt Polarity Select Register 2</i>
24	0x2017,0070	<i>GPIO Interrupt Polarity Select Register 3</i>
25	0x2017,0078	<i>GPIO Edge-triggered Interrupt Status Register 0</i>
26	0x2017,007C	<i>GPIO Edge-triggered Interrupt Status Register 1</i>
27	0x2017,0080	<i>GPIO Edge-triggered Interrupt Status Register 2</i>
28	0x2017,0084	<i>GPIO Edge-triggered Interrupt Status Register 3</i>
29	0x2017,008C	<i>GPIO Double-edge-triggered Enable Register 0</i>
30	0x2017,0090	<i>GPIO Double-edge-triggered Enable Register 1</i>
31	0x2017,0094	<i>GPIO Double-edge-triggered Enable Register 2</i>
32	0x2017,0098	<i>GPIO Double-edge-triggered Enable Register 3</i>

8.8.5 Register Definition

GPIO Direction Select Register 0~3 are used to select the direction of GPIO ports. When one bit is set to 1, its corresponding port is in output mode; when one bit is set as 0, its corresponding port is in input mode. GPIO ports are set to input mode by default.

- GPIO Direction Select Register 0 (Address: 0x2017,0000)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	gpio_oe0

Bits	Attributes	Reset	Description
			GPIO[31]
[30:28]	RO	3'h0	Reserved
[27:16]	RW	12'h0	gpio_oe0 GPIO[27:16]
[15:12]	RO	4'h0	Reserved
[11:0]	RW	12'h0	gpio_oe0 GPIO[11:0]

● GPIO Direction Select Register 1 (Address: 0x2017,0004)

Bits	Attributes	Reset	Description
[31:27]	RO	5'h0	Reserved
[26]	RW	1'h0	gpio_oe1 GPIO[58]
[25]	RO	1'h0	Reserved
[24:23]	RW	2'h0	gpio_oe1 GPIO[56:55]
[22:21]	RO	2'h0	Reserved
[20:10]	RW	11'h0	gpio_oe1 GPIO[52:42]
[9:6]	RO	4'h0	Reserved
[5:0]	RW	6'h0	gpio_oe1 GPIO[37:32]

● GPIO Direction Select Register 2 (Address: 0x2017,0008)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15]	RW	1'h0	gpio_oe2 GPIO[79]
[14]	RW	1'h0	gpio_oe2 GPIO[78]
[13:8]	RO	6'h0	Reserved
[7:1]	RW	7'h0	gpio_oe2 GPIO[71:65]
[0]	RO	1'h0	Reserved

● GPIO Direction Select Register 3 (Address: 0x2017,000C)

Bits	Attributes	Reset	Description
[31:7]	RO	25'h0	Reserved
[6:2]	RW	5'h0	gpio_oe3 GPIO[102:98]
[1:0]	RO	2'h0	Reserved

GPIO Output Data Value Register 0~3 hold the real-time output value from GPIO[X].

● GPIO Output Data Value Register 0 (Address: 0x2017,0014)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	gpio_out0

Bits	Attributes	Reset	Description
			GPIO[31]
[30:28]	RO	3'h0	Reserved
[27:16]	RW	12'h0	gpio_out0 GPIO[27:16]
[15:12]	RO	4'h0	Reserved
[11:0]	RW	12'h0	gpio_out0 GPIO[11:0]

● GPIO Output Data Value Register 1 (Address: 0x2017,0018)

Bits	Attributes	Reset	Description
[31:27]	RO	5'h0	Reserved
[26]	RW	1'h0	gpio_out1 GPIO[58]
[25]	RO	1'h0	Reserved
[24:23]	RW	2'h0	gpio_out1 GPIO[56:55]
[22:21]	RO	2'h0	Reserved
[20:10]	RW	11'h0	gpio_out1 GPIO[52:42]
[9:6]	RO	4'h0	Reserved
[5:0]	RW	6'h0	gpio_out1 GPIO[37:32]

● GPIO Output Data Value Register 2 (Address: 0x2017,001C)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15]	RW	1'h0	gpio_out2 GPIO[79]
[14]	RW	1'h0	gpio_out2 GPIO[78]
[13:8]	RO	6'h0	Reserved
[7:1]	RW	7'h0	gpio_out2 GPIO[71:65]

● GPIO Output Data Value Register 3 (Address: 0x2017,0020)

Bits	Attributes	Reset	Description
[31:7]	RO	25'h0	Reserved
[6:2]	RW	5'h0	gpio_out3 GPIO[102:98]
[1:0]	RO	2'h0	Reserved

GPIO Input Data Value Register 0 ~ 3 hold the real-time input value of GPIO[X].

● GPIO Input Data Value Register 0 (Address: 0x2017,0028)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	gpio_in0

Bits	Attributes	Reset	Description
			GPIO[31]
[30:28]	RO	3'h0	Reserved
[27:16]	RW	12'h0	gpio_in0 GPIO[27:16]
[15:12]	RO	4'h0	Reserved
[11:0]	RW	12'h0	gpio_in0 GPIO[11:0]

● GPIO Input Data Value Register 1 (Address: 0x2017,002C)

Bits	Attributes	Reset	Description
[31:27]	RO	5'h0	Reserved
[26]	RW	1'h0	gpio_in1 GPIO[58]
[25]	RO	1'h0	Reserved
[24:23]	RW	2'h0	gpio_in1 GPIO[56:55]
[22:21]	RO	2'h0	Reserved
[20:10]	RW	11'h0	gpio_in1 GPIO[52:42]
[9:6]	RO	4'h0	Reserved
[5:0]	RW	6'h0	gpio_in1 GPIO[37:32]

● GPIO Input Data Value Register 2 (Address: 0x2017,0030)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15]	RW	1'h0	gpio_in2 GPIO[79]
[14]	RW	1'h0	gpio_in2 GPIO[78]
[13:8]	RO	6'h0	Reserved
[7:1]	RW	7'h0	gpio_in2 GPIO[71:65]

● GPIO Input Data Value Register 3 (Address: 0x2017,0034)

Bits	Attributes	Reset	Description
[31:7]	RO	25'h0	Reserved
[6:2]	RW	5'h0	gpio_in3 GPIO[102:98]
[1:0]	RO	2'h0	Reserved

GPIO Interrupt Enable Register 0 ~ 3 are used to enable/disable the interrupt function of GPIO[X]. When one bit is set to 1, the interrupt function of its corresponding GPIO port is enabled; when one bit is set to 0, the interrupt function of corresponding GPIO port is disabled.

● GPIO Interrupt Enable Register 0 (Address: 0x2017,003C)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	gpio_int_en0 GPIO[31]
[30:28]	RO	3'h0	Reserved
[27:16]	RW	12'h0	gpio_int_en0 GPIO[27:16]
[15:12]	RO	4'h0	Reserved
[11:0]	RW	12'h0	gpio_int_en0 GPIO[11:0]

● GPIO Interrupt Enable Register 1 (Address: 0x2017,0040)

Bits	Attributes	Reset	Description
[31:27]	RO	5'h0	Reserved
[26]	RW	1'h0	gpio_int_en1 GPIO[58]
[25]	RO	1'h0	Reserved
[24:23]	RW	2'h0	gpio_int_en1 GPIO[56:55]
[22:21]	RO	2'h0	Reserved
[20:10]	RW	11'h0	gpio_int_en1 GPIO[52:42]
[9:6]	RO	4'h0	Reserved
[5:0]	RW	6'h0	gpio_int_en1 GPIO[37:32]

● GPIO Interrupt Enable Register 2 (Address: 0x2017,0044)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15]	RW	1'h0	gpio_int_en2 GPIO[79]
[14]	RW	1'h0	gpio_int_en2 GPIO[78]
[13:8]	RO	6'h0	Reserved
[7:1]	RW	7'h0	gpio_int_en2 GPIO[71:65]

● GPIO Interrupt Enable Register 3 (Address: 0x2017,0048)

Bits	Attributes	Reset	Description
[31:7]	RO	25'h0	Reserved
[6:2]	RW	5'h0	gpio_int_en3 GPIO[102:98]
[1:0]	RO	2'h0	Reserved

GPIO Interrupt Mode Select Register 0 ~ 3 are used to select the interrupt mode of GPIO[X]. When one bit is set to 0, the interrupt of its corresponding GPIO port is level-triggered; when one bit is set to 1, the interrupt of its corresponding GPIO port is edge-triggered.

● GPIO Interrupt Mode Select Register 0 (Address: 0x2017,0050)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	gpio_int_mode0 GPIO[31]
[30:28]	RO	3'h0	Reserved
[27:16]	RW	12'h0	gpio_int_mode0 GPIO[27:16]
[15:12]	RO	4'h0	Reserved
[11:0]	RW	12'h0	gpio_int_mode0 GPIO[11:0]

● GPIO Interrupt Mode Select Register 1 (Address: 0x2017,0054)

Bits	Attributes	Reset	Description
[31:27]	RO	5'h0	Reserved
[26]	RW	1'h0	gpio_int_mode1 GPIO[58]
[25]	RO	1'h0	Reserved
[24:23]	RW	2'h0	gpio_int_mode1 GPIO[56:55]
[22:21]	RO	2'h0	Reserved
[20:10]	RW	11'h0	gpio_int_mode1 GPIO[52:42]
[9:6]	RO	4'h0	Reserved
[5:0]	RW	6'h0	gpio_int_mode1 GPIO[37:32]

● GPIO Interrupt Mode Select Register 2 (Address: 0x2017,0058)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15]	RW	1'h0	gpio_int_mode2 GPIO[79]
[14]	RW	1'h0	gpio_int_mode2 GPIO[78]
[13:8]	RO	6'h0	Reserved
[7:1]	RW	7'h0	gpio_int_mode2 GPIO[71:65]

● GPIO Interrupt Mode Select Register 3 (Address: 0x2017,005C)

Bits	Attributes	Reset	Description
[31:7]	RO	25'h0	Reserved
[6:2]	RW	5'h0	gpio_int_mode3 GPIO[102:98]
[1:0]	RO	2'h0	Reserved

GPIO Interrupt Polarity Select Register 0 ~ 3 are used to configure the interrupt input polarity of GPIO[X]. For level-triggered mode, when one bit is set to 0, the interrupt input polarity of its corresponding GPIO port is active high; when one bit is set to 1, the interrupt input

polarity of its corresponding GPIO port is active low. For edge-triggered mode, when one bit is set to 0, the interrupt of its corresponding GPIO port is triggered at the rising edge; when one bit is set to 1, the interrupt of its corresponding GPIO port is triggered at the falling edge.

● GPIO Interrupt Polarity Select Register 0 (Address: 0x2017,0064)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	gpio_pol_sel0 GPIO[31]
[30:28]	RO	3'h0	Reserved
[27:16]	RW	12'h0	gpio_pol_sel0 GPIO[27:16]
[15:12]	RO	4'h0	Reserved
[11:0]	RW	12'h0	gpio_pol_sel0 GPIO[11:0]

● GPIO Interrupt Polarity Select Register 1 (Address: 0x2017,0068)

Bits	Attributes	Reset	Description
[31:27]	RO	5'h0	Reserved
[26]	RW	1'h0	gpio_pol_sel1 GPIO[58]
[25]	RO	1'h0	Reserved
[24:23]	RW	2'h0	gpio_pol_sel1 GPIO[56:55]
[22:21]	RO	2'h0	Reserved
[20:10]	RW	11'h0	gpio_pol_sel1 GPIO[52:42]
[9:6]	RO	4'h0	Reserved
[5:0]	RW	6'h0	gpio_pol_sel1 GPIO[37:32]

● GPIO Interrupt Polarity Select Register 2 (Address: 0x2017,006C)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15]	RW	1'h0	gpio_pol_sel2 GPIO[79]
[14]	RW	1'h0	gpio_pol_sel2 GPIO[78]
[13:8]	RO	6'h0	Reserved
[7:1]	RW	7'h0	gpio_pol_sel2 GPIO[71:65]

● GPIO Interrupt Polarity Select Register 3 (Address: 0x2017,0070)

Bits	Attributes	Reset	Description
[31:7]	RO	25'h0	Reserved
[6:2]	RW	5'h0	gpio_pol_sel3 GPIO[102:98]
[1:0]	RO	2'h0	Reserved

GPIO Edge-triggered Interrupt Status Register 0 ~ 3 are used to display/clear the edge-triggered interrupt status of GPIO[X]. When one bit is set to 0, the edge-triggered interrupt of its corresponding GPIO port is not generated; when one bit is set to 1, the edge-triggered interrupt of its corresponding GPIO port has been generated. The edge-triggered interrupt status of GPIO[X] is cleared by reading.

● GPIO Edge-triggered Interrupt Status Register 0 (Address: 0x2017,0078)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	gpio_eint_sta0 GPIO[31]
[30:28]	RO	3'h0	Reserved
[27:16]	RW	12'h0	gpio_eint_sta0 GPIO[27:16]
[15:12]	RO	4'h0	Reserved
[11:0]	RW	12'h0	gpio_eint_sta0 GPIO[11:0]

● GPIO Edge-triggered Interrupt Status Register 1 (Address: 0x2017,007C)

Bits	Attributes	Reset	Description
[31:27]	RO	5'h0	Reserved
[26]	RW	1'h0	gpio_eint_sta1 GPIO[58]
[25]	RO	1'h0	Reserved
[24:23]	RW	2'h0	gpio_eint_sta1 GPIO[56:55]
[22:21]	RO	2'h0	Reserved
[20:10]	RW	11'h0	gpio_eint_sta1 GPIO[52:42]
[9:6]	RO	4'h0	Reserved
[5:0]	RW	6'h0	gpio_eint_sta1 GPIO[37:32]

● GPIO Edge-triggered Interrupt Status Register 2 (Address: 0x2017,0080)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15]	RW	1'h0	gpio_eint_sta2 GPIO[79]
[14]	RW	1'h0	gpio_eint_sta2 GPIO[78]
[13:8]	RO	6'h0	Reserved
[7:1]	RW	7'h0	gpio_eint_sta2 GPIO[71:65]

● GPIO Edge-triggered Interrupt Status Register 3 (Address: 0x2017,0084)

Bits	Attributes	Reset	Description
[31:7]	RO	25'h0	Reserved
[6:2]	RW	5'h0	gpio_eint_sta3 GPIO[102:98]

Bits	Attributes	Reset	Description
[1:0]	RO	2'h0	Reserved

GPIO Double-edge-triggered Enable Register 0 ~ 3 are used to enable/disable double-edge-triggered function of GPIO[X]. When one bit is set to 0, single edge-triggered function of its corresponding GPIO port is enabled; when one bit is set to 1, double edge-triggered function of its corresponding GPIO port is enabled.

- GPIO Double-edge-triggered Enable Register 0 (Address: 0x2017,008C)

Bits	Attributes	Reset	Description
[31]	RW	1'h0	gpio_dedge_en0 GPIO[31]
[30:28]	RO	3'h0	Reserved
[27:16]	RW	12'h0	gpio_dedge_en0 GPIO[27:16]
[15:12]	RO	4'h0	Reserved
[11:0]	RW	12'h0	gpio_dedge_en0 GPIO[11:0]

- GPIO Double-edge-triggered Enable Register 1 (Address: 0x2017,0090)

Bits	Attributes	Reset	Description
[31:27]	RO	5'h0	Reserved
[26]	RW	1'h0	gpio_dedge_en1 GPIO[58]
[25]	RO	1'h0	Reserved
[24:23]	RW	2'h0	gpio_dedge_en1 GPIO[56:55]
[22:21]	RO	2'h0	Reserved
[20:10]	RW	11'h0	gpio_dedge_en1 GPIO[52:42]
[9:6]	RO	4'h0	Reserved
[5:0]	RW	6'h0	gpio_dedge_en1 GPIO[37:32]

- GPIO Double-edge-triggered Enable Register 2 (Address: 0x2017,0094)

Bits	Attributes	Reset	Description
[31:16]	RO	16'h0	Reserved
[15]	RW	1'h0	gpio_dedge_en2 GPIO[79]
[14]	RW	1'h0	gpio_dedge_en2 GPIO[78]
[13:8]	RO	6'h0	Reserved
[7:1]	RW	7'h0	gpio_dedge_en2 GPIO[71:65]

● GPIO Double-edge-triggered Enable Register 3(Address: 0x2017,0098)

Bits	Attributes	Reset	Description
[31:7]	RO	25'h0	Reserved
[6:2]	RW	5'h0	gpio_dedge_en3 GPIO[102:98]
[1:0]	RO	2'h0	Reserved

8.8.6 Programmer's Guide

GPIO[X] of AK3918AV100 series processors have four working modes: output mode, input & level-triggered interrupt mode, input & single edge-triggered interrupt mode and input & double edge-triggered interrupt mode. Table 8-10 GPIO Interrupt Type Setting Table summarizes the configuration information of interrupt modes of GPIO ports and more detailed information can be found in Chapter 8.8.6.1~ Chapter 8.8.6.4 .

Table 8-10 GPIO Interrupt Type Setting Table

Interrupt Type	gpio_int_mode	gpio_pol_sel	gpio_dedge_en
Rising edge-triggered	1	0	0
Falling edge-triggered	1	1	0
Double edge-triggered	1	-	1
Level-triggered, asserted HIGH	0	0	-
Level-triggered, asserted LOW	0	1	-

NOTE

- 1) The “ - ” sign means “ don't care ”.
- 2) Only the corresponding bits of GPIO[X] in [GPIO Direction Select Register X](#) are set to 0 and [GPIO Interrupt Enable Register X](#) are set to 1, can the GPIO interrupt types be configured.

8.8.6.1 Setting GPIO[X] to Output Mode

Programmers should operate according to the following steps.

- 1) Configure the corresponding bits of GPIO[X] in [GPIO Direction Select Register x](#) (Address: 0x2017,0000 ~ 0x2017,000C) as 1.
- 2) Configure the corresponding bits of GPIO[X] in [GPIO Output Data Value Register x](#) (Address: 0x2017,0014 ~ 0x2017,0020) as 0/1 accordingly.

8.8.6.2 Setting GPIO[X] to Input & Level-triggered Interrupt Mode

Programmers should operate according to the following steps.

- 1) Configure the corresponding bits of GPIO[X] in [GPIO Direction Select Register X](#) (Address: 0x2017,0000 ~ 0x2017,000C) as 0 to set the input mode.
- 2) Configure the corresponding bits of GPIO[X] in [GPIO Interrupt Mode Select Register x](#) (Address: 0x2017,0050 ~ 0x2017,005C) as 0 to select level-triggered interrupt mode of the corresponding GPIO ports.
- 3) Configure the corresponding bits of GPIO[X] in [GPIO Interrupt Polarity Select Register x](#) (Address: 0x2017,0064 ~ 0x2017,0070) as 0/1 to select high/low level-triggered interrupt mode of the corresponding GPIO ports.
- 4) Configure the corresponding bits of GPIO[X] in [GPIO Interrupt Enable Register x](#) (Address: 0x2017,003C ~ 0x2017,0048) as 1 to enable the interrupt function of the corresponding GPIO ports. After a level-triggered interrupt has been generated, read the value of the

corresponding bits in *GPIO Input Data Value Register* x (Address: 0x2017,0028~0x2017,0034) to decide which GPIO port has triggered an interrupt.

8.8.6.3 Setting GPIO[X] to Input & Single Edge-triggered Interrupt Mode

Programmers should operate according to the following steps.

- 1) Configure the corresponding bits of GPIO[X] in *GPIO Direction Select Register* X (Address: 0x2017,0000 ~ 0x2017,000C) as 0 to set the input mode.
- 2) Configure the corresponding bits of GPIO[X] in *GPIO Interrupt Mode Select Register* x (Address: 0x2017,0050 ~ 0x2017,005C) as 1 to select edge-triggered interrupt mode of the corresponding GPIO ports.
- 3) Set the corresponding bits of GPIO[X] in *GPIO Double-edge-triggered Enable Register* x (Address: 0x2017,008C ~ 0x2017,0098) as 0 to enable single edge-triggered function of GPIO[X].
- 4) Configure the corresponding bits of GPIO[X] in *GPIO Interrupt Polarity Select Register* x (Address: 0x2017,0064 ~ 0x2017,0070) as 0/1 to select rising/falling edge-triggered mode of the corresponding GPIO ports.
- 5) Configure the corresponding bits of GPIO[X] in *GPIO Interrupt Enable Register* x (Address: 0x2017,003C ~ 0x2017,0048) as 1 to enable the interrupt function of the corresponding GPIO ports. After a single edge-triggered interrupt has been generated, read the corresponding bits in *GPIO Edge-triggered Interrupt Status Register* x (Address: 0x2017,0078 ~ 0x2017,0084) to decide which GPIO port has triggered an interrupt.

8.8.6.4 Setting GPIO[X] to Input & Double Edge-triggered Interrupt Mode

Programmers should operate according to the following steps.

- 1) Configure the corresponding bits of GPIO[X] in *GPIO Direction Select Register* X (Address: 0x2017,0000 ~ 0x2017,000C) as 0 to set the input mode.
- 2) Configure the corresponding bits of GPIO[X] in *GPIO Interrupt Mode Select Register* x (Address: 0x2017,0050 ~ 0x2017,005C) as 1 to select edge-triggered interrupt mode of the corresponding GPIO ports.
- 3) Set the corresponding bits of GPIO[X] in *GPIO Double-edge-triggered Enable Register* x (Address: 0x2017,008C ~ 0x2017,0098) as 1 to enable double edge-triggered function of GPIO[X].
- 4) Configure the corresponding bits of GPIO[X] in *GPIO Interrupt Enable Register* x (Address: 0x2017,003C ~ 0x2017,0048) as 1 to enable the interrupt function of the corresponding GPIO ports. After a double edge-triggered interrupt has been generated, read the corresponding bits in *GPIO Edge-triggered Interrupt Status Register* x (Address: 0x2017,0078 ~ 0x2017,0084) to decide which GPIO port has triggered an interrupt.